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Modlin et al.

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(54) **CHEMICAL COMPOSITIONS AND METHODS OF USE**

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(51) **Int. Cl.**

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(58) **Field of Classification Search**

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See application file for complete search history.

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(57) **ABSTRACT**

The present invention is directed to methods for detecting a melanoma, methods for determining whether a melanoma is stable or progressive, methods for evaluating the extent of surgery resection in a subject having a melanoma, and methods for determining a response by a subject having a melanoma to a therapy.

17 Claims, 6 Drawing Sheets

Specification includes a Sequence Listing.

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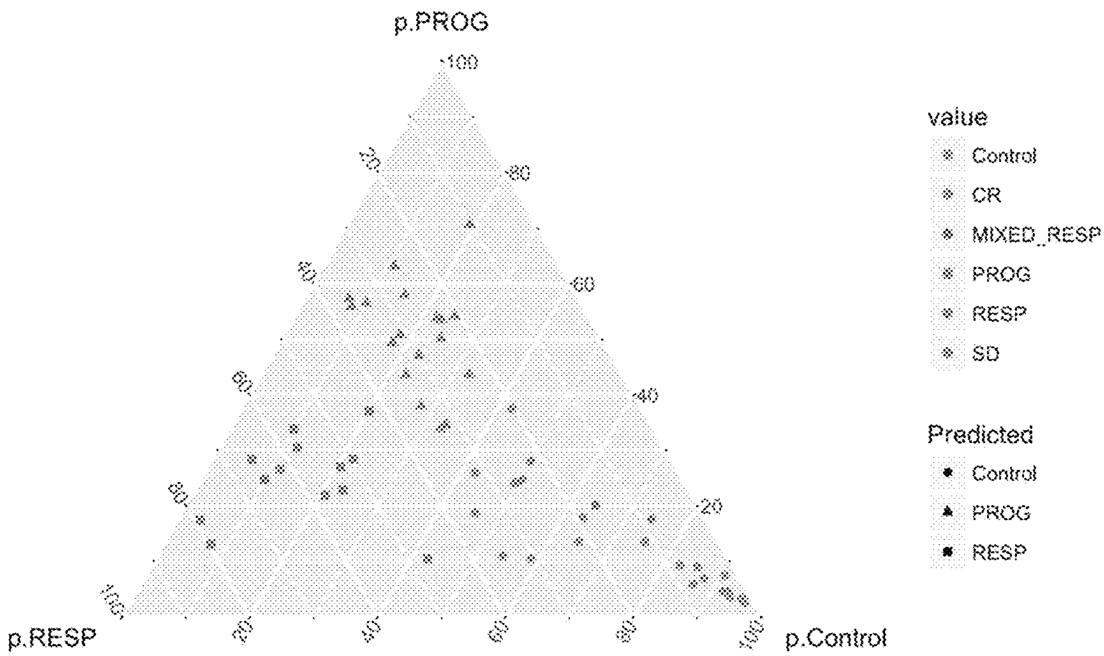

p.PROG
100
20
80
40
60
60
40
80
20
100
p.RESP
20
40
60
80
100
p.Control
value
Control
CR
MIXED_RESP
PROG
RESP
SD
Predicted
Control
PROG
RESP

FIG. 1

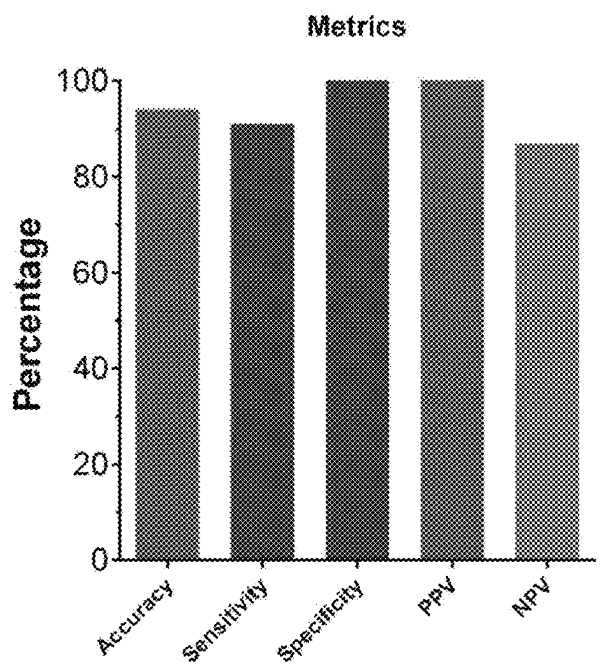
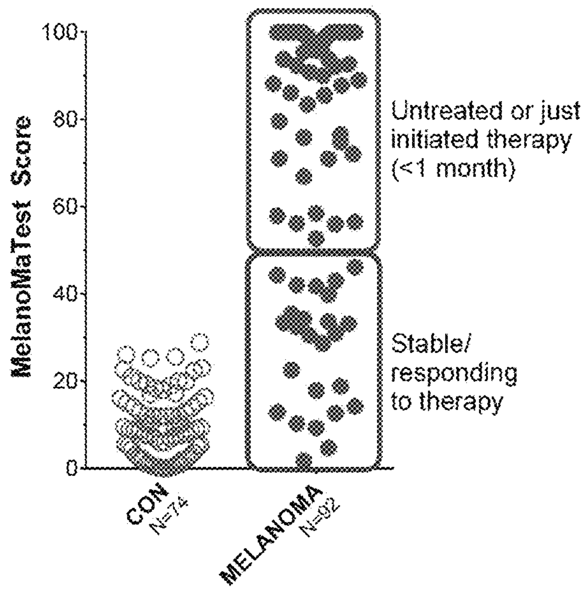
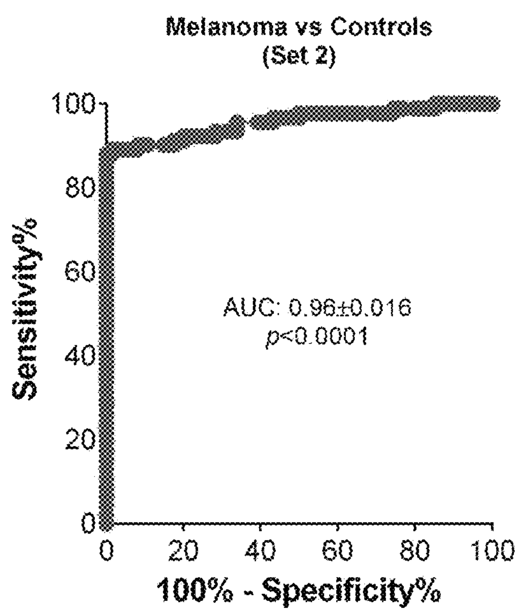

Metrics
Percentage
100
80
60
40
20
0
Accuracy
Sensitivity
Specificity
PPV
NPV

FIG. 2



MelanoMaTest Score
100
80
60
40
20
0
Untreated or just initiated therapy (<1 month)
Stable/ responding to therapy
CON N=74
MELANOMA N=92

FIG. 3A



Melanoma vs Controls (Set 2)
Sensitivity%
100
80
60
40
20
0
AUC: 0.96±0.016
p<0.0001
0
20
40
60
80
100
100% - Specificity%

FIG. 3B

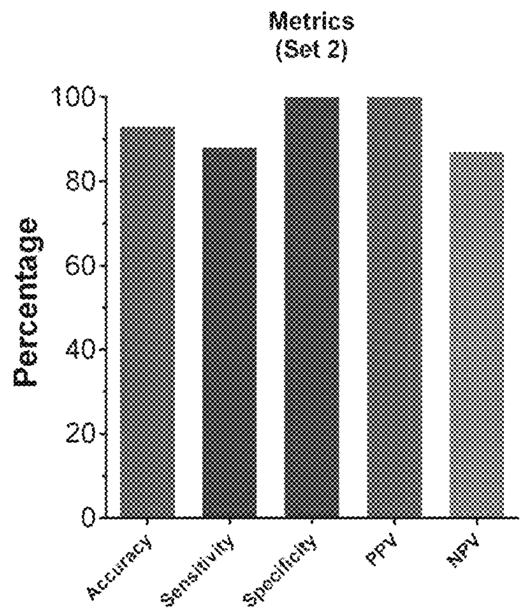

Metrics
(Set 2)
Percentage
100
80
60
40
20
0
Accuracy
Sensitivity
Specificity
PPV
NPV

FIG. 3C

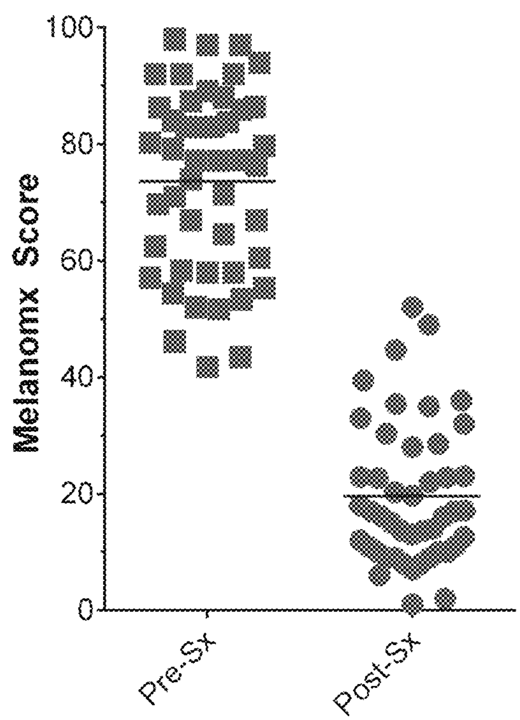

Melanomx Score
100
80
60
40
20
0
Pre-Sx
Post-Sx

FIG. 4A

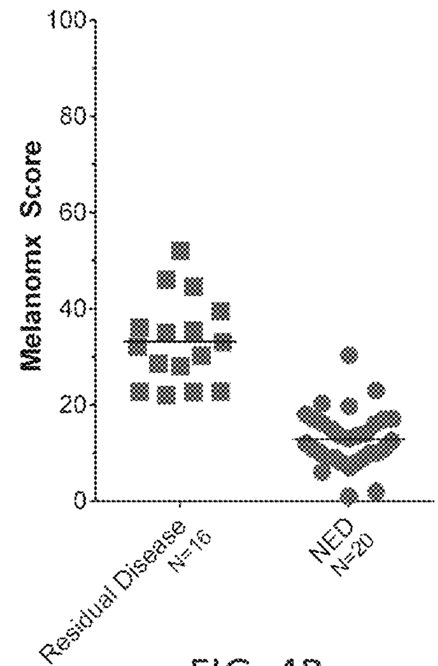

100
80
60
40
20
0
Melanomx Score
Residual Disease
N=16
NED
N=20

FIG. 4B

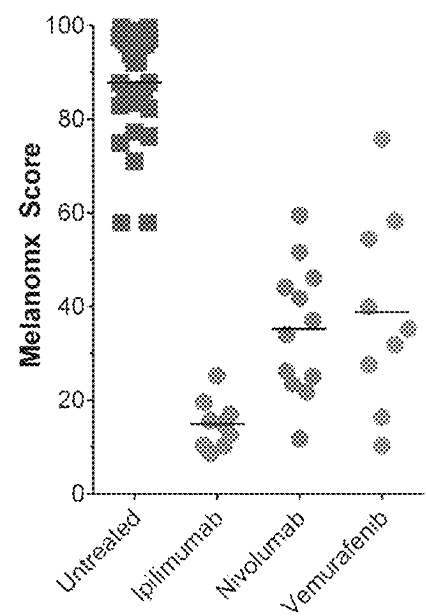
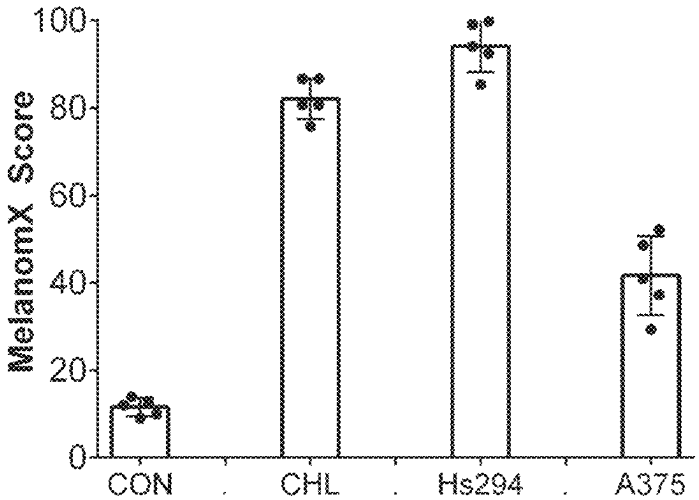
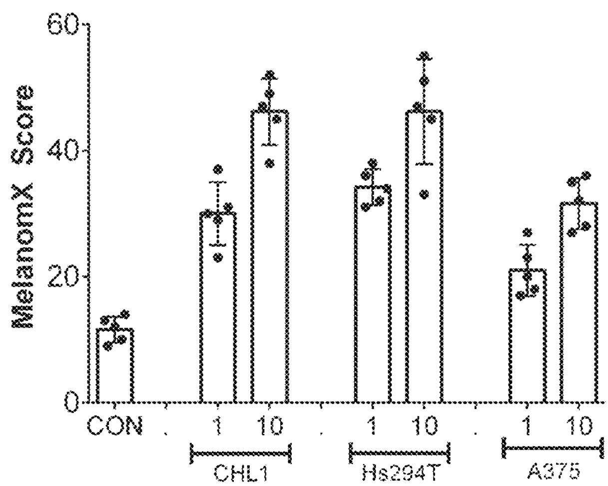

100
80
60
40
20
0
Melanomx Score
Untreated
Ipilimumab
Nivolumab
Vemurafenib

FIG. 5



100
80
60
40
20
0
MelanomX Score
CON
CHL
Hs294
A375

FIG. 6A



60
40
20
0
MelanomX Score
CON
1
10
1
10
1
10
CHL1
Hs294T
A375

FIG. 6B

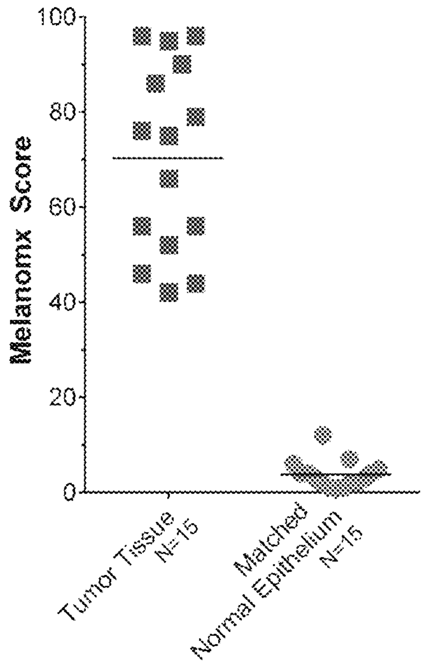

100
80
60
40
20
0
Melanomx Score
Tumor Tissue
N=15
Matched
Normal Epithelium
N=15

FIG. 7A

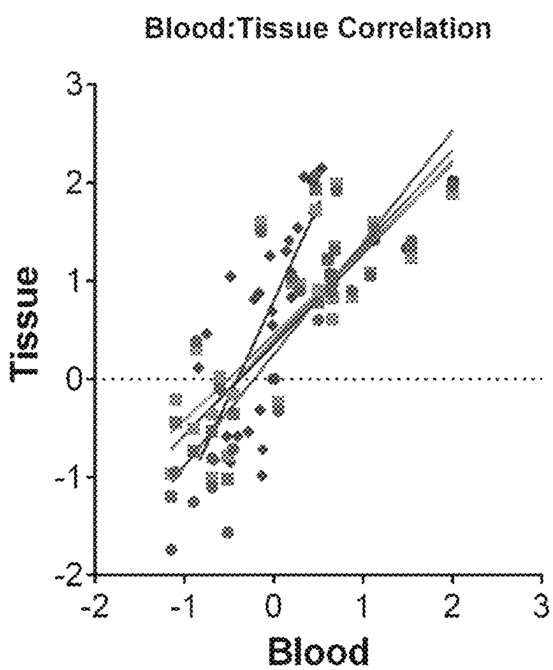

Blood:Tissue Correlation
3
2
1
0
-1
-2
Tissue
-2
-1
0
1
2
3
Blood

FIG. 7B

CHEMICAL COMPOSITIONS AND METHODS OF USE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 15/984,935, filed on May 21, 2018, which claims the benefit of and priority to U.S. Provisional Application No. 62/511,058, filed on May 25, 2017. The contents of each of the aforementioned patent application are hereby incorporated by reference in their entireties for all purposes.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted in ASCII format via EFS-Web and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Sep. 28, 2021, is named "LBIO-001_C01US_SeqList.txt" and is about 272,331 bytes in size.

FIELD OF THE INVENTION

The present invention relates to melanoma detection.

BACKGROUND OF THE INVENTION

Melanoma is a common (~24-35/100,000 incidence—US), highly aggressive, skin cancer with an incidence that continues to rise. The most common, cutaneous melanomas, are associated with UV exposure and immune dysregulation. As a group, melanoma is known to carry the highest mutational burden (>10 mutations/Mb). Major mutations include BRAF (~50%), N-Ras (~20%) and NF-1 (~5%), which together, comprise 75% of all mutations. Melanomas are addicted to MAPK pathway activation, regardless of whether tumors exhibit mutations in genes coding for proteins in this pathway. This provides the rationale for targeted therapy e.g., BRAF v600E agents, in this tumor. Other gain-of-function and loss-of-function mutations e.g., in RASopathy genes and amplification of cyclin D1/cdk4 and/or mutation/loss of the tumor suppressor PTEN, also characterize the tumor. This makes melanoma one of the most aggressive and therapy-resistant cancers.

Five-year survival rates range from 95-100% for stage I, 65-93% for stage II, to 41-71% and 9-28% for stage III and IV, respectively. Surgery, immunotherapy and targeted therapies provide the basis for management, with chemotherapy and radiation as adjuncts. Surgery, however, has a critical role in melanoma care (diagnosis, cure and palliation). Sentinel lymph node biopsy has become widespread, as it provides prognostic information. Melanoma, however, lacks a clinically useful non-invasive e.g., blood-based biomarker of disease activity to help guide patient management by providing predictive or prognostic information.

Blood-based factors include lactate dehydrogenase (LDH), detecting mutations in circulating tumor (ct) DNA, measurements of circulating tumor cells (CTCs) and circulating mRNA. LDH is typically used to identify aggressive tumor behavior and predict recurrence but its metrics are very low e.g., 30-50% accurate. It is also non-specific for melanoma. Mutations in target genes, like BRAF, can be detected in the blood in ctDNA but its utility as an indicator of therapeutic efficacy is limited e.g., 45-70% accurate. CTCs do not appear to be an accurate marker in melanomas and there is no consensus as to their clinical utility. Circulating microNAs (miRNA) have been detected but there is no evidence yet for clinical usefulness.

For melanomas, there are no multi-mRNA circulating biomarkers that function as a diagnostic or as a prognostic for disease recurrence. A biomarker that can be used to monitor the efficacy of surgery or drug therapy in melanomas is currently lacking.

SUMMARY OF THE INVENTION

Among other things, disclosed herein is a 28-gene expression tool for melanoma. It has high sensitivity and specificity (>95%) for the detection of melanoma and can differentiate aggressive untreated disease from stable, treated disease.

One aspect of the present disclosure relates to a method for detecting a melanoma in a subject in need thereof, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a first predetermined cutoff value; and (5) producing a report, wherein the report identifies the presence of a melanoma in the subject when the score is equal to or greater than the first predetermined cutoff value or identifies the absence of a melanoma in the subject when the score is below the first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100.

In some embodiments, the method further comprises treating the subject identified as having a melanoma with surgery or drug therapy.

Another aspect of the present disclosure relates to a method for determining whether a melanoma in a subject is stable or progressive, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA,

HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a second predetermined cutoff value; and (5) producing a report, wherein the report identifies that the melanoma is progressive when the normalized expression level is equal to or greater than the second predetermined cutoff value or identifies that the melanoma is stable when the normalized expression level is below the second predetermined cutoff value, wherein the second predetermined cutoff value is 50 on a scale of 0 to 100.

Another aspect of the present disclosure relates to a method for evaluating the extent of surgical resection in a subject having a melanoma, comprising: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject after the surgical resection by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a third predetermined cutoff value; and (5) producing a report, wherein the report identifies that the surgical resection does not remove the entire melanoma when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the surgical resection removes the entire melanoma when the normalized expression level is below the third predetermined cutoff value, wherein the third predetermined cutoff value is 20 on a scale of 0 to 100.

In some embodiments, the report further identifies that the risk of melanoma recurrence is high when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the risk of melanoma recurrence is low when the normalized expression level is below the third predetermined cutoff value.

Yet another aspect of the present disclosure relates to a method for determining a response by a subject having a melanoma to a therapy, comprising: (1) determining a first expression level of at least 28 biomarkers from a first test sample from the subject at a first time point by contacting the first test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the 28 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) determining a second expression level of the at least 28 biomarkers from a second test sample from the subject at a second time point by contacting the second test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the second time point is after the first time point and after the administration of the therapy to the subject; (3) comparing the first expression level with the second expression level; and (4) producing a report, wherein the report identifies that the subject is responsive to the therapy when the second expression level is significantly decreased as compared to the first expression level.

In some embodiments, the first time point is prior to the administration of the therapy to the subject. In some embodiments, the first time point is after the administration of the therapy to the subject. In some embodiments, the therapy comprises an immunotherapy or a targeted therapy (e.g., a BRAF inhibitor).

In some embodiments, the at least one housekeeping gene is selected from the group consisting of ALG9, SEPN, YWHAQ, VPS37A, PRRC2B, DOPEY2, NDUFB11, ND4, MRPL19, PSMC4, SF3A1, PUM1, ACTB, GAPD, GUSB, RPLP0, TFRC, MORF4L1, 18S, PPIA, PGK1, RPL13A, B2M, YWHAZ, SDHA, HPRT1, TOX4, and TPT1.

In some embodiments, the at least one housekeeping gene comprises TOX4 and TPT1.

In some embodiments, the normalized expression level is obtained by: (1) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and (3) averaging the first normalized expression level and the second normalized expression level to obtain the normalized expression level.

In some embodiments, the method can have a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%. In some embodiments, the method has a sensitivity of greater than 90%. In some embodiments, the method has a specificity of greater than 90%.

In some embodiments, the biomarker is RNA, cDNA, or protein. When the biomarker is RNA, the RNA can be reverse transcribed to produce cDNA, and the produced cDNA expression level is detected. In some embodiments, the expression level of the biomarker is detected by forming a complex between the biomarker and a labeled probe or primer. When the biomarker is RNA or cDNA, the RNA or cDNA can be detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer. When the biomarker is protein, the protein can be detected by forming a complex between the protein and a labeled antibody. In some embodiments, the label is a fluorescent label.

In some embodiments, the test sample is blood, serum, plasma, or neoplastic tissue.

In some embodiments, the first predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects free of a neoplastic disease. The second predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas are being adequately controlled by therapies like immune therapy. The third predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas have been completely removed by surgery and they are considered "disease free." In some embodiments, each reference sample can be blood, serum, plasma, or non-neoplastic tissue.

In some embodiments, the subject in need thereof is a subject diagnosed with a melanoma, a subject having at least one melanoma symptom, or a subject having a predisposition or familial history for developing a melanoma. In some embodiments, the subject is a human.

In some embodiments, the algorithm is XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, or mlp.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a graph showing visualization of the melanoma score as a system of three contributors to the clinical picture—Control, Response, and Progression. Samples towards each of the corner represent pure representations of each clinical group. Samples in the middle are in the area of both algorithmic and clinical uncertainty.

FIG. **2** is a graph showing the metrics for the test in the test set ranged from 87-100%.

FIGS. **3**A-**3**C are a set of graphs showing the evaluation of the circulating melanoma gene test (Melanomx) in test set 2. Values were significantly higher in melanoma samples than in controls (FIG. **3**A). Patients who were responding to therapy had values similar to controls. Receiver operator curve analysis of test set 2 identifying the AUC for differentiating melanoma from controls was >0.95 (FIG. **3**B). The metrics for the test ranged from 78-92% (FIG. **3**C).

FIGS. **4**A-**4**B are a set of graphs showing the effect of surgery on the Melanomx. Levels were significantly decreased by surgery ($p<0.0001$) (FIG. **4**A). Values in the NED (no evidence of disease after surgery) group were significantly lower than in those with residual disease after surgery ($p=0.0007$) (FIG. **4**B).

FIG. **5** is a graph showing the effect of therapy on the Melanomx score. Levels were significantly decreased by immunotherapy (ipilimumab) or a BRAF inhibitor (Vemurafenib).

FIGS. **6**A-**6**B are a set of graphs showing Melanomx score in 3 different melanoma cell lines. FIG. **6**A identifies the cell lines demonstrate elevated expression—Melanomx score ranging from 40 (A375) to 95 (Hs294). FIG. **6**B identifies that spiking these cells into blood from a subject that does not have a melanoma, resulted in detectable gene expression and scores. A minimum of 1 cell/ml of blood could be consistently identified.

FIGS. **7**A-**7**B are a set of graphs showing expression in tumor tissue and its correlation with blood samples collected at the same time. In FIG. **7**A, the Melanomx score ranged 40-97 in melanoma tumor tissue. In contrast, normal epithelium exhibited values <20. In FIG. **7**B, gene expression in tumor tissue is compared to matched blood samples. This is highly concordant (correlation ~0.80).

DETAILED DESCRIPTION OF THE INVENTION

The details of the invention are set forth in the accompanying description below. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, illustrative methods and materials are now described. Other features, objects, and advantages of the invention will be apparent from the description and from the claims. In the specification and the appended claims, the singular forms also include the plural unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All patents and publications cited in this specification are incorporated herein by reference in their entireties.

Early signs of melanoma include changes to the shape or color of existing moles or, in the case of nodular melanoma, the appearance of a new lump anywhere on the skin. At later stages, the mole may itch, ulcerate or bleed. Visual inspection is the most common diagnostic technique. Melanoma can be divided into the following types: lentigo maligna, lentigo maligna melanoma, superficial spreading melanoma, acral lentiginous melanoma, mucosal melanoma, nodular melanoma, polypoid melanoma, and desmoplastic melanoma.

Measurements of circulating melanoma transcripts—the Melanomx-identify melanomas and decreases in the Melanomx score correlate with the efficacy of therapeutic interventions such as immunotherapy and targeted therapy. Targeted gene expression profile of RNA can be isolated from the peripheral blood of patients with melanoma. This expression profile is evaluated in an algorithm and converted to an output (prediction). It can identify active disease, provide an assessment of treatment responses, or predict risk of relapse, in conjunction with standard clinical assessment and imaging.

In one aspect, the present disclosure provides a method for detecting a melanoma in a subject in need thereof, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a first predetermined cutoff value; and (5) producing a report, wherein the report identifies the presence of a melanoma in the subject when the score is equal to or greater than the first predetermined cutoff value or identifies the absence of a melanoma in the subject when the score is below the first predetermined cutoff value, wherein the first predetermined cutoff value is 20 on a scale of 0 to 100.

In some embodiments, the at least one housekeeping gene is selected from the group consisting of ALG9, SEPN, YWHAQ, VPS37A, PRRC2B, DOPEY2, NDUFB11, ND4, MRPL19, PSMC4, SF3A1, PUM1, ACTB, GAPD, GUSB, RPLP0, TFRC, MORF4L1, 18S, PPIA, PGK1, RPL13A, B2M, YWHAZ, SDHA, HPRT1, TOX4, and TPT1.

In some embodiments, the at least one housekeeping gene comprises TOX4 and TPT1. In some embodiments, the normalized expression level is obtained by: (1) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and (3) averaging the first normalized expression level and the second normalized expression level to obtain the normalized expression level.

Among the provided methods are those that are able to classify or detect a melanoma. In some embodiments, the provided methods can identify or classify a melanoma in a human blood sample. In some examples, the methods can provide such information with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

The agents can be any agents for detection of the biomarkers, and typically are isolated polynucleotides or isolated polypeptides or proteins, such as antibodies, for example, those that specifically hybridize to or bind to the at least 29 biomarkers.

The biomarker can be RNA, cDNA, or protein. When the biomarker is RNA, the RNA can be reverse transcribed to produce cDNA (such as by RT-PCR), and the produced cDNA expression level is detected. The expression level of the biomarker can be detected by forming a complex between the biomarker and a labeled probe or primer. When the biomarker is RNA or cDNA, the RNA or cDNA detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer. The complex between the RNA or cDNA and the labeled nucleic acid probe or primer can be a hybridization complex.

When the biomarker is protein, the protein can be detected by forming a complex between the protein and a labeled antibody. The label can be any label, for example a fluorescent label, chemiluminescence label, radioactive label, etc. The protein level can be measured by methods including, but not limited to, immunoprecipitation, ELISA, Western blot analysis, or immunohistochemistry using an agent, e.g., an antibody, that specifically detects the protein encoded by the gene.

In some embodiments, the methods are performed by contacting the test sample with one of the provided agents, more typically with a plurality of the provided agents, for example, a set of polynucleotides that specifically bind to the at least 29 biomarkers. In some embodiments, the set of polynucleotides includes DNA, RNA, cDNA, PNA, genomic DNA, or synthetic oligonucleotides. In some embodiments, the methods include the step of isolating RNA from the test sample prior to detection, such as by RT-PCR, e.g., QPCR. Thus, in some embodiments, detection of the melanoma biomarkers, such as expression levels thereof, includes detecting the presence, absence, or amount of RNA. In one example, the RNA is detected by PCR or by hybridization.

In some embodiments, the polynucleotides include sense and antisense primers, such as a pair of primers that is specific to each of the at least 29 biomarkers. In one aspect of this embodiment, the detection of the at least 29 biomarkers is carried out by PCR, typically quantitative or real-time PCR. For example, in one aspect, detection is carried out by producing cDNA from the test sample by reverse transcription; then amplifying the cDNA using the pairs of sense and antisense primers that specifically hybridize to the panel of at least 28 biomarkers, and detecting products of the amplification.

The test sample can be any biological fluid obtained from the subject. Preferably, the test sample is blood, serum, plasma or neoplastic tissue.

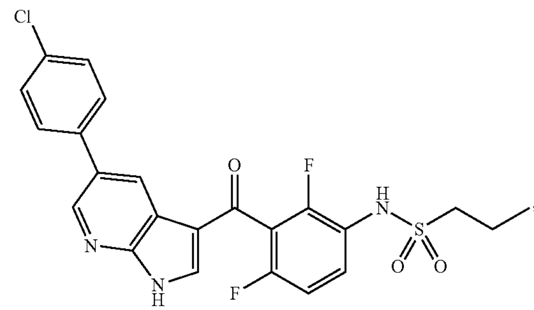
The first predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects free of a neoplastic disease. Each reference sample can be any biological fluid obtained from a subject not having, showing symptoms of or diagnosed with a neoplastic disease. In some embodiments, the reference sample is blood, serum, plasma, or non-neoplastic tissue.

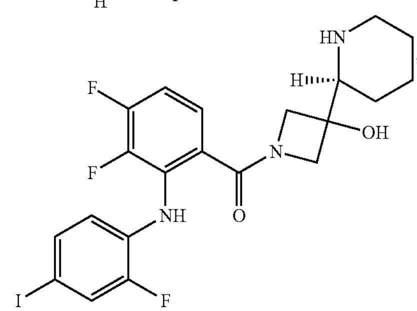
The subject in need thereof can be a subject diagnosed with a melanoma, a subject having at least one melanoma symptom or a subject having a predisposition or familial history for developing a melanoma. The subject can be any mammal. Preferably, the subject is human. The terms "subject" and "patient" are used interchangeably herein.

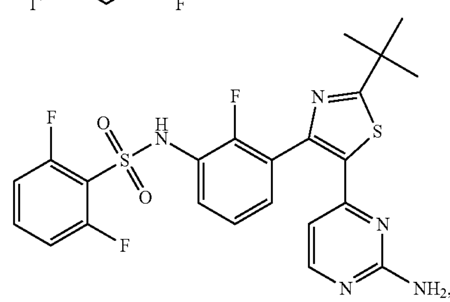
The score is the Melanomx score, which has a scale of 0 to 100. The Melanomx score is the product of a classifier built from predictive classification algorithms, e.g., XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, or mlp. The algorithm analyzes the data (i.e., expression levels) and then assigns a score.

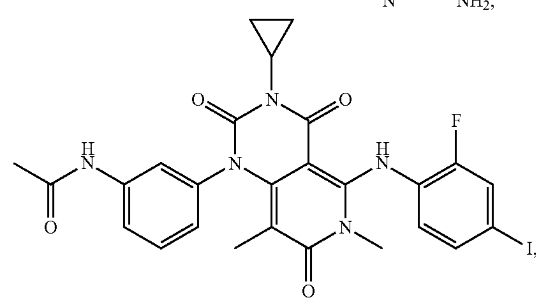
The method can further include treating the subject identified as having a melanoma with surgery, drug therapy, radiation therapy, or a combination thereof. The drug therapy can be an immunotherapy, a targeted therapy, a chemotherapy, or a combination thereof. In some embodiments, the drug therapy includes an immunotherapy. Examples of immunotherapies for treating a melanoma include, but are not limited to, Imlygic (T-VEC), Yervoy in combination with Opdivo, Opdivo (nivolumab), Keytruda (pembrolizumab), Yervoy (ipilimumab), Interleukin-2 (IL-2), and Interferon alpha 2-b. In some embodiments, the drug therapy includes a targeted therapy such as a BRAF inhibitor. Examples of targeted therapies for treating a melanoma include, but are not limited to, Zelboraf in combination with Cotellic (cobimetinib), Tafinlar in combination with Mekinist, Tafinlar (dabrafenib), Mekinist (trametinib), and Zelboraf (vemurafenib). In some embodiments, the drug therapy includes a chemotherapy. In some embodiments, the chemotherapy includes dacarbazine.

Accordingly, the present disclosure provides a method of treating melanoma in a human subject in need thereof, the method comprising: a) determining the expression level of 29 biomarkers from a biological fluid sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the 29 biomarkers, wherein the 29 biomarkers comprise each of atlastin GTPase 1 (ATL1), ATPase H+ transporting V0 subunit d1 (ATP6V0D), chromosome 1 open reading frame 21 (C1ORF21), CASP8 and FADD like apoptosis regulator (CFLAR), CFLAR antisense RNA 1 (CFLAR-AS1), calcineurin like EF-hand protein 1 (CHP1), DEAD-box helicase 55 (DDX55), dystrophin (DMD), DnaJ heat shock protein family (Hsp40) member C9 (DNAJC9), enolase superfamily member 1 (ENOSF1), Fanconi anemia complementation group L (FANCL), Holliday junction recognition protein (HJURP), major histocompatibility complex, class II, DO alpha (HLA-DOA), major histocompatibility complex, class II, DR alpha (HLA-DRA), heterogeneous nuclear ribonucleoprotein A3 pseudogene 1 (HNRNPA3P1), interleukin 23 subunit alpha (IL23A), IQ motif containing GTPase activating protein 1 (IQGAP1), NFYC pseudogene (LOC494127), uncharacterized LOC646471 (LOC646471), loss of heterozygosity, 12, chromosomal region 2 (non-protein coding) (LOH12CR), PBX homeobox interacting protein 1 (PBXIP1), ring finger protein 5 (RNF5), SERTA domain containing 2 (SERTAD2), solute carrier family 35 member G5 (SLC35G5), spermatogenesis associated serine rich 2 like (SPATS2L), tudor domain containing 7 (TDRD7), TXK tyrosine kinase (TXK), YY2 transcription factor (YY2), and at least one housekeeping gene; b) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; c) inputting each normalized expression level into an algorithm to generate a score, wherein the algorithm is the product of a trained classifier built from at least one predictive classification algorithm; d) determining that the human subject has melanoma by determining that the score is greater than or equal to a first predetermined cutoff value, wherein the first predetermining cutoff value is 20 on a scale of 0 to 100; and e) administering to the human subject at least one therapeutically effective amount of at least one of:









or any combination thereof.

The present disclosure also provides a method for determining whether a melanoma in a subject is stable or progressive, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1,

ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a second predetermined cutoff value; and (5) producing a report, wherein the report identifies that the melanoma is progressive when the normalized expression level is equal to or greater than the second predetermined cutoff value or identifies that the melanoma is stable when the normalized expression level is below the second predetermined cutoff value, wherein the second predetermined cutoff value is 50 on a scale of 0 to 100.

The second predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanomas are being adequately controlled by therapies like immune therapy.

Surgical resection is a procedure that removes melanoma tissues from the subject in need thereof. The present disclosure also provides a method for evaluating the extent of surgical resection in a subject having a melanoma, including: (1) determining the expression level of at least 29 biomarkers from a test sample from the subject after the surgical resection by contacting the test sample with a plurality of agents specific to detect the expression of the at least 29 biomarkers, wherein the at least 29 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, and at least one housekeeping gene; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) inputting each normalized expression level into an algorithm to generate a score; (4) comparing the score with a third predetermined cutoff value; and (5) producing a report, wherein the report identifies that the surgical resection does not remove the entire melanoma when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the surgical resection removes the entire melanoma when the normalized expression level is below the third predetermined cutoff value, wherein the third predetermined cutoff value is 20 on a scale of 0 to 100.

The third predetermined cutoff value can be derived from a plurality of reference samples obtained from subjects whose melanoma disease has been completely removed by surgery and they are considered "disease free."

When it is determined that the surgical resection does not remove the entire melanoma, the subject is at risk of melanoma recurrence. Accordingly, in some embodiments, the report further identifies that the risk of melanoma recurrence is high when the normalized expression level is equal to or greater than the third predetermined cutoff value or identifies that the risk of melanoma recurrence is low when the normalized expression level is below the third predetermined cutoff value.

The present disclosure also provides a method for determining a response by a subject having a melanoma to a therapy, comprising: (1) determining a first expression level of at least 28 biomarkers from a first test sample from the subject at a first time point by contacting the first test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the 28 biomarkers comprise ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (2) determining a second expression level of the at least 28 biomarkers from a second test sample from the subject at a second time point by contacting the second test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the second time point is after the first time point and after the administration of the therapy to the subject; (3) comparing the first expression level with the second expression level; and (4) determining that the subject is responsive to the therapy when the second expression level is significantly decreased as compared to the first expression level.

In some embodiments, the methods can predict treatment responsiveness to, or determine whether a patient has become clinically stable following, or is responsive or non-responsive to, a melanoma treatment, such as a surgical intervention or drug therapy (for example, an immunotherapy or targeted therapy). In some cases, the methods can do so with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%. In some cases, it can differentiate between treated and untreated melanoma with a specificity, sensitivity, and/or accuracy of at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

In some embodiments, the first and second test samples can be of the same type. In some embodiments, the first and second test samples can be of different types.

In some embodiments, the therapy can be a drug therapy. The drug therapy can be an immunotherapy, a targeted therapy, a chemotherapy, or a combination thereof. In some embodiments, the therapy can be a radiation therapy.

In some embodiments, the first time point is prior to the administration of the therapy to the subject. In some embodiments, the first time point is after the administration of the therapy to the subject. The second time point can be a few days, a few weeks, or a few months after the first time point. For example, the second time point can be at least 1 day, at least 7 days, at least 14 days, at least 30 days, at least 60 days, or at least 90 days after the first time point.

In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 10% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 20% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 30% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 40% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 50% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 60% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 70% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 80% less than the first expression level. In some embodiments, the second expression level is significantly decreased as compared to the first expression level when the second expression level is at least 90% less than the first expression level.

In some embodiments, the method further comprises determining a third expression level of the at least 28 biomarkers from a third test sample from the subject at a third time point by contacting the third test sample with a plurality of agents specific to detect the expression of the at least 28 biomarkers, wherein the third time point is after the second time point. The method can further comprise creating a plot showing the trend of the expression level change.

The present disclosure also provides an assay comprising: (1) determining the expression level of biomarkers consisting essentially of the following 30 biomarkers from a test sample from a patient diagnosed of a melanoma or a subject suspected of having a melanoma: ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, TOX4, and TPT1, wherein the expression level is measured by contacting the test sample with a plurality of agents specific to detect the expression of the 30 biomarkers; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (4) averaging the first normalized expression level and the second normalized expression level to obtain a normalized expression level for each biomarker; (5) inputting each normalized expression level into an algorithm to generate a score; and (6) comparing the score with a first predetermined cutoff value.

The present disclosure also provides an assay comprising: (1) determining the expression level of biomarkers consisting of the following 30 biomarkers from a test sample from a patient diagnosed of a melanoma or a subject suspected of having a melanoma: ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, YY2, TOX4, and TPT1, wherein the expression level is measured by contacting the test sample with a plurality of agents specific to detect the expression of the 30 biomarkers; (2) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (3) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; (4) averaging the first normalized expression level and the second normalized expression level to obtain a normalized expression level for each biomarker; (5) inputting each normalized expression level into an algorithm to generate a score; and (6) comparing the score with a first predetermined cutoff value.

The sequence information of the melanoma biomarkers and housekeepers is shown in Table 1.

TABLE 1

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
ATL1	NM_001127713.1	AGTTTGAGGC TGCACTGCAC ACCTGCACAA CCTGCCTTCT ACTTAGTTCT TCTGAGACAT TTCTGAAAGT CTGAATTCCT AGGACTGCTC AATGACCTTT GTCCCTGTTG GGCACACGCA GTGTCTCATC GCTGGTATTG CACCTTTAAT GAGACCAGGA GTTCCGCAAA AGTAAAACAA AGGGGTCACA GACTGGTTCT GGTTCTACCA CTTCCTCAGT ATGTGCTCTG GGACAAAACA GCATATTTGG CAACATCTCG GTGTCCTTAT CTGCAGCTTG AAGAGGCGCC ACAGCAACAT CCTCAGAGTC TGAGCGAACT GCGCCCAGCG CGGGCACGGA GCCTCCCACC GCCAGCAACC TGCGGCCCCG GAGAAGGCAG CGAGCGCAGT GACAGCGCCT CACCGCCACC AGCTCCTGGA CCACCATGGC CAAGAACCGC AGGGACAGAA ACAGTTGGGG TGGATTTTCG GAAAAGACAT ATGAATGGAG CTCAGAAGAG GAGGAGCCAG TGAAAAAGGC AGGACCAGTC CAAGTCCTCA TTGTCAAAGA TGACCATTCC TTTGAGTTAG ATGAAACTGC ATTAAATCGG ATCCTTCTCT CGGAGGCTGT CAGAGACAAG GAGGTTGTTG CTGTATCTGT TGCTGGAGCA TTTAGAAAAG GAAAATCATT CCTGATGGAC TTCATGTTGA GATACATGTA CAACCAGGAA TCAGTTGATT GGGTTGGAGA CTACAATGAA CCATTGACTG GTTTTTCATG GAGAGGTGGA TCTGAGCGAG AGACCACAGG AATTCAGATA TGGAGTGAAA TCTTCCTTAT CAATAAACCT GATGGTAAAA AGGTTGCAGT GTTATTGATG GATACTCAGG GAACCTTTGA TAGTCAGTCA ACTTTGAGAG ATTCAGCCAC AGTATTTGCC CTTAGCACAA TGATCAGCTC AATACAGGTA TATAACTTAT CCCAAAATGT CCAGGAGGAT GATCTTCAGC ACCTCCAGCT TTTCACTGAG TATGGCAGAC TGGCAATGGA GGAAACATTC CTGAAGCCAT TTCAGAGTCT GATATTTCTT GTTCGAGACT GGAGTTTCCC ATACGAATTT TCATATGGAG CCGATGGTGG TGCCAAATTC TTGGAAAAAC GCCTCAAGGT CTCAGGGAAC CAGCATGAAG AACTACAGAA CGTCAGAAAA CACATCCATT CCTGTTTCAC CAACATTTCC TGTTTTCTGC TACCTCATCC TGGCTTAAAA GTAGCTACCA ATCCAAACTT TGATGGAAAA TTGAAAGAAA TAGATGATGA ATTCATCAAA AACTTGAAAA TACTGATTCC TTGGCTACTT AGTCCCGAGA GCCTAGATAT TAAAGAGATC AATGGGAATA AAATCACCTG CCGGGGTCTG GTGGAGTACT TCAAGGCTTA TATAAAGATC TATCAAGGTG AAGAATTACC ACATCCCAAA TCCATGTTAC AGGCCACAGC AGAAGCTAAC AATTTAGCAG CCGTGGCAAC TGCCAAGGAC ACATACAACA AAAAAATGGA AGAGATTTGT GGTGGTGACA AACCATTTCT GGCCCCAAAT GACTTGCAGA CCAAACACCT GCAACTTAAG GAAGAATCTG TGAAGCTATT CCGAGGGGTG AAGAAGATGG GTGGGGAAGA ATTTAGCCGG CGTTACCTGC AGCAGTTGGA GAGTGAAATA GATGAACTTT ACATCCAATA TATCAAGCAC AATGATAGCA AAAATATCTT CCATGCAGCT CGTACCCCAG CCACACTGTT TGTAGTCATC TTTATCACAT ATGTGATTGC TGGTGTGACT GGATTCATTG GTTTGGACAT CATAGCTAGC CTATGCAATA TGATAATGGG ACTGACCCTT ATCACCCTGT GCACTTGGGC ATATATCCGG TACTCTGGAG AATACCGAGA GCTGGGAGCT GTAATAGACC AGGTGGCTGC AGCTCTGTGG GACCAGGCTT TGTACAAGCT TTACAGTGCA GCAGCAACCC ACAGACATCT GTATCATCAA GCTTTCCCTA CACCAAAGTC GGAATCTACT GAACAATCAG AAAAGAAAAA AATGTAATGC AAATTTTAAG AAATACAGGT GCATGACCAA TTGTCAATTA AATATTCAGT TTTATGTCTC CATGCAAACA TTCAAAGTGC TTCCATCAGA ACGGAGTAAA ATACTAAACA CCTCTGAAGA CTGCAAACTG GATTAGTTCT TTTACTTCAG TGTTTAATAA GCAGATGTAT GTATGCATGG TTATACTATT TTGTTAACAT GTACAATTTC CTGATTTTTC TTCAAAAATG CTGTTATAAA GTATTTGTCT ATTTATGATA ACAGTACACG TGTTCTGCTT GAATTTACTA AATTCTACTA CTGGGTTATA ATTAAATCAT GTGATATTCC ACGTTTGGAT ATGCTCATTT AATTTCTACA GAAAAAATTT TAAATTATTT CACATTAGCC ATTTGTTAAA ACACAGCATC ATAACTCAGC AGGCTGGATT TAATCTGTAT CATCTTATAT ATATCACAAT CTTATTTTTA AGCACATTTT AGAGTTCCTT AGTTGCTTTA TCAAAAACCA GATATTGCTT TTACATGGTT TAATAGAATA TAAACCTCTT GATAAAAAAT GCACAAAAAA TCACTTTGTA TATGTGAGTT TCACTGCATT GTATATTTTT TCATTTGGTA CACAAAGAAT GTATTCTTCA TAGGTTTATT CTTTTAATAT GTGAACTATT ATTAAAGTTT ACTCTGGTTC CTAAGATTAA AAACAAATGC TTACTGAATT TGAAAAAAAA A	1

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
ATP6V0D	NM_004691.4	ACTTGACAGC CCGCTGAGGA CGCAGCGTCA GCTGACCTGG GGAGTCGCGA TTCGTGCCGG CCGGTCCTGG TTCTCCGGTC CCGCCGCTCC CGCAGCAGCC ATGTCGTTCT TCCCGGAGCT TTACTTTAAC GTGGACAATG GCTACTTGGA GGGACTGGTG CGCGGCCTGA AGGCCGGGGT GCTCAGCCAG GCCGACTACC TCAACCTGGT GCAGTGCGAG ACGCTAGAGG ACTTGAAACT GCATCTGCAG AGCACTGATT ATGGTAACTT CCTGGCCAAC GAGGCATCAC CTCTGACGGT GTCAGTCATC GATGACCGGC TCAAGGAGAA GATGGTGGTG GAGTTCCGCC ACATGAGGAA CCATGCCTAT GAGCCACTCG CCAGCTTCCT AGACTTCATT ACTTACAGTT ACATGATCGA CAACGTGATC CTGCTCATCA CAGGCACGCT GCACCAGCGC TCCATCGCTG AGCTCGTGCC CAAGTGCCAC CCACTAGGCA GCTTCGAGCA GATGGAGGCC GTGAACATTG CTCAGACACC TGCTGAGCTC TACAATGCCA TTCTGGTGGA CACGCCTCTT GCGGCTTTTT TCCAGGACTG CATTTCAGAG CAGGACCTTG ACGAGATGAA CATCGAGATC ATCCGCAACA CCCTCTACAA GGCCTACCTG GAGTCCTTCT ACAAGTTCTG CACCCTACTG GGCGGGACTA CGGCTGATGC CATGTGCCCC ATCCTGGAGT TTGAAGCAGA CCGCCGCGCC TTCATCATCA CCATCAATTC TTTCGGCACA GAGCTGTCCA AAGAGGACCG TGCCAAGCTC TTTCCACACT GTGGGCGGCT CTACCCTGAG GGCCTGGCGC AGCTGGCTCG GGCTGACGAC TATGAACAGG TCAAGAACGT GGCCGATTAC TACCCGGAGT ACAAGCTGCT CTTCGAGGGT GCAGGTAGCA ACCCTGGAGA CAAGACGCTG GAGGACCGAT TCTTTGAGCA CGAGGTAAAG CTGAACAAGT TGGCCTTCCT GAACCAGTTC CACTTTGGTG TCTTCTATGC CTTCGTGAAG CTCAAGGAGC AGGAGTGTCG CAACATCGTG TGGATCGCTG AATGTATCGC CCAGCGCCAC CGCGCCAAAA TCGACAACTA CATCCCTATC TTCTAGCGTC CTGGCCCAAG GCTCTCAATT GCACTCTTTG TGTGTGTGTG TGTGTGTGTG CGCGTGTGTG TGCGTGTGTG TGTATGTGGT CTGTGACAAG CCTGTGGCTC ACCTGCCTGT CCGGGGTGTA GTACGCTGTC CTAGCGGCTG CCCAGTTCTC CTGACCCTCT TAGAGACTGT TCTTAGGCCT GAAAAGGGGC TGGGCACCCC CCCCCACCAA GGATGGACGA AGACCCCCTC CAGAGCAAGG AGGCCCCCTC AGCCCTGTGG TTACAGCCGC TGATGTATCT AAGAAGCATG TCACTTTCAT GTTCCTCCCT AACTCCCTGA CCTGAGAACC CTGGGGCCTG GGGGCAGTTT GAGCCTCCTC TCCCTTCTGT GGGTCGCTCC CAGAGCCATG GCCCATGGGA AGGACAGAGT GTGTGTGTCC TTGGGGCCTG GGGGGATGTT GCTCCTCAGC TCCCTCCCTC AGCCCTGCCC CTCTGAGACA ATAAAACTGC CCTCTCTAAG GCCAACTGTC AAAAAAAAAA AAAAAAAA	2
C1ORF21	NM_030806.3	CCCTCCCTCG CTCGCTCCTC GCAAGCTCCC GCTCGCTCCC TGCCCACTCC CGGGGGGACG TTCCGTGCCG CGGCCGCCGC GGCCGCTGCT TCTTTCACAC TTTAGTTGGG AGCTGCGCGC CGCGCTCAGT TACTGGAGAG CTGGCCGCGC GCCGCCGCCT CCCGCACGCT TGCACGCGGG CCCGGCTTCG GGGTTTTGGG TTCTTACTCC AAGCGGCGGG GAGGAGGGGG AGCCCCGGAC ACACTGTGGG GAGGAGGAGG AAGAAGAGGA GGAGGGAGGA AGAAAAAAGA CGAGGAGGAC AGGGGCGGGG GGCGGGAGGC TTGCCACCTT CAGCCCCCCC GCGAACGCCC AAGGTGCACA CATCTTGACC AACTCAGCAG CAAGGTGGAT TTTCTTTGTG TTTAAAGAAA AAAAATGTCC CTGTGTCTGT AGAGATGATT TGCAGTTCAG CCCGGCTGAA GCTGACCGAA TGAGACTATG GGCTGTGCCT CCGCCAAGCA TGTTGCCACT GTTCAAAATG AAGAGGAAGC CCAGAAAGGG AAAAACTACC AGAACGGAGA TGTGTTTGGC GATGAGTATA GGATCAAACC AGTGGAAGAG GTCAAATACA TGAAAAATGG GGCAGAAGAA GAGCAGAAAA TAGCAGCCAG GAACCAAGAA AACTTGGAAA AAAGTGCCAG CTCAAATGTA AGACTTAAAA CTAATAAAGA GGTTCCGGGA TTAGTTCATC AACCCAGAGC AAACATGCAC ATCTCTGAAA GCCAACAAGA ATTCTTCAGA ATGCTGGATG AAAAAATTGA AAAGGGTCGG GATTACTGTT CGGAAGAAGA GGATATCACA TAGCACCAAT TTTACCACTC AAACCAGGAG CTACTACTGT GTAAATAGGT TACACCCCAG TTGAAATCTT TGCAAAGGTC GGTTCTATTC AGCGAACAGC ACTATAGCAA AAGAAGATCG TTCCATATTG TACGCCCCAT TAAATTACAG TGTTTCTTAA TGAACTTGCA AAGGAATATT GCTAAAAACA AACAAAAAAA ACTGTTATCG AACTTTCTTT GTTGCTGCTA GTTAAAACTT GTTGCAACTT TTCACTTCTC TTGTGTCCAG GTATGCAGCA AAATTCTGCA ATTTCACCTT AAAGATACTG TTGGTTTTAC AGATGCTCTC CAACCTATTT TCTATAAGAT GAGGTAGTGG	3

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TGAACTCAGA TAACAAACTT CTCTTCTAAA CTGGTTCTGC	
		TTCTAAGACA AGCATCTCCT GCCCTCTCTC CTTCCTCCCC	
		ATCTCTCGCA CGCAGTCTAG AGATGGACTG AGCCTTGCTT	
		CTCACTGGCA GTGTTGAGCT TTGGAGATGG GATGGTTGCT	
		ATGCCAAGCC TTGTTTCTCT GCTCAGAAGA AGTAGAGAAG	
		CTATTATCAA TTAAAAGCAT GCTGTGATGT GACTCCTGGA	
		AGTGACGTAG GAGTGAGTGG CAGGTTGTTT GATTTAATAG	
		GTATCTTAAT CAAGAATTAA GCTTGCAACA TTGGCTTTGC	
		TCAGATGCAG ATGGAAGTGT GATCACAATC ATTTTGAATC	
		CCTCTTCCTC ACTTTTTTTT CTAAGAAAAT AAACATTTTA	
		CTGTTTTTAT GGATCCTTGT CTTCTCCCAT TCATCCAGCT	
		CAGTGTTTTA AGATGATCCT GGGTGCAGAA GTTGAGCCCT	
		CCTTTGCATT GACACTGATA ATTAGCCTAT AGGGCTCCCT	
		ACCCTTCCAT TAAGAATCTA CCAAGCATTA GCAAGGCTGA	
		AAGTGGTCTA AGAGGTGAGG GGACATCCTA TGACTTTTTA	
		GGAAGGCCTG AAACCACCTT GTTACCTTTC ATTTTGTTAG	
		CAAATAAACC ATCCTATTTT GTAACTCTCC CCCTTCAAAA	
		TGCTACATGA GCCTTGCCAC TTCCTTTTTC TCTTACTTCC	
		AGCACACTAG ACATAGCAAA AGTGTTTGCC TACTCAAAAA	
		CATAATACTT TTATGCTGAT GATGGTATTT GGAGATGTGA	
		AAGCCAAAAG CCCCTGGCAG TGGTGGGGAA TGTTGACTGA	
		GTGTTCAGCA GAGTTTATTT TTCCATACTA TATGAAGAGA	
		ATGATCTCTT CTCAAAGACA GAAGTGATAT TTTTAACAAA	
		TATTGTCACA AGTAAATAGC AATCAAAAGG AGAAAATAAC	
		TTTTGTATTT TTTTAATGTG TTTGATAGCT TTGACGAGGG	
		TTCTCTTTGT TACTTTCAGG GGAGGGCATC CTATTAAATG	
		CCACGCCAGC AGTCCGGGTC TGGGTTTGTC CCACAAAATC	
		ACAGGAGCAC TGTATGTTCC TCTCTTTTGG AGTTGTGACT	
		TTGAAGGGCC TCAATATTAG CCACACTGCC GCCTGCAGAA	
		GGTGGAGAGT TAAGATGTTC TATGTCAATT TGCTCTTGCC	
		GAAAAGATGA GCCTCGATTT TAAAATCTAT CCACATCCAA	
		CTGATGGCAC CATTGATGTG CAAATAATGA GATTCCCTAT	
		CTCCTTTTAG ACCTGGGACG GCAAAAGGGA AGGGAAGGAA	
		ACTTAGCAGA GTGCTATTGA CTATAGATTC ACATATTAGC	
		AACAAAATCC CGTAATTCTT TTGGCCAACA GCAGCTATTT	
		TGGGGAGCAG CTGTGGCTGT TACATAAATA GAGATGCAGC	
		CAAAATTTTA GGCCTTTTAT CCTGCTTCTA GCAGAAAAAT	
		GCAGGGAGAG TCAAGTAGTC TAGGGTTTCA GGTTGCCTCC	
		CCTCATATGG TTTTTGGCCA AGTGACTAAA ACAGTTTTCC	
		ACAACTGTAA ACAAACTGCT AAGCCCCACC TCAAACTTGT	
		TCACTGGGGA CTTTGCTTAC CGTTCTGTGG GTGACCTTTT	
		CCGGGATTTC TTGTTCTTAT CAAGCAAGAA TTAAGCACAT	
		GCTAAACGTC TTCCATTTGA CTTCTCTACT CGGTGTCTCA	
		GACAGTGTCT TCCCAGAAAA CCACCACCCT CTACCCAAAG	
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		ACAGTTTTGA CATGTTATAC TTGCGCATAG ATCCAAGCGT	
		TTCTTGGGAA CCTGACTTTT GAGTGTTTAA TAAAGCCGGA	
		AGTGGTGTTG CCCTGAACCA GCAGATTTTC ACCTGGGTTC	
		TGGCTCCGGT GTTTAACACT GGATACATCT TTGATGTGCG	
		AAAGTGAGTT CATCTTCAGA CACATTTGGT ACATCCAGAA	
		ATAGATCCAA GAAATGGGGT GGTTGAGTGG GTCCGCACGA	
		AATGCTTGAT TATGTCAGCA ACACCCAACA CTGTCTGTTT	
		TCCATTTGTT GGTTTTAATC ATAAAATTGT CAAGTGATTC	
		GTGTTTGTAC TTTATTTTTT TGTGCCTTCT GAAAGGATCT	
		AAAACAAAAA TATTTTGCCT TTTTTTCCCC ACGTGTATCT	
		GAACATTAAG CAGATTGGCT CAGACACAAT GAAAAGGATA	
		ATCCAATGTA CGTGCTGGTG CACTCTGCTA GTTGTTATCT	
		CTGTAGGGCT CAGGAAGCTG GAAGGAGGAA GGGAGGGTAA	
		GTGGCCTGGT GAGTGGAGGT AGAAAAATGA TGAGAAATGA	
		ACTGAGAGCA TTAAGCAGAG AGGGTTGATA GGCTGGCCGT	
		GTCCGGGGTG AAATTGGAAA TCCAGCTGCC TAGTGGCCAG	
		TGGGTGGGGC AAGACTGTCA ACGAGATTTA CAGCTGGCTT	
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		TTCAGCAGTG TGTGCACAGC TTTCTTTTGG TTGGCAGAGA	
		TTCAGGATCA TGGAGTACTG CTCTCATAAT TGAAGACGTG	
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		AGCGATGCTT CCTAGCCCCA GGCTTGCAGA GAACACAGAG	
		TGGTGTTGTG GTCTATTTAG GGACAAAGAA GGTATAAAGT	
		CCAGAGATGA GAAAACTGGG TCAGCCCTCA GAAACTGCAG	
		CAGCCACGCA CACAGAAGCC TGCTGGAAGA CAGGTCTCTC	
		TCCGTCCACA GTGCCCATCA TCTGAGCCTG GGCTGGGATG	
		ACTCAACTTA GCAAAGACGG ACCCAGGAGG AGTGCTGGTT	
		CTTCAGTCTT TGTACTGGCC CCATCCTCTC CTCACTGTAA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		GGTCCAGGGA TGATGCAGCC CCCTTCTTGG TCCCATGTGA	
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		TATAACTTTA TCCTCTTCCC AGCCTCATCC CTGTTTTTCT	
		GTTAGGGCAA GACTCTTCCA TAAGCCTGCT AAAAAACAGA	
		GATGATACCT CTTACAAACT TTACCTCATA GCCTGTGAAG	
		CAGGTTGGCA TGTGGATTAC AAGTCCTGCT TTGACACTGG	
		GCAAGAATTT AAGATTGTTC TATCTCTACT AGTCATAGAA	
		AAGAAACATT GTTAAACATG TTGAGTTTTA AAGGAAGAAA	
		TATTTTCAAA TTCTTAATCC AAGAAAATAC TGAGTTGGAA	
		TCTTAGACTT CGGGACTCTG ACACGTTCTT TATGAAAGGC	
		AAAATAATTG GTATCTAAAG TTCTCTCCTT CCTGCCTCCC	
		CTAAAGAAAA AGGATATTAG ATTGCACACT ATAATTTTAC	
		ATAAGATCTG CCTTCCACAC TTCCCTGCTG GAAGGCATTC	
		TCAGAGCTTT ATGTCTTCGT ACCTCTCAGA GATTGGACTT	
		TTTCTTGTTT AAAACCCCAA CCAAAAAAAA TAATAAGGCA	
		TGATTGGTGG GGAGGGAATG TGTATTTAGG GGCATAATAA	
		AAAGGTGGCT CGAAGCAGGA ACTTTGGCCT CATGGTGTCA	
		TGGGTGGATG CCTGGACTCC AGTGTGCCTG TGAGGGGCTG	
		GGTTAGGCAG TCGGCTGTCA CACTCACATG TGCCTGCAAT	
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		TTTTGCCAAT GAATGGATGT ATTTTTCCAA GGGGGGAATA	
		GTATCCTTGA CTTTGGCAGT CACCTTTTTG TATGTCTCTA	
		GAAAGGGGTC AAAAAACTAT GGTAAAGATG AGGCTTATGA	
		GTGAATACCT CTGGGACAAA CCTTAGGACT CACAAGCTAT	
		GCCATGTTTT TCAGGAGACT CTTGTACCTT ATCTGGAATC	
		TAATCTTGGG AGAAGAGGAA AAAGGAGCTA ATATTTGTCA	
		TTTATACTCA CTCTGTGCCA GATACTGTGC TAGGCATTTT	
		ATAATTGTTT TGTGTCATCC TCATGATAAT CCTGTGAAGT	
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		AAGGAAAGAT GTCATTCAAG TATTTTTCAA ATTCTTTTTA	
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		CCTATCTGAG CTGCACACAC ACCAGGGGAG GCCACTGGCT	
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		TGCAACCAGA ACAGGCTTTC GAGTGCTGTG ATTTCTTTCC	
		TGGGTTCCCA AGTCTTGTTG TTTCATCACA AACTGTATCT	
		TTTTAAGGTT AAAAGTCTTG ACCTTCATGG GGGTCTGGGA	
		CAATCCGATC TCCAAGCATG GAGGAAAGGC AATGCCTGGA	
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		TTTCGAGACC TTACATGTTT TAGATAGTCA TGTAATGACT	
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		CCCAAGAACT CCTCAGAGTT CCTCTTCTTG CTTCTTGAAT	
		CACATCCTCT AAAGATGACT CATGTCTCCA TAGCAACTGT	
		TAAAGGTGCT GCCTAGACGG GACCCGCTCC CACCTTTACT	
		TACTTTGCTC GAAGGAACCA AAACAATGTC CTTGTGTAAA	
		GGGGCTGTCA TATCCAGTTT TCCTTTGAAA TCTGGCCCCC	
		AAGATCCTGC TTCTTTCTAA CCTGGCAGCT GTCATGTCCC	
		TTCAAGATGA TGTGGGAAAT GGCCCTTACA ACTTGGGAAC	
		CACAGAAATT GCTGTATTTC GGGAAGATTC ACCTCTAAAC	
		TGAAGGCTTC ATTCTGATAG TGTCTGCCCT CTCTACCCTG	
		ATTTCGCCCT TCTTTGCTTC CATTTTTAGC CCAAGGCTTT	
		GAATTTGATT GAGTAAGACT TAGAGGCAGT ATAAAGAACA	
		CCATAAACTT AGGCAGAGGT CCCTTAGGGT CTCTAGAGTT	
		GAAAATAATT CTACAGCCTT AGGGGGACCT CTTGGCATTG	
		ACTCTAAAGG GAGAGAATAG CCCCTGTGTC CTGGCATTTC	
		AGTCTAGACC TTCAAGGACT GTTCTCTCTT GACAGGCAAG	
		CAAGCAAAGA AAGTTTTGCA ATAGATTTCA AGCCAGTTTT	
		TCCATTCAAA CCAAGATGCA AATTCATAAA ATTACTCTTT	
		TCCTGGAATA GATCCAGGCA GCTGCCTTAT TAGAACTTTA	
		GATTCGGATC TATTTTCTTA ACACACATAC ACATACACGC	
		GCATACATAC ATATACAGAG AGATACGTGG AGAAAGGAAA	
		TTTACTCTAT CATTGCAATA CTTCAAGAAA GAGCTGTATT	
		TTGCCTTTCT GTAATCTCCA AGATAGTGTC TAGGAAAGTA	
		ATAGTATAAC TATAGGGATA CCGAAACAGG AAAAACCAGC	
		CATCACTCTT GAGAAAGTTT GAGTTCGACT CACATGGGAG	
		AATCGAGGTC TGCTACTCGT CTTGCTTTGT GCCCCATCTG	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TGCTAGCCGG GGTGTTGTGG CTGACATCCT TTCCTGGCCT TACACACATA ATAGACACAT CCCTAACGGC GTGTGCCTGG TCCAGCCACA TACAGCCACC ACATGTGTCA CACACTGTCC CCCTCATCCA TGTGGACTTG ACTGGCATTT CAGCAGCTCC ACTGGGATGC TCTAACCCCA GTGTGTGGAG TTGGGGTCCC TTCATCTAGG TTGACCCAGG TATAGCATTT TTAGCATTGC CTTTCCAGTC TTGATGATTC ATTCATTGAA CTCATTTATT TCTGGAGCCC CTGGTACACT CCAGGCACTG CGCTAATTGC CAGCAAAGCA CAACTGAACT AAATCCACCT TCAAGGAACC TAGCCATAAC GAGGGAGGCA GCATGGAAGT ACCCTACAGG GGAAAGTCCT GAGTGCTGTG GGAGCATCTC ACCGTGGCAG CCAGCCCAGT TTTGGCAATC AGGGGCTTCC TGAAAGAGGT GACATCAAAG CCCGGATGTG TCAGAGGACT GAGGGAGAGT GTTACTAAAG GACTTTCAGG CTGAGAGGAT AGCACAGGAC TCAGCCCAAA GGAGGGCCAG TGTGGACTGT CCAGGGCCAG CCTGCAGTAC AGAGGCTGGA GCTTGGACTT GTAGAGGGAG AGAGAAGAGC AAGGGACGTG GACGGGGCAG TGAGCCAGGC TAGCCACAGA GGGTTCCCGG GGCTTTGCTG GGGATTCAGG GAGCATAAAT AAGAGCTTTA GGTGGTGCTG TGTCCTCTGC AGCCCACTGC TGAGGTCCTC CAGACAGGTA AGGTGTGGTC ACAATCAGGG CCGGGGTTTC CCTGCTCACT GCGGCAGTGC AGGGGTGCTT GCTGAGATGA TTCATCCCAG GGTGTCCTCT GTCCCTTACC CAGCCCCAAC TCCTCTTCCT CTGCCAAAAG CTATTTGAAT TCAAGGACTT TAACCTGGGC CGGATCTGGT TTGGAGACAA AGGGGACAGC TCTGGGTCAG CATGACCTTC TTTAGAGCCA CTAAGGCGAA AAATACCGTT TGGGACCAGG CTGGCCTAGA CCCAGGGATG AGAATGCACC CTAAAATAAA TATACGGGAA GCAGCAGAGG GCTTCCCTGT CTAGTGTGTG ATCCTAACTA AAGGCAGCTC TCTTGGACAG CCTTCCCCTG GATTAGGTCA CATACACCTG GTGGCCAAGC CTCTGCTGGG TCCCAAATAC ACACCCGAGT CCTGCCAAAG AAAGGAGATT TTTAAAAAGC ACAGACAAAT TGTATGCAAG TGGAAAATAC CCATAGGCCT AGACAGCTGT GGAGGGAAGA CCTCGTGGGT ACCTGGAGGC TGCCAGAGCT GGGAGCTCTG CAGGTATGAG TCAGGGAAGG CTCAGAGACA AGCAGAATCT CTCTATGGAG ACAACTTGCA GTGCCTTTTA GGTTTTCCAA ATAACCTCGG AGTTCAGAGC ATTGGGTTTT TTTCTCCCCT CCCCACCCCC AGAAAAATAA TTAGAAAAAT GTTTAGGAGA AAGGAAAAGA ATTAGATGCA TCAGAATACC AGCTATAAGC CAACACTGTT TCCAGAAACT CAAGAAAAAG CTCAAACAGA AGACAGTTCC CCTGAGAGGC TGGAGGCGTT GGTGCTGAAG GCAATTTTCC TAGCTAAGGG GCACTGGGCC TTGCTGCACC TTGGGGCTGA CCTTTTTTGC AAAACACCCA CCCCTGCCCT CCTGGCATAC TCAACAGCAA CGCCAGCTTT CTGGACCCTT GGAAAGATGT TAGCTCAAAC ACCCACTTTT TCCAGATCTT CCTCTTGCTC TTCACTGAGG AATTTGTAAT TCTGAGGCTA GCGATGCCCA CTCGGATATT CCGCAGGCCC AGGTGTTTAG ATTAGAATTT GTCCAGCGGT AATCCTGATG CTGGAAACCA ACAAACATTT GGCCTCATAT TCACCCATTT AAAAACTAGA GCCCCTGGCA GGTCCCCTTA GGGCCATGTG TTCATGGAAT ATAAGCCAAG TTTGCCTTAG GCTTGTTCAT GGAATATAAG CCAAGTTTAC CTCTCCCCAT TTTCTGCCCT GGCCCACTTC CCACTCACCT CCACCTCATT GCCAGGAAGG GATCAAAATG CCTCCATGCC AGTTGTTAAT GGCTACATAT TTGCCCTTCC CAAGGGTATT TGCATTTTAT TTAGGAACAT GGCCTTATAT TCAAGGAAAA TCTAGCATCA AGATTACGAG GCATCACCTC TCAATCAGGT CTGGGAGGTA TCTTGGGGCA TTGCTCTTCT GAACACCTGC AGAGGCTTCC TCAGGTGAGT GTGGGAGCCC GGAAGGGTGG CCTCCCTAAC CACTCTGCCT GCACATGAAT TCTCCAAAGC AGTGGGCCCC CATCTGTTTC AATTACACAT GCCTGTCAGC AAAAACTTCG TGAGATGCAC TCTCTCTGTG TGTTTATTAA TTTATTTAAA GCATATATCC CTTTACTTTT GTACTACTAT ATTAGGCACA TTATAAAAAG TATACAGCAT AGAAACTTTA AATGAATAAG ACACAAAATA TTATAAACAG AGGTTCTGGC ATTTTCTCTC TGAACTCCTG AGGGGGACCT TGGGCACCTC CTGGTATGTG CACACCCCAC TTTGAACACC CTTGCTTTAG TGCAAATCAG CACGCCTAAA TAGCATCCCA AACCACTGTG CGACATTTGG CCTGCAGAAT AAGAAAGTCC TAAGGCAGAA TCCCCAGGAA CTTCAAATCT GGGAGATGAG GAAAGAAAAC TCACTCACAA ACAACATAAA TGTAAATAAT TCAAAACCCT AAAGGAGAGC TGTCCCTAGA CAGTAGTGGC TACAGGTGCT AAGCAAAAGT CAGGTACACT GGACAGAGCC ATGCGATTGA TTGTTCATCT TCCCTCTCTG GGTCTCCAGA AAAGGACACT GGCAATATCC CCAGCTCTCT CCTGATTGGA ATTCTTTGAA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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CFLAR	NM_001127183.2	GACCACGCAT GGAGAATTTT ACCACCCAGA GACACGCGAG TGGCCCTGTG CAAGTTTCAA CTGCGCGGGG GGCGGGGAAT TCCGCAGACA GGATTCTAGA ATTTATGTCT TGTGTGGTAA CATTTCAGCC GGTGGGTGGC GGGGATTAGG CGTGAAGCGG TTCAGCAGGC AGAGGTTCTC GGACGCCCTC CGGCGAAGCC ACCTGTTGAT GCTTTTGACT TTCTGTCCTT GTTCCTCGTC CCATCTGGAG CATTTCCAAT TCTGGTTTTG CGGAGCAGCA GGTCTGAGCT TGTCCGGCGA GGGTGGGAGT TGGTCCCGGC GGAGATCCAG TGGGAAGAGC CGGCGGCTGC CCGGGCAACT CCCCCACTGG AAAGGATTCT GAAAGAAATG AAGTCAGCCC TCAGAAATGA AGTTGACTGC CTGCTGGCTT TCTGTTGACT GGCCCGGAGC TGTACTGCAA GACCCTTGTG AGCTTCCCTA GTCTAAGAGT AGGATGTCTG CTGAAGTCAT CCATCAGGTT GAAGAAGCAC TTGATACAGA TGAGAAGGAG ATGCTGCTCT TTTTGTGCCG GGATGTTGCT ATAGATGTGG TTCCACCTAA TGTCAGGGAC CTTCTGGATA TTTTACGGGA AAGAGGTAAG CTGTCTGTCG GGGACTTGGC TGAACTGCTC TACAGAGTGA GGCGATTTGA CCTGCTCAAA CGTATCTTGA AGATGGACAG AAAAGCTGTG GAGACCCACC TGCTCAGGAA CCCTCACCTT GTTTCGGACT ATAGAGTGCT GATGGCAGAG ATTGGTGAGG ATTTGGATAA ATCTGATGTG TCCTCATTAA TTTTCCTCAT GAAGGATTAC ATGGGCCGAG GCAAGATAAG CAAGGAGAAG AGTTTCTTGG ACCTTGTGGT TGAGTTGGAG AAACTAAATC TGGTTGCCCC AGATCAACTG GATTTATTAG AAAAATGCCT AAAGAACATC CACAGAATAG ACCTGAAGAC AAAAATCCAG AAGTACAAGC AGTCTGTTCA AGGAGCAGGG ACAAGTTACA GGAATGTTCT CCAAGCAGCA ATCCAAAAGA GTCTCAAGGA TCCTTCAAAT AACTTCAGGC TCCATAATGG GAGAAGTAAA GAACAAAGAC TTAAGGAACA GCTTGGCGCT CAACAAGAAC CAGTGAAGAA ATCCATTCAG GAATCAGAAG CTTTTTTGCC TCAGAGCATA CCTGAAGAGA GATACAAGAT GAAGAGCAAG CCCCTAGGAA TCTGCCTGAT AATCGATTGC ATTGGCAATG AGACAGAGCT TCTTCGAGAC ACCTTCACTT CCCTGGGCTA TGAAGTCCAG AAATTCTTGC ATCTCAGTAT GCATGGTATA TCCCAGATTC TTGGCCAATT TGCCTGTATG CCCGAGCACC GAGACTACGA CAGCTTTGTG TGTGTCCTGG TGAGCCGAGG AGGCTCCCAG AGTGTGTATG GTGTGGATCA GACTCACTCA GGGCTCCCCC TGCATCACAT CAGGAGGATG TTCATGGGAG ATTCATGCCC TTATCTAGCA GGGAAGCCAA AGATGTTTTT TATTCAGAAC TATGTGGTGT CAGAGGGCCA GCTGGAGGAC AGCAGCCTCT TGGAGGTGGA TGGGCCAGCG ATGAAGAATG TGGAATTCAA GGCTCAGAAG CGAGGGCTGT GCACAGTTCA CCGAGAAGCT GACTTCTTCT GGAGCCTGTG TACTGCGGAC ATGTCCCTGC TGGAGCAGTC TCACAGCTCA CCATCCCTGT ACCTGCAGTG CCTCTCCCAG AAACTGAGAC AAGAAAGAAA ACGCCCACTC CTGGATCTTC ACATTGAACT CAATGGCTAC ATGTATGATT GGAACAGCAG AGTTTCTGCC AAGGAGAAAT ATTATGTCTG GCTGCAGCAC ACTCTGAGAA AGAAACTTAT CCTCTCCTAC ACATAAGAAA CCAAAAGGCT GGGCGTAGTG GCTCACACCT GTAATCCCAG CACTTTGGGA GGCCAAGGAG GGCAGATCAC TTCAGGTCAG GAGTTCGAGA CCAGCCTGGC CAACATGGTA AACGCTGTCC CTAGTAAAAA TACAAAAATT AGCTGGGTGT GGGTGTGGGT ACCTGTATTC CCAGTTACTT GGGAGGCTGA GGTGGGAGGA TCTTTTGAAC CCAGGAGTTC AGGGTCATAG CATGCTGTGA TTGTGCCTAC GAATAGCCAC TGCATACCAA CCTGGGCAAT ATAGCAAGAT CCCATCTCTT TAAAAAAAAA AAAAAAGGAC AGGAACTATC TTACTCAATG TATTAGTCAT GTTTCTCTAG AGGGACAGAA CTAATAGGAT ACATGTATAT AAAAAGGGGA GTTTATTAAG GAGTATTGAC TCACATGATC ACAGGGTTAG GTCCCACAAT AGGTCATCTG CAAGCAAGGA AGCCAATTCA AGTCCCAAAG CTGAAGAACT TGGAGTCCAA TGTTTGAGGG CAGGAAGCAT TCAGCATGAG AGAAAGATGG AGGCCAGAAG ACTACACCAG TCTAGTCTTT CCATGTTTTG CCTGCTTTTA TTCTGGCAGT GCTGGCAGCT GATTAGATGG TGCCCACCCA GATTGAGGAT GGTCTGCCTT TCCCAGTCCA CTGACTCAAA TGTTAAATCT CCTTTGGCAG CACCCTCACA GATGTACCCG GGAACACTTT GCATCCTTCT	4

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TATGACAGAA GGAAAGTGTG TGAGAGAGAT CTTAACTACC TAGCAGCTCT AGCTGCCATC TTGAACCATG AAGATACGGG CCACACGTAG GGGTAGCTGG GTAGTGAGCA GCAAGAAGCC TTGTTGGATG AGGGCACGAA GGAGCAGAAT CACTGGAATC ACTGTGTCAG CCCTAATTAC CTACCTCTGG ACTTTTATGT GAGGGGAAAA AAAATTGACA GTTTATATTT ATCTCAACCT AGTTAACCCA AGTGATGCAT TGTTATGAGA TTAAAATGTT TGGAGGCCGG GTGCGGTGGC TCACGCCTAT AATCCCAGCC CTTTGGGAGG CCAAGGCGGG CGGATCACGA GGTCAGGAGA TCAAGACCAT CCTGGCTAAC ATGTAAAACC CCGTCTCTAC TAAAAATACA AAAAATTAGC CAGGCGTTGT GGCGGTCGCC TGTAGTCCCT GCTATTTGGG AGGCCGAGGC AAGAGAACGG CATGAACCTG GGAGGTGGAG CTTGCAGCGA GCTGAGATCT TGCCACTGCA CTCCAGCCTG GGCGACAGTG CGAGACTCTG TCTCAAAAAT AAATAAATAA ATAAATAATA AATAAAATGT TTGGAATGTT GGCTTCATCC CTGGGATGCA AGGCTGGTTC AACATACGCA AATCAAGAAA CATAATTCAT CACATAAACA GAACTAAAGA CAAAAACCAC ATGATTATCT CAATAGATAC AGAAAAGGCC TTCAATAAAA TTCAACGTTG CTTCATGTTA AAAACTCTCA ATAAACTAGG TATTGATGGA AAATATCTCA AAATAATAAC CATTTATGAC AAACCCACAG CCATTATCAT ACTGAATGGG CAAAAGCTGG AAGCATTCCC CTTGAAAACT GGCACAAGAC AGGGATGCCG TCTCACCACT CCTATTTAAC ATAGTATTGG AAGTTCTGGC CAAGAAAATC AGGCAAGAGA AACAAATAAG GGGTATTCAA ATAGGAAAAG AGGAAGTAAA ACTGTGTTTG CAGATGACAT GATACTATAT CTAGAAAACC CCATTATCTC CACCCAAAAG TTCCTTAAGC TGATAAGCAA CTTCAGCAAA GTCTCAGGAT ACAAAATCAA TGTGCAGAAA TCACAAGCAT TCTATACACC AACAATACAC AAGCAGAGAG CCAAATCATG AATGAACTCC CATTCACAGT TGCTAGAAAG AGAATAAAAT ACCTAGGAAT ACAGCTAATA AGATGTGAAG GATCTCTTCA AGGAGAACTA CAAACCACTG CTCAAGGAAA TAAGAGAGGA CACAAATGAA AAAACATTCC ATTCTCGTGG ATAGGAAGAA TCAATATCAT GAAAATGGCC ATACTACCCA AAGTAATTTA TAGGTTCATT GCTATTCCCA TTAAACTACT ATTGACATTC TTCACAGAAT TAGAAAAAAA CTACTTTAAA ATTCAAATGG AACCAAAAAA GAGCCCGTAT AACCAAGACA ACAATAAGCA AAAAGAACAA AGCTGGAAGC ATCACACTAC CCAACTTCAA AGTATACTGC AAGGCTACAG TAGCCAAAAT GGCATGGTAC TGGTACAAAA ACAGACACAT AGACCAATGG AACAGAATAG AGACCAGAGA AAGAAGACCA CACATCTACA GCCATCTGAT CATCGACAAA CCTGACAAAA ACAAGCAATG GGGAAAAGAT TCCCTATTTA ATAAATGGTG CTGGGAAAAC TGGCTAGCCA TATGCAGAAA ATTGAAACTG ACCCCTTCCT TACACCTTAT ACAAAAATTA ACTCAAGATT AAAGACTTAA TGTAAAACCT AAAACTATAA AAACCCTAGA AGAAAATCTA TTTAATACCA TTCAAGACAT AGGCACAAGC AAAGGTTTCA TGACAAAAAC ATCAAAAGCA ATTGCAACAA AAGCAAAAAT TACAAATGGG ATCTAATTAA ACTAAAGAGC TCCTGCACAG CAAAAGAAAC TATCATTAGA GTGAACAGGC AACCTACAGA ATGGGAGAAC ATTTTTGCAA TCTATCCATC TGACAAAGGT CTAATATCCA GAACCTACAA GGAACTTAAA ACAAATTTAC AAGGAAAAAA ACAACCCCAT CAAAAAGTGG ACAAAGGACA TGAACAGACA CTTCTCAAAA GAAGACATTT ATGTGGCCAA CAAACATATA AAAAAAAGCT CAACCTTACT GATCATTAGA GAAATGCAAA GGAGAACCAC AATGAGATAC CATCTCATGC CGGTCAGAAT GGTGATTATT AAAAAGTCAA AAAACAACAG ATGCTGGCGA GGCTGTGGAG AAGTAGGAAC ACTTTTACAT TGTTGGTGGG AATGTAAATT AGTTCAACCG TTGTGGAAGT GTGTGTGGCT ATTCCTCAAA GATCTAGAAC TAGAAATACT ATTTGTCCCA GCAATCCCAT TACTGGGTAT ATACCCAAAG GAATATAAAC CATTTTATTA TAAAGATACA TGCACATTTT TGTTCATTGC AGCACTCTTC ACAATAGCAA AGACACAATA GCAAATGCCC ATCAAAGATA GACTGGATAA AGAAAATGTG GTACATATAC ACCATGGAAT ACTGTGCAGT GCAGCCATTA CAGCTTTTGG TGATACAGTG AATCAGATTT TTCATTAATT CTTTTAATTG GTTATTACTG AACGTGAAAA AGTAATGTTT GTATTGAAAT CTTGAGTCTG GCCATGTTTC TATTTTAAAT TCATAAAGAA TTCTAACAAG AGGAATTCCA AGAATGTCAT AAATGGATGT TTCTCCATGG ATGAAGGAAC TGTTTTATTC ACTTGCTGAT AATTCAGCCT AATCCAGTTT GACATCATAT AGATAAGTAG TTGAATTATG GATTTAAAAT ACATATCATT TTCTAACTCC AAAGGTAATA CTTATTTAAA TGGTTTTGAA AATATAGAAA GGCACAATTT CTTTTTAAAT CTGTTATTCT	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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CFLAR-AS1	NR_040030.1	CATGCATACA GCCCAGTCTA ACGTACAGGA TCAAAAAGAG ACCTTATTTC GGCTGGCAAT TACCTTACCC GCTTGTTCAT TATCTGACTC CTGCACTCCT CCCAAGTGTT CAATGTTAAC CAGATGGCAA AGTTAACCAG ACAGAGGAAA TCCACATAAT TACTCTAGAT GAAGCCCCCT AACTGCGGAT TAGAAGGAAA AGAGATGAAG GGTGCTGCAC TAATAATAAC AACCATTTTC AACCAAGGCA GGATTCTGGG CCTAGAATCT TCAGTGACAA ATTACCAGGA GAGGAGGAGG GAAATGAAGA GCTGGAGGCC TAGGGAGATG GTGTGAGGAG AGGAGACTGT GCCTTTCCAC TTCTTAATAA TTAATGTTCT CCTGCTGCTC TGAGTCTGCA TCACCAGTGA TTCAGTTTGA GTCAGACATC CAACAACAAC TGAGCAAAAG AATCATCTCA GATCCAAGCA ATGGTGACGC GCAGATAGAA GAGACTGAGG CCAACGACCA AAGCAAAGGA CAGCAGAACT GACTGACACA GCTCAGAAAA TATCCAAAGC TCTGTATGCT CTGGAAAGGA AGACAAGAAA TGCTAAAGAC TCTGGATATT GCTCTGTTTC TATAAGGTCC CACCCATATA CCATCTCTTC TTTTGGATTT CCTTTGGCAG GGTTCCCTTG TATGGAAGTT CTCAGGAGAA CAACACTCTC AGAGTCCCAC TGTCTCAAAG AATTCTTAAT GGACAGCTAT ACTTGGCCCC AAGGTTTCAA CCTCACACTG AAGAAAGTGA CATAAGAAAA CACAAGTCCC AGTGCTATCA GGGGCTATCT GGAAAAGCTG	5

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CTAGACTCAC CCTGAAGTTA TTTGAAGGAT CCTTGAGACT	
		CTTTTGGATT GCTGCTTGGA GAACATTCCT GTAACTTGTC	
		CCTGCTCCTT GAACTATAAA AGCAGAAGGG AGGTACTGAG	
		TTAAAACTTA CAAATTATTA TCAAATAAAT CTTCAATAGT G	
CHP1	NM_007236.4	ACCACCCCTG GGTTCCCTCC CGGGTCCGCA GTGGAAACAC	6
		TGCCCTCTCC CTTCTTGACC CCTAGCCCTT CCTTCCCTCC	
		CTCCTTCCCT CCTGTCGCCG TCTCTTCTGG CGCCGCTGCT	
		CCCGGAGGAG CTCCCGGCAC GGCGATGGGT TCTCGGGCCT	
		CCACGTTACT GCGGGACGAA GAGCTCGAGG AGATCAAGAA	
		GGAGACCGGC TTTTCCCACA GTCAAATCAC TCGCCTCTAC	
		AGCCGGTTCA CCAGCCTGGA CAAAGGAGAG AATGGGACTC	
		TCAGCCGGGA AGATTTCCAG AGGATTCCAG AACTTGCCAT	
		CAACCCACTG GGGGACCGGA TCATCAATGC CTTCTTTCCA	
		GAGGGAGAGG ACCAGGTAAA CTTCCGTGGA TTCATGCGAA	
		CTTTGGCTCA TTTCCGCCCC ATTGAGGATA ATGAAAAGAG	
		CAAAGATGTG AATGGACCCG AACCACTCAA CAGCCGAAGC	
		AACAAACTGC ACTTTGCTTT TCGACTATAT GATTTGGATA	
		AAGATGAAAA GATCTCCCGT GATGAGCTGT TACAGGTGCT	
		ACGCATGATG GTCGGAGTAA ATATCTCAGA TGAGCAGCTG	
		GGCAGCATCG CAGACAGGAC CATTCAGGAG GCTGATCAGG	
		ATGGGGACAG TGCCATATCT TTCACAGAAT TTGTTAAGGT	
		TTTGGAGAAG GTGGATGTAG AACAGAAAAT GAGCATCCGA	
		TTTCTTCACT AAAGGAGACC AAACTGTTCC TTGCGGTCTA	
		GTATTTAAGA ACTGGAACTT GAAAGTCCTC CTTCTACCAA	
		CTCCACCTCC ACCCCCTCAT TCCCCTTCTC CCAAAGTACT	
		ACTGCTGTTG CATGACAACC CCAAATATGT TCTGTCAACA	
		CAAACCTGCC TTTGGTGTAT AAACAGGGCA TTACAGAATG	
		GTACACCCTA TATATTTCTG TTCAGTATCC ATTCACTAGT	
		TCTTCATTTA TAAATATCAT CTTCCCCATT CTGCTGCTGA	
		ATGCCACACA TCCATCCAGT CTGAGAAAGT GAGAGAGGCA	
		ATCATGCCAA GAACAAGCCA GCAAAGCTCT TTCACCAGAT	
		GTAGACTGTA GCCCTGCTGC CTTCCCTCCA GCGAGTCTGC	
		CAGCATGCTT CTTCATCCTT TTTATATGTT CTTTGCTTCC	
		TACTTCCCTG TCTTCCAACA TACTGTTCAC TTACTCTGGC	
		AGTCTTTCTG CTTTTCATTA AGCCTCAAAA TCTCCTCTGT	
		TCTACTTGGC ACCACAAGCT ATGTCCTATA TATGTATTTC	
		TGACTTGGCA GGATAGTTCA GGGGTCTGGC AGTTTTTATT	
		TACCTTCATT ATTAAATGGG CCTCTGGGAT GTTGCCTCTT	
		CAGGAGCTTT TTGGTAATCA ATACTTCTCT CAGAAGTATG	
		AGACCATCCT CTGCACTCTG CTCTGTCATC AAAGGCTGCT	
		GGGTGGAGAT ACCCTTTTTG AAAGGTGGCC TTGGTGAGAG	
		GTATGGAGCC AAGTCTTCTA GGTTGCTTGC CCACATCACT	
		CTATCTCTGG CCTCTGATTC TCAACTTTGT ACCTGTGTGG	
		CTCCTCTTGT TAGTGCAATG TTGACTGTTG AAAAAGCAGC	
		AGTATGCTTA CAGGTTTGCT TAGTTTGGGG ACACCGTTAC	
		CACCAGAATG GCTGCTCTGA CAATATGCCT AGGGACTTTC	
		TCATGGCTTT TATTTAATAA GGAGGCTGGG CACCCTATAA	
		AGCCTCATGC ATTCACACCT TTGCAGCATG GTTTATGCCT	
		CAGTGTTATG TGCACTGGAA TGTTTTCCAC TTCACATTTC	
		CAAGTAGAAA TATTAGTGTT ACGGAAGTGC CTAATATCCC	
		AGTCCAAATT TTTTTTTTTT TTTTTTTTTT TTTTTGAGAC	
		AGAGTCTTGC TCTGTCACCC AGGCTGGAGT GCAGTGGTGC	
		GATCGCTCAC TGCAACCTCA GCCTCCTGGA TTTAAGTGAT	
		TCTCCTGCCT CAGCCTCCCA AGTAGCTGGG ATTACAGGTG	
		TGCACCACCA TGCCCGGCTA ATTTTTTGTA TTTTTAGTGG	
		AGACAGGGTT TCACCATGTT GGCCAGGCTG GTCTCGAACT	
		CCTGACCTCG TGATCCGCCT GCCTCAGCCT CCCAAAGTGC	
		TGGGATTACA GGTGTGAGCC ACCACGCCTG GCCCCAGTCC	
		AAAATATTTA AAGATTGTTT CCTTAGTGTC TTGAAGTTTT	
		GCACAAAATT CTTTTTTTTG AGATGGAGTC TCACTCTGTC	
		ACCCAGGCTG GAGTGCAGTG GCGTGATCTT GGCTCACTGC	
		AACCTCTGCC TCCTGGGTTC AAGCAATTCT CCCACCTCAG	
		CCTCCCAAGT AGCTGGGATT ACAGACGTGT GCCACCATAC	
		CTGGGTAATT TTTGCATTTT TAGTGGAGAG GGAGTTTCAC	
		CATGTTGGCC AGGTTGGTCT TGAACTCCTG ACCTCAGGTG	
		ATCCTCCTGC CTCGGCCTCC CAAAGTGCTG GGATTACAGG	
		CATGAGCCAC CGTGCTCAGC CGCAAAATTC TTTATGAATT	
		TTACACTTGG CAAATGTTAA TGACGGAAGC CATAGTCTGC	
		TCCTAATACA TGTCCAAAGC ATTGACTGTT GTGTCATTAG	
		CTGCCTGGTT ACATTAGCTC CCTGGCTTCT TGTTTAGACC	
		ACTGCTAATC CCTTAAAAAC AAGAGGTCTG GCACTAGTAG	
		CACAACCTAA GGTGGCATTA CAGATCTTTG AGCGAGCCAC	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AGCAACTTTT CTGCCAAGTC AGCTTAGTTT AGACTTCAGT GAATCAGGCT ATTGCTATCC TAATGTATGT CTCTATGAGT GTATTTAGCC ACACATCTGC CCTTGGTTGA CTTTCTGACT CATTGCTTGC TTGCTTGTTT CCTTGCTTTG GAAAACTATT GAAGATTGCT AAAAAATACC ACTGCAAAGT GATGGAAAAG GGTGGAGAAC AGGGGAGTAG CCAGGCTGGA TGGCTCAAAT ATAAATGAAT GAGGAATTCT TTATGAAGTA TCAGTCAGAT TTTATGATTA AGTGATGTAA TATAGGAATT ATGTAAAAGG GAAGAATGTC TGATACTGAT CTATTAGAGA GGTACTTTAG AGGCTTCTTG ATTGGCATAA AGTTCCTAAG GTTATAGATT TTCCCCCCTT TTGGCTGTAT AGCAAAGTGT TTTAATCCAC GGTTGTGCCT TATTGTTCCA TTAAAATTGT ATCTTCGATC CATCAATAAA TACTTGTGGT TGAAACAAAA	
DDX55	NM_020936.2	AGCGGAAGTG CTCGTTGGGG GTGCACAAGG CGCGTTCGAG CAGCGGCGAC CGACGCGGCG AAGGAGCGCG CCATGGAGCA TGTGACAGAG GGCTCCTGGG AGTCGCTGCC TGTGCCGCTG CACCCGCAGG TGCTGGGCGC GCTGCGGGAG CTGGGCTTCC CGTACATGAC GCCGGTGCAG TCCGCAACCA TCCCTCTGTT CATGCGAAAC AAAGATGTCG CTGCAGAAGC GGTCACAGGT AGTGGCAAAA CACTCGCTTT TGTCATCCCC ATCCTGGAAA TTCTTCTGAG AAGAGAAGAG AAGTTAAAAA AGAGTCAGGT TGGAGCCATA ATCATCACCC CCACTCGAGA GCTGGCCATT CAAATAGACG AGGTCCTGTC GCATTTCACG AAGCACTTCC CCGAGTTCAG CCAGATTCTT TGGATCGGAG GCAGGAATCC TGGAGAAGAT GTTGAGAGGT TTAAGCAACA AGGTGGGAAC ATCATTGTGG CCACTCCAGG CCGCTTGGAG GACATGTTCC GGAGGAAGGC CGAAGGCTTG GATCTGGCCA GCTGTGTGCG ATCCCTGGAT GTCCTGGTGT TGGATGAGGC AGACAGACTT CTGGACATGG GGTTTGAGGC AAGCATCAAC ACCATTCTGG AGTTTTTGCC AAAGCAGAGG AGAACAGGCC TTTTCTCTGC CACTCAGACG CAGGAAGTGG AGAACCTGGT GAGAGCGGGC CTCCGGAACC CTGTCCGGGT CTCAGTGAAG GAGAAGGGCG TGGCAGCCAG CAGTGCCCAG AAGACCCCCT CCCGCCTGGA AAACTACTAC ATGGTATGCA AGGCAGATGA GAAATTTAAT CAGCTGGTCC ATTTTCTTCG CAATCATAAG CAGGAGAAAC ACCTGGTCTT CTTCAGCACC TGTGCCTGTG TGGAATACTA TGGGAAGGCT CTGGAAGTGC TGGTGAAGGG CGTGAAGATT ATGTGCATTC ACGGAAAGAT GAAATATAAA CGCAATAAGA TCTTCATGGA GTTCCGCAAA TTGCAAAGTG GGATTTTAGT GTGCACTGAT GTGATGGCCC GGGGAATTGA TATTCCTGAA GTCAACTGGG TTTTGCAGTA TGACCCTCCC AGCAATGCAA GTGCCTTCGT GCATCGCTGC GGTCGCACAG CTCGCATTGG CCACGGGGGC AGCGCTCTGG TGTTCCTCCT GCCCATGGAA GAGTCATACA TCAATTTCCT TGCAATTAAC CAAAAATGCC CCCTGCAGGA GATGAAGCCC CAGAGAAACA CAGCGGACCT TCTGCCAAAA CTCAAGTCCA TGGCCCTGGC TGACAGAGCT GTGTTTGAAA AGGGCATGAA AGCTTTTGTG TCATATGTCC AAGCTTATGC AAAGCATGAA TGCAACCTGA TTTTCAGATT AAAGGATCTT GATTTTGCCA GCCTTGCTCG AGGTTTTGCC CTGCTGAGGA TGCCCAAGAT GCCAGAATTG AGAGGAAAGC AGTTTCCAGA TTTTGTGCCC GTGGACGTTA ATACCGACAC GATTCCATTT AAAGATAAAA TCAGAGAAAA GCAGAGGCAG AAACTCCTGG AGCAACAAAG AAGAGAGAAA ACAGAAAATG AAGGGAGAAG AAAATTCATA AAAAATAAAG CTTGGTCAAA GCAGAAGGCC AAAAAAGAAA AGAAGAAAAA AATGAATGAG AAAAGGAAAA GGGAAGAGGG TTCTGATATT GAAGATGAGG ACATGGAAGA ACTTCTTAAT GACACAAGAC TCTTGAAAAA ACTTAAGAAA GGCAAAATAA CTGAAGAAGA ATTTGAGAAG GGCTTGTTGA CAACTGGCAA AAGAACAATC AAGACAGTGG ATTTAGGGAT CTCAGATTTG GAAGATGACT GCTGATTCCA GTGCCACAGA TGAACCCACA AGGACATAGC TGTTCCCTAA CTTGGTGGAT GGCTCCAGTT TGCTTTTAAC GAAAATCACA ACTTCAGGAG ACATCTGAAA AGAATGATGT CTCTGAAAGC TGTCCTTTCA GATGAGGGAG AAATGAAGGA TTTCACACTT CAGAATATTT TACTAAAAAC ATTCCAGTCT TGGCCGGGTG CGGTGGCTCC TGCCTATAAT CCCAGCACTT TGGGAGGCTG AGGCAGGAGG ATCACTTGAG CCCAGGAGTT CAAGACCAGC CTGGGAACAC AGCGAGACCC TCTCATTAAA AACAACAAAA CAAAACAATT CCAGTCTTGG AGTAGTCTAA CAGAAGAAAA TGTAAAATTA TTTGAGTGTA AATAATAGAT GTCAGTATTT ATCATGATGG GTCACATATA GACATATGTA CATATTATAT ATATATATAT ATATATATAT ATATATATAT ATATATATAT ATAAGCTCTT TTTTCTGAGG CTATTTTATA GTTATTTTTA	7

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CAGGGGTGGC TAATGTGCTG GGGTTTTTCT GTGTTAATAG	
		TCACAGTATT GTTTTATTGG TGAATAGCTG AAAAACAGAG	
		GGATTAAGTC ATATTCCGGG AAAGAGAATT ATAGTTTTTA	
		TGCCTCCTGT TGAATAAATG GTGTCCTGAT TGCCTGGG	
DMD	NM_000109.3	CGTTAAATGC AAACGCTGCT CTGGCTCATG TGTTTGCTCC	8
		GAGGTATAGG TTTTGTTCGA CTGACGTATC AGATAGTCAG	
		AGTGGTTACC ACACCGACGT TGTAGCAGCT GCATAATAAA	
		TGACTGAAAG AATCATGTTA GGCATGCCCA CCTAACCTAA	
		CTTGAATCAT GCGAAAGGGG AGCTGTTGGA ATTCAAATAG	
		ACTTTCTGGT TCCCAGCAGT CGGCAGTAAT AGAATGCTTT	
		CAGGAAGATG ACAGAATCAG GAGAAAGATG CTGTTTTGCA	
		CTATCTTGAT TTGTTACAGC AGCCAACTTA TTGGCATGAT	
		GGAGTGACAG GAAAAACAGC TGGCATGGAA GATGAAAGAG	
		AAGATGTTCA AAAGAAAACA TTCACAAAAT GGGTAAATGC	
		ACAATTTTCT AAGTTTGGGA AGCAGCATAT TGAGAACCTC	
		TTCAGTGACC TACAGGATGG GAGGCGCCTC CTAGACCTCC	
		TCGAAGGCCT GACAGGGCAA AAACTGCCAA AAGAAAAAGG	
		ATCCACAAGA GTTCATGCCC TGAACAATGT CAACAAGGCA	
		CTGCGGGTTT TGCAGAACAA TAATGTTGAT TTAGTGAATA	
		TTGGAAGTAC TGACATCGTA GATGGAAATC ATAAACTGAC	
		TCTTGGTTTG ATTTGGAATA TAATCCTCCA CTGGCAGGTC	
		AAAAATGTAA TGAAAAATAT CATGGCTGGA TTGCAACAAA	
		CCAACAGTGA AAAGATTCTC CTGAGCTGGG TCCGACAATC	
		AACTCGTAAT TATCCACAGG TTAATGTAAT CAACTTCACC	
		ACCAGCTGGT CTGATGGCCT GGCTTTGAAT GCTCTCATCC	
		ATAGTCATAG GCCAGACCTA TTTGACTGGA ATAGTGTGGT	
		TTGCCAGCAG TCAGCCACAC AACGACTGGA ACATGCATTC	
		AACATCGCCA GATATCAATT AGGCATAGAG AAACTACTCG	
		ATCCTGAAGA TGTTGATACC ACCTATCCAG ATAAGAAGTC	
		CATCTTAATG TACATCACAT CACTCTTCCA AGTTTTGCCT	
		CAACAAGTGA GCATTGAAGC CATCCAGGAA GTGGAAATGT	
		TGCCAAGGCC ACCTAAAGTG ACTAAAGAAG AACATTTTCA	
		GTTACATCAT CAAATGCACT ATTCTCAACA GATCACGGTC	
		AGTCTAGCAC AGGGATATGA GAGAACTTCT TCCCCTAAGC	
		CTCGATTCAA GAGCTATGCC TACACACAGG CTGCTTATGT	
		CACCACCTCT GACCCTACAC GGAGCCCATT TCCTTCACAG	
		CATTTGGAAG CTCCTGAAGA CAAGTCATTT GGCAGTTCAT	
		TGATGGAGAG TGAAGTAAAC CTGGACCGTT ATCAAACAGC	
		TTTAGAAGAA GTATTATCGT GGCTTCTTTC TGCTGAGGAC	
		ACATTGCAAG CACAAGGAGA GATTTCTAAT GATGTGGAAG	
		TGGTGAAAGA CCAGTTTCAT ACTCATGAGG GGTACATGAT	
		GGATTTGACA GCCCATCAGG GCCGGGTTGG TAATATTCTA	
		CAATTGGGAA GTAAGCTGAT TGGAACAGGA AAATTATCAG	
		AAGATGAAGA AACTGAAGTA CAAGAGCAGA TGAATCTCCT	
		AAATTCAAGA TGGGAATGCC TCAGGGTAGC TAGCATGGAA	
		AAACAAAGCA ATTTACATAG AGTTTTAATG GATCTCCAGA	
		ATCAGAAACT GAAAGAGTTG AATGACTGGC TAACAAAAAC	
		AGAAGAAAGA ACAAGGAAAA TGGAGGAAGA GCCTCTTGGA	
		CCTGATCTTG AAGACCTAAA ACGCCAAGTA CAACAACATA	
		AGGTGCTTCA AGAAGATCTA GAACAAGAAC AAGTCAGGGT	
		CAATTCTCTC ACTCACATGG TGGTGGTAGT TGATGAATCT	
		AGTGGAGATC ACGCAACTGC TGCTTTGGAA GAACAACTTA	
		AGGTATTGGG AGATCGATGG GCAAACATCT GTAGATGGAC	
		AGAAGACCGC TGGGTTCTTT TACAAGACAT CCTTCTCAAA	
		TGGCAACGTC TTACTGAAGA ACAGTGCCTT TTTAGTGCAT	
		GGCTTTCAGA AAAAGAAGAT GCAGTGAACA AGATTCACAC	
		AACTGGCTTT AAAGATCAAA ATGAAATGTT ATCAAGTCTT	
		CAAAAACTGG CCGTTTTAAA AGCGGATCTA GAAAAGAAAA	
		AGCAATCCAT GGGCAAACTG TATTCACTCA AACAAGATCT	
		TCTTTCAACA CTGAAGAATA AGTCAGTGAC CCAGAAGACG	
		GAAGCATGGC TGGATAACTT TGCCCGGTGT TGGGATAATT	
		TAGTCCAAAA ACTTGAAAAG AGTACAGCAC AGATTTCACA	
		GGCTGTCACC ACCACTCAGC CATCACTAAC ACAGACAACT	
		GTAATGGAAA CAGTAACTAC GGTGACCACA AGGGAACAGA	
		TCCTGGTAAA GCATGCTCAA GAGGAACTTC CACCACCACC	
		TCCCCAAAAG AAGAGGCAGA TTACTGTGGA TTCTGAAATT	
		AGGAAAAGGT TGGATGTTGA TATAACTGAA CTTCACAGCT	
		GGATTACTCG CTCAGAAGCT GTGTTGCAGA GTCCTGAATT	
		TGCAATCTTT CGGAAGGAAG GCAACTTCTC AGACTTAAAA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		ATCAAACAAG CCTCAGAACA ACTGAACAGC CGGTGGATCG	
		AATTCTGCCA GTTGCTAAGT GAGAGACTTA ACTGGCTGGA	
		GTATCAGAAC AACATCATCG CTTTCTATAA TCAGCTACAA	
		CAATTGGAGC AGATGACAAC TACTGCTGAA AACTGGTTGA	
		AAATCCAACC CACCACCCCA TCAGAGCCAA CAGCAATTAA	
		AAGTCAGTTA AAAATTTGTA AGGATGAAGT CAACCGGCTA	
		TCAGGTCTTC AACCTCAAAT TGAACGATTA AAAATTCAAA	
		GCATAGCCCT GAAAGAGAAA GGACAAGGAC CCATGTTCCT	
		GGATGCAGAC TTTGTGGCCT TTACAAATCA TTTTAAGCAA	
		GTCTTTTCTG ATGTGCAGGC CAGAGAGAAA GAGCTACAGA	
		CAATTTTTGA CACTTTGCCA CCAATGCGCT ATCAGGAGAC	
		CATGAGTGCC ATCAGGACAT GGGTCCAGCA GTCAGAAACC	
		AAACTCTCCA TACCTCAACT TAGTGTCACC GACTATGAAA	
		TCATGGAGCA GAGACTCGGG GAATTGCAGG CTTTACAAAG	
		TTCTCTGCAA GAGCAACAAA GTGGCCTATA CTATCTCAGC	
		ACCACTGTGA AAGAGATGTC GAAGAAAGCG CCCTCTGAAA	
		TTAGCCGGAA ATATCAATCA GAATTTGAAG AAATTGAGGG	
		ACGCTGGAAG AAGCTCTCCT CCCAGCTGGT TGAGCATTGT	
		CAAAAGCTAG AGGAGCAAAT GAATAAACTC CGAAAAATTC	
		AGAATCACAT ACAAACCCTG AAGAAATGGA TGGCTGAAGT	
		TGATGTTTTT CTGAAGGAGG AATGGCCTGC CCTTGGGGAT	
		TCAGAAATTC TAAAAAAGCA GCTGAAACAG TGCAGACTTT	
		TAGTCAGTGA TATTCAGACA ATTCAGCCCA GTCTAAACAG	
		TGTCAATGAA GGTGGGCAGA AGATAAAGAA TGAAGCAGAG	
		CCAGAGTTTG CTTCGAGACT TGAGACAGAA CTCAAAGAAC	
		TTAACACTCA GTGGGATCAC ATGTGCCAAC AGGTCTATGC	
		CAGAAAGGAG GCCTTGAAGG GAGGTTTGGA GAAAACTGTA	
		AGCCTCCAGA AAGATCTATC AGAGATGCAC GAATGGATGA	
		CACAAGCTGA AGAAGAGTAT CTTGAGAGAG ATTTTGAATA	
		TAAAACTCCA GATGAATTAC AGAAAGCAGT TGAAGAGATG	
		AAGAGAGCTA AAGAAGAGGC CCAACAAAAA GAAGCGAAAG	
		TGAAACTCCT TACTGAGTCT GTAAATAGTG TCATAGCTCA	
		AGCTCCACCT GTAGCACAAG AGGCCTTAAA AAAGGAACTT	
		GAAACTCTAA CCACCAACTA CCAGTGGCTC TGCACTAGGC	
		TGAATGGGAA ATGCAAGACT TTGGAAGAAG TTTGGGCATG	
		TTGGCATGAG TTATTGTCAT ACTTGGAGAA AGCAAACAAG	
		TGGCTAAATG AAGTAGAATT TAAACTTAAA ACCACTGAAA	
		ACATTCCTGG CGGAGCTGAG GAAATCTCTG AGGTGCTAGA	
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		AATCAGATTC GCATATTGGC ACAGACCCTA ACAGATGGCG	
		GAGTCATGGA TGAGCTAATC AATGAGGAAC TTGAGACATT	
		TAATTCTCGT TGGAGGGAAC TACATGAAGA GGCTGTAAGG	
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		ATTCATTGAC AAGCAGTTGG CAGCTTATAT TGCAGACAAG	
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		GAAGAAACAT AATCAGGGGA AGGAGGCTGC CCAAAGAGTC	
		CTGTCTCAGA TTGATGTTGC ACAGAAAAAA TTACAAGATG	
		TCTCCATGAA GTTTCGATTA TTCCAGAAAC CAGCCAATTT	
		TGAGCAGCGT CTACAAGAAA GTAAGATGAT TTTAGATGAA	
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		GGCAGCTACA GATATGGAAT TGACAAAGAG ATCAGCAGTT	
		GAAGGAATGC CTAGTAATTT GGATTCTGAA GTTGCCTGGG	
		GAAAGGCTAC TCAAAAAGAG ATTGAGAAAC AGAAGGTGCA	
		CCTGAAGAGT ATCACAGAGG TAGGAGAGGC CTTGAAAACA	
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		GTCTTCTGAA TAGTAACTGG ATAGCTGTCA CCTCCCGAGC	
		AGAAGAGTGG TTAAATCTTT TGTTGGAATA CCAGAAACAC	
		ATGGAAACTT TTGACCAGAA TGTGGACCAC ATCACAAAGT	
		GGATCATTCA GGCTGACACA CTTTTGGATG AATCAGAGAA	
		AAAGAAACCC CAGCAAAAAG AAGACGTGCT TAAGCGTTTA	
		AAGGCAGAAC TGAATGACAT ACGCCCAAAG GTGGACTCTA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		AAAATCCTGA GATCCCTGGA AGGTTCCGAT GATGCAGTCC	
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		TGAACTTCGG AAAAAGTCTC TCAACATTAG GTCCCATTTG	
		GAAGCCAGTT CTGACCAGTG GAAGCGTCTG CACCTTTCTC	
		TGCAGGAACT TCTGGTGTGG CTACAGCTGA AAGATGATGA	
		ATTAAGCCGG CAGGCACCTA TTGGAGGCGA CTTTCCAGCA	
		GTTCAGAAGC AGAACGATGT ACATAGGGCC TTCAAGAGGG	
		AATTGAAAAC TAAAGAACCT GTAATCATGA GTACTCTTGA	
		GACTGTACGA ATATTTCTGA CAGAGCAGCC TTTGGAAGGA	
		CTAGAGAAAC TCTACCAGGA GCCCAGAGAG CTGCCTCCTG	
		AGGAGAGAGC CCAGAATGTC ACTCGGCTTC TACGAAAGCA	
		GGCTGAGGAG GTCAATACTG AGTGGGAAAA ATTGAACCTG	
		CACTCCGCTG ACTGGCAGAG AAAAATAGAT GAGACCCTTG	
		AAAGACTCCA GGAACTTCAA GAGGCCACGG ATGAGCTGGA	
		CCTCAAGCTG CGCCAAGCTG AGGTGATCAA GGGATCCTGG	
		CAGCCCGTGG GCGATCTCCT CATTGACTCT CTCCAAGATC	
		ACCTCGAGAA AGTCAAGGCA CTTCGAGGAG AAATTGCGCC	
		TCTGAAAGAG AACGTGAGCC ACGTCAATGA CCTTGCTCGC	
		CAGCTTACCA CTTTGGGCAT TCAGCTCTCA CCGTATAACC	
		TCAGCACTCT GGAAGACCTG AACACCAGAT GGAAGCTTCT	
		GCAGGTGGCC GTCGAGGACC GAGTCAGGCA GCTGCATGAA	
		GCCCACAGGG ACTTTGGTCC AGCATCTCAG CACTTTCTTT	
		CCACGTCTGT CCAGGGTCCC TGGGAGAGAG CCATCTCGCC	
		AAACAAAGTG CCCTACTATA TCAACCACGA GACTCAAACA	
		ACTTGCTGGG ACCATCCCAA AATGACAGAG CTCTACCAGT	
		CTTTAGCTGA CCTGAATAAT GTCAGATTCT CAGCTTATAG	
		GACTGCCATG AAACTCCGAA GACTGCAGAA GGCCCTTTGC	
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		CCTGCAGATT ATTAATTGTT TGACCACTAT TTATGACCGC	
		CTGGAGCAAG AGCACAACAA TTTGGTCAAC GTCCCTCTCT	
		GCGTGGATAT GTGTCTGAAC TGGCTGCTGA ATGTTTATGA	
		TACGGGACGA ACAGGGAGGA TCCGTGTCCT GTCTTTTAAA	
		ACTGGCATCA TTTCCCTGTG TAAAGCACAT TTGGAAGACA	
		AGTACAGATA CCTTTTCAAG CAAGTGGCAA GTTCAACAGG	
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		TCTATCCAAA TTCCAAGACA GTTGGGTGAA GTTGCATCCT	
		TTGGGGGCAG TAACATTGAG CCAAGTGTCC GGAGCTGCTT	
		CCAATTTGCT AATAATAAGC CAGAGATCGA AGCGGCCCTC	
		TTCCTAGACT GGATGAGACT GGAACCCCAG TCCATGGTGT	
		GGCTGCCCGT CCTGCACAGA GTGGCTGCTG CAGAAACTGC	
		CAAGCATCAG GCCAAATGTA ACATCTGCAA AGAGTGTCCA	
		ATCATTGGAT TCAGGTACAG GAGTCTAAAG CACTTTAATT	
		ATGACATCTG CCAAAGCTGC TTTTTTTCTG GTCGAGTTGC	
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		GAAGCATCCC CGAATGGGCT ACCTGCCAGT GCAGACTGTC	
		TTAGAGGGGG ACAACATGGA AACTCCCGTT ACTCTGATCA	
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		ATCTAAATGA TAGCATCTCT CCTAATGAGA GCATAGATGA	
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		CACCTCTCCC CAGAGTCCCC GGGATGCTGA GCTCATTGCT	
		GAGGCCAAGC TACTGCGTCA ACACAAAGGC CGCCTGGAAG	
		CCAGGATGCA AATCCTGGAA GACCACAATA AACAGCTGGA	
		GTCACAGTTA CACAGGCTAA GGCAGCTGCT GGAGCAACCC	
		CAGGCAGAGG CCAAAGTGAA TGGCACAACG GTGTCCTCTC	
		CTTCTACCTC TCTACAGAGG TCCGACAGCA GTCAGCCTAT	
		GCTGCTCCGA GTGGTTGGCA GTCAAACTTC GGACTCCATG	
		GGTGAGGAAG ATCTTCTCAG TCCTCCCCAG GACACAAGCA	
		CAGGGTTAGA GGAGGTGATG GAGCAACTCA ACAACTCCTT	
		CCCTAGTTCA AGAGGAAGAA ATACCCCTGG AAAGCCAATG	
		AGAGAGGACA CAATGTAGGA AGTCTTTTCC ACATGGCAGA	
		TGATTTGGGC AGAGCGATGG AGTCCTTAGT ATCAGTCATG	
		ACAGATGAAG AAGGAGCAGA ATAAATGTTT TACAACTCCT	
		GATTCCCGCA TGGTTTTTAT AATATTCATA CAACAAAGAG	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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DNAJC9	NM_015190.4	GGATGCGCGG CGTGGCCACG CCCCTTCAGT GCTTGTGACG CAGGCGCCCT GGGCTTTTTG GGCGCGAAAA AGAAGCAGTC CTGGGTTGTA CCCGGCGCAG CTGGGAGCGG CTGCTTCCTC CGGGGTCGTA TCTCCGCCCG GCATGGGGCT GCTGGACCTT TGCGAGGAAG TGTTCGGCAC CGCCGACCTT TACCGGGTGC TGGGCGTGCG ACGCGAGGCC TCCGACGGCG AGGTCCGACG AGGCTACCAC AAGGTGTCCC TGCAGGTACA CCCGGACCGG GTGGGTGAGG GCGACAAGGA GGACGCCACC CGCCGCTTCC AGATCCTGGG AAAAGTCTAT TCCGTTCTCA GTGACAGAGA	9

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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FANCL	NM_001114636.1	AGCGGACTGC GCATGTGCAG GACCCAGCAG GTCTAGAGCT TTTCTGTGTT TCTCCGGACT TCGAGCCATG GCGGTGACGG AAGCGAGCCT GTTGCGCCAG TGCCCCCTGC TTCTGCCCCA GAACCGGTCG AAAACCGTGT ATGAGGGATT CATCTCGGCT CAGGGAAGAG ACTTCCACCT TAGGATAGTG TTGCCTGAAG ATTTACAACT GAAGAATGCA AGATTATTAT GTAGTTGGCA GCTGAGAACA ATACTTAGTG GATACCATCG AATAGTACAA CAGAGAATGC AGCACTCTCC TGATCTAATG AGCTTTATGA TGGAGTTGAA GATGCTTTTG GAAGTTGCCT TAAAGAATAG ACAAGAGCTG TATGCACTAC CTCCTCCTCC CCAGTTCTAC TCAAGCCTTA TTGAAGAGAT AGGAACTCTT GGTTGGGATA AACTTGTGTA TGCGGATACC TGCTTCAGTA CCATCAAGTT AAAAGCAGAA GATGCTTCTG GTAGAGAGCA TTTAATCACT CTCAAGTTGA AGGCAAAGTA TCCTGCAGAA TCACCAGATT ATTTTGTGGA TTTTCCTGTT CCATTTTGTG CCTCCTGGAC ACCTCAGGTA AATTCTCCTC AGAGCTCCTT AATAAGCATT TATAGTCAGT TTTTGGCAGC AATAGAATCA CTAAAGGCAT TCTGGGATGT TATGGATGAA ATCGATGAGA AGACCTGGGT ACTTGAGCCA GAAAAACCTC CACGGAGTGC AACAGCACGC AGAATTGCAT TAGGTAATAA TGTTTCCATA AATATAGAGG TAGACCCCAG GCATCCTACT ATGCTTCCTG AGTGCTTCTT TCTTGGAGCT GACCATGTGG TAAAACCCCT GGGAATTAAG CTGAGCAGGA ACATACATTT GTGGGATCCA GAAAATAGTG TGTTACAAAA TTTGAAAGAT GTTTTAGAAA TTGATTTTCC AGCTCGTGCT ATCCTGGAAA AATCTGATTT TACTATGGAT TGTGGAATTT GTTATGCTTA TCAACTTGAC GGTACCATTC CTGATCAAGT GTGTGATAAT TCTCAGTGTG GACAACCTTT CCATCAAATA TGCTTATATG AGTGGCTGAG AGGACTACTA ACTAGTAGAC AGAGTTTTAA CATCATATTT GGTGAATGTC CATATTGTAG TAAGCCAATT ACCTTAAAAA TGTCTGGAAG GAAACACTGA AATAAGAATA CAACATTTCG GTGAAGAGCT GGAAACTTAA AAAATTATCA AAAGGAATTT TGGTATCATC TTCAGAGAAA AAATAAAGCA AGAAATACTA ACATCAAAAG GACAGGTATG ATGATGCGAT AATAATAAAC ATCTGCGTTT GTCTCTTCAC TAAGAGTAAA CTGGGAAATT GTAGGCCAAA GTCCAGTTGA ACTTTCTAAG TCTGTGATCC CCGTGCTGAC TGTGGAAGTG TATTTATACC AAGATGGAGA TCTTGACTTC TTGAATATAT CTGGACTGGT AAAATCTTGA TGAGGCTCAT AAAATGAGTT TGGGAATTGT GTATAGCTGA TTTTTTGTGG GAAACTGTTT ACTTCATTCA AAGGTTCTTG AGACTCTTGA TATTTCTGTC TTCTCCTTGT GCTTTCCTAT GGAAAAAATA CATATATAGT TTAGTTTGTT AGACGTGAGT TATCCAAGTA TTTATTTTGT GTAGTGTGTA AGAATGCTAA ATAAAATGTT ATACAAGATC AAAAAAAAAA AAAAAAAAAA AAA	11
HJURP	NM_018410.4	CTATTTGAGT TTGTGGCGCG CGAGGCCCTG CAGTCCGGGT TGGCGCTTGG GTACTGGCTG GGTCCGATGC TGGGTACGCT GCGCGCCATG GAGGGCGAGG ACGTGGAAGA CGACCAGCTG CTGCAGAAGC TCAGGGCCAG TCGCCGCCGC TTCCAGAGGC GCATGCAGCG GCTGATAGAG AAGTACAACC AGCCCTTCGA GGACACCCCG GTGGTGCAAA TGGCCACGCT GACCTACGAG ACGCCACAGG GATTGAGAAT TTGGGGTGGA AGACTAATAA AGGAAAGAAA CGAAGGAGAG ATCCAGGACT CCTCCATGAA GCCCGCGGAC AGGACAGATG GCTCCGTGCA AGCTGCAGCC TGGGGTCCTG AGCTTCCCTC GCACCGCACA GTCCTGGGAG CCGATTCAAA AAGCGGTGAG GTCGATGCCA CGTCAGACCA GGAAGAGTCA GTTGCTTGGG CCTTAGCACC TGCAGTGCCT CAAAGCCCTT TGAAAAATGA ATTAAGAAGG AAATACTTGA CCCAAGTGGA TATACTGCTA CAAGGTGCAG AGTATTTTGA GTGTGCAGGT AACAGAGCTG GAAGGGATGT ACGTGTGACT CCGCTGCCTT CACTGGCCTC ACCTGCCGTG CCTGCCCCCG GATACTGCAG TCGTATCTCC AGAAAGAGTC CTGGTGACCC AGCGAAACCA GCTTCATCTC CCAGAGAATG GGATCCTTTG	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		ACCATCAGTG ACCTGTACGC AGGGATGCTG CACTCCATGA	
		GCCGGCTGTT GAGCACAAAG CCATCAAGCA TCATCTCCAC	
		CAAAACGTTC ATCATGCAAA ACTGGAACTC CAGGAGGAGG	
		CACAGATATA AGAGCAGGAT GAACAAAACA TATTGCAAAG	
		GAGCCAGACG TTCTCAGAGG AGCTCCAAGG AGAACTTCAT	
		ACCCTGCTCT GAGCCTGTGA AAGGGACAGG GGCATTAAGA	
		GATTGCAAGA ACGTATTAGA TGTTTCTTGC CGTAAGACAG	
		GTTTAAAATT GGAAAAAGCT TTTCTTGAAG TCAACAGACC	
		CCAAATCCAT AAGTTAGATC CAAGTTGGAA GGAGCGCAAA	
		GTGACACCCT CGAAGTATTC TTCCTTGATT TACTTCGACT	
		CCAGTGCAAC ATATAATCTT GATGAGGAAA ATAGATTTAG	
		GACATTAAAA TGGTTAATTT CTCCTGTAAA AATAGTTTCC	
		AGACCAACAA TACGACAGGG CCATGGAGAG AACCGTCAGA	
		GGGAGATTGA AATCCGATTT GATCAGCTTC ATCGGGAATA	
		TTGCCTGAGT CCCAGGAACC AGCCTCGCCG GATGTGCCTC	
		CCGGACTCCT GGGCCATGAA CATGTACAGA GGGGGTCCTG	
		CGAGTCCTGG TGGCCTTCAG GGCTTAGAAA CCCGCAGGCT	
		GAGTTTACCT TCCAGCAAAG CAAAAGCAAA AAGTTTAAGT	
		GAGGCTTTTG AAAACCTAGG CAAAAGATCT CTGGAAGCAG	
		GTAGGTGCCT GCCCAAGAGC GATTCATCTT CATCACTTCC	
		AAAGACCAAC CCCACACACA GCGCAACTCG CCCGCAGCAG	
		ACATCTGACC TTCACGTTCA GGGAAATAGT TCTGGAATAT	
		TTAGAAAGTC AGTGTCACCC AGCAAAACTC TTTCAGTCCC	
		AGATAAAGAA GTGCCAGGCC ACGGAAGGAA TCGTTACGAT	
		GAAATTAAAG AAGAATTTGA CAAGCTTCAT CAAAAGTATT	
		GCCTCAAATC TCCTGGGCAG ATGACAGTGC CTTTATGTAT	
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		CAAACAGAAG GCTTCTTAGG AAAATTAAAT CCAGACCCTC	
		ACTTCCAGGG TTTCCAGAAG TTGCCATCAT CACCCCTGGG	
		GTGCAGAAAA AGTCTACTGG GCTCAACTGC AATTGAGGCT	
		CCTTCATCTA CATGTGTTGC TCGTGCCATC ACGAGGGATG	
		GCACGAGGGA CCATCAGTTC CCTGCAAAAA GACCCAGGCT	
		ATCAGAACCC CAGGGCTCCG GACGCCAGGG CAATTCCCTG	
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		AGAGAACACG TCTTACAGGA TGGAAGAGAA AAGTGATTTC	
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		GTATTTTTGT TTTGTTTTTG ATTCTTGAGA CTGTGAGGAC	
		TTGGTTGACT TCTCTGCCCT TAAAGTAAAT ATTAGTGAAA	
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		CGGTGGCACA ATCTTGGCTC ACTGCAACTT CCGCCTCCCG	
		GGTTCAAGCG ATTCTTCTGC CTCAGCCTCC CGAGTAGCTG	
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		AGTTTTAGTA GAGACGGGGT TTCACCATAT TGACCAGGCT	
		GGTCTCGAGC TCCTGGCCTC GTGATCCATC TGCCTCGGCC	
		TCCCAAAGTG CTGGGATTAT AGGCGTGAGC CACTGCTCCC	
		AGCCGTGTGT TCTTTTTTAA ATTTAGATAT GTCCAGAGAA	
		TCCTCTCTCC TGTTTCCCAT TTCATTCGAG AATATTGTTT	
		GCTTGTGAGA CGTAAGTTCG AGCCCTGCAT GCAATGACCC	
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		GACCCTCCTG AGCCCGCAGG AGGCAGGGGC CACCAAGGCT	
		GACCACATGG GCTCCTACGG ACCCGCCTTC TACCAGTCTT	
		ACGGCGCCTC GGGCCAGTTC ACCCATGAAT TTGATGAGGA	
		ACAGCTGTTC TCTGTGGACC TGAAGAAAAG CGAGGCCGTG	
		TGGCGTCTGC CTGAGTTTGG TGACTTTGCC CGCTTTGACC	
		CGCAGGGCGG GCTGGCCGGC ATCGCCGCAA TCAAAGCCCA	
		TCTGGACATC CTGGTGGAGC GCTCCAACCG CAGCAGAGCC	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		GCTTCTATTC CCAGCCTGAC CATTTGTTCC GCAAGTTCCA	
		CTACCTGCCC TTCGTGCCCT CAGCCGAGGA CGTCTATGAC	
		TGCCAGGTGG AGCACTGGGG CCTGGATGCG CCACTCCTCA	
		GGCATTGGGA GCTCCAGGTG CCTATTCCAC CACCAGATGC	
		CATGGAGACC CTGGTCTGTG CCCTGGGCCT GGCCATCGGC	
		CTGGTGGGCT TCCTCGTGGG CACCGTCCTC ATCATCATGG	
		GCACATATGT GTCCAGTGTC CCCAGGTAAT GATCCTTCTG	
		AGAGAAATGA CTTGTGGGAG ACACCCTGCA GATCCTCATG	
		GGTTTGTGAC AGCCCCTGCG TGCTCAGTGC CCTTTAAGTG	
		CATCCCGCTG TGCTGACTTT GAGTGGGATC AACATCTGTC	
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		CACACACACA TTCTTGCTCT ACCCAAAGCT CTGGCTGGCA	
		GCACTAAATG CTTTGGTGGT GTTTGCACTG TGTCCTTTCC	
		AGGCCTTGGC CAGTTCTTCC AGGGGTGAGG CATGTGGTGC	
		TGGGGATTGG CAGCCATCCT GGGGCCCACA CAGGTGTGTC	
		TTGCTCCATT TGGCCCATTG TGTGTTACTT TGTGAATGAG	
		CCATTTCACA TGGACTTCAT GAAATTTGCC TCCTGAGTTC	
		AGGTTTACCC TGAAAGGGAT GCAGATTATC CTGTTCCTCA	
		CGACCCCCTC AGCTAACAAC AGTTCTGAAG GGTGCTGGGA	
		CAGGACAGGC TCATGGGGAC TCCACTCCTG CCTGGGTTTA	
		CTCTGTATGA AGAGGCCACT GGTATCCTGC CATGATGTTA	
		TCTCCTTTTT CTACTTTTCC CTAGAGTCCC ATGCATGATA	
		AAGAGAGGCC CAAGGCTTGG ATAAGGTGGC CACTTCCCTC	
		AGTGGAGTCA GTCATGTTAG GTAGGAGGTG GTAGAGTCGG	
		TCTGCGAGGT ATCTCGTAAG AGGGGAGGTC CACCTAGACA	
		CACTCTAAAT ATGTGGCCTA GAAGATTTTG GTCTACTTTT	
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		ATACCACATA GTTTATGTCA GGACCAAAAT GAGCAATACA	
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		TACTGAATTA GAATCTCGGG AGTGCAGGGG GCAGCAACAG	
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		AGGCTGGAGT GCCATGGCAT GATCTCAGCT CACTGCAACC	
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		CTGAGTAGCT GGGACTACAG GTGTGCGCCA CTACACTCAG	
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		TTGGCCAGGA TGGCCTCCAT CTCTTGACCT CGTGATCCAC	
		CCACCTTCCC TCCCAAAGTA CTGGAATTAC AGGCATTAGC	
		CACTGTGCCC AGCCGAGGCT GTCATTTTTA ACCGGCTCTG	
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		TGGTAACTGG AACCCAGTGA CGTAATCAGG TGCCATCGGG	
		GGTCATGGGA AAGGGGGATC CCCAAGGTCT GAGGTGGACT	
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		GCCAATCAAG GTACAAGTAA ATAGAGGCAA AATGAGGGTT	
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		ATTTTAGCCA GGCCTCCCAG GAACAGAACT CGGATCCGAA	
		AAGCCCAGAG AAGCTAGAGC TGCCCCTCCA ACACTCTCGG	
		ATCCACATGG TCTGTGTTCT CTAGACCCCC CTGCATGTTA	
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		ACCCTCACAA CCCACAGAGG CTGTGTCTCC TAGTCACAGC	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TTATCTGTTG TAGCTTTGTC TTATTCTTTT TCTTTTCATT ACATCAGGTA TATTGCCCTG TAAATTGTGG TAGTGGTACC AGGAATAAAA AATTAAGGAA TTTTTAACTT TTCAATATCT GTGTAGTTCA GTTTTTCTAC ATTTTAGTAC AGAAACTTTA ACAAAATGCA GTTTTGAAGG TGTTTCCTTG TGAGTTAACA AGTAAAGAAG ATCAATTGTT AATTACTATT TTGTATAAAT TTTGCTAAAG TTAACTGTAA AGAAACACCT GCTGACTTGC AGTTTAAGGG GAATCTATTC TCCCCATTTC CAAACCATGA TATGAATGGA CGCCGACATG TGGAGAGAAC AGATAATTTG TGTGTTTGCA ATGTGTGTTT TAGGTAAATA GGATTGGGTA TTTAAATTAG CATTTGTGAA TTTAATAGCA TTAAGATTAC CTTCAAATAA AAAAGTCTCA AAATTTCTTT TTGGTTTTTG TGCACTTTCT TTTAAAATGT AATCACATGA TTTTAGTGTG TTAGACTTGC TGAGTCCTAG CTGTGTTTAG AACATCTCCA TTCTACATTT ACCTTGGTCA AATTTGAACT GCTGCCATAG GTTTTGGGTG TAAAGAATGT TTACTGCCCT CCATTTAAAT TCTGAAAAGG TATAGTGGAT GTTTTCCCTC TCCTACATTA GAAACCATTC TTAAAAACTT TTCAAAATAT AGAACCATTA AGCCTGCTAT ATCTGAGCAA ATTAGTGGGT ACCTTTTTTC CTTTTTTAAA GCACAAGAGG CCCATAAATC TTGAGTTATT TGCATTAGTT TACATTTTTT GATACAACTT TTCAGACCAA GAGAATAAAA ATCATGCGTT ATTAAACCCC TAGCTGGCTG GCATGCTTTC CTGTTTGTAC TGTATACATT TTGCTGGATG AAACCAAGGA TAGTTTAGGT ATAATTGTCC AAAATAACCT AACTGCAGCA GAAATGTAGG ACAGTTGCTT AGTACAGGCT TCTCACTTCC TACAGACCTG AATTCAAATT TGGATAGTCT GAGTTATTAA ATTCCCAAAG ACAAAGAACA CACTCTTATT TCTTGTGTAT ATTTCAACAT AAATCATGTT GTTACCAATT TGTTGGGAAG GCCCTGGTTG AGAAGAGTTT TAGATAATAA GGCTGTATAT ATATAGATAT ATATAGATAT ATACCAATGT CTATATATAG AGATATTTTA TATATATATA TACAGGTATA TATATGTGTG TGTATATATA TAGGTATATA CATATATACA TATATATATA TATATATATG GATATATACC CATGTCTACT GTTTTGCTTC AGCTAGTGCT TACAATTTCA TTCAAGTCCT GAGTATGTGT CCTGCTGTTA CTCCTTCTTT GGTAGTTGAA CGTTGAATTC AAGTCTTTCC TTCTGTTTTA AGAAGTACTA AGCAAACAAG CAATAAAAAG GGGAATGGCG CATGCTAGTG TTTGAATATG CTCTCTTGTT GCTCTAATTC TGTGCCTCCG TGCATTAATA TTTGGATGCA TGCAATGCCA GCATGGAAAT TGGCCT	
IL23A	NM_016584.2	AAAACAACAG GAAGCAGCTT ACAAACTCGG TGAACAACTG AGGGAACCAA ACCAGAGACG CGCTGAACAG AGAGAATCAG GCTCAAAGCA AGTGGAAGTG GGCAGAGATT CCACCAGGAC TGGTGCAAGG CGCAGAGCCA GCCAGATTTG AGAAGAAGGC AAAAAGATGC TGGGGAGCAG AGCTGTAATG CTGCTGTTGC TGCTGCCCTG GACAGCTCAG GGCAGAGCTG TGCCTGGGGG CAGCAGCCCT GCCTGGACTC AGTGCCAGCA GCTTTCACAG AAGCTCTGCA CACTGGCCTG GAGTGCACAT CCACTAGTGG GACACATGGA TCTAAGAGAA GAGGGAGATG AAGAGACTAC AAATGATGTT CCCCATATCC AGTGTGGAGA TGGCTGTGAC CCCCAAGGAC TCAGGGACAA CAGTCAGTTC TGCTTGCAAA GGATCCACCA GGGTCTGATT TTTTATGAGA AGCTGCTAGG ATCGGATATT TTCACAGGGG AGCCTTCTCT GCTCCCTGAT AGCCCTGTGG GCCAGCTTCA TGCCTCCCTA CTGGGCCTCA GCCAACTCCT GCAGCCTGAG GGTCACCACT GGGAGACTCA GCAGATTCCA AGCCTCAGTC CCAGCCAGCC ATGGCAGCGT CTCCTTCTCC GCTTCAAAAT CCTTCGCAGC CTCCAGGCCT TTGTGGCTGT AGCCGCCCGG GTCTTTGCCC ATGGAGCAGC AACCCTGAGT CCCTAAAGGC AGCAGCTCAA GGATGGCACT CAGATCTCCA TGGCCCAGCA AGGCCAAGAT AAATCTACCA CCCCAGGCAC CTGTGAGCCA ACAGGTTAAT TAGTCCATTA ATTTTAGTGG GACCTGCATA TGTTGAAAAT TACCAATACT GACTGACATG TGATGCTGAC CTATGATAAG GTTGAGTATT TATTAGATGG GAAGGGAAAT TTGGGGATTA TTTATCCTCC TGGGGACAGT TTGGGGAGGA TTATTTATTG TATTTATATT GAATTATGTA CTTTTTTCAA TAAAGTCTTA TTTTTGTGGC TAAAAAAAA	16
IQGAP1	NM_003870.3	GGACCCCGGC AAGCCCGCGC ACTTGGCAGG AGCTGTAGCT ACCGCCGTCC GCGCCTCCAA GGTTTCACGG CTTCCTCAGC AGAGACTCGG GCTCGTCCGC CATGTCCGCC GCAGACGAGG TTGACGGGCT GGGCGTGGCC CGGCCGCACT ATGGCTCTGT CCTGGATAAT GAAAGACTTA CTGCAGAGGA GATGGATGAA	17

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AGGAGACGTC AGAACGTGGC TTATGAGTAC CTTTGTCATT TGGAAGAAGC GAAGAGGTGG ATGGAAGCAT GCCTAGGGGA AGATCTGCCT CCCACCACAG AACTGGAGGA GGGGCTTAGG AATGGGGTCT ACCTTGCCAA ACTGGGGAAC TTCTTCTCTC CCAAAGTAGT GTCCCTGAAA AAAATCTATG ATCGAGAACA GACCAGATAC AAGGCGACTG GCCTCCACTT TAGACACACT GATAATGTGA TTCAGTGGTT GAATGCCATG GATGAGATTG GATTGCCTAA GATTTTTTAC CCAGAAACTA CAGATATCTA TGATCGAAAG AACATGCCAA GATGTATCTA CTGTATCCAT GCACTCAGTT TGTACCTGTT CAAGCTAGGC CTGGCCCCTC AGATTCAAGA CCTATATGGA AAGGTTGACT TCACAGAAGA AGAAATCAAC AACATGAAGA CTGAGTTGGA GAAGTATGGC ATCCAGATGC CTGCCTTTAG CAAGATTGGG GGCATCTTGG CTAATGAACT GTCAGTGGAT GAAGCCGCAT TACATGCTGC TGTTATTGCT ATTAATGAAG CTATTGACCG TAGAATTCCA GCCGACACAT TTGCAGCTTT GAAAAATCCG AATGCCATGC TTGTAAATCT TGAAGAGCCC TTGGCATCCA CTTACCAGGA TATACTTTAC CAGGCTAAGC AGGACAAAAT GACAAATGCT AAAAACAGGA CAGAAAACTC AGAGAGAGAA AGAGATGTTT ATGAGGAGCT GCTCACGCAA GCTGAAATTC AAGGCAATAT AAACAAAGTC AATACATTTT CTGCATTAGC AAATATCGAC CTGGCTTTAG AACAAGGAGA TGCACTGGCC TTGTTCAGGG CTCTGCAGTC ACCAGCCCTG GGGCTTCGAG GACTGCAGCA ACAGAATAGC GACTGGTACT TGAAGCAGCT CCTGAGTGAT AAACAGCAGA AGAGACAGAG TGGTCAGACT GACCCCCTGC AGAAGGAGGA GCTGCAGTCT GGAGTGGATG CTGCAAACAG TGCTGCCCAG CAATATCAGA GAAGATTGGC AGCAGTAGCA CTGATTAATG CTGCAATCCA GAAGGGTGTT GCTGAGAAGA CTGTTTTGGA ACTGATGAAT CCCGAAGCCC AGCTGCCCCA GGTGTATCCA TTTGCCGCCG ATCTCTATCA GAAGGAGCTG GCTACCCTGC AGCGACAAAG TCCTGAACAT AATCTCACCC ACCCAGAGCT CTCTGTCGCA GTGGAGATGT TGTCATCGGT GGCCCTGATC AACAGGGCAT TGGAATCAGG AGATGTGAAT ACAGTGTGGA AGCAATTGAG CAGTTCAGTT ACTGGTCTTA CCAATATTGA GGAAGAAAAC TGTCAGAGGT ATCTCGATGA GTTGATGAAA CTGAAGGCTC AGGCACATGC AGAGAATAAT GAATTCATTA CATGGAATGA TATCCAAGCT TGCGTGGACC ATGTGAACCT GGTGGTGCAA GAGGAACATG AGAGGATTTT AGCCATTGGT TTAATTAATG AAGCCCTGGA TGAAGGTGAT GCCCAAAAGA CTCTGCAGGC CCTACAGATT CCTGCAGCTA AACTTGAGGG AGTCCTTGCA GAAGTGGCCC AGCATTACCA AGACACGCTG ATTAGAGCGA AGAGAGAGAA AGCCCAGGAA ATCCAGGATG AGTCAGCTGT GTTATGGTTG GATGAAATTC AAGGTGGAAT CTGGCAGTCC AACAAAGACA CCCAAGAAGC ACAGAAGTTT GCCTTAGGAA TCTTTGCCAT TAATGAGGCA GTAGAAAGTG GTGATGTTGG CAAAACACTG AGTGCCCTTC GCTCCCCTGA TGTTGGCTTG TATGGAGTCA TCCCTGAGTG TGGTGAAACT TACCACAGTG ATCTTGCTGA AGCCAAGAAG AAAAAACTGG CAGTAGGAGA TAATAACAGC AAGTGGGTGA AGCACTGGGT AAAAGGTGGA TATTATTATT ACCACAATCT GGAGACCCAG GAAGGAGGAT GGGATGAACC TCCAAATTTT GTGCAAAATT CTATGCAGCT TTCTCGGGAG GAGATCCAGA GTTCTATCTC TGGGGTGACT GCCGCATATA ACCGAGAACA GCTGTGGCTG GCCAATGAAG GCCTGATCAC CAGGCTGCAG GCTCGCTGCC GTGGATACTT AGTTCGACAG GAATTCCGAT CCAGGATGAA TTTCCTGAAG AAACAAATCC CTGCCATCAC CTGCATTCAG TCACAGTGGA GAGGATACAA GCAGAAGAAG GCATATCAAG ATCGGTTAGC TTACCTGCGC TCCCACAAAG ATGAAGTTGT AAAGATTCAG TCCCTGGCAA GGATGCACCA AGCTCGAAAG CGCTATCGAG ATCGCCTGCA GTACTTCCGG GACCATATAA ATGACATTAT CAAAATCCAG GCTTTTATTC GGGCAAACAA AGCTCGGGAT GACTACAAGA CTCTCATCAA TGCTGAGGAT CCTCCTATGG TTGTGGTCCG AAAATTTGTC CACCTGCTGG ACCAAAGTGA CCAGGATTTT CAGGAGGAGC TTGACCTTAT GAAGATGCGG GAAGAGGTTA TCACCCTCAT TCGTTCTAAC CAGCAGCTGG AGAATGACCT CAATCTCATG GATATCAAAA TTGGACTGCT AGTGAAAAAT AAGATTACGT TGCAGGATGT GGTTTCCCAC AGTAAAAAAC TTACCAAAAA AAATAAGGAA CAGTTGTCTG ATATGATGAT GATAAATAAA CAGAAGGGAG GTCTCAAGGC TTTGAGCAAG GAGAAGAGAG AGAAGTTGGA AGCTTACCAG CACCTGTTTT ATTTATTGCA AACCAATCCC ACCTATCTGG CCAAGCTCAT TTTTCAGATG CCCCAGAACA AGTCCACCAA GTTCATGGAC TCTGTAATCT TCACACTCTA CAACTACGCG TCCAACCAGC GAGAGGAGTA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CCTGCTCCTG CGGCTCTTTA AGACAGCACT CCAAGAGGAA	
		ATCAAGTCGA AGGTAGATCA GATTCAAGAG ATTGTGACAG	
		GAAATCCTAC GGTTATTAAA ATGGTTGTAA GTTTCAACCG	
		TGGTGCCCGT GGCCAGAATG CCCTGAGACA GATCTTGGCC	
		CCAGTCGTGA AGGAAATTAT GGATGACAAA TCTCTCAACA	
		TCAAAACTGA CCCTGTGGAT ATTTACAAAT CTTGGGTTAA	
		TCAGATGGAG TCTCAGACAG GAGAGGCAAG CAAACTGCCC	
		TATGATGTGA CCCCTGAGCA GGCGCTAGCT CATGAAGAAG	
		TGAAGACACG GCTAGACAGC TCCATCAGGA ACATGCGGGC	
		TGTGACAGAC AAGTTTCTCT CAGCCATTGT CAGCTCTGTG	
		GACAAAATCC CTTATGGGAT GCGCTTCATT GCCAAAGTGC	
		TGAAGGACTC GTTGCATGAG AAGTTCCCTG ATGCTGGTGA	
		GGATGAGCTG CTGAAGATTA TTGGTAACTT GCTTTATTAT	
		CGATACATGA ATCCAGCCAT TGTTGCTCCT GATGCCTTTG	
		ACATCATTGA CCTGTCAGCA GGAGGCCAGC TTACCACAGA	
		CCAACGCCGA AATCTGGGCT CCATTGCAAA AATGCTTCAG	
		CATGCTGCTT CCAATAAGAT GTTTCTGGGA GATAATGCCC	
		ACTTAAGCAT CATTAATGAA TATCTTTCCC AGTCCTACCA	
		GAAATTCAGA CGGTTTTTCC AAACTGCTTG TGATGTCCCA	
		GAGCTTCAGG ATAAATTTAA TGTGGATGAG TACTCTGATT	
		TAGTAACCCT CACCAAACCA GTAATCTACA TTTCCATTGG	
		TGAAATCATC AACACCCACA CTCTCCTGTT GGATCACCAG	
		GATGCCATTG CTCCGGAGCA CAATGATCCA ATCCACGAAC	
		TGCTGGACGA CCTCGGCGAG GTGCCCACCA TCGAGTCCCT	
		GATAGGGGAA AGCTCTGGCA ATTTAAATGA CCCAAATAAG	
		GAGGCACTGG CTAAGACGGA AGTGTCTCTC ACCCTGACCA	
		ACAAGTTCGA CGTGCCTGGA GATGAGAATG CAGAAATGGA	
		TGCTCGAACC ATCTTACTGA ATACAAAACG TTTAATTGTG	
		GATGTCATCC GGTTCCAGCC AGGAGAGACC TTGACTGAAA	
		TCCTAGAAAC ACCAGCCACC AGTGAACAGG AAGCAGAACA	
		TCAGAGAGCC ATGCAGAGAC GTGCTATCCG TGATGCCAAA	
		ACACCTGACA AGATGAAAAA GTCAAAATCT GTAAAGGAAG	
		ACAGCAACCT CACTCTTCAA GAGAAGAAAG AGAAGATCCA	
		GACAGGTTTA AAGAAGCTAA CAGAGCTTGG AACCGTGGAC	
		CCAAAGAACA AATACCAGGA ACTGATCAAC GACATTGCCA	
		GGGATATTCG GAATCAGCGG AGGTACCGAC AGAGGAGAAA	
		GGCCGAACTA GTGAAACTGC AACAGACATA CGCTGCTCTG	
		AACTCTAAGG CCACCTTTTA TGGGGAGCAG GTGGATTACT	
		ATAAAAGCTA TATCAAAACC TGCTTGGATA ACTTAGCCAG	
		CAAGGGCAAA GTCTCCAAAA AGCCTAGGGA AATGAAAGGA	
		AAGAAAAGCA AAAAGATTTC TCTGAAATAT ACAGCAGCAA	
		GACTACATGA AAAAGGAGTT CTTCTGGAAA TTGAGGACCT	
		GCAAGTGAAT CAGTTTAAAA ATGTTATATT TGAAATCAGT	
		CCAACAGAAG AAGTTGGAGA CTTCGAAGTG AAAGCCAAAT	
		TCATGGGAGT TCAAATGGAG ACTTTTATGT TACATTATCA	
		GGACCTGCTG CAGCTACAGT ATGAAGGAGT TGCAGTCATG	
		AAATTATTTG ATAGAGCTAA AGTAAATGTC AACCTCCTGA	
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		CGTTTGCTGC CAGCCCAGAA GGATGAAGGA AAGAAGCACC	
		TCACAGCTCC TTTCTAGGTC CTTCTTTCCT CATTGGAAGC	
		AAAGACCTAG CCAACAACAG CACCTCAATC TGATACACTC	
		CCGATGCCAC ATTTTTAACT CCTCTCGCTC TGATGGGACA	
		TTTGTTACCC TTTTTTCATA GTGAAATTGT GTTTCAGGCT	
		TAGTCTGACC TTTCTGGTTT CTTCATTTTC TTCCATTACT	
		TAGGAAAGAG TGGAAACTCC ACTAAAATTT CTCTGTGTTG	
		TTACAGTCTT AGAGGTTGCA GTACTATATT GTAAGCTTTG	
		GTGTTTGTTT AATTAGCAAT AGGGATGGTA GGATTCAAAT	
		GTGTGTCATT TAGAAGTGGA AGCTATTAGC ACCAATGACA	
		TAAATACATA CAAGACACAC AACTAAAATG TCATGTTATT	
		AACAGTTATT AGGTTGTCAT TTAAAAATAA AGTTCCTTTA	
		TATTTCTGTC CCATCAGGAA AACTGAAGGA TATGGGGAAT	
		CATTGGTTAT CTTCCATTGT GTTTTTCTTT ATGGACAGGA	
		GCTAATGGAA GTGACAGTCA TGTTCAAAGG AAGCATTTCT	
		AGAAAAAAGG AGATAATGTT TTTAAATTTC ATTATCAAAC	
		TTGGGCAATT CTGTTTGTGT AACTCCCCGA CTAGTGGATG	
		GGAGAGTCCC ATTGCTAAAA TTCAGCTACT CAGATAAATT	
		CAGAATGGGT CAAGGCACCT GCCTGTTTTT GTTGGTGCAC	
		AGAGATTGAC TTGATTCAGA GAGACAATTC ACTCCATCCC	
		TATGGCAGAG GAATGGGTTA GCCCTAATGT AGAATGTCAT	
		TGTTTTTAAA ACTGTTTTAT ATCTTAAGAG TGCCTTATTA	
		AAGTATAGAT GTATGTCTTA AAATGTGGGT GATAGGAATT	
		TTAAAGATTT ATATAATGCA TCAAAAGCCT TAGAATAAGA	
		AAAGCTTTTT TTAAATTGCT TTATCTGTAT ATCTGAACTC	
		TTGAAACTTA TAGCTAAAAC ACTAGGATTT ATCTGCAGTG	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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LOC494127	NR_036691.1	GCAGTGCCCG ACTCCGCAGG AGCGCCAGGG CGGCTCCTGC TCTTCCTGGA CTCCCTGAAG AGGCGTTTGT CGAAATGTCC ACAGAAGGAG GATTTGGTGG TACTAGCCGC AGTGATGCCC AGCAAAGCCT AAAGTCCTTC TGGCTTCGGG TCATGGAAGA AATCTGGAAT TTAGCAGTGA AAGATTTCTG AATGCAGGAA CTCCCACTGG CTCGTATTAA GAAGATTATG AAACTGGATG AAGATGTGAA GATGATCAGT GCAGAGGCCC CTGTGCTCTT TGCCAGGGCA GCCCAGATTT TTTATCACAG AGTTGACTCT TCGAGCCTGG ATTCACACAG AGGATAACAA CTGCCGGACT TATGTCGCCA TGGCAATTAC GAAATTTGAT CAATTGGATT TTCTCATCGA TATTGTTCTA AGAGATGAAC TGAAACCTCC AAAGTGTCAG GAGGAGGTGC TGCAGTCTGT AACTCCTGCT GAGCCAGTCC AATACTATTT CACGCTGGCT CAGCAGCCCA CCGCCCGTCC AAGTCCAGGG ACAGCAGCAA GGCCAGACCA CCGCCAGCTC CATGACCACC ATGCAGCCTG GGCAGATCAT CATCGCACAG CTTCAGCAGG GCCAGACCAC GCCCGTGACG ACGCAGGTTG GAGAAGGTCA GCAGGTGCAG ATTGTCCAGG CCCAGCCACA GGGTCAAGCC CAGTAGGCCC AGAGTGGTAC TGGATGGACC GTGCAGGTGA TGTAGCAGAT CCTCACTAAC ACAGGAGAGA TCCAGCAGAT CCCGGTGCAG CTGAAGGCTG GCCAGCTGCA GTGTATCCGC TTAGCCCAGT CTGTATCAGG CACCCACGTT CTGCAGGGAC AGATCCAGAC ACTTGCCACC AGCGCTCAAC CGATTACACA GACAGAGGTC CAGCAAAGAC AGCAGTAGTT CAGCCAGTTC ACAGATGGAC AGCAGCTCTA CCAGATCCAG CAAGTATCCA TACCTGCGGG CCAGGACCTG CCCAGCCCAT GTTTTTCCAG TCAGTCAACC AGCCCTCTGA TGGGCAGGCC CCCCGGGTGA CTGGCGGCTG AGGGCCGGAG CTGGCAAGGC CGAGGACACT CAACACAATT TTTGCCGTAC AGCCCCAGGT CATGAACACA GCCTTCTTCC CCAGAGGACC CGGCCGACCT CAGCTCCTCC TGCAGGCTAG GACAATGGCG CACTAGGCCT CATGCCTGGG GGCCGAGATT CTCCAACAGA AAGATGCAAT ATTTTTTGTT TCCTTTTTTC TCCAAGGAAT CAATATTTCA ATATGTTGAG CTGTGTGTCC AATGCTATGA AATTAAAATA TTAAATCACA AAAAAAAAAA AAAA	18
LOC646471	NR_024498.1	AATGAATCCA ACTACTGTCC TTGTCCTCTC CTCTCTAGGT TTGTGGAGAC TGAATTCCAC AAACCTAGGG ACAGGGCACT CTCTAGGGGC AAGTCAACTC TCAATTATAG TGAGGGCAGG TTCCCCAGTT GCCAGCCTAC ACCCTGGCCA GCCACCCAAG GGAATACCTG CTGCTGCTAA GGCAGTCAAT GTTGGGAGGG TCAGGGAAGG GGAGAGGAAG TAGCTGAGTG TAGAGATTAT CCAGGCTTTC CTTCCCGTCC TCTGTACAGA AAGGCAGACA TACACTGACT CCTGAAGTGC CCCAGGATCA TAGTTGGTTC TTCTTAGGGG GAGGGAGTGA AACGGTGGCC TTGCTCCAAT TCCAGGCCTC TGGAGAAAGG AGTGCTCAAC TGGAGAGTCT	19

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CAGAACCTTT TACAGTACTT CAGCAGGCCC AGCACCCAAC CTTCCCCACC ATACCCCTGC CCTGGGAATG CCCCTTACCA CAGTGGCTTC CATTCCTGAC ATTTGAACCA GCTCCTGCAC TTGAAGGACA AGGCAGCCAC TGGGCAGCTC CGGGAGGCGG GAGGCGTGAC TCCTTCATAT CAGGCCATCT GGAGGAGCAC CAGATTTCCC TCTTGTGAAA CAACCCCCCA CAGTGGGGAT TTAATGGTCA CCACACAAGC CCCAAAAAAC CAGGCCCTGC AGAGCCATGA CTTCACAGGG GAGAAAACCA GCCCAAGAGG CTGCCCTCAC CCAACACTTA GCTCTAAGTC CAGACTCTTA AAGGTTCTGT TAAAGATCTG CGGCCAGGCG CAGTGGCTCA CATCTGTAAT CCCAGCACTT TGGGAGGCCG AGGTGGGCGG ATCACGAGGT CAGGAGAGCG AGACCATCCT GGCTAACACA GTGAAGCCCT GTCTCTACTA AAAATACAAA AAATTAGCCA GGTGTGGTGG CGGGCACCTG TAGTACCAGC TACTTGGGAG GCTGAGGCAG GAGAATGGCG TGAACCCAGG AGGCAGAGCT TGCAGTGAGC CGAGATCACG CCACTGCACT CCAACCTGGG TGACAGAGCG AGACTCCGTC TCAAAAAAAA AAAAAAAAAA AAAAGATTTG CCTAGTGAGC TTGGCCAACC CAAGAGCCTT CCTGGGCTCT CAGGTTTCTT GAGGCAAAAC TTCTAAAGAT GCAAATAGAA TCCTTGGTAG CTACAAAGCT CATCTAGCCA AATTATAAGT ACTATCCAAG CCCAGGATTC ATTCTTGAAG ATGAGATGTA AAAGCACTGC TGGGCCCTTG CAAAGGGAGA GATCAGAGAA AATAGGAGAA CGAACACCAG CTCATGCCAA TCTCAGTCTC ATGAGCTACA GAGAAAAAGA GCCAAAAAAG TTTATCCTGG CTCAGCCACC TTCTGGCTTT GTAGCTACAG GCAAGTTATT TAACTTTTCT GAATGTTGGT TTTTCCATCT GGAAAAAGGA GACAGGAAAA GCTGCCACTG GAGGCTACCT TAAGGGTCAA TGAGTGAAAT GGGGTTGAGT GCTTAATGAA CTGTTAAAAC ATGACACAGA TGGGTGAGGT GACATCTGAC TGCTCTTCAC AGCAGTCCTT GCATGGTGAG CAGATGCAAT CAGTGTGCTC GTTCATTCAC TTGTTCATCA AACACCAACA GTACCCACTT GTGCCAGGCC TTGTGCTGGG CACCGTAGGG GCTATGGAAG GAGTAAGACA CAGTCATGCT CTCAAGGAAC TCGCTGTCCA GCAGAGTGAT AGGACATGAA TACATACTTC TAGCAAAGGC TGAGTAGCAT TAGGCAGGTG GCCAGATGAG GTCAGAGCTC TGGCCCTAGT GCCCTGTAGT AAGTTGCGAG GGAGTAGGAA ATGAAGACAA AACAGCAAGA GCACAGGTCC AGCACAAGAA GGTGAAGGAC ATGGAAAGAG GAGCAGAGCA TGATAAAGCT ACAAAAGTAG GGGGCAAGTC ATGAGGTCCG GGTCACTGGG GAGCCCTGGA CAGGCTCTGA ATGGGAGATA GACTAGACTG ACAAAGCAGC TCTTTCTGGA GACCTCCTGT GGAGGGTCCC CTCTTCTACC CTTGGGAGAC GAGGTGTCAT TCACTTCACG ATACACATGG GGAAAGGAAG TTGGACAGGG GAAGGCACTT GTCTAAGGTC ACTCAGTGGT CATGCCAGTG CAAAGCACCC AAAGCCTGGT CCAGGTCCTG CCACCTCCTG CGTGGGCTCC ATCTTCTCGG CTCAGTTCAA GCCTTTATTA TCAGAGTCCC TCTGTGTGCC AGGCTCTGTG CTGAATGTCC GTGTCGGCCA CAAATTGCAC TAACGGTGCT TACTGCCACT TCCTCTCCGA CCTTACCTCC AAAGACCCAG CCGGGCCTAA CCCTCACACA CAGCCAGAGG AATCCTTCCC AAACACAGTT AATCACATCG CTCATCCGTT TAAAACTGGC TTACTAGAGC CAACACAGGA TAAAATCCAC AATCAGTCCC AAACCCATCT TTCTCTGCCG TCTCTAGACC ACATTCACTG AGCACCAAGG CCTGCACGCT TCGCAGCCAG CCCTCCTGCT CTCCCCCTTG CCCTCCAGCC ACACTGGCCC TCAACCCCCA ACACTGGGCC CACCTCTGAG CCCTGCATCA AACACCAGCA CCCCCTTCGC CGAGCTCATT CTTCATCATT CCTCAGAGCT CAGCTTAAGC ATCACCTTCC TTCCCTGCGA CAGGGTCAGG TCCTTTCCTG TGCGCTCCGC AGCACCCTGC CTGTCCCTCG GAGTCCTCGC GAGAGGAACT AGATAACTGT GATCCCCGCG CACAGCACGG TGTCTGCCTA GGACACAGTG AGCGCTTCAA CGCGCTGAAC TAAACGAATG AAGGGGAACA CCAACTCAGC TGTCACCCCC GCCCCCAAGT GTCACTGGCT AGAAGATCTT CCCCCGGTTC GCCGGCAGCA CCGGTGGGTC TTGTCACCTC TGCGCCCCGC AGCCCTCCGG AGGCTGCAGT TGCTGTCTGC TGACAGGCGT CCCCTCAGGA GAACGGGAGC TCCTCCAGCG CAGGGCCCTG CTCAGCACCC CTGGTCCCGG GGGCCCCCAC GGAGGCCGCC GCGCTCAGTG AAACTTTGCC GCGCGCAGCA GTGGCCGGAG ACCCGCGCGG ACCCTCTCCC CGCGGGGACC CGCCAGGGCG AGTCGCCCCT CCCCCGCGGC AGACGACCCG ACCTGGGCCA AGGTCGGCCA AGTGCTCGCC GCCTGCCCAG CGGCCCTTGG ACCCACAGGA CCCGGACCCG AGTGCGCGGA AGCCTCCGAG CCGCGGCCAG GTCTCAGAAT GCGCGGCGGG GCAGCCCGGC CCGTCAAGCA GCGAAGCCGA AACTAGAAGC CTGAGGTCTG	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GCTTGGTCAC TGTTTGCTGG GGAACTCCGA CGAGCAACTA CCCTTCGGGG ACTCAGTTTC CCCCGGGCCA AGTGAGACGG ACAGCTGCGG CTCCCGCGAG AACTCGCGCG GGAACAGGAG CTGCGGGCTC CGCACAGCGC CCGGCACGTA GACCCAGTCT ACTAACGGGC TGGAAGGTCG CGCTCCCGCC TCCGCGCCTC CCTAGAACCT TTCTCACCTC AAGCTCGGGC TCCCGGACTC TCTCACCTCA CGGCCCCTCC TTCCACCTTG GGCCGCCCAG CCTCAACCGC CCT	
LOH12CR	NR_024061.1	GGCTACGCAA GACCTTCAGT TCCGGATTAG GAGGCCCCGC CCCCCGGCCC GAGGGAGGGG CGGAGAGACC CGCTCCTGCG ACTTAGGGCG ATGCCACCTT AAAGGGCTTG ACCTCCTCGA AGCCAGAACT GCGGAAGAGG ATGGAGAAAG AAACGCTAGA TAGACCCAGG ATTCGAACCC ACAACAGCTT GGATGCAAAG CTCATTTTGA ATTCTGAAGA GCTGGGAGTT TAGCAGTGGA CGACCAAACA AAAAAATACC AGAAGACGGT GAAAGCAGCA GCTGGAAATG AGAGAAAAAT TAAGAAATAG AAAATGCTGT TTCATGTATT AGGGAGTGCA CATCTTTAAA GAGAAGTGAA GAACCTTTTT GGCTAGGCGC TTTCTGGGGA CCTTGGCAGT TATCAACAAG TCACTCTGAA ACTAACAGAA GGTGAGGAGG GAGTAATCCA GAAAGGAAAC TGCCATTTTT GCTAGGGCAA GAAAGTAGAC CTAGCAATGC AATGCCATCA AGGAGTTAGC TGGCTCTAGC GTTCTCCGAA CTTTGCAACT CATTTTATAT TACATTGTCT GCGTCAAGAA ATTTCAAGTA AATGCCTTGA GATTTTATAT GTATAAAATG TAGTCTCTGG CCAGGCGGGG TGGCTCACGC CTGTAATCCC AGCACTTTGG GAGGCTGAGG CGAGTAGATC ACGATGTCAG GAGATCAAGA CCATCCTGGC TAACATGGTG AAACCCCGTC TCTACTAAAA AATACAAAAA ATTAGCCGGG CATGATGGTG GGCGCCTGTA GTCCCAGCTA CTCGGGAGGC TGAGACAGGA GAATGGTGTG AACCCGGGAG GCAGAGCTTG CAGTGAGCCG AGATCGCGCC ACTGCACTCC AGCCTGGGCG ACAGAGCGAG ACTCCGTCTC AAAAAAAAAA AAAAAAAAAA AAGTCTCAGA AGAAGTGTCA AGTTGACTAA ATGATTTTTA AGTCCCTTCC ACTCTACTAT ATCACAATAA TTTCCCCTAT TTTACCTTAT TTAATTTTCA CAACAACCCT GCAGCCTGTA AGTAGGTGAC TCCTCCATGG TCAGCAGGTG GTCAGTGGTC GAACTATTTA GGACTATCTG ATGCTCTTTC CTCTGCTCAC CATTGCAATA CAATGAGAAA AATGCAAATG AATCTAAGTA CAGTGTTCTC TGGGAATATG AAGATAGAAC TAAAAACTGC CTGAGAGATC AGGGCAGGCT TCTAGAAAGT TGTGTCTAAC TAGTCTTGAG GTTTACATAA ACCACACCAG GCTGATGGGA CTGGGAAGGG CCCACTAAGC AGTGAGACCA TTCCCTTTTG GAGAGTCCTT GCATCCCCTC CCAGATTTCC TCTTTGAGGG GAAGGTGAGA GAGGAGGTAA AAGGGGTGAG GAATGGAGAA TAACTCATTC TGGTTCTTGT TTCCCCTTTT CCACATAAAA GTATATTTGT CTTGTGTTCC ATACACCAGT CCATACTGAT GTGATGGTGT TTTTTATGCT TCCTTTTGAA TAAACATTTC ATTCTTAA	20
PBXIP1	NM_020524.3	GGTCAGTTTC TGGTCACATG ATTTTCTTCT CGGGCTGCAA ACAAAGGGAA GCCTGCAACA AGTTAAGCTG AAGACCGAAG CAAGAGCTGG TTCAGGTGGC AGCCACAGCA GCCTCAGGGA CCTCAGCAAC TATGGCCTCC TGCCCAGACT CTGATAATAG CTGGGTGCTT GCTGGCTCCG AGAGCCTGCC AGTGGAGACA CTGGGCCCGG CATCCAGGAT GGACCCAGAA TCTGAGAGAG CCCTGCAGGC CCCTCACAGC CCCTCCAAGA CAGATGGGAA AGAATTAGCT GGGACCATGG ATGGAGAAGG GACGCTCTTC CAGACTGAAA GCCCTCAGTC TGGCAGCATT CTAACAGAGG AGACTGAGGT CAAGGGCACC CTGGAAGGTG ATGTTTGTGG TGTGGAGCCT CCTGGCCCAG GAGACACAGT AGTCCAGGGA GACCTGCAGG AGACCACCGT GGTGACAGGC CTGGGACCAG ACACACAGGA CCTGGAAGGC CAGAGCCCTC CACAGAGCCT GCCTTCAACC CCCAAAGCAG CTTGGATCAG GGAGGAGGGC CGCTGCTCCA GCAGTGACGA TGACACCGAC GTGGACATGG AGGGTCTGCG GAGACGGCGG GGCCGGGAGG CCGGCCCACC TCAGCCCATG GTGCCCCTGG CTGTGGAGAA CCAGGCTGGG GGTGAGGGTG CAGGCGGGGA GCTGGGCATC TCCCTCAACA TGTGCCTCCT TGGGGCCCTG GTTCTGCTTG GCCTGGGGGT CCTCCTCTTC TCAGGTGGCC TCTCAGAGTC TGAGACTGGG CCCATGGAGG AAGTGGAGCG GCAGGTCCTC CCAGACCCCG AGGTGCTGGA AGCTGTGGGG GACAGGCAGG ATGGGCTAAG GGAACAGCTG CAGGCCCCAG TGCCTCCTGA CAGTGTCCCC AGCCTGCAAA ACATGGGTCT TCTGCTGGAC AAGCTGGCCA AGGAGAACCA GGACATCCGG CTGCTGCAGG CCCAGCTGCA	21

TABLE 1-continued

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
Melanoma Biomarker/Housekeeper Sequence Information			
		GGCCCAAAAG GAAGAGCTTC AGAGCCTGAT GCACCAGCCC AAAGGGCTAG AGGAGGAGAA TGCCCAGCTC CGGGGGGCTC TGCAGCAGGG CGAAGCCTTC CAGCGGGCTC TGGAGTCAGA GCTGCAGCAG CTGCGGGCCC GGCTCCAGGG GCTGGAGGCC GACTGTGTCC GGGGCCCAGA TGGGGTGTGC CTCAGTGGGG GTAGAGGCCC ACAGGGTGAC AAGGCCATCA GGGAGCAAGG CCCCAGGGAG CAGGAGCCAG AACTCAGCTT CCTGAAGCAG AAGGAACAGC TGGAGGCTGA GGCACAGGCA TTAAGGCAAG AGTTAGAGAG GCAGCGACGG CTGCTGGGGT CTGTACAGCA GGATCTGGAG AGGAGCTTGC AGGATGCCAG CCGCGGGGAC CCAGCTCATG CTGGCTTGGC TGAGCTGGGC CACAGATTGG CCCAGAAACT GCAGGGCCTG GAGAACTGGG GCCAGGACCC TGGGGTCTCT GCCAATGCCT CAAAGGCCTG GCACCAGAAG TCCCACTTCC AGAATTCTAG GGAGTGGAGT GGAAAGGAAA AGTGGTGGGA TGGGCAGAGA GACCGGAAGG CTGAGCACTG GAAACATAAG AAGGAAGAAT CTGGCCGGGA AAGGAAGAAG AACTGGGGAG GTCAGGAGGA CAGGGAGCCA GCAGGAAGGT GGAAGGAGGG CAGGCCAAGG GTGGAGGAGT CGGGGAGCAA GAAGGAGGGC AAGCGACAGG GCCCGAAGGA ACCCCCAAGG AAAAGTGGTA GCTTCCACTC CTCTGGAGAA AAGCAGAAGC AACCTCGGTG GAGGGAAGGG ACTAAGGACA GCCATGACCC CCTGCCATCC TGGGCAGAGC TGTTGAGGCC CAAGTACCGG GCACCCCAGG GCTGCTCAGG TGTGGACGAG TGTGCCCGGC AGGAGGGCCT GACTTTCTTT GGCACAGAGC TAGCCCCAGT GCGGCAACAG GAGCTGGCCT CTCTGCTAAG AACATACTTG GCACGGCTGC CCTGGGCTGG GCAGCTGACC AAGGAGCTAC CCCTCTCACC TGCTTTCTTT GGTGAGGATG GCATCTTCCG TCATGACCGC CTCCGCTTCC GGGATTTTGT GGATGCCCTG GAGGACAGCT TGGAGGAGGT GGCTGTGCAA CAGACAGGTG ATGATGATGA AGTAGATGAC TTTGAGGACT TCATCTTCAG CCACTTCTTT GGAGACAAAG CACTGAAGAA GAGGTCAGGG AAGAAGGACA AGCACTCACA GAGCCCAAGA GCTGCGGGGC CCAGGGAGGG GCACAGCCAT AGCCACCACC ACCACCACCG GGGCTGACAC CCTGCCCCAC AGGGAATGGC CTTGGCCTGG CCCAGCCCAA GATCCCAGCG TTATCTAACT CCTGGAGGGT GGACTCTGTC CTGGCTTGTT TGGTGTCCTC AGATATCTTT CACACAGTAG AGCAAAATCA CCAGCCCTGC ACTGATGTCA CTTTATGTAG AAAAAGGCCT TAGCTGGACC TGCGTTGCCG TCTATGCAAA TGCATGCAAA TACTCCAGGC CCTGGGATGT GGGCTTGTGT TTTGTCACTG TGAAGGGGGA GATGGGAGAG GAGCCTGTTT TGGGGTGGGG TCTGGGGAAG GCAATCTGAT TCTGAAGCTA AAGAGCTTTC ATCCTCTTGA GTGTATGTCC CCATAGTGGG CCCCTTGACC CACATGCTGA CCGGTGCCTT GGGATTTGAC TAGAGTTGCT GGCTCGAGGC CCAGCACGAG GACTTACCCT GGGGTTTTGT TAGGTTTGGA AGCAGCTGTC CCTAGGGGGT GAAGTCCCCC CCCTTTTTTT TTTTACCCCT GCTTCTCCCA CGGCTTCACC TCCCTATGTG AACTGTAGAC TCAGATCCCA ATAAAGTGCT GTTGCAGCTA TGATGCTAGG TGGTTTCTAA GCACAGGGGA CACCCCACAC CCCCTGCCTG AATGGATGGG TCCATCCCAG GCACTGGTAC TTGCCCCCTT GTTCTGTATC CCCCTTTGCC CTTGCCTTGC CCTTCCAACA AACCCTAGGC CCTTGAGAAG CTGATACTTC TCCTTTTGCT CACAGCTGCC TTGGCCCCAC CCCTGGGAGA TGTAGCAAAT TGAGTGTGGG TTTTGGAGTC TGAGCCTCAG GCTCAAATCC AGGCCAAGTG ATCTTGGGCA AGTTAATCTC TGGGAACTTT GGGTTTCTTA TCCTCAAAAA AGGCGATGGA AGGGCTGGGG AAGTGATTAA ATAAAAGCAA CGCAAGAAAA AAAAAAAAAA AAAAA	
RNF5	NM_006913.3	AATAGTGATT AGGAAACCTT GAAGCCTGCC CAACGATCGT GGGCAGGAGG TGGTTTCTGG TTTGTTGGGG CGTGTGTATG TGTATTTGGG GGGACTGAAG GGTACGTGGG GCGAAACAAA ACCGGCCATG GCAGCAGCGG AGGAGGAGGA CGGGGGCCCC GAAGGGCCAA ATCGCGAGCG GGGCGGGGCG GGCGCGACCT TCGAATGTAA TATATGTTTG GAGACTGCTC GGGAAGCTGT GGTCAGTGTG TGTGGCCACC TGTACTGTTG GCCATGTCTT CATCAGTGGC TGGAGACACG GCCAGAACGG CAAGAGTGTC CAGTATGTAA AGCTGGGATC AGCAGAGAGA AGGTTGTCCC GCTTTATGGG CGAGGGAGCC AGAAGCCCCA GGATCCCAGA TTAAAAACTC CACCCCGCCC CCAGGGCCAG AGACCAGCTC CGGAGAGCAG AGGGGGATTC CAGCCATTTG GTGATACCGG GGGCTTCCAC TTCTCATTTG GTGTTGGTGC TTTTCCCTTT GGCTTTTTCA CCACCGTCTT CAATGCCCAT GAGCCTTTCC GCCGGGGTAC AGGTGTGGAT CTGGGACAGG GTCACCCAGC	22

TABLE 1-continued

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CTCCAGCTGG CAGGATTCCC TCTTCCTGTT TCTCGCCATC TTCTTCTTTT TTTGGCTGCT CAGTATTTGA GCTATGTCTG CTTCCTGCCC ACCTCCAGCC AGAGAAGAAT CAGTATTGAG GGTCCCTGCT GACCCTTCCG TACTCCTGGA CCCCCTTGAC CCCTCTATTT CTGTTGGCTA AGGCCAGCCC TGGACATTGT CCAGGAAGGC CTGGGGAGGA GGAGTGAAGT CTGTGCATAG ATGGGAGAGC CTTCTGCTCA GAGGCTCACT CAGTAACGTT GTTTAATTCT CTGCCCTGGG GAAGGAGGAT GGATTGAGAG AATGTCTTTC TCCTCTCCTA AGTCTTTGCT TTCCCTGATT TCTTGATTTG ATCTTCAAAG GTGGGCAAAG TTCCCTCTGA CTCTTCCCCC ACTCCCCATC TTACTGATTT AATTTAATTT TTCACTCCCC AGAGTCTAAT ATGGATTCTG ACTCTTAAGT GCTTCCGCCC CCTCACTACC TCCTTTAATA CAAATTCAAT AAAAAAGGTG AAATATAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAA	
SERTAD2	NM_014755.2	CTCCTGCACG GCGAGTGCTG GAGCACGACG TACCGCTCGC TCGGTCAGGG CGCCCCCTCC GCCCGCCTCC TGCTTCCTCC TCCGCTGCCT GCCGCCGCCG CCTCCACCAT TGTATAATGC TCGGGGCGCG CAGGCAGAGA ACGGCGGAGT CTTAGCTTCA GCCTCGCCTG CTGCCCGCTC CCCGGCGCCA CCCTCGGGCC CCTGGAGCGG GGCACTCCGC ATGGAGCGGG AGTAGCTGAG GAGTGGGCGG AAACCCCTCC TGATGCGTTA GTTCCCAGGT GGAGCTGCAT GTGATATATG TTGGGTAAAG GAGGAAAACG GAAGTTTGAT GAGCATGAAG ATGGGCTGGA AGGCAAAATC GTGTCTCCCT GTGACGGTCC ATCCAAGGTG TCTTACACCT TACAGCGCCA GACTATCTTC AACATTTCCC TTATGAAACT CTATAACCAC AGGCCCCTGA CAGAGCCCAG CTTGCAAAAG ACCGTTTTAA TTAACAACAT GTTGAGGCGG ATCCAGGAGG AACTCAAACA GGAAGGCAGC CTGAGGCCCA TGTTCACCCC CTCCTCCCAG CCCACCACCG AGCCCAGCGA CAGCTACCGA GAGGCCCCGC CGGCCTTCAG CCACCTGGCG TCCCCGTCCT CCCACCCCTG CGACCTCGGA AGCACTACGC CCCTGGAGGC CTGCCTCACC CCGGCCTCAC TGCTCGAGGA CGACGATGAC ACGTTTTGCA CCTCCCAGGC CATGCAGCCC ACGGCTCCCA CCAAACTGTC ACCTCCAGCC CTCTTGCCAG AAAAGGACAG TTTCTCCTCT GCCTTGGACG AGATCGAGGA GCTCTGTCCC ACATCTACCT CCACAGAGGC GGCCACGGCT GCGACTGACA GTGTGAAAGG GACCTCCAGC GAGGCTGGCA CCCAGAAACT CGACGGTCCT CAAGAGAGCC GCGCAGATGA CTCAAAACTG ATGGACTCTC TGCCTGGGAA TTTTGAAATA ACGACGTCCA CGGGTTTCCT GACAGACTTG ACCCTGGATG ACATCCTGTT TGCTGACATT GATACGTCCA TGTATGATTT TGACCCCTGC ACTTCCTCAT CAGGGACAGC CTCAAAAATG GCCCCTGTGT CTGCCGACGA CCTCCTCAAA ACTCTGGCTC CTTACAGCAG TCAGCCTGTC ACCCCAAGTC AGCCTTTCAA AATGGACCTC ACAGAGCTGG ACCACATCAT GGAGGTGCTT GTTGGGTCCT AAGACCCAGG GACCCAGCGA CTATGCCCAC CCAGACCCCA GAGCGTTCCC ATAACCCTGA CAGTTCTCCA CACTGTGCAT GCACCCTTGC TTGCCTTTTT CAGAGAAAAA GAAAATTTTA CAACAGGATC ACACTAGTTT TTGCTTTGAG CAGAGTTGGA GTGCCTTCAT CCAAGTATGA CCACTTTTAA TACACTTTTT TGAGTGGTTC CTCAGAGACC TACTACCCTG GTATAGGAAA GAATCCATTT GAAGACAATG TTGCAATGTT GAATGACAAA AATAAACAGT TCAAGTGAAG CACAAGGATT AAGTTGGAAA AGCTGTAAAT TGCATGTGCA TATTTGTCTA TTTTTTCTAT AAGTTTTATT GCAAGAGGTA AAGAAGAAAA CTATATATAT ATATCTTATT TAGATAATCT CAGTACCTTT TCTGGCATTT TTGCCCTGTA TAGGTTGACT TGGCAATTCG GCCTTTTTAG AGGCATTAAC TACTCCTCGT AAGTGTTGCA TTTACATGGC TGTTTAGAAA ACTGCTGCCC AAATTTATTT TATATTTTTG TACAGATTCT GCAGTTTATG ATATTGTTTT TCTAAAAACA AATGCTGTTT ATACATATGA GATAGCTATT TTGATAGGAT TTGCTCACAT AGTTCCTGCA AACTTCAGAT GTACAAGTTG CACTTGTACT TTTATAGAGT TGTAATGTTT TATATGTGTA TGGTGCAAGA GAAAATTGGA TCAAATCAAT CTGCAGTTGA TGTCCCCAAA TGCAAACACA GGCACACACA TGCACACACC CATAAACACA CACACAGTGC TTTAAGAAAG GGCCAGGTGA TATCACACCC AAATTTCACA AGCACTGACC CCCTGGCACC AACACCCGCC AGTACTGTGA CTTCCAAAGC CAGAGCCACA TGTGCTCATC AAACTTGCAT TAAGCAGTTG GCGGGAGATG GCTGTGGAGC TGGGGGTTTA AGTGATGGTT CTCTTTTGCT CCCTCTTTTG AGGGTAAAGC TACTGTCTTT CTTAAGAGTG TATTTATGCC AAGTTTGCGC TTTTAATTGT TTTTATTTTG	23

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TTTTTTAATG AAAACCCAGA TCTTTCCTTT TTGGCATAAT TTTTATGATG ACCTGAAATT TTACATCCGA ACAAAATTTT ACATCCGAAA AGCAACCAAC TTCTTCATGG AACTCAGCCC TGTTGCAATG CTTAGGGCCC TTAAAGAAGA AAATCTCCCC AGAAGGCATC CATCATGTTG CTTAATTGTC TTCTGCAGCT TCCTTTCCCT AGAGCTTTCC CTGTGTTGCT AAGAGCTGAA AATGGCATCT TCGTGATCAC CACAGTGAGC TTGGCTCGCC TCGGCCGGCC CGGGATGCAC TCTTACAACA TGTGTGACTC TTGAACCTGG AGTTCATCAC ATTACGTCAC AGCTTCCCAT CTGGTTGCTT TCCTGAGTCA GCTACTTCAC ACTTGTCAAG GCTGTTTTAC CCCAAAACTC AGACAGGACT TTCTATGCAT GTTTTCCCTC CTCCCCCCAA TTCCCCCCCC CATCACCTTA TCTCCCAGGA CACACTTGAG AAGTAGCTTT TTATTCCTAG TGGTGTACAT TTAATTTTAA AAAGGTTGCA ATGTATCATG CTTGTTGCCG AAACTGTTTA TGGCCTTCTT GTTTCAGTTT TTTCTTTTCT TCCAATGGTA CTTTAGCTGT TGAGTGCAGG TTACAACCTA TATTGTTATG CAGATGGCTT CTTTAGGAAT AACTTTTATA TTTATTTAAA AATTTTTAAA TTATGGGATG TTTTGTTGTT GTTGTTGTCT TTGTTGTTGG TCATTTGTCA ATATTCAGTC ACCAATTCTG CTCACTTCTT GCCATGGATA AAATTGGGTC TTTCTGGCTA ATTAAAAAAG ACAACTTTAT AAAATGGCAC TTTAAGCAAG CCATAGTTAG TTTTATTTTT GTAATGCACA TGGCAAAGCA AAGACGTTTG TGATGAAGGA ACTGCTCATC TAAGCAAAAG ATTTGAGTAT GATATGATAA AGGCTTTCTA CATTCTAATT TACTTTTTCC CCCCACTTGA ATGTGTTTTA AAGGCTAATT ATCAGCTCAG TAGAGCAGTG AGAAACTGAT CAAATTGCAC TTGTTCTCCT ACAAGCAACC TCCACGCAGA CACCTCGTAC TGCTACAGGT GTGTCATTTC CTTTAATAGG ACCAGGGACC ATGTAACTGA GGTGAGGGTT GTAGTAGATG CTTCCAGTGT CAGTATGCCT GTTAATTTTA AGAGCTTCCC TTTCTTGCAG AGAACAAGTC TGCCCAGATT CCATGCTTTC TATAACTGGA GGACCTGGCA AACCTGCCGC ATGCTGCACA CATCTACCTA CGTACACATA TACAATAGTA TTGATGATTC TGAACAATAA CAGGGTAAAA CAGTTGGTTT GCCATTGTTA AAAACTGATT TACAGTAACT TACAACAACT GTACTTTTGT TGGATTAGCA AATCATGTGT TTAAACAAAT CCCATATGTT GGGCAACAGT TCAAATAAGC ACGGAGAAGT GTTGCCCAAA CTTGGTTCTC TGACTCTTAT GTATTTGTAA GGCTGGGCTT CAAAATCAAA ACAAAAACCC CAAAAACAGC AGGCAAATGC TTTTTAACTC TGACACCGTT GCCATAAATC CCTGATACTC AAAGTCTAAC AAGAAAGACA TGGAAAATTA GCAGCCCATT TTCAGAAAGA TCAAAATGAT CTAGGGTTCT AATTGCTTTT GCATCCTATT CTTACAAAGT GATGTCCCAA CAGGGAACAG TAGGAGCTGG AGTGGGATCT CCAAGTCCCA GTTTGAGTGT GGGATGTGCT TCCAGCAGTG CCTTCCCTTT ATGAAAGACA TCACATGGCA TCCAGGGCCA GGCAGGCAGC TTGAGGTGCC TTTACGAGAA AACCGAGCTG GGGCTGGGAG AGGACAGTTA TTGACACTGA TGTGCAATGA AGTGACAAGA TGAGAGCAGA ATCGTAAGAG CTTTGAATTT GAAGTGAGTT TTTTTCCCCC CATAAGTTAT TTATTCCTTT TTCTGTGTA AATATATTTA TTTTACTGTG GAGCGCTAAC ATCTGGATCG TAACATGTGC AGAATGTATG GTAGGAATGT ATTCTCTTGT AGGAATGTAA ATCTGTATTA AAAGGGGGTC CAAGCCAGGC CCCCAGGTCT TCTCATTGTA TGCACAGTCC GCATTCATTT TTACTCTTCT CTAATATGGG TCTATTTGAA ATATGCAAAA GGTATGAGGA ATGTTTTAAT ACCTCCAAAT TTTTAAGAAA AGCATCAAAG GGTTGATATT TTTTAAAGTT TTTTTAGTAG CACTTTCTCT GGATGACAGA AGGAGCAACC ACATGGGCAC CCTTGTTCAT ACCAAAGGGT GAGCAGTGGC CAGAGCCTCC TCTGCACCTC TCGAGTGTCT TTACCAATTG AGCTTTTTAT CGCCATAGCC CCTTGGAGTG CCCCAGCTGC CCTGAGGTCA ATCAAGGAAA ATTTCTTAAT GAAATAAGCT CCAAAGAGCC AAAGTATCAA CTTACAGATC GTTTTTAAAG CTTAAATTTA TGAACCACCT TTGTGGTAAA CAATGAATTA TGAATACCGC AGGGCAGCCT TCTTAAATGA CAAATGTAAA AAAAAAAAAA AAAAAGACTC TACTTCGTGC AGCAATTGCT ACTCTATACG AATTGTCTTA ATTTGAAAAC CTTGCTGTTA CAAATTGGAC CTTTATACAT TTTCTGAAAA CAATGAAAAG AGTATATTTA ACCTTTTCTG GCTGTAAATG GTTACCTTCC TGTAACTGCC CCGCACCTGG AGGCATGGAG TTGTGTGCAT CCTGCTTATG TACAATTGTT TTCAGTGTTT CTAAGAATGA GTCTGAATGG TTCTTGAAAA TTAGCCAGGA TCAAATGCTA TTGCAGACAA AGCCAATAAA AAGTTGGACT TCTTTTGGGG ATAACAAGTT TTGGAAGAGA AATGCAGGCC ATATGTGCGC ATGACCGAGA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TTTTGAAAAA AGATGTACAT AGTGACATGT TTGGTGCATG GTTTTTGAGG AGGGCTTTTG TCAAAAAGGA GGTATAACCT TTCCCCCACA GACCTGAGAG CTGTGCCTTT TCTATGCAAT ATTACAGACG TTACATCGGA ACCCAGATGG CTGTATTCAC ATGTAGGTTT GGGCTGTAAT CTAAACAATT GGACAGATTA AATGTACATG GAAATGAGCA GTCTTACTTT TGTAGTTTTA TATTATACAA TAAACAGTTA AAAGATGAAA AAAAAAAAAA AAAAA	
SLC35G5	NM_054028.1	TACCACCTCA GGAGAGTTCC AGGGAAGAAC CCCACCCGCA CTCCAATGAG GTCACAATGG CTGGAGCTCT GAGGGGCCCA GGCTCCCTGA GCCAGGAGGA GAGGAGAAAG TCCAAGGAAA GATGGCTGGC AGTCACCCCT ACTTCAACCT GCCTGACTCC ACACACCCAT CGCCGCCCTC CGCTCCACCC AGCCTCCGCT GGCACCAGCG CTGCCAGCCC TCTGGTGCCA CCAATGGCCT GCTGGTGGCC CTGCTGGGTG GGGGCCTGCC TGCTGGCTTC GTGGGCCCCC TTTCTCGTAT GGCTTACCAG GGTTCCAACC TGCCCTCGCT GGAGCTGCTC ATCTGTCGAT GCCTCTTCCA CCTCCCTATT GCCCTGCTAC TTAAACTGCG TGGCGACCCC CTTCTGGGAC CTCCTGACAT CCGAGGCTGG GCCTGCTTCT GTGCCCTGCT CAACGTCCTC AGCATTGGAT GTGCCTACAG TGCAGTTCAG GTGGTGCCCG CTGGCAACGC TGCCACTGTT CGCAAAGGTT CTTCCACCGT ATGCTCCGCT GTCCTCACCC TCTGCCTTGA GAGCCAGGGT CTCGGTGGCT ACGAGTGGTG TGGACTGTTG GGCAGCATCC TAGGACTAAT CATCATTCTG GGACCTGGAC TCTGGACACT ACAGGAGGGG ACCACAGGTG TCTACACCAC CCTGGGCTAT GTGCAGGCTT TCCTGGGAGG CCTGGCGCTG TCCCTGGGGC TTCTGGTCTA TCGTTCTCTG CACTTTCCCT CCTGCCTCCC AACAGTGGCC TTCCTATCTG GCTTGGTGGG GCTGCTGGGC TGTGTGCCAG GCCTCTTTGT GCTGCAGACC CCCGTGTTGC CCAGTGACCT CCTGAGTTGG AGTTGTGTGG GGGCAGAGGG GATCCTCGCC TTGGTCTCCT TCACATGTGT GGGCTATGCG GTCACCAAGG CCCACCCTGC CCTGGTGTGC GCTGTCCTGC ATTCCGAGGT GGTTGTGGCC CTTATACTGC AGTATTATAT GCTCCATGAG ACTGTGGCAC TTTCTGACAT CATGGGGGCA GGGGTTGTGC TGGGCAGCAT TGCCATCATT ACAGCCCGGA ACCTCAGCTG TGAGAGGACA GGGAAGGTGG AGGAGTGAGA TAGAACTTGG GAGCCCGGGG GTTGGGAGGG ACAGGGATAA ATAAAGACAA AGACTGAAGA C	24
SPATS2L	NM_001100422.1	AGTGCTGAGG GAGCAAAGTT CATTTTCTCG GGTAGGAGAA GATGATTCTC TTGCAACACG TGCGGATTGT GACAAAATCT TTCATTAACA AGGGGAGTTT CGGTGAAGTG GAGGTTTGGG GAAAGGCGAG GAAGTCGGTC TGGAGCAAGC AAGCAAAGTG CGGAAGCTGT ACTGGGATTC TTCTAGAAAG TGGGGTGGGA AAGGAGCTAG GGAGGGCGTG TGGAGGGACG AGATCTGTGT CAGAACGTGC GTGTGAGCGG ATACAAAACC CGAGAGAGGC GTGAGCAGCG CTGTGTTTGC GAGCGGGAGC GAGGGGCGCC GGCTGGGGTG TGTGCTCCTG AGCTCTTCAG AAACCAGGCT GCTTTCAGGA ACATTGCTGT GGATTCCCAG GGCCTATTCC ACTAGAAGCA AGATGGCTGA ACTCAATACT CATGTGAATG TCAAGGAAAA GATCTATGCA GTTAGATCAG TTGTTCCCAA CAAAAGCAAT AATGAAATAG TCCTGGTGCT CCAACAGTTT GATTTTAATG TGGATAAAGC CGTGCAAGCC TTTGTGGATG GCAGTGCAAT TCAAGTTCTA AAAGAATGGA ATATGACAGG AAAAAAGAAG AACAATAAAA GAAAAAGAAG CAAGTCCAAG CAGCATCAAG GCAACAAAGA TGCTAAAGAC AAGGTGGAGA GGCCTGAGGC AGGGCCCCTG CAGCCGCAGC CACCACAGAT TCAAAACGGC CCCATGAATG GCTGCGAGAA GGACAGCTCG TCCACAGATT CTGCTAACGA AAAACCAGCC CTTATCCCTC GTGAGAAAAA GATCTCGATA CTTGAGGAAC CTTCAAAGGC ACTTCGTGGG GTCACAGAAG GCAACAGACT ACTGCAACAG AAACTATCCT TAGATGGGAA CCCCAAACCT ATACATGGAA CAACAGAGAG GTCAGATGGC CTACAGTGGT CAGCTGAGCA GCCTTGTAAC CCAAGCAAGC CTAAGGCAAA AACATCTCCT GTTAAGTCCA ATACCCCTGC AGCTCATCTT GAAATAAAGC CAGATGAGTT GGCAAAGAAA AGAGGCCCAA ATATTGAGAA ATCAGTGAAG GATTTGCAAC GCTGCACCGT TTCTCTAACT AGATATCGCG TCATGATTAA GGAAGAAGTG GATAGTTCCG TGAAGAAGAT CAAAGCTGCC TTTGCTGAAT TACACAACTG CATCATTGAC AAAGAAGTTT CATTAATGGC AGAAATGGAT AAAGTTAAAG AAGAAGCCAT GGAAATCCTG ACTGCTCGTC AGAAGAAAGC AGAAGAACTA AAGAGACTCA CTGACCTTGC CAGTCAGATG GCAGAGATGC AGCTGGCCGA ACTCAGGGCA	25

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GAAATTAAGC ACTTTGTCAG CGAGCGTAAA TATGACGAGG	
		AGCTCGGGAA AGCTGCCCGG TTTTCCTGTG ACATCGAACA	
		GCTGAAGGCC CAAATCATGC TCTGCGGAGA AATTACACAT	
		CCAAAGAACA ACTATTCCTC AAGAACTCCC TGCAGCTCCC	
		TGCTGCCTCT GCTGAATGCG CACGCAGCAA CCTCTGGGAA	
		ACAGAGTAAC TTTTCCCGAA AATCATCCAC TCACAATAAG	
		CCCTCTGAAG GCAAAGCGGC AAACCCCAAA ATGGTGAGCA	
		GTCTCCCCAG CACCGCCGAC CCCTCTCACC AGACCATGCC	
		GGCCAACAAG CAGAATGGAT CTTCTAACCA AAGACGGAGA	
		TTTAATCCAC AGTATCATAA CAACAGGCTA AATGGGCCTG	
		CCAAGTCGCA GGGCAGTGGG AATGAAGCCG AGCCACTGGG	
		AAAGGGCAAC AGCCGCCACG AACACAGAAG ACAGCCGCAC	
		AACGGCTTCC GGCCCAAAAA CAAAGGCGGT GCCAAAAATC	
		AAGAGGCTTC CTTGGGGATG AAGACCCCCG AGGCCCCGGC	
		CCATTCTGAA AAGCCCCGGC GAAGGCAGCA CGCTGCAGAC	
		ACCTCGGAGG CCAGGCCCTT CCGGGGTAGT GTCGGTAGGG	
		TTTCACAGTG CAATCTCTGC CCCACGAGAA TAGAAGTTTC	
		CACAGATGCA GCAGTTCTCT CAGTCCCGGC TGTGACGTTG	
		GTGGCCTGAG CTAGGAGGAA AAAGAGCAGT TTTCACTCAG	
		TTTTGGTTCC CTGCCCGAGG TGCTGACCCA ATTCGCTGCC	
		AAAAGAGTGT CAATCAGAAT ATACAAATCC CGTATGGTTG	
		TGTCATCCTC TCTTAATCAT TTTTACTAAT TCTAATAATC	
		AGCTCTAGCT TGCTTCATAA TTTTCATGGC TTTGCTTGAT	
		CTGTTGATGC TTTCTCTCAT CAAGACTTTG CAGCATTTTA	
		GCCAGGCAGT ATTTACTCAT TATTAGGAAA ATCAAGATGT	
		GGCTGAAGAT CAGAGGCTCA GTTAGCAACC TGTGTTGTAG	
		CAGTGATGTC AGTCCATTGA TTGTCTTTAG AGAGTTAATG	
		TTACAAAAAA GAATTCTTAA TAATCAGACA AACATGATCT	
		GCTGAGGACA CATGCGCTTT TGTAGAATTT AACATCTGGT	
		GTTTTTCTGA AAAAATATAT ATACATATAT TGCTTTATTT	
		GAAACAAATT AAAATATGCT GCATTTGACA CCTGGCTAGT	
		TTCTTTTATT GATACCCACC TAGTTATTGA ATGTACTGTT	
		TAGTGCTTTC AAAAAAAACT TTAGAGACTA GAGGTTGTGG	
		TGCAAAGCTG TGTACAATAA ATACTGTTTC TGTTGAGACA	
		AGTACTCTTT CAGGAAAATA TATATATGCC CTTTCAATTA	
		GATTACACAA ATAGATGGAT ATGCACCCTG ATCATCTTAG	
		ACACACCATG TGGCAGTTGG GCAGTTGGAA ACATTGGTTG	
		CAAGTACTAC AAACCTCAGC TGAGCCTAGC TTCAACAATA	
		AGAATTTATT GGTTCAGTGA GTGAAAAGTC CAGTTGGTCC	
		AATTTGATCA GGATTTCAGC TCCCATCTCT TTTTTTTTTT	
		TCTTTTTTCT TTTTCTTTTT TTGAGACAGT CTTGCTTTGT	
		CACCCAGGTG GAGTGCAATG GCATGATCTC GGCTCACTGC	
		AACCTCCGCC TCCTGGGTTC AAGCCATTCT CCTGCCCGGC	
		CTTCCGAGTA GCTGGGATTA CAGGCATGTG CCACCACGCC	
		CTGCTAATTT TTATATTATT AGTACAGACA GGATTTCACC	
		ATGTTGTCCA GGCTGGTCTC AAACCCCTGA CCTCAGGTGA	
		TCCACCTGCT TTGGCCTCCG AAACTGCCGG AATTACAGGC	
		ATGAGCCACC GTACCCGGCC AGCTCCCATC TCTTAATTAT	
		CTTAGCTCTC CTTTCCTCCT TGGGTTGACA TTGCATTCAG	
		AATTGTGGCA AAGGGCCACC ATGCTCTTAA GACTCAAGTC	
		TATTTTCCAC ACTGTCCAGA GGAGAGAAGT TATCCTAGTA	
		GCTTCTACAC AGAGCAAGTG TCTTTCTCAG AAATCCCAGC	
		AAAGGTCTTG CATACCATTG CTGGTAGGCC TGTCTCTTAA	
		ACCAATCATG AAAGGGGGGA GAGAAGTGAA TGGGATGAAC	
		TAATTTGCAT AGACTAATTA GAGCCCACCC CTGGAGCCTG	
		GGTGTGGCCA TCTTCCCAGA GTTCCTGAGC TAGATGGAGA	
		AGGGGTACTT CTCAGAAAGG GAAAATGGAT ACCCAATGGC	
		CCAAACCCAA AATAGCCCCA GTTCCCCTAA CTTTGACTAC	
		AGGGCAGTCC AGTTTGGGTG CCGCTTCCGT TGCACTCACA	
		TGTCCTACAT ATCTGTTGAC TACTCACAAG TGCAAATGCT	
		TATTCTCAAC TCAACATTAA CATTTTTTCT GGCAACCCAG	
		GTTCACTGGT TCTCTTCCAC AGAGCGGCCC TGAGCAGCTG	
		AGCCTGCAAG CCACGCAAGC ATCTGTTTCT TCTTTTGCCA	
		AGTACAGGAG GATGTTTGCT CTCTCTGTAG AGAGCTTTCT	
		GAGGTCTCTG GGTGTACCCA GAGATTTAAT AGAAATTCTT	
		AACGTTAAGT CACATTCCAG GAAGGAAGGA AGAGTTGTTC	
		GTTCAAATAA GAAAGATAAA TGTTCGGCAC TGTAGGCCCT	
		GTTTACCCCA TCTGAGGCCC TGAATTCATA TATTACAAGA	
		CGGAAGGATT TTGCACAGTT TTTTATGTAG CAAGATTTTG	
		CTCACCACTG AAAAATGTCA GTGTAAATGT GACCGCTTTA	
		AAGATGAGTC AAGTAATTCT TGGAACAGGG AAAAAAATGA	
		ATTTGCCAGG TCAGGAGTTC ACCTGCCTTT GTCAGAGTTG	
		AACCCAACCA CTCTTGACCT CGACTCACTC CCTTAGGGTT	
		AAGAAAGCCC AAACACATTC CTGAGCACAG AGCAAACACT	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TDRD7	NM_001302884.1	AGAGCCGAGG CCAGGCTGCC CTCGAAGCGG GGCGGGGCGA AGCGGGGCGG GGCCGAGCAG GGCGGGGCGG GGGCTTGAGG TGATTCCCAA GCCGCGGGGC GGCTCCGGTG GTGCGGGGAA ACCGAAAGTG GGCGGCGGCC GCGGCGGGGC CCCTGGCGGA GACGGCGGCA GGAGCTGGGC CCAGAGACGC GGGGACGGGC CGTGGGCCCC CGGAACGAGA TTACCTGCTA TGCCATGGCC TGCACAGAAA CTGCAAGAAT TGCTCAGCTT GTGGCTCGTC AAAGGAGTTC TAAAAGGAAA ACCGGGCGTC AAGTTAATTG TCAGATGAGA GTGAAGAAAA CCATGCCATT TTTTCTAGAA GGAAAACCAA AAGCAACCCT CAGACAACCA GGATTTGCTT CAAATTTTTC TGTTGGCAAA AAACCTAATC CAGCACCGTT AAGAGACAAA GGAAACTCTG TTGGAGTTAA GCCTGATGCT GAAATGTCTC CTTATATGCT ACACACAACT CTTGGAAATG AAGCATTCAA AGACATTCCA GTGCAAAGGC ATGTGACCAT GTCCACCAAC AACAGGTTTA GCCCAAAGGC GTCCCTTCAA CCACCTTTGC AGATGCATCT CTCAAGAACC TCTACTAAGG AAATGAGTGA TAATTTAAAT CAGACTGTTG AAAAACCCAA TGTCAAGCCT CCTGCCTCTT ACACTTATAA AATGGATGAG GTTCAAAATC GCATAAAGGA AATACTAAAC AAGCATAACA ATGGCATTTG GATATCTAAG CTTCCACATT TTTACAAAGA GTTATATAAA GAAGACCTTA ATCAAGGAAT TTTACAACAG TTTGAACACT GGCCTCATAT TTGCACGGTG GAGAAACCTT	26

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		AAAGCAGCTT TTGAGAAGTG AACTGGATAC TGAGAAAGTA	
		CCTCTATCCC CACTACCTGG TCCCAAACAA ACACCACCGT	
		TGAAAGGGTG TCCAACAGTT ATGGCAGGAG ACTTTAAAGA	
		AAAAGTGGCA GACCTGCTGG TGAAATACAC AAGTGGCCTT	
		TGGGCCAGTG CACTTCCGAA AGCATTTGAG GAAATGTACA	
		AAGTGAAATT CCCTGAGGAT GCCTTAAAAA ATCTTGCCTC	
		ACTTTCTGAT GTATGCAGCA TAGACTACAT TTCTGGAAAT	
		CCCCAGAAGG CCATTCTCTA TGCTAAACTT CCATTGCCCA	
		CTGACAAAAT CCAAAAGGAT GCAGGGCAAG CACATGGTGA	
		TAATGATATC AAGGCTATGG TTGAACAAGA GTATTTGCAG	
		GTAGAAGAAA GCATTGCTGA AAGTGCTAAT ACCTTTATGG	
		AGGACATAAC AGTTCCTCCT TTAATGATTC CAACTGAAGC	
		ATCACCATCT GTATTGGTGG TTGAACTGAG CAACACAAAT	
		GAAGTGGTTA TCAGGTATGT GGGCAAAGAC TATTCTGCTG	
		CTCAGGAATT AATGGAAGAT GAGATGAAGG AATATTACAG	
		TAAGAATCCT AAGATCACAC CAGTCCAGGC TGTGAATGTT	
		GGGCAGTTGC TGGCCGTAAA TGCCGAGGAG GACGCCTGGT	
		TACGGGCACA GGTCATCTCA ACAGAAGAGA ACAAAATAAA	
		GGTATGCTAT GTTGACTATG GTTTTAGTGA AAATGTTGAA	
		AAAAGCAAAG CATACAAATT AAACCCGAAG TTTTGTTCAC	
		TCTCATTTCA AGCTACAAAA TGTAAGCTTG CAGGCTTGGA	
		AGTCCTAAGC GATGACCCTG ATCTAGTGAA GGTGGTTGAA	
		TCTTTAACTT GTGGAAAGAT CTTTGCAGTG GAAATACTTG	
		ACAAAGCTGA CATTCCACTT GTTGTTCTGT ACGATACCTC	
		AGGAGAAGAT GATATCAATA TCAATGCCAC CTGCTTGAAG	
		GCTATATGTG ACAAGTCACT AGAGGTTCAC CTGCAGGTTG	
		ACGCCATGTA CACAAATGTC AAAGTAACTA ATATTTGCTC	
		TGATGGGACA CTCTACTGCC AGGTGCCTTG TAAGGGTCTG	
		AACAAGCTCA GTGACCTTCT ACGTAAGATA GAGGACTACT	
		TCCATTGCAA GCACATGACC TCTGAGTGCT TTGTTTCATT	
		ACCCTTCTGT GGGAAAATCT GCCTCTTCCA TTGCAAAGGA	
		AAATGGTTAC GAGTAGAGAT CACAAATGTT CACAGCAGCC	
		GGGCTCTTGA TGTTCAGTTC CTGGACTCTG GCACTGTGAC	
		ATCTGTAAAA GTGTCAGAGC TCAGGGAAAT TCCACCTCGG	
		TTTCTACAAG AAATGATTGC AATACCACCT CAGGCCATTA	
		AGTGCTGTTT AGCAGATCTT CCACAATCTA TTGGCATGTG	
		GACACCAGAT GCAGTGCTGT GGTTAAGAGA TTCTGTTTTG	
		AATTGCTCGG ACTGTAGCAT TAAGGTTACA AAAGTGGATG	
		AAACCAGAGG GATCGCACAT GTTTATTTAT TTACCCCTAA	
		GAACTTCCCT GACCCTCATC GCAGTATTAA TCGCCAGATT	
		ACAAATGCAG ACTTGTGGAA GCATCAGAAG GATGTGTTTT	
		TGAGTGCCAT ATCCAGTGGA GCTGACTCTC CCAACAGCAA	
		AAATGGCAAC ATGCCCATGT CGGGCAACAC TGGAGAGAAT	
		TTCAGAAAGA ACCTCACAGA TGTCATCAAA AAGTCCATGG	
		TGGACCATAC GAGCGCTTTC TCCACAGAGG AACTGCCACC	
		TCCTGTCCAC TTATCAAAGC CAGGGGAACA CATGGATGTG	
		TATGTGCCTG TGGCCTGTCA CCCAGGCTAC TTCGTCATCC	
		AGCCTTGGCA GGAGATACAT AAGTTGGAAG TTCTGATGGA	
		AGAGATGATT CTATATTACA GCGTGTCTGA AGAGCGCCAC	
		ATAGCAGTGG AGAAAGACCA AGTGTATGCT GCAAAAGTGG	
		AAAATAAGTG GCACAGGGTG CTTTTAAAAG GAATCCTGAC	
		CAATGGACTG GTATCTGTGT ATGAGCTGGA TTATGGCAAA	
		CACGAATTAG TCAACATAAG AAAAGTACAG CCCCTAGTGG	
		ACATGTTCCG AAAGCTGCCC TTCCAAGCAG TCACAGCTCA	
		ACTTGCAGGA GTGAAGTGCA ACCAGTGGTC TGAGGAGGCT	
		TCTATGGTGT TTCGAAATCA TGTGGAGAAG AAACCTCTGG	
		TGGCACTGGT GCAGACAGTC ATTGAAAATG CTAACCCTTG	
		GGACCGGAAA GTAGTGGTCT ACTTAGTGGA CACATCGTTG	
		CCAGACACCG ATACCTGGAT TCATGATTTT ATGTCAGAGT	
		ATCTGATAGA GCTTTCAAAA GTTAATTAAT GACTGCCTCT	
		GAAACCTTGA CAACTAATTC AGATTTTTTA GCAATAACAA	
		AATGTAGTAG GCTTAAAAAA AATCTTAACT CTGCTACATG	
		GCTCTGACTG CTGTGGGGGA TTGAAAAGAA TATGCTTATG	
		TTTGATGAAA GATATTTAAC AAGTTTTGTT TTAACAGAGT	
		TGACTTTTCA AAGAAAATTG TACTTGAATT ATTACTATAA	
		TATTAGAATA AAAATGTTTA TCAATATAAA AAAAAAAAAA	
		AAAAAAAAAA AA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
TOX4 (house-keeping gene)	NM_001303523.1	AGCAGAGAGA ACACACGTCC TTGCGGAAGT GACGGCAGTT CCGAGTCCAG TGGGGGCGGT GGGAGCGATG AGGGTCTGAG ACGGTGGGAG CGGTTGTGTG AAGATGGAGA CATTCCATAC ACCAAGCTTG GGTGATGAGG AATTTGAAAT CCCACCTATC TCCTTGGATT CTGATCCCTC ATTGGCTGTC TCAGATGTGG TTGGCCACTT TGATGACCTG GCAGACCCTT CCTCTTCACA GGATGGCAGT TTTTCAGCCC AGTATGGGGT CCAGACATTG GACATGCCTG TGGGCATGAC CCATGGCTTG ATGGAGCAGG GCGGGGGGCT CCTGAGTGGG GGCTTGACCA TGGACTTGGA CCACTCTATA GGAACTCAGT ATAGTGCCAA CCCACCTGTT ACAATTGATG TACCAATGAC AGACATGACA TCTGGCTTGA TGGGGCATAG CCAGTTGACC ACCATTGATC AGTCAGAACT GAGTTCCCAG CTGGGTTTGA GCCTAGGGGG TGGCACCATC CTGCCACCTG CCCAGTCACC TGAAGATCGT CTTTCAACCA CCCCTTCACC TACTAGTTCA CTTCACGAGG ATGGTGTTGA GGATTTCCGG AGGCAACTTC CCAGCCAGAA GACAGTCGTG GTGGAAGCAG GGAAAAAGCA GAAGGCCCCA AAGAAGAGAA AAAAGAAAGA TCCTAATGAA CCTCAGAAAC CAGTTTCAGC ATATGCTTTA TTCTTTCGTG ATACACAGGC TGCCATCAAG GGACAGAATC CTAATGCCAC TTTTGGTGAG GTTTCAAAAA TTGTGGCCTC CATGTGGGAT AGTCTTGGAG AGGAGCAAAA ACAGGTATAT AAGAGGAAAA CTGAGGCTGC CAAGAAAGAG TATCTGAAGG CACTGGCTGC TTACAAAGAC AACCAGGAGT GTCAGGCCAC TGTGGAAACA GTGGAATTGG ATCCAGCACC ACCATCACAA ACTCCTTCTC CACCTCCTAT GGCTACTGTT GACCCAGCAT CTCCAGCACC AGCTTCAATA GAGCCCCCTG CCCTGTCCCC ATCCATTGTT GTTAACTCCA CCCTTTCATC CTATGTGGCA AACCAGGCAT CTTCTGGAGC TGGGGGTCAG CCCAATATCA CCAAGTTGAT TATTACCAAA CAAATGTTGC CCTCTTCTAT TACTATGTCT CAAGGAGGGA TGGTTACTGT TATCCCAGCC ACAGTGGTGA CCTCCCGGGG GCTCCAACTA GGCCAAACCA GTACAGCTAC TATCCAGCCC AGTCAACAAG CCCAGATTGT CACTCGGTCA GTGTTGCAGG CAGCAGCAGC TGCTGCTGCT GCTGCTTCTA TGCAACTGCC TCCACCCCGA CTACAGCCCC CTCCATTACA ACAGATGCCA CAGCCCCCGA CTCAGCAGCA AGTTACCATT CTGCAGCAGC CTCCTCCACT CCAGGCCATG CAACAGCCTC CACCTCAGAA AGTTCGAATC AATTTACAGC AACAGCCTCC TCCTCTGCAG ATCAAGAGTG TGCCTCTACC CACTTTGAAA ATGCAGACTA CCTTAGTCCC ACCAACTGTG GAAAGTAGTC CTGAGCGGCC TATGAACAAC AGCCCTGAGG CCCATACAGT GGAGGCACCT TCTCCTGAGA CTATCTGTGA GATGATCACA GATGTAGTTC CTGAGGTTGA GTCTCCTTCT CAGATGGATG TTGAATTGGT GAGTGGGTCT CCTGTGGCAC TCTCACCCCA GCCTCGATGT GTGAGGTCTG GTTGTGAGAA CCCTCCCATT GTGAGTAAGG ACTGGGACAA TGAATACTGC AGCAATGAGT GTGTGGTGAA GCACTGCAGG GATGTATTCT TGGCCTGGGT AGCCTCTAGA AATTCAAACA CAGTGGTGTT TGTGAAATAG TCCTTCCTGT TCTCCAAGCC AGTGAAGAGT TATCTGCTGG GAAAGTGTCC AAGAGCCTGT TTTTGAAACA CAAGCTGGGC TTCTGGTAGT GCCTCATCAC AACCCATGAT GGCTGTTCAT GTTTCACCCC TTTTCTTCCT TCAGCAGAGG CCAGGCTATG GAGCAGGGCC ACTGAATTTG CTGTAATCTG GAGATGCTTT TTACTTTCAA CCATAAGCGG TAATAGCAGA GGAAAGGGTG AAGGGAGTCT GGGCAAGCAA AGCATAGAGA TGGTGGGGTG GTGGTGGGGT TGAAGAAACT TGTTGGTATA ATTGTCATAG GACTTGCCTA AAATATTATT AAAATTACGG GAGTGTACTC AGCTTTGAGC CTAGGAGAAA ATGCCACTGT GTGCATCCAT TTTAAAGGGT TCCCTCATAA AAAAATGTTA TTCCCCATTA TCACATCAGT ACACTGCTTT GAAAACAAAA CTTTTCAACA TGGGCATACT GGGCTACATG GAAAATGACA TCACCCAGGA GTGATTTCTC TTTATATATA TTATTTCTGC AGTTACCATC CTTATCTGAG TTATCACAGT TCATGAATCT AAGAGGCGGA ACTCTACATC ATTAGTAAGA GGTTCCACCA AAGTCTAAAG TTGTATTCAC TTGTGTTTGA TGAACTATCT TTAAAAGACC ATAGGTCTAT CATTATTTCT TAGACATAAT CTAAAGAAAA ACAGACTAGA GAAGCCACCT GGTTGTAACA GAATAAGCAG AAGTTTACAG CATGATAGTC CAAGTGGTGA TAACTTTAAA TAAAACTCAA ATTTTTACTG TTTGTAGACA GGAATGCTGT CCTAGAGAAC CTCCTCCTCA ACCAGCTACG TACATAGTTT TATCCTATGC ATTCCTGTTT TCTGTGTGTT TTTTGTTTTT TTTTTTTTTT TTTTTTTTTG AGACAGAGTC TCGCTCTGTC ACCCAGGCTG GAGTGCAGTG GTGCGACCTC AGCTCACTGA AACCTCTGCC TCCCGGGTTC	27

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AAGCGATTCT CCTGCATCAG CCTCCCGAGT AGCTAGGATT ACAGGCGCCC GCCACTACGC CCAGCTAATT TGTGGTATTT TTAGTAGAGA CAGGGTTTCA CCATGTTGGC CAGGCTGGTC TCGAACTCCT GACCTCATGA TCCGCCCGCC TTGACCTCCC AAAGTGCTGG GATTACAGGC ATGAGCCACC GCACCCAGCC TGCATTCCTG TTTTTTTAAT GGTTTTGGAG GGTAGCAGTA GAGATGGGGT CTCACTATGT TGCCCAGTCT AGTCTTGAAC TCCTGGGCTA CAGTTACCCT CCTACCTCGG CTTCCCAAAG TGCTCGGATT ACAGGTGTGA GCCACTGTGC CTAGCCTATA ATGATCATTT TAATGTTTCC CATGCACTCA TTTAGTTTGA ACCTTCACAG CAACCCAATG AGGTAATACT CCCATTTCAC ATATAATACT GAGAGATGAG TTGCACAAGA TTATACACTG TTAAGTAGCA GAGCCAGAAT GGACTTCAGA ATCCCAACTA CAATACAAAT GTTTATTTAA ATAAAGAAGA AAGCTATTGT ACAAATATCA CTCTTCAGGT TTAGCTTACA GAGCCATGGC TATGGATTCT TAGCTCTGTA AGGAAGTGCT TCTATAAATT CTTAGGTTTA GAGATGATAC CATCTGGGTA CCTTTGCTTG AACCGTGCAA CCACATCTGG GTCTAGTAGG TGGATCCCAT CCAGTTGGTT TCCAAGGGTG ATCCTGAAAC AGTGTAAAAG GAGGGGCAAA CCAGAAATCC TGGAATTAGA GGGTTTAATA TTGTTAAAAA ATGCATACCA AATGAAGACT GCCTATCATC ATATCAAATA TGCCAATTCT AAAAAGAGCT TAACATTAGA ATAGTATATG GTAGAATTAC TAGTTCAGAA TTGGCATAGA TTCTGGTGTT AAAATAGACT GGATCTGTAT TATCTGAGGG TTAGTAACTA ATGCTTAGCC AGGCCTGCTT CACAGAGTTG CTACCAGGGA GTATTCTTTG GATAAGCAAA ATGCTAGCAG CATGTGTTTT AAGCTCTGTT AAGGGGTGAA AGATGTAATT ATTGACAGAT TAAATAGATA ACTTCGTAAC CACCAGGGGG CAGATTCAAT ACATCACAGA ATGGCTGAGG AAGATCCTTG GGTTGTGAAG AGAGTAGAAA CCCTAGGGAG CAGTGCTTTT GGGTCCTAGA ACCTGTTGAG TTTCTAATGA ATATTTGTAG AATCTCATAA AACAGTTTAA ATACAAGCTT AAGTGGCTTA TGAATCCTGT GAAGCTCATT TATGGACTAG TGTAAAACAA TGTGAAGCTC TACTAAGTTC TGTCCTTAAT CATAAATAAT AGCCCCTTGA GGACTAGCCT GTTCTCTGGT CACCTTACCA GTTGGGTTGC ACATTGTGTG GTCGTCCAAA TAACTCAATC TTGCGAGTGC CAGGAGATAG TCTTTCAATC ATGCCATAGA TTTCATCTGG TTTATGACTG GTGGAACGAA CCTAGGAAAT AAAAACTAGC TGCTTTTTAA GTTACACAAG AAAAAA	
TPT1 (house-keeping gene)	NM_001286272.1	CTTCGTGCCA CGTCACCGCC TGCGTCGCTT CCGGAGGCGC AGCGGGCGAT GACGTAGAGG GACGTGCCCT CTATATGAGG TTGGGGAGCG GCTGAGTCGG CCTTTTCCGC CCGCTCCCCC CTCCCCCCGA GCGCCGCTCC GGCTGCACCG CGCTCGCTCC GAGTTTCAGG CTCGTGCTAA GCTAGCGCCG TCGTCGTCTC CCTTCAGTCG CCATCATGAT TATCTACCGG GACCTCATCA GCCACGATGA GATGTTCTCC GACATCTACA AGATCCGGGA GATCGCGGAC GGGTTGTGCC TGGAGGTGGA GGGGAAGATG GTCAGTAGGA CAGAAGGTAA CATTGATGAC TCGCTCATTG GTGGAAATGC CTCCGCTGAA GGCCCCGAGG GCGAAGGTAC CGAAAGCACA GTAATCACTG GTGTCGATAT TGTCATGAAC CATCACCTGC AGGAAACAAG TTTCACAAAA GAAGCCTACA AGAAGTACAT CAAAGATTAC ATGAAATCAA TCAAAGGGAA ACTTGAAGAA CAGAGACCAG AAAGAGTAAA ACCTTTTATG ACAGGGGCTG CAGAACAAAT CAAGCACATC CTTGCTAATT TCAAAAACTA CCAGTTCTTT ATTGGTGAAA ACATGAATCC AGATGGCATG GTTGCTCTAT TGGACTACCG TGAGGATGGT GTGACCCCAT ATATGATTTT CTTTAAGGAT GGTTTAGAAA TGGAAAAATG TGATGCAAAA GAAAGAAATC CCTGCGCTTT CTGTCTGTCT TTGTGGCGGC CCAGATTGAA TTGGGGAATA CATCTTTAGC CTGGAAATGT AGGCTGCATG TTAATGGTAA TGTAACTTTT GCAGTGTAAT GTTTGAAAAA TATTAATGTA GTTTTTGCTT TTACAGTAAC AAATGTGGCA ATTATTTTGG ATCTATCACC TGTCATCATA ACTGGCTTCT GCTTGTCATC CACACAACAC CAGGACTTAA GACAAATGGG ACTGATGTCA TCTTGAGCTC TTCATTTATT TTGACTGTGA TTTATTTGGA GTGGAGGCAT TGTTTTTAAG AAAAACATGT CATGTAGGTT GTCTAAAAAT AAAATGCATT TAAACTCATT TGAGAGAATG CCTTTTAGTT TAATGCATAT TTAAACTAAA TTGATCCTGT AGTGTTCCTG GAGAAGCTAG AGCCTGATTG TAGGCTACTA CTCATCAATT AACTTCTACA GTGGAGACTA CTTCTGGGAC TGGAATATAA AAAGAATCA AAGGTTCTGA TTTTGAGTTG CAATAAAGGG AAAGACCATG CTCATAGCAG TGCCAACATC TGAAGTGTGG AGCCTTACCC ATTTCATCAC CTACAACGGA	28

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CCTAACTAAC TCTCAGCCAT GGCCTCCAAC GAAGATTTCT	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CATCCTACAA TGCACAGACA GCCCCACGAA ACACATAATT	
		ATTTGGCTAA AATGCCAATA GTGTCAGGTG CAGGGGCTCA	
		GGCCTGTAAT GTCAACACTT TGGGAGGCCG AGGTGGGTGG	
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		TTTTGCGATG TCTTCTGGCT TTTGTGAGCT GTATTTGTGA	
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		TCTGTGGCGC TGTGGCTCTC TCTGCACTTC AGAAATGTTA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		AGCATGAGGT CTACGGTGAC CATTTGGGCA AAGCTCCAGT	
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		GAAATAAGAA TTCACTGTGA TTATGTAGTC TTCTGGCTAG	
		TATCAGGCAG CTCTGCTTTT AATTTGGTTA ATTTTATTTT	
		CTCTGAAGAG GGAGAAGAGG TACAATTTAA TCTTGGCCTC	
		CACAAGCATA TTAAAGCTCA CGTGTTAATC AGTGCATTCT	
		TATGCTCCTA CATTAAATGC CTTGGGTAAA TGGATAAATG	
		GACATGTGCC CAGCTTTAAT TTTTTTTGCA ACAGAAAGAT	
		CAGACTTCCG TATGGCATCG TTGGATTTCA GAGGCTTTCT	
		GGTGTATCTG TAAATCTGAA TGTTGCCTTC TGCCAGTCTG	
		TATAACCAGG TGATTCATGC TGCAAATGAA ATCAGGAAGC	
		AGTAAAGTGT TAAAGCAAGA GTATTGTCCA ATTCACTTGT	
		CTTCCTGATC CTTGTACTTT ATTTCACGTG TCGGTGTTTA	
		CATTACATAC TTATATTTCC TGTGAAAGAA AGAGTTAAAT	
		AAATTGTAGC AGTTTGA	
SEPN	NM_031475.2	AGCGGAGCGC CAGGCAGCGC GGAGCGGAGG CCAGGCCCAC	32
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		ACCGCGGGCT CCTCTGGCCC GCAAGAACAC GTGCATGGCG	
		TCCTGGGGAA GGCGCTGAGT GCGGAGTCGC GGCGCCGCAC	
		GCGGCACCAT GGCCCTGGAG CAGGCGCTGC AGGCGGCGCG	
		GCAGGGCGAG CTGGACGTGC TGAGGTCGCT GCACGCCGCA	
		GGCCTCCTGG GGCCCTCGCT GCGCGACCCG CTGGACGCGC	
		TGCCCGTGCA CCACGCGGCC CGCGCTGGGA AGCTGCACTG	
		TCTGCGCTTC CTGGTGGAGG AAGCCGCCCT CCCCGCCGCG	
		GCCCGCGCCC GCAACGGCGC CACACCGGCC CACGACGCCT	
		CCGCCACCGG CCACCTCGCC TGCCTGCAGT GGCTGCTGTC	
		GCAGGGCGGC TGCAGAGTGC AGGACAAAGA CAATTCTGGT	
		GCCACAGTCT TGCATCTGGC TGCCCGCTTC GGCCACCCCG	
		AGGTGGTGAA CTGGCTCTTG CATCATGGCG GTGGGGACCC	
		CACCGCGGCC ACAGACATGG GCGCCCTGCC TATCCACTAC	
		GCTGCCGCCA AAGGAGACTT CCCCTCCCTG AGGCTTCTCG	
		TCGAGCACTA CCCTGAGGGA GTGAATGCCC AAACCAAGAA	
		CGGTGCCACG CCCCTGTACC TGGCGTGCCA GGAGGGCCAC	
		CTGGAGGTGA CCCAGTACCT GGTGCAGGAA TGCGGCGCAG	
		ACCCGCACGC GCGCGCCCAC GACGGCATGA CCCCGCTGCA	
		CGCCGCGGCG CAGATGGGCC ACAGCCCAGT CATCGTGTGG	
		TTGGTGAGCT GCACCGACGT GAGCCTGTCC GAGCAGGACA	
		AAGACGGCGC CACCGCCATG CACTTCGCGG CGAGCCGCGG	
		CCACACCAAG GTGCTCAGCT GGCTGCTGCT GCACGGCGGG	
		GAGATCTCGG CTGACCTGTG GGGCGGGACC CCGCTGCACG	
		ACGCCGCCGA GAACGGGGAG CTAGAGTGCT GCCAGATCCT	
		GGTAGTGAAC GGCGCGGAGC TGGACGTCCG CGACCGCGAC	
		GGGTACACGG CCGCCGACCT GTCGGACTTC AACGGCCACA	
		GCCACTGCAC CCGCTACCTG CGCACGGTGG AGAACCTGAG	
		CGTGGAGCAC CGCGTGCTTT CCCGGGATCC ATCCGCAGAG	
		CTGGAGGCTA AGCAGCCGGA TTCAGGCATG TCCTCACCCA	
		ATACCACGGT GTCGGTCCAG CCGCTGAACT TTGACCTCAG	
		CTCGCCTACC AGCACCCTCT CCAACTACGA CTCCTGCTCC	
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		GCCCGCCAGC CCGCGCGCGC CGGCTGCCCG CGCCTCGGCC	
		CTGCCGCCCG CGGCTCACTC GAAGGCCCCT CCGCTCCCCC	
		GCAGGCGGCG CTGCTTCCTG GGAACCATGT TCCTAACGGC	
		TGCGCCGCGG ACCCCAAGGC GTCCAGGGAG CTGCCACCGC	
		CGCCCCCACC GCCGCCGCCG CCCCTGCCGG AGGCCGCGAG	
		TTCGCCACCG CCGGCCCCGC CTCTGCCCCT CGAGAGCGCT	
		GGCCCTGGCT GCGGGCAGCG CCGCTCCTCC TCGTCCACCG	
		GCAGCACCAA GTCTTTCAAC ATGATGTCCC CGACGGGCGA	
		CAACTCGGAG CTACTGGCTG AGATTAAGGC AGGCAAGAGC	
		CTGAAGCCGA CGCCCCAGAG CAAGGGGCTG ACCACAGTGT	
		TCTCAGGCAT CGGGCAGCCG GCCTTCCAGC CCGATTCGCC	
		GCTGCCTTCT GTGTCACCTG CACTGTCACC AGTCCGGAGC	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CCCACACCGC CAGCTGCGGG GTTTCAGCCG CTGCTCAATG GAAGCTTGGT TCCCGTGCCG CCCACTACTC CTGCGCCGGG AGTGCAGCTG GACGTGGAGG CTCTCATCCC CACGCACGAT GAGCAGGGCC GGCCCATCCC CGAGTGGAAG CGCCAGGTGA TGGTGCGCAA GATGCAGCTG AAGATGCAGG AGGAGGAGGA GCAGAGGCGG AAGGAGGAGG AGGAGGAGGC CCGGCTGGCC AGCATGCCCG CCTGGAGGCG GGACCTCCTG CGGAAGAAGC TGGAAGAAGA GAGGGAGCAG AAGCGGAAAG AGGAGGAGCG ACAGAAGCAG GAGGAGCTGC GGCGGGAGAA GGAACAGTCA GAGAAGCTGC GGACGCTGGG CTACGATGAG AGCAAGCTGG CGCCCTGGCA GCGACAGGTC ATCCTGAAGA AGGGGGACAT CGCTAAGTAC TAGAGGCCGC AGACTCCTGT CCGCAGCCTC GCAGCTCCGT GGGGCCCTCC GCCCCAGCCC CAGCCAGCCA GGCCCTGGTG GAAAGGCTGG GAGCCGCACA GCCCTCCCCT CCTGCGCTGG AAACCCTCCC TGACCCCCAC CCTGGCCCCC CGTATCCCCA GCCCTTGGCA ACACTGGAGT GCACACGCCG CCACGGTTGC CCAGAAAAAG TGCCCAAGCT GCTGACGCAA ACAACAACAA ATGCTGCTTA TTTGCATGCC GACTTACATA TATTTGCATG TTCGTTGACT ATCAAAGAGT GCAGAGCTCT CCCCAGCCCC GTGGGTGGTG ACTTTGTTTT CCTGCGGGGC TCAGCCCCCT CCAGGATGCA GCCCCCTCCC CCGCACCCCG GAACCGGCGT CGCTGGCGCA TCCTGGGTGG AGGCAGGCCC CGAGCTCGGG GAAGGGGTTT TCCCTTCCTC TCTGACCCAG ATCTGCGCGC GGCCTAGCCC GGGCCTCATT TCTTATCCCC GCCAAGGGTT TCCTCTCAGT CATTTGTTTA CCAGAAACAT GAAAACTGCC TGTCTGGCCG GGCCGCACTT GTGGCCCCCG GGACCCCACC TCTGGCCCCA CCTCCCTCAA GTCTGCGCCC CGTCCCCAGC CAGACCCACT CGCTGCCGGG ACCCTTTCAC TGCCCCGGTG GAGTGAATAG AGGATGAGGG GCCCTGACCC TGTGTCTCCA ACTGCTGCAC CCCATCCCGA CCCTGTCTCC GCCACCTCGC AGCCCCATTA AAGCGCTCTC ATCTGGGCTC CGGTTCACTC A	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GATAGAAGTA GCATGTTGCT GCCAGAATAC AAGCATTGCT TTTGGCAAAT TAAAGTGCAT GTCATTTCTT AATACACTAG AAAGGGGAAA TAAATTAAAG TACACAAGTC CAAGTCTAAA ACTTTAGTAC TTTTCCATGC AGATTTGTGC ACATGTGAGA GGGTGTCCAG TTTGTCTAGT GATTGTTATT TAGAGAGTTG GACCACTATT GTGTGTTGCT AATCATTGAC TGTAGTCCCA AAAAAGCCTT GTGAAAATGT TATGCCCTAT GTAACAGCAG AGTAACATAA AATAAAAGTA CATTTTATAA ACCATTTACT ATGGCTTTGT AACAATTGCA TACCCATATT TTAAGGGACA GGTGAATTTA CTACTTTCTA AAGTTTATTG ATACTTCCCT TTTATGTAAA ATGTAGTAGT GATACCTATA TTTCCACATT GTGCATTGTG ACACACTTGT CTAGGGATGC CTGGAAGTGT ATAAAATTGG ACTGCATTTC TTAGAGTGTT TTACTATAGA TCAGTCTCAT GGGCCATCTC TTCCTCAGAT GTAAATGATA TCTGGTTAAG TGTTATATGG AATAAAGTGG ACATTTTAAA ACTAGCAAAG TTAAAAAAAA AAAAAAAAAA AA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		AGTGTTCTTG TATACTACAT TGAGCAGTTT GCTTCTATAC	
		CGTGTCACAA AATTCATGTA TTTCTTGAGA AGCCCTAAAA	
		GCTCATAAAG GAAAATGCCG TGAACTATGT AGCTCAGGCT	
		TGGTAAGGTG CCATCTAAAT TACAAAACAA ACTAATGCAT	
		AATTTTGCTT AAATTTCATC CCAGTATGAT TGTCTTCCCA	
		ACACCAGCAT ATAGTATAGA TTGTCTGTCT TTTTTATATT	
		TTTTAGTTCT TCCTGTACAT GTTTTTGGCA ATAAAGTTAT	
		AGGAAGAACA AAATTATTTT GTTAGAATTA AAACATGCTT	
		AATATTTAGT CTGTTTGTGG AGGGCAGGTA TTCACGTGGA	
		CTGAGATACA ATGTTGGATA CAGAAAATAA CTTTCATTGT	
		CTTCCTGACA CTGTGCTAAG GACATGCTGT TAAAGCTTCA	
		AAGTGACCAG ATGAGGAAGG AATAATTAAT TATTACTCCT	
		GATTTGTAGA TAACTGAGGT AAGAGTGTTT CAAATTTATG	
		ATAGTCTTTT GGGTATTCAG AAACCTTTCC TTATACTGCA	
		CTGGCCACCA GAGCTTAATT TTCCCAGCAG TTACAGCAAT	
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		TCCTAGAAAC CAGGTAGGTG TATCCCATAA CAAGGGAGGA	
		GCATACCACA GCCCCTCATT TGATTAATTC ATTTGATCTA	
		TCTATGTTAT TAAGTACCTA CTAGGAATAA GGCATTGTGG	
		AAATACTATA CAAAGATAAA CATTGTTTAG ATGCTTATCT	
		ACTTTCCTTT TCACCAGAAA AACAGAAAAA AAAGAAACAT	
		TTTCTTACAG AGTAAAAATG TTCTACATAA TCACATGAGT	
		AGTTCATCTC AGTGTTTTTT ATTCTTTAAA GTTGAACTAT	
		CCCAGTTTCA TTCTATACCA TTCATTGGAT AACCTTGTTA	
		CAACCCAGTC ATGAAACAGA GCAGTGTGAT CAGTTATCTG	
		CATTTAACAA ATAGACAAAT CAGTTTACAT AAAGGTTATG	
		TATGTCACCC ACGATGAAAA GAATCTGCAT TTGAATATGC	
		CCGTATGAAT GTGGGTTCTG TTTTTGCAAC AGAGATTAAG	
		TGACCATTTT TTCTAATTTT ATGGCTATAT ATTTTCTTCA	
		TAAAAATTGG TCACATCGGA GAAGCAGTGC CACAGGAAAA	
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		ACTAGGCTAA TCAGTGTACG AATTTGTCAT AGGTAGAGAT	
		TTAAAGGTTA ATATCTTAAA ATAGAAGAAA ATTCTAAATC	
		AATCAATCAG TGAGATATAA ACTAAACAGA CCCACTTCAA	
		AGTTGAAAGA AATTTCTAGG CATAAATTGA GACTAGGAAA	
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		TAAATTGAAG GACAGCTCAG ATTCATTTTT AGGAGAAGAA	
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		TGGTTTCTGT CTCTGATCAT CACCTTCCAT TCTATAAAAA	
		GCTCAGTTAC TGATTTGCTG GGTCATGGTC AAAATTCTTA	
		CCTATTTATT TCATATCAAC TTTAAAAAAT AAATTACTTG	
		CATTCTATAT ATTACTAATT GGGAAGTAAT ATGCCTCAAA	
		TCAGTTTTAT ACTGGATTAT TCCCTATGCT TTAAACCACT	
		GCTCTCAATA AAACACTTCC TGATTAATGT TTGATTATTA	
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		TTAGCCATGC TCTGAAGAAT CTGTGAAAGT ACAGTAAAGT	
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		AAATTATAAT GTCTGTCTTG TAAAAAAGTT GAGGGGACTA	
		AAAGTTTATG ACTCTGATAT GGAAGTTGTC ATATTAAAAA	
		ACTACATTTT AAAACATCAA ATATTTATAC TATTTGCTTT	
		TCAAATAAAA GCATAGTGCT GTTTGGCATA	
PRRC2B	NM_013318.3	GCAGATCGGG AGCGGTGCCG AGAAAAATTT CCTTACTAGA	35
		TGACATTTCA TCGCAATGTC CGATCGTTTG GGGCAAATTA	
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		CCTGTTTGAT AAGTATAAAG GAAAATCAGT AGACGCGATT	
		AGATCCTCAG TTATTCCTAG ACATGGCTTA CAGAGTCTTG	
		GGAAAGTTGC TGCAGCCCGG CGCATGCCAC CGCCTGCAAA	
		CCTGCCAAGC TTGAAGTCTG AAAACAAAGG AAACGACCCC	
		AACATCGTGA TAGTACCCAA GGACGGGACG GGATGGGCAA	
		ACAAGCAGGA TCAGCAAGAC CCAAAGAGTT CCAGTGCGAC	
		GGCCTCTCAG CCGCCGGAGT CGCTGCCGCA GCCGGGTTTG	
		CAGAAATCTG TCTCCAATTT GCAGAAACCG ACACAGTCAA	
		TCAGTCAGGA GAATACAAAT TCAGTGCCAG GTGGACCAAA	
		GTCATGGGCA CAGCTGAATG GAAAGCCAGT AGGACACGAA	
		GGTGGTTTAA GGGGCTCAAG CCGACTGTTA TCCTTCTCTC	
		CCGAGGAATT TCCGACGCTG AAAGCAGCTG GAGGGCAGGA	
		CAAGGCTGGC AAAGAAAAGG GCGTCTTAGA TCTGTCGTAT	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GGAGGGAGGG CGGTGGGCGA CACATAATTT CTGCCACGTC TCTGAGCACC TCCCCAACTG AGCTGGGCAG CAGGAACTCG AGTACGGGAG ATGGAGCCCC CTCCTCGGCA TGTACCAGCG ATTCTAAGGA CCCCTCTCTC CGCCCGGCTC AGCCTGTCCG AAAAGGGGCT TCACAGTTCA TGGGAAATGT ATACCACCCA CCTACATACC ATGACATGCT TCCTGCTTTT ATGTGTTCGC CGAAGTCATC AGAAAACCAG GGTACAGTGG AACGAGGCTC TTTTCCCCTT CCTCAGCTCC GCCTTGAACC TCGAGTTCCT TTTAGACAGT TCCAGATGAA TGACCAAGAC GGAAAAGAAA ACAGGCTGGG ATTGTCTCGC CCACTCCGCC CACTAAGGCA GCTGGTGGAG CGGGCACCAC GGCCCACCAT TATCAATGCG GAAAACCTGA AGGGCCTTGA CGATCTGGAC GCCGATGCCG ATGATGGCTG GGCAGGCCTC CATGAAGAAG TGGACTATTC TGAGAAACTG AAGTTCAGTG ATGATGAAGA GGAGGAAGAA GTTGTGAAGG ACGGCAGGCC AAAGTGGAAC AGTTGGGACC CTAGGAGGCA GCGGCAGTTG TCAATGAGCT CTGCAGACAG TGCGGACGCT AAGCGGACTC GAGAGGAAGG GAAGGACTGG GCTGAAGCAG TGGGTGCGTC CCGTGTGGTC CGAAAGGCGC CAGACCCTCA GCCACCGCCC AGGAAGCTTC ATGGCTGGGC ACCAGGCCCT GACTACCAGA AGTCATCAAT GGGCAGCATG TTCCGGCAAC AGTCCATCGA GGACAAGGAG GACAAGCCCC CACCAAGGCA GAAGTTCATT CAGTCAGAGA TGTCCGAGGC GGTGGAGCGA GCCCGAAAGC GCCGGGAAGA AGAGGAGCGC CGAGCCCGGG AGGAGAGGCT GGCCGCCTGT GCTGCCAAAC TCAAGCAGCT GGACCAGAAG TGTAAGCAGG CACGAAAGGC AGGTGAGGCC CGGAAGCAGG CAGAGAAGGA AGTGCCCTGG TCTCCAAGTG CTGAGAAGGC ATCTCCCCAG GAAAACGGCC CTGCTGTCCA CAAAGGCTCC CCAGAATTCC CTGCCCAAGA GACCCCCACC ACATTCCCAG AAGAGGCACC CACAGTGTCC CCAGCAGTGG CACAGAGCAA CAGCAGTGAG GAAGAGGCCA GAGAGGCTGG GTCCCCTGCA CAGGAGTTCA AGTATCAGAA GTCCCTTCCT CCCCGATTCC AGCGCCAGCA GCAGCAACAA CAGCAGGAGC AGCTGTACAA GATGCAGCAC TGGCAGCCGG TGTACCCCCC GCCGTCCCAC CCCCAGCGCA CCTTTTACCC ACACCACCCC CAGATGTTGG GCTTCGATCC CAGGTGGATG ATGATGCCTT CCTACATGGA CCCACGTATC ACGCCCACTC GGACCCCGGT GGACTTCTAC CCCTCCGCCC TGCATCCCTC AGGACTGATG AAGCCCATGA TGCCCCAGGA GTCCCTCAAT GGGACAGGCT GTCGCTCTGA GGATCAGAAC TGTGTGCCCC CACTCCAAGA AAGAAAAGTG ACCCCCATCG ACTCACCCCC TGTGTGGAGC CCAGAGGGCT ACATGGCACT GCAGAGCAAG GGCTACCCGC TCCCGCACCC GAAGTCGAGT GACACCTTGG CTATGGACAT GCGTGTCAGG AATGAAAGCT CTTTCTCTGC CTCACTCGGA AGGGCAGGGG GCGTAAGTGC TCAGCGCGAT CTCTTTGAGG AGAGAGGGGA GGAGTACTTG AGTGCTTTTG ACAAGAAGGC CCAAGCAGAC TTTGACAGCT GTATCTCTTC TCAAAGAATA GGCCAGGAGC TTTTGTTTCC ACCCCAAGAA AATGTTCAGG ATGCAGGTGC TCCTGGGGGT CACACCCAAA ACCTCAGGTG TTCCCCATTG GAGCCTGACT TTGTCCCAGA TGAGAAAAAG CCAGAGTGTG GCAGTTGGGA TGTTAGCCAC CAGCCAGAGA CCGCTGACAC AGCCCATGGT GTTGAGCGGG AGACACCCCG GGAGGGGACG GCCTTTAACA TCTCCTCCTG GGACAAGAAC GGGAGCCCCA ACAAACAGCC ATCCTCGGAG CCTGAATGGA CTCCCGAGCC CCGGAGCTCC AGCAGCCAGC ACCCGGAGCA GACGGGCAGG ACCCGGAGGT CGGGACCCAT CAAGAAACCA GTCCTGAAAG CCCTCAAGGT GGAAGACAAG GAGAAGGAGC TTGAGAAGAT TAAGCAGGAG CTAGGGGAGG AGAGTACCCG GCTGGCCAAG GAGAAGGAGC AGAGCCCCAC GGCAGAAAAG GATGAGGACG AAGAGAACGA TGCCTCTCTG GCCAACTCCT CCACCACCAC TTTGGAGGAC AAAGGCCCTG GCCATGCCAC TTTTGGCCGC GAGGCCACCA AATTTGAAGA GGAGGAGAAA CCTGACAAGG CCTGGGAAGC CAGACCCCCA CGAGAGTCCA GCGATGTTCC CCCCATGAAG AGAAATAACT GGATCTTTAT TGATGAGGAG CAAGCCTTTG GGGTCAGAGG ACAGGCCCGG GGCCGGGGCC GTGGTTTCAG AGAGTTCACT TTTCGTGGTC GGCCTGCTGG CGGAAATGGG AGCGGCCTCT GTGGTGGGGG GGTCCTGGGG GCCCGCAGCA TCTACTGCAG CAGTCAGCGC AGCGGCCGTG GCCGGGGCCT GCGAGAGTTT GCGCGGCCAG AGGACTGCCC CAGAGCCAAG CCCCGACGGA GAGTTGCCAG TGAGACCCAT AGCGAGGGCT CAGAGTATGA AGAACTTCCC AAGCGCCGCC GGCAGAGGGG CTCCGAGAAC GGGAATGAAG GCTCGCTCCT GGAGAGGGAG GAGAGCACCT TGAAGAAGGG CGACTGCAGA GATTCTTGGC GGTCCAACAA GGGGTGCTCT GAGGACCACA GCGGTCTAGA TGCCAAGAGC	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CGAGGCCCTC GGGCCTTTGG GCGAGCCCTC CCTCCCCGGC TGAGCAATTG CGGGTATGGA CGGAGAACCT TCGTCTCCAA AGAGTCACCC CACTGGCAGA GCAAAAGTCC AGGCAGCTCT TGGCAGGAAT ATGGCCCTTC CGACACATGC GGATCCCGGC GACCTACAGA CAGAGACTAT GTCCCAGATT CCTACAGACA CCCTGACGCA TTTGGTGGCC GGGGCTTTGA GGACAGCCGC GCGGAGGACA AGAGATCCTT CTTCCAAGAT GAACACGTGG CAGATTCTGA AAATGCAGAG AACCGGCCCT TCAGGAGAAG GCGCCCCCCA CGCCAAGATA AGCCCCCTCG ATTCCGGCGC CTCCGGCAAG AGCGGGAGTC CCTGGGCCTG TGGGGACCCG AGGAGGAGCC CCACCTGCTG GCAGGTCAGT GGCCAGGCAG GCCCAAACTG TGTTCTGGGG ACAAGAGTGG CACTGTGGGC CGCAGGTCCC CTGAGCTCTC CTACCAGAAC TCCTCCGATC ACGCCAATGA GGAGTGGGAG ACGGCCTCCG AAAGCAGCGA CTTCAGCGAG CGGCGGGAGC GGCGGGAAGG CCCTGGGTCC GAGCCCGACT CCCAGGTGGA TGGTGGCCTG TCGGGGGCTA GTTTGGGTGA GAAGAAGGAG CTGGCCAAGA GGAGCTTCTC CAGTCAGAGA CCCGTGGTTG ACAGACAGAG CCGAAAGCTG GAGCCGGGAG GGTTTGGGGA GAAGCCCGTT AGGCCAGGTG GTGGTGACAC CTCCCCTCGC TATGAGAGCC AACAGAATGG GACGCCTTTG AAAGTGAAAA GATCCCCAGA CGAGGCCTTG CCTGGAGGTC TTAGTGGCTG CAGCAGTGGG AGTGGCCACT CCCCCTATGC CCTGGAGCGG GCAGCCCATG CCAGTGCTGA CCTTCCCGAA GCCTCCAGTA AAAAGGCAGA GAAGGAGGCC AAGTTGGCTG CTCCGAGGGC AGGTGAACAG GGAGAGGCCA TGAAACAGTT TGACCTGAAC TATGGAAGTG CCATCATTGA AAATTGCGGG TCCAGCCCCG GGGAGGAGAG TGAGGTGGGT TCTATGGTGG GCGAAGGCTT CATCGAAGTC CTGACCAAGA AGCAGCGCCG CCTGCTGGAG GAAGAGAGAA GAAAGAAGGA GCAGGCCGTG CAGGTGCCTG TCAAAGGTCG AGGCCTTTCC TCCCGTATTC CTCCTCGATT TGCAAAAAAG CAGAACAACT TATGTCTGGA GCAAGGTGAC GTGACCGTGC CTGGCAGCAG CCTGGGCACT GAGATCTGGG AGAGCAGCAG CCAGGCTCTC CCTGTGCAGG CCCCAGCCAA CGACTCCTGG AGGAAAGCTG TCACTGCCTT CAGCAGCACC GAGACTGGCT CTGCGGAGCA GGGTTTTAAG AGCAGCCAGG GAGATAGTGG CGTTGACTTG AGTGCCGAGT CTCGGGAGTC GTCTGCGACC TCCTCGCAGC GCAGCTCCCC ATATGGGACT CTGAAGCCAG AGGAGATGAG CGGGCCCGGC CTGGCGGAAC CCAAGGCCGA CAGCCACAAG GAGCAGGCTC CAAAGCCATC TGAGCAGAAG GATTCAGAAC AAGGCTCTGG ACAGAGCAAG GAGCACAGAC CAGGACCCAT CGGCAACGAG CGTTCTCTGA AAAACAGAAA GGGCTCGGAG GGGGCCGAGC GGCTGCAAGG GGCTGTCGTC CCGCCTGTTA ACGGGGTGGA GATTCACGTG GACTCCGTGC TGCCTGTGCC ACCCATTGAA TTTGGAGTCA GTCCAAAAGA CTCCGATTTC AGCTTGCCAC CTGGTTCTGC CTCTGGTCCT ACTGGGAGTC CAGTTGTTAA ACTTCAGGAT GCCTTGGCCA GTAATGCAGG GTTAACACAG AGTATCCCCA TCCTGCGGCG GGACCATCAC ATCCAGAGGG CCATCGGTCT CTCCCCAATG TCCTTCCCCA CCGCCGACCT TACTCTGAAG ATGGAGTCTG CGCGCAAGGC TTGGGAAAAC TCCCCCAGTT TGCCGGAGCA GAGCTCTCCA GGCGGCGCTG GCTCAGGCAT CCAGCCTCCA TCCTCTGTGG GTGCCTCCAG CGGGGTCAAC TACAGCTCCT TCGGTGGAGT GTCCATGCCA CCCATGCCTG TGGCCTCTGT AGCACCTTCT GCTTCTATGC CAGGCAGCCA CCTCCCGCCC CTGTACCTGG ATGGCCATGT GTTTGCAAGT CAGCCCCGGC TGGTTCCTCA AACGATACCT CAGCAGCAGA GTTACCAACA GGCCGCCGCT GCCCAGCAGA TCCCGATCTC CCTTCACACA TCTCTGCAGG CACAAGCTCA GCTTGGACTG AGGGGTGGGC TTCCTGTGTC CCAGTCCCAG GAGATCTTCA GCTCCTTGCA GCCCTTCAGA TCTCAGGTGT ACATGCACCC CAGCCTGTCA CCGCCCAGCA CCATGATCCT CTCTGGGGGC ACAGCCTTGA AGCCTCCATA CTCGGCGTTC CCAGGCATGC AGCCCTTGGA GATGGTGAAG CCGCAGTCTG GCTCACCCTA CCAGCCCATG AGCGGGAACC AAGCCCTGGT CTACGAGGGC CAGCTCAGCC AGGCTGCTGG CCTGGGTGCC TCCCAGATGT TGGACTCCCA GCTCCCACAG CTGACCATGC CACTGCCTCG GTACGGCTCC GGGCAGCAGC CACTGATCCT GCCCCAGTCT ATTCAGCTGC CACCTGGGCA GAGCCTCTCC GTTGGGGCCC CCCGAAGGAT TCCTCCGCCC GGGTCCCAGC CGCCAGTCCT GAACACCAGC AGAGAGCCCT CTCAGATGGA GATGAAAGGC TTCCACTTTG CCGACAGTAA ACAGAATGTC CCTTCAGGAG GCCCCGTGCC ATCGCCACAG ACCTACAGGC CTAGCTCTGC TAGCCCCAGT GGGAAGCCCT CTGGATCAGC AGTTAACATG GGCTCTGTGC AGGGACACTA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CGTGCAACAG GCAAAACAAC GAGTGGATGA GAAACCCAGC CTGGGAGCCG TGAAGCTGCA GGAGGCCCCC TCGGCTGCCT CCCAGATGAA GCGAACCGGA GCGATCAAGC CTCGGGCTGT CAAAGTGGAG GAGAGTAAGG CCTGACAGTG CCTGGCTGCC ACCTCGCCTC TCCCTACTGA GGACGGTGCC GCCATGCGGC CTCGACACAG CCGACACTCG GGAGCCTCAC CAGATCCACC GTCCAAATGC GTGGCCCAGA CTGAGAGACC TCCCTCCTCT CCACTCCCGA AAGCTCCGTT GTCAACCAGC TTGCACCCGT GGATATATGG CATTGACCCG CTTGCTTTGA TACGAAACAA AAAAGCAGAC GACTCCTTCA TCCCATCTGC TCCTACCGTG ACTGTGGAGT GACGCCTCCT GTGCAGTGCA GATTTGCCCT CCCTGCCTCC TCCCTGTCCT GCCGCGCAGC CAGGGCGCCT TCTCAGCAGT GCTTCCGGCC CAGCCGCCCA TCCCTAGGCA CAGTGATTTG GCAGCAGGGT CATTTTACTT TGAGGCTTTT TGTTTTAAAA TGTAGCCAAG GTTTTTACAA AGGGGAAAGG AAAAGAAAAC AAAAACGCAA GCTCCATGTG TATAGCTGAA CTTTTATATG TTTCTTGCCA GCCCCTCCGC TCCCTTCCAT CTCTAGCCTC TGTCCTGTTT AGTTTGATAC GTCACTGCAG TACCTTAAGA GGTGACTCTT AAGAATGCAT CCCCTCCTGA TTCCTCAGCT GGTTCACCCT TGAGGTTATT TGCAAAAAGA AAAGGAGGTT CTTGAGGGCA CCGATTGCGA GCATTCTGGT GCCTGGCTCC CCGCCTGGGA AGCGATGGGG TGCTCAGAGC AGCAGGCAGG TTGGGGGAGG GGGGGGGTCA TAGTTGGGTT CCAGCTCCTG GCTTGATGAG CCCAGGGCGC TTACAGGCAG CCCATGAAGT TGATGACAGT TTTAGCATGA GAATCACACA GGGTCCCTGT CCTGGGCTCC TCTAAAGCCA GTGGATGTGC TGGGCACCAG AGACAAATCA TGGAGATGGC TGCTGGTGGC TCCCAGGTTG GCCCAGATGG GGTGAGCTGA CATACCACAG GCCCATCCCA GGCCCCGTGG GCTCTGCTTC TGGGGCTCCA TACCCTGCCC TGCAGGGGTG CTGTGTTTTT CACACATTTC TTTCCCTGAA GCCTTCTGTA ACCTGTCATT TTCCTTCCTT CCTCTTCCGG AGCCTGCTGC TTTCTCTGGA CCTGTCTCCA CCTCCCACAC AGCTCATCGT GAACACCACT TGGTGATGGA GGGAGTGGAC CCGTGTGTGG TCCCCAAGTG AGGCCACTGG GAGTTTGTCC TTTTCCTCCT TTGCTTCACT CCCAGCAGCA GACCCAGGTT GTCAGGACAG GAGGGCCTGA GCTAAGCAGT AGGCATCAGT CTCGTTTGTC TTCAGACGGC GGGGGCAGGT CCAGGGTGAG GCTGGGTGGA GGGCTGACCA AGGTCCAAAG GGCCTGCGCA GCCTCCGGGA GGGCAGCTTC TCCAGCCAGA GGCTTGTGTG AGCCATCGTG TGCTGGGCTT GTTTTTAAGT AAGAAACAAG GAAATCACTC CAGATTCTGT CATTCCAAGG AAAGGGAAGG GGACAGTTCA GGTTTCTCAG CTGTTCTTAG GGGTCACTGA GCGTCTACCT CCTCCTCCAG AGGAGGCTGG CTCAGAACAC CTAGAGGAGG GGGCCGGGGA TGCACCCCCC ACCAGAGGCT GCCTTCAGCG TCTCACGGGT GCAGGACAGC GCTCAGGCTT GGGCTCTAAG CTCTGTGTCT AGTGTAGAAC ATGGGGAAGG AGCATCTTAG GAACTGCTGA AGTAACTTCT TACTGCTCTC ACAATTCTAA GGAAGCGGGA GAACGGCCTC CTACCAACAG CGCCCACCCC AGAGCTGCCT GGGAAAGGGC AGTTTTACTG AAAGGTGCTT TACTGTTCAC CTGCATCTTT CAGCAGCTCC CCTCCTGCCC TCACCTGGTC TTTTCCCTCT TTATCCCAAG CCTTTATGCT TGAGTCCCTT CCCCAGGGGC TGCCCACCCG ACAGTTCCAG GCATTCCCTA CCTGAGCTTC TTGTCTGCTT TTCCTTCTCC CACTGCAAGC GGCTGCTTGT GGGGCCTGGG ATGAGCCCTC TCTGTCCCCA CCGGCCCTCC TTGCCAAGCC ATTCCTGGGT GAGTTCAGGC CTGCGGGAGC CACACATTCA TCTCCACCTG GACACTTGAG CCGCATGGCC AGACCCCTCC CACCTGATGC GGTGGTGCGT GTGATTTGTC AAAAGAAAGC CTTCTGGATG CTGTTAAGAT GTACCCTTCA GGTGAACCTG GTATCAGACC CACAGTACTT GCTGTTTGAG AAAAAATAAA AACAAAAAGG TCACCTGTTC TCCAGCCCTT TTCTCTTACC TGGTATTTCC TTCCTTTCTC CTCCCCCACC CCAAATAAAA AAACAAAAAA CACTAGAATT TATTTATATG TATTGATGTT GTAGGTCTAG GTGAAAAAAA AAGAAGTAAA TGTTTCACTG CTCTATTTAT ATATAATGTC TGAATTAATT CTGTGCAGGA AAGGCCAGGA AATTGCATGT GAAGTTCGGT GCAGTCACCA CCTGTGTGTG ACCTGAGCTG CAGTCTCTTC GCTGAGATGC AGGTTTTAAA TGAGACTTGG GGGGCTGAGG GCAGGCCTCA GGCCTCCCAG CGCCCCAACC CCTCCTTGGT CTAATGAAAT GCAGTTCTTA GTGCAGAGAT GTTTTAAGGT GCAATATATC TCTTCCTTTC CCGTGGTTTT AGAGCCAAGC TCAAGGTAGT AGGACGTAGG GTCTTATTTT GTTTTCAAAC CCCCATCCTC AGAGCGCAGA TACATGCAGA GGCTTCTGCC AGGATACCAC GGGGCCTTAG TGGGAACAGG TGGAGACCAG	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CACTTCCCTT TCCTGCTGCT GAGGTAGGGA TTGGGGGGTC AGAACCCACT CACTTTTGCC TGTTAAAGTT GCCCTCCTGA CGCTGGCAGC TCTGCCTTGG TCACTGGGGA TGCGGCTCGT TGCTCAGCCA CCAGTGGCCT TGCGGTATTG TCCACCATCC ACTAGAGTGG GATGAAGTCC AGAGTGTGGG TATACATCTC AGATGCCCAT CTACCCACTG GGGACTTCAA TGCCAGCTGC ATTTGGTTTG GTTTTCTTAA CTGTTGGCTT CTCCCCACAG CGTTTTTTGT TTTTTTTTAA ACATTCATAT TGTTTTCAAA CTTGGAATTC ATAGACACTC TGGCTCTAGG TTCCTTAAGG GGGAAAACAA AAGATGACTT TATTTCACAT TCAAGAAAAT CAGTTCAGTT CCAAAGCTGT GGTCCTTCCA GCCACTTCTA GGGACACTGG GGAACCTTGT TAAACGTTGA CATCAGTGCT CTCCAGCCGT GCTGTCACCC TCCTATCTTC TGGATCTGCC TTCGCGATGG TCAGTGACAG CTTCTGGAAG CTGAGCACAC ACAGGTGCAC AGCCATGCTG TGGTCTGGCC TGCTACGGCA GCATGGCAGC TCTGGTGGAG CCTTCTCCCT TGCCATTTGG TTCCCCTGTG CCAAGTAGCT GCAGGCTGCC CCTCAAATCT TCATTTGTCC CTTTTCACTT CCTGCAGAAC AAGCCTGGGT TAGAGGGTCT GCTGGAAATG GCCTTTGAAG ACCAAGGATA CCAGGATGTG TGCACTCTGT CGTGTTCTGT GATGAATGGG AAACGTAGGC TTCCAGAAAG CCAGCTCTCT TCTGAAATGT GACGGACCTA AGCAGGAAGT CATCCAGGAC AGGAGTGGCT CAGTGTTGGG GATGGACGCT GTCGCCCAGC CATGCTCCAC CAGGGCCACC AATGTGTAGT TGGCTGGTGG TCTTCGGGCA TGTGAGACCT GCTCTTCACT GTTTCCACCC CACTTGGTGG CCTCCAGGAT GGTAGTGGCA CCCTCAGAGC CCCATCTTCA GCATGTTCTG AAGCCTCAGA GTGGAAATTC CTGCTAAGGC TCTGTGTGGA CGCCTTTCTC CCGTGATCTA AAGGGGACAC TGTACTCAAG CTTTTGACCT CATGCCTTGT GTAGTAAAAA AGGATTTGGG GGTTTTGTTT GGTTCCTGAG AGGGTTGTGT TTTGTTTTTG TTTCCTTTTG TTTATGTTTT GGCCTTTCCT CTTTGTCTTT CCATGTAGAC CAGATATTTG AAAGGGCAGA CGATGGCTAG AGGTGTAATG TGCAGCTTGT TTATACGGTA TTTTGGGAAA CTTACCTTGG ATGGGAAATC GAATCGTGGA TTCACCAGGC CGGTGCTGGC ACACTCACCC TCGCCCTTTC CCTCCGGTTC AGTACCTATT GTTTCTCCTT TCAAATATGT GATTGTACTA GCTCTTTCCA TATGAAAGAA TTCTCCTTAT TTAAATAAAA AAAGTTTAAA AA	
DOPEY2	NM_005128.3	TCCCACAGTG CCTGGCCCAG AAGCCTTGCT AAATATTTGA ACAGGATTGC CCAATACTTT TCTGCTGTGA GAATGTAAGA TGGATCCAGA AGAGCAGGAG CTCTTAAATG ATTACAGATA CAGAAGCTAC TCTTCAGTGA TTGAAAAGGC TTTGAGAAAT TTTGAGTCCT CGAGTGAATG GGCGGATCTC ATATCTTCAC TTGGCAAACT CAACAAGGCT CTTCAGAGTA ACCTGAGGTA CTCCTTGTTG CCAAGACGGC TCCTCATCAG CAAAAGATTA GCTCAGTGTT TGCACCCTGC CCTGCCCAGT GGTGTCCACT TAAAAGCTCT GGAAACCTAC GAGATTATCT TTAAAATCGT GGGGACCAAA TGGCTGGCCA AGGACTTGTT TCTGTACAGC TGCGGGTTAT TTCCTCTCCT GGCACACGCG GCGGTGTCGG TGAGGCCGGT GCTGCTCACC CTGTACGAGA AGTACTTCCT CCCACTGCAG AAGCTGCTCC TGCCCAGTCT GCAGGCCTTC ATCGTGGGCC TGCTGCCCGG CCTTGAAGAG GGCTCCGAGA TCTCCGACAG AACGGATGCT CTGCTCCTGA GACTGTCGCT GGTGGTTGGC AAAGAGGTGT TTTACACCGC CCTCTGGGGG AGCGTCCTGG CCAGCCCGTC CATCCGCCTC CCTGCCTCAG TCTTCGTGGT GGGCCACATC AACAGGGATG CCCCCGGCCG GGAGCAGAAG TACATGCTGG GGACCAATCA CCAACTCACG GTGAAGTCTT TGCGTGCCTC CCTGTTGGAC TCAAATGTTC TTGTGCAAAG AAATAATCTG GAAATCGTTC TGTTTTTCTT CCCATTTTAT ACCTGTCTGG ATTCCAATGA GAGAGCCATC CCCCTCCTCA GATCTGACAT CGTGCGCATT CTCTCAGCCG CCACCCAGAC CCTACTGAGA AGGGACATGT CCCTGAACAG AAGACTGTAT GCATGGTTAC TAGGCTCAGA CATAAAAGGA AATACCGTTG TGCCAGAATC TGAAATCTCA AATTCTTATG AAGACCAGTC GTCTTATTTT TTTGAAAAAT ACTCCAAGGA TCTTTTAGTT GAGGGTTTGG CTGAGATATT GCATCAGAAG TTCATAGATG CTGACGTGGA GGAACGCCAT CATGCATACC TGAAGCCTTT TCGCGTCCTC ATCAGTCTGC TTGACAAGCC AGAAATAGGG CCTCAAGTGG TTGGGAATTT GTTTCTCGAA GTCATCAGGG CCTTTTATTC TTACTGCAGA GATGCCCTTG GCTCTGATCT TAAACTTAGC TACACCCAGA GTGGAAATTC GCTGATAAGT GCAATCAAGG AAAACAGAAA TGCCTCTGAG ATTGTCAAAA CGGTAAATTT GCTGATAACT TCTCTAAGCA	36

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		ATGCTTTAGA CCAGTGAAGC AGCGTTACAG CGTGAGGAAC	
		AGCGTCAGCC CTCCCCCCAC GGTCTCGGAG CTCTGCGCCC	
		TCCTGGTCTT CCTGCTGGAT GTCATTCCTT TGGAACTTTA	
		CTCTGAGGTG CAAACCCAGT ATCTCCCTCA GGTGCTCGGC	
		TGCCTGGTGC AGCCTCTTGC TGAGGACATG GAGGCCTTAA	
		GTTTACCTGA ACTCACGCAT GCCTTGAAGA CGTGTTTCAA	
		GGTGCTCAGC AAAGTCCAGA TGCCTCCTTC CTACCTCGAC	
		ACGGAGTCCA CCAGCGGAAC CTCGAGTCCA GTAAAAGGTG	
		AAAACGGCAA AATAATTTTG GAAACAAAGG CAGTGATTCC	
		CGGTGACGAA GATGCTTCGT TTCCCCCTCT GAAGTCTGAG	
		GACAGTGGGA TCGGGCTCAG TGCCTCGTCA CCGGAGCTCT	
		CTGAGCACTT GAGGGTTCCT CGAGTTTCTC TGGAAAGGGA	
		CGACGTTTGG AAGAAGGGCG GGAGCATGCA GAGGACGTTT	
		CTTTGCATCC AAGAGCTAAT CGCCAACTTT GCCAGCAAGA	
		ACATTTTTGG AGTACAGCTG ACAGCGTCAG GAGAAGAAAG	
		CAAGTCCGAG GAGCCTGCAG GGAAGAGGGA CAGGGATGGG	
		ACGCAGAGCC TGGCAGCCAA TGATTCCAGC AGGAAGAACT	
		CTTGGGAGCC CAAGCCCATC ACTGTGCCTC AGTTCAAGCA	
		GATGCTGTCA GACTTGTTCA CAGCACGAGG GTCTCCATTC	
		AAGACAAAAA GTTCAGAGTC ACCATCGTCT TCGCCCAGCA	
		GCCCTGCCAG GAAAAACGGG GGAGAATGGG ATGTTGAGAA	
		GGTGGTCATT GACCTGGGGG GTTCCAGGGA GGAACGCAGG	
		GAGGCCTTTG CCGCCGCCTG CCACCTGCTG CTGGATTGTG	
		CCACTTTCCC TGTCTACCTG TCCGAGGAAG AGACCGAGCA	
		GCTCTGTGCA ACGCTCTTCC AGCTGCCAGG AGCCGGTGAT	
		TCCAGTTTTC CATCTTGGCT GAAGTCCCTC ATGACTATTT	
		GCTGCTGTGT GACTGACTGC TACCTCCAGA ACGTGGCCAT	
		TTCCACTCTG CTGGAAGTGA TAAACCATTC CCAGTCCCTG	
		GCGCTTGTCA TTGAAGACAA GATGAAACGC TATAAGAGCT	
		CTGGACACAA CCCTTTTTTT GGCAAGCTGC AGATGGTGAC	
		GGTTCCTCCC ATTGCTCCAG GGATATTGAA AGTCATTGCA	
		GAGAAAACAG ATTTCTATCA GAGGGTGGCT CGTGTGCTTT	
		GGAATCAGCT GAACAAAGAG ACCCGGGAGC ATCACGTCAC	
		CTGCGTAGAA TTGTTCTACC GGCTGCACTG CCTGGCCCCT	
		ACGGCCAACA TCTGCGAGGA CATCATCTGC CATGCCCTCC	
		TGGACCCTGA CAAGGGAACA AGGCTGGAAG CTCTGTTTAG	
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		AGTCGAGTAA CATCTCACAA TCGCTCCTTT GATAGGTCCT	
		TGTTTGTCGT GCTGGACAGC CTGGCCTGCA CGGATGGTGC	
		CATCGGTGCG GCAGCCCAGG GCTGGCTGGT GCGTGCGCTC	
		TCCCTCGGGG ACGTGGCTCG CATCCTCGAA CCCGTGCTCC	
		TGCTGCTGCT GCAGCCAAAA ACCCAGAGAA CCTCCATCCA	
		CTGCCTCAAG CAGGAGAACT CGGCCGATGA CTTGCACCGT	
		TGGTTTAACA GGAAGAAAAC CTCTTTCAGA GAGGCATGCG	
		CAGTGCCCGA GCCTCAGGAG AGCGGCTCTG AAGAGCACCT	
		GCCTCTGAGC CAGTTCACCA CAGTGGACCG TGAAGCCATT	
		TGGGCCGAAG TGGAGAAGGA GCCCGAGAAG TACCCGCTGC	
		GAGGCGAGCT GAGCGAGGAA GAGCTGCCCT ACTACGTGGA	
		GCTTCCAGAC AGGACGGCCC ACGGCGCCCC GGACAGCAGC	
		GAGCACACCG AGTCTGCAGA TACAAGCTCC TGCCACACGG	
		ACAGCGAGAA CACGTCCTCC TTCTCCTCCC CTTCCCACGA	
		CCTGCAGGAG CTGAGCAACG AAGAGAACTG CTGTGCACCC	
		ATCCCCATGG GGGGCAGGGC GTACCCCAAG CGCTCGGCCC	
		TGCTGGCGGC CTTCCAGTCA GAAAGCTTCA AGGCTGGGGC	
		CAAGTTAAGC CTGGTGCGGG TGGACTCGGA CAAGACGCAG	
		GCTTCTGAGT CGTTCTCCAG CGACGAGGAG GCGGACTTGG	
		AGCTCCAGGC CCTCACCACA TCCAGGCTGC TAAAGCAGCA	
		GCGGGAAAGG CAGGAGGCCG TCGAGGCCTT GTTCAAGCAC	
		ATCCTGCTCT ACCTGCAGCC CTACGACTCT CGGCGGGTCC	
		TCTATGCCTT CTCGGTGCTG GAGGCTGTGC TCAAAACCAA	
		CCCTAAGGAA TTCATCGAGG CTGTGTCCAG GACTAGCATG	
		GATACCAGCT CCACCGCGCA CCTCAACCTC ATCTCCAACC	
		TCCTCGCTCG CCACCAGGAG GCCCTCATTG GCCAGAGTTT	
		CTACGGAAAG CTCCAGACCC AGGTCCCCAA CGTGTGCCCC	
		CACTCTCTGC TCCTGGAGCT GCTCACCTAC CTCTGCCTGA	
		GCTTCCTGCG CTCCTACTAC CCTTGCTATT TGAAGGTCTC	
		GCACCGAGAC ATTCTCGGCA ACCGGGACGT GCAGGTCAAA	
		AGTGTCGAGG TTTTGATCAG GATAATGATG CAGCTGGTCT	
		CAGTGGCCAA GTCTTCGGAA GGGAAGAACG TGGAGTTCAT	
		CCACAGCTTG CTGCAGAGGT GCAAAGTTCA GGAGTTTGTC	
		CTGCTCTCCC TGTCGGCGTC CATGTACACG AGCCAGAAGC	
		GCTACGGGCT GGCCACCGCC CACCACGGCA GGGCCCTGCC	
		AGAGGACAGC CTCTTTGAGG AGAGTCTCAT TAACTTGGGT	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		TGCTGAAGCT GCTGCAGGTG CTGATTGTCT TGGAACACCA	
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		CTGTCCCGGG AGTGGCAGAG AGCCCTGAAC TTCCAGCAGG	
		CCATCAGCGC CCTGCAGTAC GTGCAGCCCC ACCCCCTCAC	
		CTCCCAGGGT CTTCTGGTCT CTGCGGTGGT GAGGGGTCTG	
		CAGCCCGCCT ACGGTTACGG CATGCATCCG GCCTGGGTGA	
		GCTTGGTCAC GCATTCCTTG CCCTACTTCG GAAAGTCCCT	
		GGGCTGGACG GTGACACCCT TTGTTGTCCA GATTTGCAAA	
		AACTTGGATG ACTTGGTCAA GCAGTATGAA AGCGAATCTG	
		TGAAGCTCTC TGTCAGCACA ACCTCCAAGA GGGAAAACAT	
		TTCTCCAGAT TATCCACTCA CCCTTCTAGA AGGTCTAACG	
		ACCATTAGTC ATTTTTGTCT TTTGGAACAA GCCAACCAAA	
		ACAAAAAGAC CATGGCTGCA GGTGATCCTG CCAACTTGAG	
		GAATGCCAGA AATGCCATTT TGGAAGAGCT GCCTCGAACT	
		GTTAACACCA TGGCCCTTCT CTGGAATGTT CTCAGAAAGG	
		AGGAGACTCA AAAGAGACCT GTCGATCTCC TAGGGGCCAC	
		GAAGGGATCC TCTTCCGTTT ACTTTAAAAC CACCAAAACC	
		ATAAGACAAA AAATTTTAGA CTTCTTAAAC CCCTTGACGG	
		CCCATCTTGG GGTTCAGTTG ACAGCGGCTG TTGCGGCAGT	
		GTGGAGCAGA AAGAAAGCCC AGCGTCACAG TAAGATGAAG	
		ATTATCCCAA CGGCAAGTGC ATCCCAGCTA ACCCTTGTCG	
		ACTTGGTGTG TGCACTCAGC ACCCTGCAGA CTGACACGCT	
		GCTGCACCTG GTGAAGGAGG TGGTGAAGAG GCCACCCCAA	
		GTCAAAGGGG GTGATGAGAA ATCGCCCCTA GTGGACATTC	
		CTGTGTTGCA GTTTTGCTAT GCTTTTCTCC AAAGGCTCCC	
		AGTACCAGCC TTGCAAGAGA ACTTTTCTTC ACTGTTGGGA	
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		GGTATTTTCT GCTTCTCAGC ATGCTGAATG ACTTTGTAAC	
		AAGAACTCCC AACCTGGAAA ACAAGAAGGA CCAAAAAGAC	
		CTGCAGGAAA TCACTCAGAA AATCCTAGAA GCTGTGGGGA	
		ACATTGCCGG CTCTTCCTTG GAGCAAACCA GCTGGCTAAG	
		CAGAAACCTG GAAGTGAAGG CCCAACCTCA GGCCTCTCTA	
		GAAGAATCTG ATGCTGAGGA GGACCTGTAT GATGCTGCTG	
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		CAGCGTGCAA GCCCTCTCTC TCCTGGCAGA GGTACTGGCT	
		TCCCTCCTGG ACATGGTTTA TCGAAGTGAT GAGAAGGAGA	
		AAGCTGTGCC GTTAATCTCC CGTCTGCTTT ACTATGTTTT	
		TCCATACTTA CGCAACCACA GTGCCTACAA TGCTCCCAGC	
		TTCCGGGCTG GCGCTCAGCT GCTGAGCTCC CTGAGTGGCT	
		ATGCCTACAC AAAGCGAGCC TGGAGGAAGG AGGTCCTGGA	
		GCTGTTTCTC GACCCCGCTT TCTTTCAGAT GGATACTTCC	
		TGTGTTCATT GGAAGTCCAT TATTGACCAT CTTTTGACTC	
		ATGAGAAAAC AATGTTTAAG GATTTAATGA ACATGCAGAG	
		CAGTTCTTTG AAACTATTCT CAAGTTTTGA ACAGAAAGCC	
		ATGCTGTTAA AGCGCCAGGC TTTTGCTGTC TTCAGTGGAG	
		AACTTGATCA ATACCACCTT TACCTTCCAC TGATACAAGA	
		ACGCCTGACA GACAATCTCA GAGTTGGACA GACATCCATA	
		GTTGCTGCTC AGATGTTTCT TTTTTTCAGA GTTTTGCTGC	
		TAAGAATATC TCCTCAACAT TTGACTTCAT TGTGGCCAAT	
		AATGGTCTCT GAATTGATTC AGACATTCAC ACAGCTTGAA	
		GAAGATCTAA AAGATGAAGA TGAGTCATTG AGAAGCACCA	
		ACAAAGTAAA CAGAACGAAA GTTTCAGTCC CGGATGCAAA	
		TGGACCCTCA GTGGGGGAGA TACCCCAGAG TGAACTCATC	
		TTGTATTTAT CAGCTTGCAA ATTCTTGGAC ACAGCGCTTT	
		CTTTTCCACC TGACAAGATG CCATTATTTC AAATTTATAG	
		GTGGGCATTT ATTCCAGAAG TGGACACAGA GGGCCCTGCC	
		TTCCTGTCGG ATGTAGAGGA GAATCACCAA GAATGCAAAC	
		CCCACACTGT CAGGATTCTA GAACTTCTAA AATTAAAGTT	
		TGGGGAAATC AGTAGCTCTG ATGAGATCAC CATGAAGAGT	
		GAATTCCCGC TTCTGCGCCA ACATTCTGTT TCCAGCATCA	
		GGCAGTTGAT GCCATTCTTC ATGACTCTAA ATGGTGCATT	
		TAAGACCCAG AGACAGCTGC CTGCTGATAG CCCAGGAACT	
		CCATTCTTGG ACTTTCCTGT CACAGATAGC CCAAGGATCT	
		TAAAACAACT GGAAGAATGC ATCGAATATG ATTTTCTGGA	
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		TCCATGTATT GGTACTTTAC TGAAAACCAG GTTATATTCT	
		AAAGAAGAAA GAAGGCAGGA TAGTGCTTTT GAACAAGCCT	
		ATTTCCATTT TGAAAGTAGA TTTCAGGCTA GGTGCGGTGG	
		CTCACACCTG TAATCTCAGC ACTTTGGGAG GCCAAGGCAG	
		GCAGATCACT TGAGGTCAGG AGTTCGAGAC CAGCCTGACC	
		AACATGGTGA GACCCTGTCT CTACTAAAAA TACAAAAATT	
		AGCTGGGTGT GGTGGCGGCG CCTGTAATCC CAGCTACTTG	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AGGCTGCAGT GAGCCGAGAT CACGACACTG CACTCCAGCT GTGTGACAGA ATGAGACCAT CTCCAAAAAA AAAAAAAAGT AGATTTCAGA TAATTTACTG TTCAGCAACA GGACACACCT CCCTAAATGC CTTGTAATAT ATTTGAATCT GATTCTGCAT TTCTTCCTCA ATTTATGTAA TGAAAATAAA ATTAATATAT CATCTAACAG TAGCACAAAA TTTGTAATAT GAAGTAAAGT ATGAAGATAA TGAAGAAGTT GTTTTCTTTG TTGAAGCAGT TATATGGGTC TTTCTCAGTA TATTTCTCTT TTCTCTAAAA GTTTAAACTT ATTAAAAGAA TGTTATTTTT AACCTTTCAA AAAAAAAAA	
NDUFB11	NM_019056.6	GCTCTGGCCG GCCCCGGCGA TTGGTCACCG CCCGCTAGGG GACAGCCCTG GCCTCCTCTG ATTGGCAAGC GCTGGCCACC TCCCCACACC CCTTGCGAAC GCTCCCCTAG TGGAGAAAAG GAGTAGCTAT TAGCCAATTC GGCAGGGCCC GCTTTTTAGA AGCTTGATTT CCTTTGAAGA TGAAAGACTA GCGGAAGCTC TGCCTCTTTC CCCAGTGGGC GAGGGAACTC GGGGCGATTG GCTGGGAACT GTATCCACCC AAATGTCACC GATTTCTTCC TATGCAGGAA ATGAGCAGAC CCATCAATAA GAAATTTCTC AGCCTGGCCG AAAATGGTTG GCCCCACGAA GCCACGACAA CTGGAGGCAA AGAGGGTTGC TCAACGCCCC GCCTCATTGG AAAACCAAAT CAGATCTGGG ACCTATATAG CGTGGCGGAG GCGGGGCGAT GATTGTCGCG CTCGCACCCA CTGCAGCTGC GCACAGTCGC ATTTCTTTCC CCGCCCCTGA GACCCTGCAG CACCATCTGT CATGGCGGCT GGGCTGTTTG GTTTGAGCGC TCGCCGTCTT TTGGCGGCAG CGGCGACGCG AGGGCTCCCG GCCGCCCGCG TCCGCTGGGA ATCTAGCTTC TCCAGGACTG TGGTCGCCCC GTCCGCTGTG GCGGGAAAGC GGCCCCCAGA ACCGACCACA CCGTGGCAAG AGGACCCAGA ACCCGAGGAC GAAAACTTGT ATGAGAAGAA CCCAGACTCC CATGGTTATG ACAAGGACCC CGTTTTGGAC GTCTGGAACA TGCGACTTGT CTTCTTCTTT GGCGTCTCCA TCATCCTGGT CCTTGGCAGC ACCTTTGTGG CCTATCTGCC TGACTACAGG TGCACAGGGT GTCCAAGAGC GTGGGATGGG ATGAAAGAGT GGTCCCGCCG CGAAGCTGAG AGGCTTGTGA AATACCGAGA GGCCAATGGC CTTCCCATCA TGGAATCCAA CTGCTTCGAC CCCAGCAAGA TCCAGCTGCC AGAGGATGAG TGACCAGTTG CTAAGTGGGG CTCAAGAAGC ACCGCCTTCC CCACCCCCTG CCTGCCATTC TGACCTCTTC TCAGAGCACC TAATTAAAGG GGCTGAAAGT CTGAAAAAAA AAAAAAAA	37
ND4	NC_012920.1	ATGCTAAAACTAATCGTCCCAACAATTATATTACTACCACTGAC ATGACTTTCCAAAAAACACATAATTTGAATCAACACAACCACCC ACAGCCTAATTATTAGCATCATCCCTCTACTATTTTTTAACCAAA TCAACAACAACCTATTTAGCTGTTCCCCAACCTTTTCCTCCGACC CCCTAACAACCCCCCTCCTAATACTAACTACCTGACTCCTACCC CTCACAATCATGGCAAGCCAACGCCACTTATCCAGTGAACCACT ATCACGAAAAAAACTCTACCTCTCTATACTAATCTCCCTACAAA TCTCCTTAATTATAACATTCACAGCCACAGAACTAATCATATTT TATATCTTCTTCGAAACCACACTTATCCCCACCTTGGCTATCATC ACCCGATGAGGCAACCAGCCAGAACGCCTGAACGCAGGCACAT ACTTCCTATTCTACACCCTAGTAGGCTCCCTTCCCCTACTCATCG CACTAATTTACACTCACAACACCCTAGGCTCACTAAACATTCTA CTACTCACTCTCACTGCCCAAGAACTATCAAACTCCTGAGCCAA CAACTTAATATGACTAGCTTACACAATAGCTTTTATAGTAAAGA TACCTCTTTACGGACTCCACTTATGACTCCCTAAAGCCCATGTC GAAGCCCCCATCGCTGGGTCAATAGTACTTGCCGCAGTACTCTT AAAACTAGGCGGCTATGGTATAATACGCCTCACACTCATTCTCA ACCCCCTGACAAAACACATAGCCTACCCCTTCCTTGTACTATCC CTATGAGGCATAATTATAACAAGCTCCATCTGCCTACGACAAAC AGACCTAAAATCGCTCATTGCATACTCTTCAATCAGCCACATAG CCCTCGTAGTAACAGCCATTCTCATCCAAACCCCCTGAAGCTTC ACCGGCGCAGTCATTCTCATAATCGCCCACGGGCTTACATCCTC ATTACTATTCTGCCTAGCAAACTCAAACTACGAACGCACTCACA GTCGCATCATAATCCTCTCTCAAGGACTTCAAACTTACTCCCAC TAATAGCTTTTTGATGACTTCTAGCAAGCCTCGCTAACCTCGCC TTACCCCCCACTATTAACCTACTGGGAGAACTCTCTGTGCTAGT AACCACGTTCTCCTGATCAAATATCACTCTCCTACTTACAGGAC TCAACATACTAGTCACAGCCCTATACTCCCTCTACATATTTACC ACAACACAATGGGGCTCACTCACCCACCACATTAACAACATAA AACCCTCATTCACACGAGAAAACACCCTCATGTTCATACACCTA TCCCCCATTCTCCTCCTATCCCTCAACCCCGACATCATTACCGGG TTTTCCTCTT	38

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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MRPL19	NM_014763.3	GTAGTCTTGA CGTGAGCTAG CTGGCATGGC GGCCTGCATT GCAGCGGGGC ACTGGGCTGC AATGGGCCTA GGCCGGAGTT TCCAAGCCGC CAGGACTCTG CTCCCCCCGC CGGCCTCTAT CGCCTGCAGG GTCCACGCGG GGCCTGTCCG GCAGCAGAGC ACTGGGCCTT CCGAGCCCGG TGCGTTCCAA CCGCCGCCGA AACCGGTCAT CGTGGACAAG CACCGCCCCG TGGAACCGGA ACGCAGGTTC TTGAGTCCTG AATTCATTCC TCGAAGGGGA AGAACAGATC CTCTGAAATT TCAAATAGAA AGAAAAGATA TGTTAGAAAG GAGAAAAGTA CTCCACATTC CAGAGTTCTA TGTTGGAAGT ATTCTTCGTG TTACTACAGC TGACCCATAT GCCAGTGGAA AAATCAGCCA GTTTCTGGGG ATTTGCATTC AGAGATCAGG AAGAGGACTT GGAGCTACTT TCATCCTTAG GAATGTTATC GAAGGACAAG GTGTCGAGAT TTGCTTTGAA CTTTATAATC CTCGGGTCCA GGAGATTCAG GTGGTCAAAT TAGAGAAACG GCTGGATGAT AGCTTGCTAT ACTTACGAGA TGCCCTTCCT GAATATAGCA CTTTTGATGT GAATATGAAG CCAGTAGTAC AAGAGCCTAA CCAAAAAGTT CCTGTTAATG AGCTGAAAGT AAAAATGAAG CCTAAGCCCT GGTCTAAACG CTGGGAACGT CCAAATTTTA ATATTAAAGG AATCAGATTT GATCTTTGTT TAACTGAACA GCAAATGAAA GAAGCTCAGA AGTGGAATCA GCCATGGCTT GAATTTGATA TGATGAGGGA ATATGATACT TCAAAAATTG AAGCTGCAAT ATGGAAGGAA ATTGAAGCGT CGAAAAGGTC TTGATTCTGA GAATGAATTT GGTTAGTTGC AGAAGATACA TTGGCTCTAA GAGGATATAT TTTGAGACCA ATTTAATTTC ATTTATAAGA ACATAGTAAT TAAGTGAACT AAGCATTCAT TGTTTTATTA ATACTTTTTT TCTAAAATAA AACTTGTACA CCAGTTTATT ACTCTAAAAA GAGAATTACA CATGCCAAAT GGACCAATGT CCATTTGCTT	40

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		ATTGGAGGCA AAGCTACAAT AGAAGTCAGA GCATCACCAG AATGGTCTTT AATGAGCATG GAACCTGAGC AAAGGGAATA GGTGGGATGA ATTTTTTTTT TAATTGTGAA ACAATTCATA AGCACAATAT GATTTACAGA ATAATAAACA TTCATGTACC CACTATCAGG TTAAGAAATA GAACATTTAT TAATATGTAG GAATGTTAAG AAATAAAACA TTTAATAAGA TCTCAGAAGA CTCCAGTAAA TCTGCAATTG TATCTCTCTC CTTTTTAAAT GTAAATATCA TCTTGACTTG TTAATTATTC CCTTGCATTT CTTTTAGTTT ACTGCCAACA CATATATTCT TCAACAATAT ATTTAATTTT GAAAAACCTG AAAAAAAAAA CCTGTTAGCA AGTATAAAGG GGCAGTATTA CTATTATTGC ATGAAGGCTT CAAGGGAAAC GTTACAGTCT TTGGGTCATA GTCTGGCTTC AGCTTCCTCT GAGAGTTTAC AGAGGCCAAT TTTGAGCAAA TTCATGGCTA AGGTTATGAG TGAGTTCTGC TAAACAGAAG GCTCACCACA AGGTATCTGG CAGGATTATA CTGGGTAGCT GGATGTTGCA GAAATGTGGT TAGAGGAAGT AAACTGTTTT TTGATGCTCA CAGCATGATG AATCAAACTC TGTATCTTAG GATTAGGTTA AAACAATACC TTTGGTATGA TATGAGTGTT GTTGCTGATC CATGCAGCAT GGATTGGAAA GCTGGGGTAT AAGCACACAT GCTAAAGAAA AACATGTAAT TTGGTCCATA CTCACCTGGA TATACTGTTC CTCAGGTTAA AAAATACAGT ACTATCCTAA ATCTTGAAGG CAACTCTCAG CCTATCCATT GAGTTACCTT CAGATCTGCC CTCTGGTTCC TAGCTGTCTT GGGACTAACT TCTTTCCTGC GCTCAGCTGT TTTCTGGATT CCATGTTTTC CATTTTATTG AGTACTAACT TGTTTTGCTG CAGCACATCC TTTGGTAGCT TCTAGAGGAA GTTTGTGTGG AGGTAAAATT TTTGAGACCT TGCATGTCTC ATGTTTGATT GATACTTTAT ACGTTTAGGT AGGAGGTAAT TTTCCTTCAG GACTTTAAAA ATATTGTTGC TCCATTTTCT TTGTTTCTAT TGTTGTATTG AGAAATCCAA TGCCATTTTG ATTTCCCCAT CATAAATTTC ATGATGATGT GTCTTGGTGT GGGTCTATAT TTATCCATTG TATTGGGTTT TAGGTGAACC CTTCCAGATA GTAACTCATT TCTGTCAGTT CTGGGAAACA CTTAGCATTG GTTGATGATT TATTCTCTGC TGCTTTGTTC TCCCAACTAT TATTTGGATG TTGGATATCC AGCACTGGGT ATCTATTTTC TTACCTCCCT CCCTTGACCC CAGTCTCTGT TTTTTAGCTC TTTAGCTCAA TCTTCCAACT CTTTGCTATT GTATTTTAAA ATCTTAAGAC CCCTTCTTGA TTTGTAGAAG TTCCTTTTCT TACAACCAAA AAGCCTTTAT CTATGGATTT GTTCACAGAT AAGGGGTATT CAATATAGTG TATTTTTTTT TCATTTAAAA TTGTTTGCGC ATCTATTTCC TCCAAATTTC TTTCTGTATT TATTTTTTGT TGTCTATATT TCAGACTTTT CCAGGATATC TGATAATCTT TGGCTGTCTT CTTATGGTTG AAAGAGGGAC TAAAAAGCTT GGAAAGCCTT TGGGTTGTGG GAAGGGGCTG TCTTTAGGAT TATCTGAATG GGCTTTTTTG GGAGTCCCCT CCTCCACATG AATATTTTGG TTTTGTCAGA TTCCCTAGAA TAGAGGCTTC CAATCTCCTT CCTGGAGGGG TCTGTCCAGG AAGGAGATTG TCTAGGGGTC TGTCAGACAG CAGCTTTCAG CTACTTCCTT GATCTTTTTC ACTAATGATT ATATAGTCAT CTAACTACTG TCAACAAGTA ATAGATATCC TATCCTTCAC TTGTTTAGAT TATTTGCTGA GATAACCTCT CAAAAGAACC TCTCAAAATA AAAGGTTAAC AAGAGCCTAT ATCTTATATT TTTCTTCTCT TTATCTTGTT AGAAGATAGC TATTAAAACC TGTTCTTTTT CTGTCTTGAT AAACACACTT CAATCTTGGT AGAATGGTAG ATGGGACAGT ATATTTTAGG ACCTAAAGCT CTGCAAATGT ATGATCAGCT TGTAAGTACA GGTGCTCAAA AACATGTAAA CAATCATGCT TTTTACTCTG TAGGAATATC TTTAAAATTC TTGTGAATTT TTCCCCAGAA GTAAAGCAAA TCTTCCCCCA GAAATAAAAT TAAATGTGCA TAATCTAAAG CTTTTTTTTT TTATTGTGGT AGGATATATA TATAAAACAT AATTTGCCAT TGTAAACATT TTAAATTTAC AAGTCAGAGG CATTAATTAC ATCACAATGT TGTGAAATTA TTACTACTAT TTCCAAAATT TTCTCATCAC CCCAAACTGA AACTCTGTAA CTGTTGAGCA ATAACCTCAT TCCTGTATCT CTCCCAACCC CAGGTAACCT CAAATCTTTC TTTTTATCTT TGAGACAAGG TCTCATTCTA TCACTCAGGT AGGAGTGCAG TGGTGTGATC ATAGCTCATT GCAGCCTCAA AATCCTGGGC TCAAGCAATC CTCCTTGAGT AGCTAAGACT ATAGGCACAC ATTAACTGCG CCTGGCTGAT TTTGTTTTTT GTAGAGATGT GGTCTTGCTA TGTTTCCCAT GCTGGTCTTG AGTTCCTGGC CTCAAGCAGT CCTTAAGATT CATCCATGTT GTGGCATGTG TCAGAATTTC ATTTGTTTTT ATGACTAAAT AATATTCCAT TGTATGTATA TACATTTTGT TCATCCATCT TCTGATGAAC ACTGGGATAT GTCTACCTTT TGGCTATTGT GAATAATGCT GCAGTAAACA	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		TTGACATAAC AAGTATGTAT TTGATTGCCT GTTTCTAAGT TCTTTTGGGT ATACATCTTG AGTAGAATTG CTAGATAATG TCATGTTTTA TTTCTCTTGT GATTTCTTCT TCGATCCCCT GGTTGAGTGT GTTAATTTCT ACATGTTTAT GAATTTCCCA CTGTTTTTTT GTTATTGATT TCCAAGTTCA TTCCATTGTG ATTAGAGAAG ATACTTAGTA TGATTTTAAT GTTTTTGAGA ATTGGTGTGT GGCCTGATAG ATGGTCTGTC CTGGAGAATG TTCCTCATAC ACTTGAGCAA AATATTTATC ATGCTATTGT TGACTGTAGT TTTCTATATG TCTCTTAGGT CAAGGTGGTT TACAATGTGT TAAGGTTCTC TTTTTTTAAA AAAATTTTTG CACAGAGTAT CTTTTTCTAT GTGTTCCATG TATTTGTGTC TTTGGAGCTA TAGTCTCTTG TAGACAGCAT ATCACTATCT TGTTTTGTTT TGTTTTTTCT GTCCATTCTG CCAATTTCTG CCTTTTGATT GGAAAATTTA ATCCATTTGC ATTTAAAGTA ATTAAGGAAG GACTTTCTTC TACCATTTAA CACTTCTTCT ATATGTCATA TACTTTTTTG GCCCCTCATT TCCTCTTTAT GGCCTTCTTT TCTGTTTTTT TGTAGTGAAC TAGTCTGATT CTCTTTCCAC TCCCCTTTGT GTATATTTGT TAGATGTTTT ATTTGTGGTT GCTATGGGGA TTATAGTTAA CATCCTACAC TTAAAACAAT CTAATTTAAA CTGATACCAA TTTACCTTCA ATAGCATACA AAATCTCTAC TCCTGTAAAG CTCTGCCCCT GCCCCCCTTA TGTTATTGAT GGCACAAATT GCCTAATAAA TAATTTATAG TTATTTGTAT GAGTTTGTCT TTTAAATCAT TTAGGAAATA AAAAGTGGAG TTAGAAAACA GTATGATAGT AATACTGACT TTTATATTTG TCAATATATT TATCTTATTT TGGATCCTTA TTTCATTATA TAGATTTGAG TTACTGTCTA GTGCCCTTCC ATTTCGGCCC AAAGGATTCC CTTATGCATT TCTTGCAGGG CAAGTCTAAT TGTAATAAAC TCCCTCAGCT TTTGTTTTAT CTGAGAATGT CTTGATTTCT CCCTTATTTT TGATGGATAA TTTTGCCAGA TACATGAATT TTTGGTAACA GTATTTTTCT TTCAGCACTT TAAATATGTC ATCCCACTAC CTTCTGACTT CATGGTTTCT CATGAGATAT TAGATGTTAT AAAATTTGAG GATTCCTCAT TCTTGATGAG TCAGTTCTGT CTTATTGCTT TTCGGATTTG CTCAGCTTTT GTCTTTTGAC AGTTTGATTA TAACGCGGCT CAGTGTGGGT CTCTGAGTTT ATCCCACTTA GAGTTTGTTG AGTTTCTTGG AGTCATAGAT TTATGTCTTT TATCAAATTT TGGACATATT TGGCTATTAT TTCTTCAATT TTTTTCACTG CTTCTTTCTT TTCCTTCTGA AATATTCTTA ATGTATATGT TGGTCTGTTT GATGCTGTCT CACCAGTTTC TTAGGCTGTG TTCTCTTTTG TTCCTCAGAC TTGATTATTG CAGTTGCCCT TCTTTTTATT TTTTTCAAGT TTGTTGATTC TTCTCCCTGT TCAGATCAAC TGTTGAACTC CTCTAGTGAA TTTATTTCAG TTACTGTACT TTTCAGCTCC AAGATTTATC TTTGGTTCCT TTTTATAACG TCTGTGTCTT TATTGATATT CTCATTTTGT TCATATGTCT CTTTCTTCCT TTAGTTCTTT GTCCATGTTT TCCTTTAGCT CTTTGGGCTT ATTTAAGACA ATTGTTTAAA GTCTTTGCAT AGTAAGTCCA ATGTCTGTGT TTCTTCAGGG ATGGTTTTCA TTATTTTGTT TTCAATGAGC CATACTTTCC TGTGTCTTTG TATGCTGTCT TTTTGTTGTT GAAAACTGTA TGTTTGAACA TCATAACGTG GTGGCCCTGA AAATCAGATA TTCCCCCCTT CCTGAGAGTT AGTTTTATTT TTATTATTGA AGATTGTAGC AGTCTATTGC TACATGTGCA GTCATTTCCA AACTATTTTT GCAAAGACTG TATTCCTTCT GTGTGTCATC ACTGAAGTCT CTGTTCCTTA GTTTGTGTTT AATAGTTTGA CATAGATTTC CTTGAAAGGA GTTAAAACTA GCAGAAAAAT CTCTCTCCCA GTCTTTCCAG TCTTTGTAGA TTGGTTCTGT GCTGGGCTTT TCCATTAATA CTTAGCCAGG CTTGTACTGA GCCTAACAAT CAGGCCCAAA AGCGTAGGGT CTTTGCAGAT CTTGTCTGAG CATGCTTCTT GCTGTGTATG CACGTAGTTT TCTAAATCTC CCTGTATGTG CTGTTGAATA TTCTAATTTC CCAAAGAAAC TCCTTTGCAG CTTTTTCTCA CAGAACATAG ATGGTTTTTT GGATATCTTG ACCATAGTCT TTCGACCCAG GTGTTTGCGG TTGTTAGTTC ACCTTACACT TTTTTCAAGC ATTGCCTACT GCTTACGATG AGTGCTCTGT CAATCCTTTA AGTAGCCCCA GACAGGCTAC CAGAGACTTA AACAAGAATT TGTAAGTTCT GCTCAGCTTC CTCTAGAAAT GGGGATCAGG GTCCAAGACA GAATGCAGTT GCTGATTTCA AGACTGCTGC AACACCAGGG AGCTTGTGGG GGAAGGGCAA GCAGAAATGT CACAAAGCTT TCTTGCCATT TTAAAGTTGC CTGTTCTTGA CTCAGCATTT GCTTCATTGC TATAAACTTT TTACTGTTTT TCAGAGTTCT GATAAAATTG GCTATGCCTG TTCCTGCTTT AAAAAATATA TATATATTTT TTAGGGATTG GGGTCTCACT ATACTGACCA GGCTGGTCTT GAACTTCTGG CCTCAAGCCA TCCTCTCATT TCAGCTTCCC	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AAAGTGCTGC AATTACACGC GTGAACCACC ACACCCAGCC CCTGCTTGTT TTTCAATGTG CCTACTCCAC CATGTTGCTC AAGTATGTAT ATTTTCTAAA CTACCTTGTA GTGTTGTGAT GGGAAATAAA TCCCTGAGCC TTTTGAATAA CTCAGAGAGA TCAAAAACTT AGTTTATCCT ATTCGAAGGA TTAGAAAAAT GATATATCTT TCACTTTTTC AGGGATAGGC TCCTCATTAG AAGGCTCCTA TGTGCCGATG CTGTACAAGA CATTTCATTT CTCTTAATGT TTACAACAAG CTTGTTGCCA AGGCTGATCT TGAACTCCTG GCCTCAAACG ATCCTCCCAG CTCAGTCTCA CAAAGTGTTG GGATGTCTGG CCAACTAATG ACTATCTTAA CTCTTGTGTT TCAATGTTTA TGCCTTCTTT TATCTTGACT GATTGTATGA CTATGTCTTC TAGAACAATG TTGAACAGAA ATGGTGAGAG CAGACATCCT TGCTTTAATA TTTCACCATT ATATATGATG TTAGGTATAG ATTTTTCTCA CAGATGCCTT TTATCAGATT GAGGAATTTA TATTCCTACT TTGCCGAAAG GTTTTTGTAG TATGAGGGGG TGCTGAATTT TGTCAAACAC TTTTTCGGTA ATAATTGAGA TGATTGGTTC TGCAGTCATC GAGATGTGGA TTTTCTCCTT TATTCTGTTC GTGAGTGATT ACACTGGTTG ACTAATGTTA AAACAACCTT ACTTTCCAGG AATAAACCCT ATTATCTTTT TTATACA	
PSMC4	NM_153001.2	TGCGGGTACG GACAGCGCAT GAGCTTATGT TGAGGGCGGA GCCCAGACCA GCCCTTCGTC CTATCCTGCC CTTCCAGCAC CTCTCAGCCG TAACTTAAAC TACACTTCCC AGAAGCCTCC TCAGCCAGGG ACTTCCGTTG TCGTCAGCGG AAGCGGTGAC AGATCATCCC AGGCCACACA GAGGCCGGCT TGGTCACTAT GGAGGAGATA GGCATCTTGG TGGAGAAGGC TCAGGATGAG ATCCCAGCAC TGTCCGTGTC CCGGCCCCAG ACCGGCCTGT CCTTCCTGGG CCCTGAGCCT GAGGACCTGG AGGACCTGTA CAGCCGCTAC AAGGAGGAGG TGAAGCGAAT CCAAAGCATC CCGCTGGTCA TCGGACAATT TCTGGAGGCT GTGGATCAGA ATACAGCCAT CGTGGGCTCT ACCACAGGCT CCAACTATTA TGTGCGCATC CTGAGCACCA TCGATCGGGA GCTGCTCAAG CCCAACGCCT CAGTGGCCCT CCACAAGCAC AGCAATGCAC TGGTGGACGT GCTGCCCCCC GAAGCCGACA GCAGCATCAT GATGCTCACC TCAGACCAGA AGCCAGATGT GATGTACGCG GACATCGGAG GCATGGACAT CCAGAAGCAG GAGGTGCGGG AGGCCGTGGA GCTCCCGCTC ACGCATTTCG AGCTCTACAA GCAGATCGGC ATCGATCCCC CCCGAGGCGT CCTCATGTAT GGCCCACCTG GCTGTGGGAA GACCATGTTG GCAAAGGCGG TGGCACATCA CACAACAGCT GCATTCATCC GGGTCGTGGG CTCGGAGTTT GTACAGAAGT ATCTGGGTGA GGGCCCCCGC ATGGTCCGGG ATGTGTTCCG CCTGGCCAAG GAGAATGCAC CTGCCATCAT CTTCATAGAC GAGATTGATG CCATCGCCAC CAAGAGATTC GATGCTCAGA CAGGGGCCGA CAGGGAGGTT CAGAGGATCC TGCTGGAGCT GCTGAATCAG ATGGATGGAT TTGATCAGAA TGTCAATGTC AAGGTAATCA TGGCCACAAA CAGAGCAGAC ACCCTGGATC CGGCCCTGCT ACGGCCAGGA CGGCTGGACC GTAAAATTGA ATTTCCACTT CCTGACCGCC GCCAGAAGAG ATTGATTTTC TCCACTATCA CTAGCAAGAT GAACCTCTCT GAGGAGGTTG ACTTGGAAGA CTATGTGGCC CGGCCAGATA AGATTTCAGG AGCTGATATT AACTCCATCT GTCAGGAGAG TGGAATGTTG GCTGTCCGTG AAAACCGCTA CATTGTCCTG GCCAAGGACT TCGAGAAAGC ATACAAGACT GTCATCAAGA AGGACGAGCA GGAGCATGAG TTTTACAAGT GACCCTTCCC TTCCCTCCAC CACACCACTC AGGGGCTGGG GCTTCTCTCG CACCCCCAGC ACCTCTGTCC CAAAACCTCA TTCCCTTTTT TCTTTACCCA GGATTGGTTT CTTCAATAAA TAGATAAGAT CGAATCCATT TAATTTCTTC TTAGAAGTTT AACTCCTTTG GAGAATGTGG GCCTTGAATA GGATCCTCTG GGTCCCTCTT AATCTGACAG ATGAGCAGAC GAGGTGCATG GCCTGGGTTG CAGCTTGAGA GAACCAAAAT ATTCAAACCA GATGACTTCC AAAATGTGGG GAAAGGGATG GAAAATGAAC CTGAGATGGA GTCCTTAATC ACGGGATAAA GCCCTGTGCA TCTCCCTCAT TTCCTACAGG TAAAAGACAG TAAAGAAATT CAGGTCACAG GCCTTGGGAG TTCATAGGAA GGAGATGTCC AGTGCTGTCC AGTAGAACTT T	41
SF3A1	NM_005877.5	GGTCCCGGAA GTGCGCCAGT CGTACCTTCG CGGCCGCAAC TCGCTCGGCC GCCGCCATCT TGCGAGCTCG TCGTACTGAC CGAGCGGGGA GGCTGTCTTG AGGCGGCACC GCTCACCGAC ACCGAGGCGG ACTGGCAGCC CTGAGCGTCG CAGTCATGCC GGCCGGACCC GTGCAGGCGG TGCCCCCGCC GCCGCCCGTG CCCACGGAGC CCAAACAGCC CACAGAAGAA GAAGCATCTT	42

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CAAAGGAGGA TTCTGCACCT TCTAAGCCAG TTGTGGGGAT	
		TATTTACCCT CCTCCAGAGG TCAGAAATAT TGTTGACAAG	
		ACTGCCAGCT TTGTGGCCAG AAACGGGCCT GAATTTGAAG	
		CTAGGATCCG ACAGAACGAG ATCAACAACC CCAAGTTCAA	
		CTTTCTGAAC CCCAATGACC CTTACCATGC CTACTACCGC	
		CACAAGGTCA GCGAGTTCAA GGAAGGGAAG GCTCAGGAGC	
		CGTCCGCCGC CATCCCCAAG GTCATGCAGC AGCAGCAGCA	
		GACCACCCAG CAGCAGCTGC CCCAGAAGGT CCAAGCCCAA	
		GTAATCCAAG AGACCATCGT GCCCAAAGAG CCTCCTCCTG	
		AGTTTGAGTT CATTGCTGAT CCTCCCTCTA TCTCAGCCTT	
		CGACTTGGAT GTGGTGAAGC TGACGGCTCA GTTTGTGGCC	
		AGGAATGGGC GCCAGTTTCT GACCCAGCTG ATGCAGAAAG	
		AGCAGCGCAA CTACCAGTTT GACTTTCTCC GCCCACAGCA	
		CAGCCTCTTC AACTACTTCA CGAAGCTAGT GGAACAGTAC	
		ACCAAGATCT TGATTCCACC CAAAGGTTTA TTTTCAAAGC	
		TCAAGAAAGA GGCTGAAAAC CCCCGAGAAG TTTTGGATCA	
		GGTGTGTTAC CGAGTGGAAT GGGCCAAATT CCAGGAACGT	
		GAGAGGAAGA AGGAAGAAGA GGAGAAGGAG AAGGAGCGGG	
		TGGCCTATGC TCAGATCGAC TGGCATGATT TTGTGGTGGT	
		GGAAACAGTG GACTTCCAAC CCAATGAGCA AGGGAACTTC	
		CCTCCCCCCA CCACGCCAGA GGAGCTGGGG GCCCGAATCC	
		TCATTCAGGA GCGCTATGAA AAGTTTGGGG AGAGTGAGGA	
		AGTTGAGATG GAGGTCGAGT CTGATGAGGA GGATGACAAA	
		CAGGAGAAGG CGGAGGAGCC TCCTTCCCAG CTGGACCAGG	
		ACACCCAAGT ACAAGATATG GATGAGGGTT CAGATGATGA	
		AGAAGAAGGG CAGAAAGTGC CCCCACCCCC AGAGACACCC	
		ATGCCTCCAC CTCTGCCCCC AACTCCAGAC CAAGTCATTG	
		TCCGCAAGGA TTATGATCCC AAAGCCTCCA AGCCCTTGCC	
		TCCAGCCCCT GCTCCAGATG AGTATCTTGT GTCCCCCATT	
		ACTGGGGAGA AGATCCCCGC CAGCAAAATG CAGGAACACA	
		TGCGCATTGG ACTTCTTGAC CCTCGCTGGC TGGAGCAGCG	
		GGATCGCTCC ATCCGTGAGA AGCAGAGCGA TGATGAGGTG	
		TACGCACCAG GTCTGGATAT TGAGAGCAGC TTGAAGCAGT	
		TGGCTGAGCG GCGTACTGAC ATCTTCGGTG TAGAGGAAAC	
		AGCCATTGGT AAGAAGATCG GTGAGGAGGA GATCCAGAAG	
		CCAGAGGAAA AGGTGACCTG GGATGGCCAC TCAGGCAGCA	
		TGGCCCGGAC CCAGCAGGCT GCCCAGGCCA ACATCACCCT	
		CCAGGAGCAG ATTGAGGCCA TTCACAAGGC CAAAGGCCTG	
		GTGCCAGAGG ATGACACTAA AGAGAAGATT GGCCCCAGCA	
		AGCCCAATGA AATCCCTCAA CAGCCACCGC CACCATCTTC	
		AGCCACCAAC ATCCCCAGCT CGGCTCCACC CATCACTTCA	
		GTGCCCCGAC CACCCACAAT GCCACCTCCA GTTCGTACTA	
		CAGTTGTCTC CGCAGTACCC GTCATGCCCC GGCCCCCAAT	
		GGCATCTGTG GTCCGGCTGC CCCCAGGCTC AGTGATCGCC	
		CCCATGCCGC CCATCATCCA CGCGCCCAGA ATCAACGTGG	
		TGCCCATGCC TCCCTCGGCC CCTCCTATTA TGGCCCCCCG	
		CCCACCCCCC ATGATTGTGC CAACAGCCTT TGTGCCTGCT	
		CCACCTGTGG CACCTGTCCC AGCTCCAGCC CCAATGCCCC	
		CTGTGCATCC CCCACCTCCC ATGGAAGATG AGCCCACCTC	
		CAAAAAACTG AAGACAGAGG ACAGCCTCAT GCCAGAGGAG	
		GAGTTCCTGC GCAGAAACAA GGGTCCAGTG TCCATCAAAG	
		TCCAGGTGCC CAACATGCAG GATAAGACGG AATGGAAACT	
		GAATGGGCAG GTGCTGGTCT TCACCCTCCC ACTCACGGAC	
		CAGGTCTCTG TCATTAAGGT GAAGATTCAT GAAGCCACAG	
		GCATGCCTGC AGGGAAACAG AAGCTACAGT ATGAGGGTAT	
		CTTCATCAAA GATTCCAACT CACTGGCTTA CTACAACATG	
		GCCAATGGCG CAGTCATCCA CCTGGCCCTC AAGGAGAGAG	
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		AGACCCAGGA GCCCCCCTGA GGCTTTGCCT TGCCTGCATA	
		TTTGTTTCGC TCTTACTCAG TTTGGGAATT CAAATTGTCC	
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		TGTGTTAGGG TCTCGATTTG AGGACATTCA TTTCTTCAGC	
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		AATAAGACTT GTCTCGGTCA CCTGTGCCCT GTCCAGACTC	
		GAGGCAGTGG TAACACTGCA CAGTGCTATG TGGCTTCTCT	
		TTGAGGATTT TTGGGTTTTG TAACTAAATT CTTGCTGCCC	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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PUM1	NM_001020658.1	AGTGGGCCGC CATGTTGTCG GAGTGAAAGG TAAGGGGGAG CGAGAGCGCC AGAGAGAGAA GATCGGGGGG CTGAAATCCA TCTTCATCCT ACCGCTCCGC CCGTGTTGGT GGAATGAGCG TTGCATGTGT CTTGAAGAGA AAAGCAGTGC TTTGGCAGGA CTCTTTCAGC CCCCACCTGA AACATCACCC TCAAGAACCA GCTAATCCCA ACATGCCTGT TGTTTTGACA TCTGGAACAG GGTCGCAAGC GCAGCCACAA CCAGCTGCAA ATCAGGCTCT TGCAGCTGGG ACTCACTCCA GCCCTGTCCC AGGATCTATA GGAGTTGCAG GCCGTTCCCA GGACGACGCT ATGGTGGACT ACTTCTTTCA GAGGCAGCAT GGTGAGCAGC TTGGGGGAGG AGGAAGTGGA GGAGGCGGCT ATAATAATAG CAAACATCGA TGGCCTACTG GGGATAACAT TCATGCAGAA CATCAGGTGC GTTCCATGGA TGAACTGAAT CATGATTTTC AAGCACTTGC TCTGGAGGGA AGAGCGATGG GAGAGCAGCT CTTGCCAGGT AAAAAGTTTT GGGAAACAGA TGAATCCAGC AAAGATGGAC CAAAAGGAAT ATTCCTGGGT GATCAATGGC GAGACAGTGC CTGGGGAACA TCAGATCATT CAGTTTCCCA GCCAATCATG GTGCAGAGAA GACCTGGTCA GAGTTTCCAT GTGAACAGTG AGGTCAATTC TGTACTGTCC CCACGATCGG AGAGTGGGGG ACTAGGCGTT AGCATGGTGG AGTATGTGTT GAGCTCATCC CCGGGCGATT CCTGTCTAAG AAAAGGAGGA TTTGGCCCAA GGGATGCAGA CAGTGATGAA AACGACAAAG GTGAAAAGAA GAACAAGGGT ACGTTTGATG GAGATAAGCT AGGAGATTTG AAGGAGGAGG GTGATGTGAT GGACAAGACC AATGGTTTAC	43

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CCCAGCTGAC CAGCACCAAT GGTGCCAAGC CTGTGGAGGA	
		TTTCTCCAAC ATGGAGTCCC AGAGTGTCCC CTTGGACCCC	
		ATGGAACATG TGGGCATGGA GCCTCTTCAG TTTGATTATT	
		CAGGCACGCA GGTACCTGTG GACTCAGCAG CAGCAACTGT	
		GGGACTTTTT GACTACAATT CTCAACAACA GCTGTTCCAA	
		AGACCTAATG CGCTTGCTGT CCAGCAGTTG ACAGCTGCTC	
		AGCAGCAGCA GTATGCACTG GCAGCTGCTC ATCAGCCGCA	
		CATCGGTTTA GCTCCCGCTG CGTTTGTCCC CAATCCATAC	
		ATCATCAGCG CTGCTCCCCC AGGGACGGAC CCCTACACAG	
		CTGGATTGGC TGCAGCAGCG ACACTAGGCC CAGCTGTGGT	
		CCCTCACCAG TATTATGGAG TTACTCCCTG GGGAGTCTAC	
		CCTGCCAGTC TTTTCCAGCA GCAAGCTGCC GCTGCCGCTG	
		CAGCAACTAA TTCAGCTAAT CAACAGACCA CCCCACAGGC	
		TCAGCAAGGA CAGCAGCAGG TTCTCCGTGG AGGAGCCAGC	
		CAACGTCCTT TGACCCCAAA CCAGAACCAG CAGGGACAGC	
		AAACGGATCC CCTTGTGGCA GCTGCAGCAG TGAATTCTGC	
		CCTTGCATTT GGACAAGGTC TGGCAGCAGG CATGCCAGGT	
		TATCCGGTGT TGGCTCCTGC TGCTTACTAT GACCAAACTG	
		GTGCCCTTGT AGTGAATGCA GGCGCGAGAA ATGGTCTTGG	
		AGCTCCTGTT CGACTTGTAG CTCCTGCCCC AGTCATCATT	
		AGTTCCTCAG CTGCACAAGC AGCTGTTGCA GCAGCCGCAG	
		CTTCAGCAAA TGGAGCAGCT GGTGGTCTTG CTGGAACAAC	
		AAATGGACCA TTTCGCCCTT TAGGAACACA GCAGCCTCAG	
		CCCCAGCCCC AGCAGCAGCC CAATAACAAC CTGGCATCCA	
		GTTCTTTCTA CGGCAACAAC TCTCTGAACA GCAATTCACA	
		GAGCAGCTCC CTCTTCTCCC AGGGCTCTGC CCAGCCTGCC	
		AACACATCCT TGGGATTCGG AAGTAGCAGT TCTCTCGGCG	
		CCACCCTGGG ATCCGCCCTT GGAGGGTTTG GAACAGCAGT	
		TGCAAACTCC AACACTGGCA GTGGCTCCCG CCGTGACTCC	
		CTGACTGGCA GCAGTGACCT TTATAAGAGG ACATCGAGCA	
		GCTTGACCCC CATTGGACAC AGTTTTTATA ACGGCCTTAG	
		CTTTTCCTCC TCTCCTGGAC CCGTGGGCAT GCCTCTCCCT	
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		TCTCTTCACA TGGATCCTCT TCAAGCTTAA ACCTGGGAGG	
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		GGCGCTGAAG CCAAGTACCG CAGTGCAAGC AGCGCCTCCA	
		GCCTCTTCAG CCCGAGCAGC ACTCTTTTCT CTTCCTCTCG	
		TTTGCGATAT GGAATGTCTG ATGTCATGCC TTCTGGCAGG	
		AGCAGGCTTT TGGAAGATTT TCGAAACAAC CGGTACCCCA	
		ATTTACAACT GCGGGAGATT GCTGGACATA TAATGGAATT	
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		TCTTGAAGTG TGTGAAAGAT CAGAATGGCA ATCACGTGGT	
		TCAGAAATGC ATTGAATGTG TACAGCCCCA GTCTTTGCAA	
		TTTATCATCG ATGCGTTTAA GGGACAGGTA TTTGCCTTAT	
		CCACACATCC TTATGGCTGC CGAGTGATTC AGAGAATCCT	
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		GAGCTTCACC AGCACACAGA GCAGCTTGTA CAGGATCAAT	
		ATGGAAATTA TGTAATCCAA CATGTACTGG AGCACGGTCG	
		TCCTGAGGAT AAAAGCAAAA TTGTAGCAGA AATCCGAGGC	
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		AGGCCAGCGG AAGATCGTCA TGCATAAGAT CCGGCCCCAC	
		ATCGCAACTC TTCGTAAGTA CACCTATGGC AAGCACATTC	
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		GGCAGTGTCA CCCGCTGTTC CCTCATTCCC GCTGACCTCA	
		CTGGCCCACT GGCAAATCCA ACCAGCAACC AGAAATGTTC	
		TAGTGTAGAG TCTGAGACGG GCAAGTGGTT GCTCCAGGAT	
		TACTCCCTCC TCCAAAAAAG GAATCAAATC CACGAGTGGA	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		AAGCTTTAAA AAAAAAAGAA AAAAAGGAAA AGGTTTTTAG	
		CTCATTTGCC TGGCCGGCAA GTTTTGCAAA TAGCTCTTCC	
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		AACCTGAGAA GTTTGAATTG TAGTTAAATG ACCCCAAACT	
		GGCATTTAAC ACTGTTTATA AAAAATATAT ATATATATAT	
		ATATATATAT AATGAAAAAG GTTTCAGAGT TGCTAAAGCT	
		TCAGTTTGTG ACATTAAGTT TATGAAATTC TAAAAAATGC	
		CTTTTTTGGA GACTATATTA TGCTGAAGAA GGCTGTTCGT	
		GAGGAGGAGA TGCGAGCACC CAGAACGTCT TTTGAGGCTG	
		GGCGGGTGTG ATTGTTTACT GCCTACTGGA TTTTTTTCTA	
		TTAACATTGA AAGGTAAAAT CTGATTATTT AGCATGAGAA	
		AAAAAAATCC AACTCTGCTT TTGGTCTTGC TTCTATAAAT	
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		TAATGTACCC TAGTAGTCGA ACATACTTTT GATTGTACAA	
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		TGGTCGTAGC ATATAGTGTT GTATATTGTA AATTGTAATT	
		TCAACCAGAA GTAAATTTTT TTCTTTTGAA GGAATAAATG	
		TTCTTTATAC AGCCTAGTTA ATGTTTAAAA AGAAAAAAAT	
		AGCTTGGTTT TATTTGTCAT CTAGTCTCAA GTATAGCGAG	
		ATTCTTTCTA AATGTTATTC AAGATTGAGT TCTCACTAGT	
		GTTTTTTTAA TCCTAAAAAA GTAATGTTTT GATTTTGTGA	
		CAGTCAAAAG GACGTGCAAA AGTCTAGCCT TGCCCGAGCT	
		TTCCTTACAA TCAGAGCCCC TCTCACCTTG TAAAGTGTGA	
		ATCGCCCTTC CCTTTTGTAC AGAAGATGAA CTGTATTTTG	
		CATTTTGTCT ACTTGTAAGT GAATGTAACA TACTGTCAAT	
		TTTCCTTGTT TGAATATAGA ATTGTAACAC TACACGGTGT	
		ACATTTCCAG AGCCTTGTGT ATATTTCCAA TGAACTTTTT	
		TGCAAGCACA CTTGTAACCA TATGTGTATA ATTAACAAAC	
		CTGTGTATGC TTATGCCTGG GCAACTATTT TTTGTAACTC	
		TTGTGTAGAT TGTCTCTAAA CAATGTGTGA TCTTTATTTT	
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		AAAAAAAAAA AAAAAA	
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		ATAAAACCCA GCGGCGCGAC GCGCCACCAC CGCCGAGACC	
		GCGTCCGCCC CGCGAGCACA GAGCCTCGCC TTTGCCGATC	
		CGCCGCCCGT CCACACCCGC CGCCAGCTCA CCATGGATGA	
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		TGCAAGGCCG GCTTCGCGGG CGACGATGCC CCCCGGGCCG	
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		GAGGCCCAGA GCAAGAGAGG CATCCTCACC CTGAAGTACC	
		CCATCGAGCA CGGCATCGTC ACCAACTGGG ACGACATGGA	
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		TGAACCCCAA GGCCAACCGC GAGAAGATGA CCCAGATCAT	
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		GCATCGTGAT GGACTCCGGT GACGGGGTCA CCCACACTGT	
		GCCCATCTAC GAGGGGTATG CCCTCCCCCA TGCCATCCTG	
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		TGAAGATCCT CACCGAGCGC GGCTACAGCT TCACCACCAC	
		GGCCGAGCGG GAAATCGTGC GTGACATTAA GGAGAAGCTG	
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		TGACGGCCAG GTCATCACCA TTGGCAATGA GCGGTTCCGC	
		TGCCCTGAGG CACTCTTCCA GCCTTCCTTC CTGGGCATGG	
		AGTCCTGTGG CATCCACGAA ACTACCTTCA ACTCCATCAT	
		GAAGTGTGAC GTGGACATCC GCAAAGACCT GTACGCCAAC	
		ACAGTGCTGT CTGGCGGCAC CACCATGTAC CCTGGCATTG	
		CCGACAGGAT GCAGAAGGAG ATCACTGCCC TGGCACCCAG	
		CACAATGAAG ATCAAGATCA TTGCTCCTCC TGAGCGCAAG	
		TACTCCGTGT GGATCGGCGG CTCCATCCTG GCCTCGCTGT	
		CCACCTTCCA GCAGATGTGG ATCAGCAAGC AGGAGTATGA	
		CGAGTCCGGC CCCTCCATCG TCCACCGCAA ATGCTTCTAG	
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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GAPD	NM_002046.6	GCTCTCTGCT CCTCCTGTTC GACAGTCAGC CGCATCTTCT TTTGCGTCGC CAGCCGAGCC ACATCGCTCA GACACCATGG GGAAGGTGAA GGTCGGAGTC AACGGATTTG GTCGTATTGG GCGCCTGGTC ACCAGGGCTG CTTTTAACTC TGGTAAAGTG GATATTGTTG CCATCAATGA CCCCTTCATT GACCTCAACT ACATGGTTTA CATGTTCCAA TATGATTCCA CCCATGGCAA ATTCCATGGC ACCGTCAAGG CTGAGAACGG GAAGCTTGTC ATCAATGGAA ATCCCATCAC CATCTTCCAG GAGCGAGATC CCTCCAAAAT CAAGTGGGGC GATGCTGGCG CTGAGTACGT CGTGGAGTCC ACTGGCGTCT TCACCACCAT GGAGAAGGCT GGGGCTCATT TGCAGGGGGG AGCCAAAAGG GTCATCATCT CTGCCCCCTC TGCTGATGCC CCCATGTTCG TCATGGGTGT GAACCATGAG AAGTATGACA ACAGCCTCAA GATCATCAGC AATGCCTCCT GCACCACCAA CTGCTTAGCA CCCCTGGCCA AGGTCATCCA TGACAACTTT GGTATCGTGG AAGGACTCAT GACCACAGTC CATGCCATCA CTGCCACCCA GAAGACTGTG GATGGCCCCT CCGGGAAACT GTGGCGTGAT GGCCGCGGGG CTCTCCAGAA CATCATCCCT GCCTCTACTG GCGCTGCCAA GGCTGTGGGC AAGGTCATCC CTGAGCTGAA CGGGAAGCTC ACTGGCATGG CCTTCCGTGT CCCCACTGCC AACGTGTCAG TGGTGGACCT GACCTGCCGT CTAGAAAAAC CTGCCAAATA TGATGACATC AAGAAGGTGG TGAAGCAGGC GTCGGAGGGC CCCCTCAAGG GCATCCTGGG CTACACTGAG CACCAGGTGG TCTCCTCTGA CTTCAACAGC GACACCCACT CCTCCACCTT TGACGCTGGG GCTGGCATTG CCCTCAACGA CCACTTTGTC AAGCTCATTT CCTGGTATGA CAACGAATTT GGCTACAGCA ACAGGGTGGT GGACCTCATG GCCCACATGG CCTCCAAGGA GTAAGACCCC TGGACCACCA GCCCCAGCAA GAGCACAAGA GGAAGAGAGA GACCCTCACT GCTGGGGAGT CCCTGCCACA CTCAGTCCCC CACCACACTG AATCTCCCCT CCTCACAGTT GCCATGTAGA CCCCTTGAAG AGGGGAGGGG CCTAGGGAGC CGCACCTTGT CATGTACCAT CAATAAAGTA CCCTGTGCTC AACCAGTTAA AAAAAAAAAA AAAAAAAAA	45
GUSB	NM_000181.3	GTCCTCAACC AAGATGGCGC GGATGGCTTC AGGCGCATCA CGACACCGGC GCGTCACGCG ACCCGCCCTA CGGGCACCTC CCGCGCTTTT CTTAGCGCCG CAGACGGTGG CCGAGCGGGG GACCGGGAAG CATGGCCCGG GGGTCGGCGG TTGCCTGGGC GGCGCTCGGG CCGTTGTTGT GGGGCTGCGC GCTGGGGCTG CAGGGCGGGA TGCTGTACCC CCAGGAGAGC CCGTCGCGGG AGTGCAAGGA GCTGGACGGC CTCTGGAGCT TCCGCGCCGA CTTCTCTGAC AACCGACGCC GGGGCTTCGA GGAGCAGTGG TACCGGCGGC CGCTGTGGGA GTCAGGCCCC ACCGTGGACA TGCCAGTTCC CTCCAGCTTC AATGACATCA GCCAGGACTG GCGTCTGCGG CATTTTGTCG GCTGGGTGTG GTACGAACGG GAGGTGATCC TGCCGGAGCG ATGGACCCAG GACCTGCGCA CAAGAGTGGT GCTGAGGATT GGCAGTGCCC ATTCCTATGC CATCGTGTGG GTGAATGGGG TCGACACGCT AGAGCATGAG GGGGGCTACC TCCCCTTCGA GGCCGACATC AGCAACCTGG TCCAGGTGGG GCCCCTGCCC TCCCGGCTCC GAATCACTAT CGCCATCAAC AACACACTCA CCCCCACCAC CCTGCCACCA GGGACCATCC AATACCTGAC TGACACCTCC AAGTATCCCA AGGGTTACTT TGTCCAGAAC ACATATTTTG ACTTTTTCAA CTACGCTGGA CTGCAGCGGT CTGTACTTCT GTACACGACA CCCACCACCT ACATCGATGA CATCACCGTC ACCACCAGCG TGGAGCAAGA CAGTGGGCTG GTGAATTACC AGATCTCTGT CAAGGGCAGT AACCTGTTCA AGTTGGAAGT GCGTCTTTTG GATGCAGAAA ACAAAGTCGT GGCGAATGGG ACTGGGACCC AGGGCCAACT TAAGGTGCCA GGTGTCAGCC TCTGGTGGCC	46

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GTACCTGATG CACGAACGCC CTGCCTATCT GTATTCATTG GAGGTGCAGC TGACTGCACA GACGTCACTG GGGCCTGTGT CTGACTTCTA CACACTCCCT GTGGGGATCC GCACTGTGGC TGTCACCAAG AGCCAGTTCC TCATCAATGG GAAACCTTTC TATTTCCACG GTGTCAACAA GCATGAGGAT GCGGACATCC GAGGGAAGGG CTTCGACTGG CCGCTGCTGG TGAAGGACTT CAACCTGCTT CGCTGGCTTG GTGCCAACGC TTTCCGTACC AGCCACTACC CCTATGCAGA GGAAGTGATG CAGATGTGTG ACCGCTATGG GATTGTGGTC ATCGATGAGT GTCCCGGCGT GGGCCTGGCG CTGCCGCAGT TCTTCAACAA CGTTTCTCTG CATCACCACA TGCAGGTGAT GGAAGAAGTG GTGCGTAGGG ACAAGAACCA CCCCGCGGTC GTGATGTGGT CTGTGGCCAA CGAGCCTGCG TCCCACCTAG AATCTGCTGG CTACTACTTG AAGATGGTGA TCGCTCACAC CAAATCCTTG GACCCCTCCC GGCCTGTGAC CTTTGTGAGC AACTCTAACT ATGCAGCAGA CAAGGGGGCT CCGTATGTGG ATGTGATCTG TTTGAACAGC TACTACTCTT GGTATCACGA CTACGGGCAC CTGGAGTTGA TTCAGCTGCA GCTGGCCACC CAGTTTGAGA ACTGGTATAA GAAGTATCAG AAGCCCATTA TTCAGAGCGA GTATGGAGCA GAAACGATTG CAGGGTTTCA CCAGGATCCA CCTCTGATGT TCACTGAAGA GTACCAGAAA AGTCTGCTAG AGCAGTACCA TCTGGGTCTG GATCAAAAAC GCAGAAAATA CGTGGTTGGA GAGCTCATTT GGAATTTTGC CGATTTCATG ACTGAACAGT CACCGACGAG AGTGCTGGGG AATAAAAAGG GGATCTTCAC TCGGCAGAGA CAACCAAAAA GTGCAGCGTT CCTTTTGCGA GAGAGATACT GGAAGATTGC CAATGAAACC AGGTATCCCC ACTCAGTAGC CAAGTCACAA TGTTTGGAAA ACAGCCTGTT TACTTGAGCA AGACTGATAC CACCTGCGTG TCCCTTCCTC CCCGAGTCAG GGCGACTTCC ACAGCAGCAG AACAAGTGCC TCCTGGACTG TTCACGGCAG ACCAGAACGT TTCTGGCCTG GGTTTTGTGG TCATCTATTC TAGCAGGGAA CACTAAAGGT GGAAATAAAA GATTTTCTAT TATGGAAATA AAGAGTTGGC ATGAAAGTGG CTACTGAAAA AAAAAAAAAA AAAAAAAAAA A	
RPLP0	NM_001002.3	GTCTGACGGG CGATGGCGCA GCCAATAGAC AGGAGCGCTA TCCGCGGTTT CTGATTGGCT ACTTTGTTCG CATTATAAAA GGCACGCGCG GGCGCGAGGC CCTTCTCTCG CCAGGCGTCC TCGTGGAAGT GACATCGTCT TTAAACCCTG CGTGGCAATC CCTGACGCAC CGCCGTGATG CCCAGGGAAG ACAGGGCGAC CTGGAAGTCC AACTACTTCC TTAAGATCAT CCAACTATTG GATGATTATC CGAAATGTTT CATTGTGGGA GCAGACAATG TGGGCTCCAA GCAGATGCAG CAGATCCGCA TGTCCCTTCG CGGGAAGGCT GTGGTGCTGA TGGGCAAGAA CACCATGATG CGCAAGGCCA TCCGAGGGCA CCTGGAAAAC AACCCAGCTC TGGAGAAACT GCTGCCTCAT ATCCGGGGGA ATGTGGGCTT TGTGTTCACC AAGGAGGACC TCACTGAGAT CAGGGACATG TTGCTGGCCA ATAAGGTGCC AGCTGCTGCC CGTGCTGGTG CCATTGCCCC ATGTGAAGTC ACTGTGCCAG CCCAGAACAC TGGTCTCGGG CCCGAGAAGA CCTCCTTTTT CCAGGCTTTA GGTATCACCA CTAAAATCTC CAGGGGCACC ATTGAAATCC TGAGTGATGT GCAGCTGATC AAGACTGGAG ACAAAGTGGG AGCCAGCGAA GCCACGCTGC TGAACATGCT CAACATCTCC CCCTTCTCCT TTGGGCTGGT CATCCAGCAG GTGTTCGACA ATGGCAGCAT CTACAACCCT GAAGTGCTTG ATATCACAGA GGAAACTCTG CATTCTCGCT TCCTGGAGGG TGTCCGCAAT GTTGCCAGTG TCTGTCTGCA GATTGGCTAC CCAACTGTTG CATCAGTACC CCATTCTATC ATCAACGGGT ACAAACGAGT CCTGGCCTTG TCTGTGGAGA CGGATTACAC CTTCCCACTT GCTGAAAAGG TCAAGGCCTT CTTGGCTGAT CCATCTGCCT TTGTGGCTGC TGCCCCTGTG GCTGCTGCCA CCACAGCTGC TCCTGCTGCT GCTGCAGCCC CAGCTAAGGT TGAAGCCAAG GAAGAGTCGG AGGAGTCGGA CGAGGATATG GGATTTGGTC TCTTTGACTA ATCACCAAAA AGCAACCAAC TTAGCCAGTT TTATTTGCAA AACAAGGAAA TAAAGGCTTA CTTCTTTAAA AAGTAAAAAA AAAAAAAAAA AAAAAAAAA	47
TFRC	NM_003234.3	AGAGCGTCGG GATATCGGGT GGCGGCTCGG GACGGAGGAC GCGCTAGTGT GAGTGCGGGC TTCTAGAACT ACACCGACCC TCGTGTCCTC CCTTCATCCT GCGGGGCTGG CTGGAGCGGC CGCTCCGGTG CTGTCCAGCA GCCATAGGGA GCCGCACGGG GAGCGGGAAA GCGGTCGCGG CCCCAGGCGG GGCGGCCGGG ATGGAGCGGG GCCGCGAGCC TGTGGGGAAG GGGCTGTGGC GGCGCCTCGA GCGGCTGCAG GTTCTTCTGT GTGGCAGTTC AGAATGATGG ATCAAGCTAG ATCAGCATTC TCTAACTTGT	48

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
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		CAGTTTCAGG TGTTTAGTTG CAGACTCAGT TTGTCAGACT	
		TTAAAGAATA ATATGCTGCC AAATTTTGGC CAAAGTGTTA	
		ATCTTAGGGG AGAGCTTTCT GTCCTTTTGG CACTGAGATA	
		TTTATTGTTT ATTTATCAGT GACAGAGTTC ACTATAAATG	
		GTGTTTTTTT AATAGAATAT AATTATCGGA AGCAGTGCCT	
		TCCATAATTA TGACAGTTAT ACTGTCGGTT TTTTTTAAAT	
		AAAAGCAGCA TCTGCTAATA AAACCCAACA GATACTGGAA	
		GTTTTGCATT TATGGTCAAC ACTTAAGGGT TTTAGAAAAC	
		AGCCGTCAGC CAAATGTAAT TGAATAAAGT TGAAGCTAAG	
		ATTTAGAGAT GAATTAAATT TAATTAGGGG TTGCTAAGAA	
		GCGAGCACTG ACCAGATAAG AATGCTGGTT TTCCTAAATG	
		CAGTGAATTG TGACCAAGTT ATAAATCAAT GTCACTTAAA	
		GGCTGTGGTA GTACTCCTGC AAAATTTTAT AGCTCAGTTT	
		ATCCAAGGTG TAACTCTAAT TCCCATTTTG CAAAATTTCC	
		AGTACCTTTG TCACAATCCT AACACATTAT CGGGAGCAGT	
		GTCTTCCATA ATGTATAAAG AACAAGGTAG TTTTTACCTA	
		CCACAGTGTC TGTATCGGAG ACAGTGATCT CCATATGTTA	
		CACTAAGGGT GTAAGTAATT ATCGGGAACA GTGTTTCCCA	
		TAATTTTCTT CATGCAATGA CATCTTCAAA GCTTGAAGAT	
		CGTTAGTATC TAACATGTAT CCCAACTCCT ATAATTCCCT	
		ATCTTTTAGT TTTAGTTGCA GAAACATTTT GTGGTCATTA	
		AGCATTGGGT GGGTAAATTC AACCACTGTA AAATGAAATT	
		ACTACAAAAT TTGAAATTTA GCTTGGGTTT TTGTTACCTT	
		TATGGTTTCT CCAGGTCCTC TACTTAATGA GATAGTAGCA	
		TACATTTATA ATGTTTGCTA TTGACAAGTC ATTTTAACTT	
		TATCACATTA TTTGCATGTT ACCTCCTATA AACTTAGTGC	
		GGACAAGTTT TAATCCAGAA TTGACCTTTT GACTTAAAGC	
		AGAGGGACTT TGTATAGAAG GTTTGGGGGC TGTGGGGAAG	
		GAGAGTCCCC TGAAGGTCTG ACACGTCTGC CTACCCATTC	
		GTGGTGATCA ATTAAATGTA GGTATGAATA AGTTCGAAGC	
		TCCGTGAGTG AACCATCATT ATAAACGTGA TGATCAGCTG	
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		ATTTACAGCC TCACTTGAAT GTGTCACTAC TCACAGTCTC	
		TTTAATCTTC AGTTTTATCT TTAATCTCCT CTTTTATCTT	
		GGACTGACAT TTAGCGTAGC TAAGTGAAAA GGTCATAGCT	
		GAGATTCCTG GTTCGGGTGT TACGCACACG TACTTAAATG	
		AAAGCATGTG GCATGTTCAT CGTATAACAC AATATGAATA	
		CAGGGCATGC ATTTTGCAGC AGTGAGTCTC TTCAGAAAAC	
		CCTTTTCTAC AGTTAGGGTT GAGTTACTTC CTATCAAGCC	
		AGTACGTGCT AACAGGCTCA ATATTCCTGA ATGAAATATC	
		AGACTAGTGA CAAGCTCCTG GTCTTGAGAT GTCTTCTCGT	
		TAAGGAGATG GGCCTTTTGG AGGTAAAGGA TAAAATGAAT	
		GAGTTCTGTC ATGATTCACT ATTCTAGAAC TTGCATGACC	
		TTTACTGTGT TAGCTCTTTG AATGTTCTTG AAATTTTAGA	
		CTTTCTTTGT AAACAAATGA TATGTCCTTA TCATTGTATA	
		AAAGCTGTTA TGTGCAACAG TGTGGAGATT CCTTGTCTGA	
		TTTAATAAAA TACTTAAACA CTGAAAAAAA AAAA	
18S	X03205.1	TACCTGGTTG ATCCTGCCAG TAGCATATGC TTGTCTCAAA	49
		GATTAAGCCA TGCATGTCTA AGTACGCACG GCCGGTACAG	
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		TGGTCGCTCG CTCCTCTCCC ACTTGGATAA CTGTGGTAAT	
		TCTAGAGCTA ATACATGCCG ACGGGCGCTG ACCCCCTTCG	
		CGGGGGGGAT GCGTGCATTT ATCAGATCAA AACCAACCCG	
		GTCAGCCCCT CTCCGGCCCC GGCCGGGGGG CGGGCGCCGG	
		CGGCTTTGGT GACTCTAGAT AACCTCGGGC CGATCGCACG	
		CCCCCCGTGG CGGCGACGAC CCATTCGAAC GTCTGCCCTA	
		TCAACTTTCG ATGGTAGTCG CCGTGCCTAC CATGGTGACC	
		ACGGGTGACG GGGAATCAGG GTTCGATTCC GGAGAGGGAG	
		CCTGAGAAAC GGCTACCACA TCCAAGGAAG GCAGCAGGCG	
		CGCAAATTAC CCACTCCCGA CCCGGGGAGG TAGTGACGAA	
		AAATAACAAT ACAGGACTCT TTCGAGGCCC TGTAATTGGA	
		ATGAGTCCAC TTTAAATCCT TTAACGAGGA TCCATTGGAG	
		GGCAAGTCTG GTGCCAGCAG CCGCGGTAAT TCCAGCTCCA	
		ATAGCGTATA TTAAAGTTGC TGCAGTTAAA AAGCTCGTAG	
		TTGGATCTTG GGAGCGGGCG GGCGGTCCGC CGCGAGGCGA	
		GCCACCGCCC GTCCCCGCCC CTTGCCTCTC GGCGCCCCCT	
		CGATGCTCTT AGCTGAGTGT CCCGCGGGGC CCGAAGCGTT	
		TACTTTGAAA AAATTAGAGT GTTCAAAGCA GGCCCGAGCC	
		GCCTGGATAC CGCAGCTAGG AATAATGGAA TAGGACCGCG	
		GTTCTATTTT GTTGGTTTTC GGAACTGAGG CCATGATTAA	
		GAGGGACGGC CGGGGGCATT CGTATTGCGC CGCTAGAGGT	

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GAAATTCTTG GACCGGCGCA AGACGGACCA GAGCGAAAGC ATTTGCCAAG AATGTTTTCA TTAATCAAGA ACGAAAGTCG GAGGTTCGAA GACGATCAGA TACCGTCGTA GTTCCGACCA TAAACGATGC CGACCGGCGA TGCGGCGGCG TTATTCCCAT GACCCGCCGG GCAGCTTCCG GGAAACCAAA GTCTTTGGGT TCCGGGGGGA GTATGGTTGC AAAGCTGAAA CTTAAAGGAA TTGACGGAAG GGCACCACCA GGAGTGGAGC CTGCGGCTTA ATTTGACTCA ACACGGGAAA CCTCACCCGG CCCGGACACG GACAGGATTG ACAGATTGAT AGCTCTTTCT CGATTCCGTG GGTGGTGGTG CATGGCCGTT CTTAGTTGGT GGAGCGATTT GTCTGGTTAA TTCCGATAAC GAACGAGACT CTGGCATGCT AACTAGTTAC GCGACCCCCG AGCGGTCGGC GTCCCCCAAC TTCTTAGAGG GACAAGTGGC GTTCAGCCAC CCGAGATTGA GCAATAACAG GTCTGTGATG CCCTTAGATG TCCGGGGCTG CACGCGCGCT ACACTGACTG GCTCAGCGTG TGCCTACCCT ACGCCGGCAG GCGCGGGTAA CCCGTTGAAC CCCATTCGTG ATGGGGATCG GGGATTGCAA TTATTCCCCA TGAACGAGGA ATTCCCAGTA AGTGCGGGTC ATAAGCTTGC GTTGATTAAG TCCCTGCCCT TTGTACACAC CGCCCGTCGC TACTACCGAT TGGATGGTTT AGTGAGGCCC TCGGATCGGC CCCGCCGGGG TCGGCCCACG GCCCTGGCGG AGCGCTGAGA AGACGGTCGA ACTTGACTAT CTAGAGGAAG TAAAAGTCGT AACAAGGTTT CCGTAGGTGA ACCTGCGGAA GGATCATTA	
PPIA	NM_021130.4	GGGGCCGAAC GTGGTATAAA AGGGGCGGGA GGCCAGGCTC GTGCCGTTTT GCAGACGCCA CCGCCGAGGA AAACCGTGTA CTATTAGCCA TGGTCAACCC CACCGTGTTC TTCGACATTG CCGTCGACGG CGAGCCCTTG GGCCGCGTCT CCTTTGAGCT GTTTGCAGAC AAGGTCCCAA AGACAGCAGA AAATTTTCGT GCTCTGAGCA CTGGAGAGAA AGGATTTGGT TATAAGGGTT CCTGCTTTCA CAGAATTATT CCAGGGTTTA TGTGTCAGGG TGGTGACTTC ACACGCCATA ATGGCACTGG TGGCAAGTCC ATCTATGGGG AGAAATTTGA AGATGAGAAC TTCATCCTAA AGCATACGGG TCCTGGCATC TTGTCCATGG CAAATGCTGG ACCCAACACA AATGGTTCCC AGTTTTTCAT CTGCACTGCC AAGACTGAGT GGTTGGATGG CAAGCATGTG GTGTTTGGCA AAGTGAAAGA AGGCATGAAT ATTGTGGAGG CCATGGAGCG CTTTGGGTCC AGGAATGGCA AGACCAGCAA GAAGATCACC ATTGCTGACT GTGGACAACT CGAATAAGTT TGACTTGTGT TTTATCTTAA CCACCAGATC ATTCCTTCTG TAGCTCAGGA GAGCACCCCT CCACCCCATT TGCTCGCAGT ATCCTAGAAT CTTTGTGCTC TCGCTGCAGT TCCCTTTGGG TTCCATGTTT TCCTTGTTCC CTCCCATGCC TAGCTGGATT GCAGAGTTAA GTTTATGATT ATGAAATAAA AACTAAATAA CAATTGTCCT CGTTTGAGTT AAGAGTGTTG ATGTAGGCTT TATTTTAAGC AGTAATGGGT TACTTCTGAA ACATCACTTG TTTGCTTAAT TCTACACAGT ACTTAGATTT TTTTTACTTT CCAGTCCCAG GAAGTGTCAA TGTTTGTTGA GTGGAATATT GAAAATGTAG GCAGCAACTG GGCATGGTGG CTCACTGTCT GTAATGTATT ACCTGAGGCA GAAGACCACC TGAGGGTAGG AGTCAAGATC AGCCTGGGCA ACATAGTGAG ACGCTGTCTC TACAAAAAAT AATTAGCCTG GCCTGGTGGT GCATGCCTAG TCCTAGCTGA TCTGGAGGCT GACGTGGGAG GATTGCTTGA GCCTAGAGTG AGCTATTATC ATGCCACTGT ACAGCCTGGG TGTTCACAGA TCTTGTGTCT CAAAGGTAGG CAGAGGCAGG AAAAGCAAGG AGCCAGAATT AAGAGGTTGG GTCAGTCTGC AGTGAGTTCA TGCATTTAGA GGTGTTCTTC AAGATGACTA ATGTCAAAAA TTGAGACATC TGTTGCGGTT TTTTTTTTTT TTTTTTCCCC TGGAATGCAG TGGCGTGATC TCAGCTCACT GCAGCCTCCG CCTCCTGGGT TCAAGTGATT CTAGTGCCTC AGCCTCCTGA GTAGCTGGGA TAATGGGCGT GTGCCACCAT GCCCAGCTAA TTTTTGTATT TTTAGTATAG ATGGGGTTTC ATCATTTTGA CCAGGCTGGT CTCAAACTCT TGACCTCAGC TGATGCGCCT GCCTTGGCCT CCCAAACTGC TGAGATTACA GATGTGAGCC ACCGCACCCT ACCTCATTTT CTGTAACAAA GCTAAGCTTG AACACTGTTG ATGTTCTTGA GGGAAGCATA TTGGGCTTTA GGCTGTAGGT CAAGTTTATA CATCTTAATT ATGGTGGAAT TCCTATGTAG AGTCTAAAAA GCCAGGTACT TGGTGCTACA GTCAGTCTCC CTGCAGAGGG TTAAGGCGCA GACTACCTGC AGTGAGGAGG TACTGCTTGT AGCATATAGA GCCTCTCCCT AGCTTTGGTT ATGGAGGCTT TGAGGTTTTG CAAACCTGAC CAATTTAAGC CATAAGATCT GGTCAAAGGG ATACCCTTCC CACTAAGGAC TTGGTTTCTC AGGAAATTAT ATGTACAGTG CTTGCTGGCA GTTAGATGTC AGGACAATCT AAGCTGAGAA	50

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		AACCCCTTCT CTGCCCACCT TAACAGACCT CTAGGGTTCT TAACCCAGCA ATCAAGTTTG CCTATCCTAG AGGTGGCGGA TTTGATCATT TGGTGTGTTG GGCAATTTTT GTTTTACTGT CTGGTTCCTT CTGCGTGAAT TACCACCACC ACCACTTGTG CATCTCAGTC TTGTGTGTTG TCTGGTTACG TATTCCCTGG GTGATACCAT TCAATGTCTT AATGTACTTG TGGCTCAGAC CTGAGTGCAA GGTGGAAATA AACATCAAAC ATCTTTTCAT TATCCCTA	
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RPL13A	NM_012423.3	CACTTCTGCC GCCCCTGTTT CAAGGGATAA GAAACCCTGC GACAAAACCT CCTCCTTTTC CAAGCGGCTG CCGAAGATGG CGGAGGTGCA GGTCCTGGTG CTTGATGGTC GAGGCCATCT	52

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CCTGGGCCGC CTGGCGGCCA TCGTGGCTAA ACAGGTACTG CTGGGCCGGA AGGTGGTGGT CGTACGCTGT GAAGGCATCA ACATTTCTGG CAATTTCTAC AGAAACAAGT TGAAGTACCT GGCTTTCCTC CGCAAGCGGA TGAACACCAA CCCTTCCCGA GGCCCCTACC ACTTCCGGGC CCCCAGCCGC ATCTTCTGGC GGACCGTGCG AGGTATGCTG CCCCACAAAA CCAAGCGAGG CCAGGCCGCT CTGGACCGTC TCAAGGTGTT TGACGGCATC CCACCGCCCT ACGACAAGAA AAAGCGGATG GTGGTTCCTG CTGCCCTCAA GGTCGTGCGT CTGAAGCCTA CAAGAAAGTT TGCCTATCTG GGGCGCCTGG CTCACGAGGT TGGCTGGAAG TACCAGGCAG TGACAGCCAC CCTGGAGGAG AAGAGGAAAG AGAAAGCCAA GATCCACTAC CGGAAGAAGA AACAGCTCAT GAGGCTACGG AAACAGGCCG AGAAGAACGT GGAGAAGAAA ATTGACAAAT ACACAGAGGT CCTCAAGACC CACGGACTCC TGGTCTGAGC CCAATAAAGA CTGTTAATTC CTCATGCGTT GCCTGCCCTT CCTCCATTGT TGCCCTGGAA TGTACGGGAC CCAGGGGCAG CAGCAGTCCA GGTGCCACAG GCAGCCCTGG GACATAGGAA GCTGGGAGCA AGGAAAGGGT CTTAGTCACT GCCTCCCGAA GTTGCTTGAA AGCACTCGGA GAATTGTGCA GGTGTCATTT ATCTATGACC AATAGGAAGA GCAACCAGTT ACTATGAGTG AAAGGGAGCC AGAAGACTGA TTGGAGGGCC CTATCTTGTG AGTGGGGCAT CTGTTGGACT TTCCACCTGG TCATATACTC TGCAGCTGTT AGAATGTGCA AGCACTTGGG GACAGCATGA GCTTGCTGTT GTACACAGGG TATTTCTAGA AGCAGAAATA GACTGGGAAG ATGCACAACC AAGGGGTTAC AGGCATCGCC CATGCTCCTC ACCTGTATTT TGTAATCAGA AATAAATTGC TTTTAAAGAA AAAAAAAAAA AAAAAA	
B2M	NM_004048.2	AATATAAGTG GAGGCGTCGC GCTGGCGGGC ATTCCTGAAG CTGACAGCAT TCGGGCCGAG ATGTCTCGCT CCGTGGCCTT AGCTGTGCTC GCGCTACTCT CTCTTTCTGG CCTGGAGGCT ATCCAGCGTA CTCCAAAGAT TCAGGTTTAC TCACGTCATC CAGCAGAGAA TGGAAAGTCA AATTTCCTGA ATTGCTATGT GTCTGGGTTT CATCCATCCG ACATTGAAGT TGACTTACTG AAGAATGGAG AGAGAATTGA AAAAGTGGAG CATTCAGACT TGTCTTTCAG CAAGGACTGG TCTTTCTATC TCTTGTACTA CACTGAATTC ACCCCCACTG AAAAAGATGA GTATGCCTGC CGTGTGAACC ATGTGACTTT GTCACAGCCC AAGATAGTTA AGTGGGATCG AGACATGTAA GCAGCATCAT GGAGGTTTGA AGATGCCGCA TTTGGATTGG ATGAATTCCA AATTCTGCTT GCTTGCTTTT TAATATTGAT ATGCTTATAC ACTTACACTT TATGCACAAA ATGTAGGGTT ATAATAATGT TAACATGGAC ATGATCTTCT TTATAATTCT ACTTTGAGTG CTGTCTCCAT GTTTGATGTA TCTGAGCAGG TTGCTCCACA GGTAGCTCTA GGAGGGCTGG CAACTTAGAG GTGGGGAGCA GAGAATTCTC TTATCCAACA TCAACATCTT GGTCAGATTT GAACTCTTCA ATCTCTTGCA CTCAAAGCTT GTTAAGATAG TTAAGCGTGC ATAAGTTAAC TTCCAATTTA CATACTCTGC TTAGAATTTG GGGGAAAATT TAGAAATATA ATTGACAGGA TTATTGGAAA TTTGTTATAA TGAATGAAAC ATTTTGTCAT ATAAGATTCA TATTTACTTC TTATACATTT GATAAAGTAA GGCATGGTTG TGGTTAATCT GGTTTATTTT TGTTCCACAA GTTAAATAAA TCATAAAACT TGATGTGTTA TCTCTTA	53
YWHAZ	NM_003406.3	CTTTCTCCTT CCCCTTCTTC CGGGCTCCCG TCCCGGCTCA TCACCCGGCC TGTGGCCCAC TCCCACCGCC AGCTGGAACC CTGGGGACTA CGACGTCCCT CAAACCTTGC TTCTAGGAGA TAAAAAGAAC ATCCAGTCAT GGATAAAAAT GAGCTGGTTC AGAAGGCCAA ACTGGCCGAG CAGGCTGAGC GATATGATGA CATGGCAGCC TGCATGAAGT CTGTAACTGA GCAAGGAGCT GAATTATCCA ATGAGGAGAG GAATCTTCTC TCAGTTGCTT ATAAAAATGT TGTAGGAGCC CGTAGGTCAT CTTGGAGGGT CGTCTCAAGT ATTGAACAAA AGACGGAAGG TGCTGAGAAA AAACAGCAGA TGGCTCGAGA ATACAGAGAG AAAATTGAGA CGGAGCTAAG AGATATCTGC AATGATGTAC TGTCTCTTTT GGAAAAGTTC TTGATCCCCA ATGCTTCACA AGCAGAGAGC AAAGTCTTCT ATTTGAAAAT GAAAGGAGAT TACTACCGTT ACTTGGCTGA GGTTGCCGCT GGTGATGACA AGAAAGGGAT TGTCGATCAG TCACAACAAG CATACCAAGA AGCTTTTGAA ATCAGCAAAA AGGAAATGCA ACCAACACAT CCTATCAGAC TGGGTCTGGC CCTTAACTTC TCTGTGTTCT ATTATGAGAT TCTGAACTCC CCAGAGAAAG CCTGCTCTCT TGCAAAGACA GCTTTTGATG AAGCCATTGC TGAACTTGAT ACATTAAGTG AAGAGTCATA CAAAGACAGC ACGCTAATAA TGCAATTACT	54

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GAGAGACAAC TTGACATTGT GGACATCGGA TACCCAAGGA GACGAAGCTG AAGCAGGAGA AGGAGGGGAA AATTAACCGG CCTTCCAACT TTTGTCTGCC TCATTCTAAA ATTTACACAG TAGACCATTT GTCATCCATG CTGTCCCACA AATAGTTTTT TGTTTACGAT TTATGACAGG TTTATGTTAC TTCTATTTGA ATTTCTATAT TTCCCATGTG GTTTTTATGT TTAATATTAG GGGAGTAGAG CCAGTTAACA TTTAGGGAGT TATCTGTTTT CATCTTGAGG TGGCCAATAT GGGGATGTGG AATTTTTATA CAAGTTATAA GTGTTTGGCA TAGTACTTTT GGTACATTGT GGCTTCAAAA GGGCCAGTGT AAAACTGCTT CCATGTCTAA GCAAAGAAAA CTGCCTACAT ACTGGTTTGT CCTGGCGGGG AATAAAAGGG ATCATTGGTT CCAGTCACAG GTGTAGTAAT TGTGGGTACT TTAAGGTTTG GAGCACTTAC AAGGCTGTGG TAGAATCATA CCCCATGGAT ACCACATATT AAACCATGTA TATCTGTGGA ATACTCAATG TGTACACCTT TGACTACAGC TGCAGAAGTG TTCCTTTAGA CAAAGTTGTG ACCCATTTTA CTCTGGATAA GGGCAGAAAC GGTTCACATT CCATTATTTG TAAAGTTACC TGCTGTTAGC TTTCATTATT TTTGCTACAC TCATTTTATT TGTATTTAAA TGTTTTAGGC AACCTAAGAA CAAATGTAAA AGTAAAGATG CAGGAAAAAT GAATTGCTTG GTATTCATTA CTTCATGTAT ATCAAGCACA GCAGTAAAAC AAAAACCCAT GTATTTAACT TTTTTTTAGG ATTTTTGCTT TTGTGATTTT TTTTTTTTTG ATACTTGCCT AACATGCATG TGCTGTAAAA ATAGTTAACA GGGAAATAAC TTGAGATGAT GGCTAGCTTT GTTTAATGTC TTATGAAATT TTCATGAACA ATCCAAGCAT AATTGTTAAG AACACGTGTA TTAAATTCAT GTAAGTGGAA TAAAAGTTTT ATGAATGGAC TTTTCAACTA CTTTCTCTAC AGCTTTTCAT GTAAATTAGT CTTGGTTCTG AAACTTCTCT AAAGGAAATT GTACATTTTT TGAAATTTAT TCCTTATTCC CTCTTGGCAG CTAATGGGCT CTTACCAAGT TTAAACACAA AATTTATCAT AACAAAAATA CTACTAATAT AACTACTGTT TCCATGTCCC ATGATCCCCT CTCTTCCTCC CCACCCTGAA AAAAATGAGT TCCTATTTTT TCTGGGAGAG GGGGGGATTG ATTAGAAAAA AATGTAGTGT GTTCCATTTA AAATTTTGGC ATATGGCATT TTCTAACTTA GGAAGCCACA ATGTTCTTGG CCCATCATGA CATTGGGTAG CATTAACTGT AAGTTTTGTG CTTCCAAATC ACTTTTTGGT TTTTAAGAAT TTCTTGATAC TCTTATAGCC TGCCTTCAAT TTTGATCCTT TATTCTTTCT ATTTGTCAGG TGCACAAGAT TACCTTCCTG TTTTAGCCTT CTGTCTTGTC ACCAACCATT CTTACTTGGT GGCCATGTAC TTGGAAAAAG GCCGCATGAT CTTTCTGGCT CCACTCAGTG TCTAAGGCAC CCTGCTTCCT TTGCTTGCAT CCCACAGACT ATTTCCCTCA TCCTATTTAC TGCAGCAAAT CTCTCCTTAG TTGATGAGAC TGTGTTTATC TCCCTTTAAA ACCCTACCTA TCCTGAATGG TCTGTCATTG TCTGCCTTTA AAATCCTTCC TCTTTCTTCC TCCTCTATTC TCTAAATAAT GATGGGGCTA AGTTATACCC AAAGCTCACT TTACAAAATA TTTCCTCAGT ACTTTGCAGA AAACACCAAA CAAAAATGCC ATTTTAAAAA AGGTGTATTT TTTCTTTTAG AATGTAAGCT CCTCAAGAGC AGGGACAATG TTTTCTGTAT GTTCTATTGT GCCTAGTACA CTGTAAATGC TCAATAAATA TTGATGATGG GAGGCAGTGA GTCTTGATGA TAAGGGTGAG AAACTGAAAT CCCAAACACT GTTTTGTTGC TTGTTTTATT ATGACCTCAG ATTAAATTGG GAAATATTGG CCCTTTTGAA TAATTGTCCC AAATATTACA TTCAAATAAA AGTGCAATGG AGAAAAAAAA AAA	
SDHA	NM_004168.3	ACTGCAGCCC CGCTCGACTC CGGCGTGGTG CGCAGGCGCG GTATCCCCCC TCCCCCGCCA GCTCGACCCC GGTGTGGTGC GCAGGCGCAG TCTGCGCAGG GACTGGCGGG ACTGCGCGGC GGCAACAGCA GACATGTCGG GGGTCCGGGG CCTGTCGCGG CTGCTGAGCG CTCGGCGCCT GGCGCTGGCC AAGGCGTGGC CAACAGTGTT GCAAACAGGA ACCCGAGGTT TTCACTTCAC TGTTGATGGG AACAAGAGGG CATCTGCTAA AGTTTCAGAT TCCATTTCTG CTCAGTATCC AGTAGTGGAT CATGAATTTG ATGCAGTGGT GGTAGGCGCT GGAGGGGCAG GCTTGCGAGC TGCATTTGGC CTTTCTGAGG CAGGGTTTAA TACAGCATGT GTTACCAAGC TGTTTCCTAC CAGGTCACAC ACTGTTGCAG CACAGGGAGG AATCAATGCT GCTCTGGGGA ACATGGAGGA GGACAACTGG AGGTGGCATT TCTACGACAC CGTGAAGGGC TCCGACTGGC TGGGGGACCA GGATGCCATC CACTACATGA CGGAGCAGGC CCCCGCCGCC GTGGTCGAGC TAGAAAATTA TGGCATGCCG TTTAGCAGAA CTGAAGATGG GAAGATTTAT CAGCGTGCAT TTGGTGGACA GAGCCTCAAG TTTGGAAAGG	55

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information

Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		GCGGGCAGGC CCATCGGTGC TGCTGTGTGG CTGATCGGAC TGGCCACTCG CTATTGCACA CCTTATATGG AAGGTCTCTG CGATATGATA CCAGCTATTT TGTGGAGTAT TTTGCCTTGG ATCTCCTGAT GGAGAATGGG GAGTGCCGTG GTGTCATCGC ACTGTGCATA GAGGACGGGT CCATCCATCG CATAAGAGCA AAGAACACTG TTGTTGCCAC AGGAGGCTAC GGGCGCACCT ACTTCAGCTG CACGTCTGCC CACACCAGCA CTGGCGACGG CACGGCCATG ATCACCAGGG CAGGCCTTCC TTGCCAGGAC CTAGAGTTTG TTCAGTTCCA CCCTACAGGC ATATATGGTG CTGGTTGTCT CATTACGGAA GGATGTCGTG GAGAGGGAGG CATTCTCATT AACAGTCAAG GCGAAAGGTT TATGGAGCGA TACGCCCCTG TCGCGAAGGA CCTGGCGTCT AGAGATGTGG TGTCTCGGTC CATGACTCTG GAGATCCGAG AAGGAAGAGG CTGTGGCCCT GAGAAAGATC ACGTCTACCT GCAGCTGCAC CACCTACCTC CAGAGCAGCT GGCCACGCGC CTGCCTGGCA TTTCAGAGAC AGCCATGATC TTCGCTGGCG TGGACGTCAC GAAGGAGCCG ATCCCTGTCC TCCCCACCGT GCATTATAAC ATGGGCGGCA TTCCCACCAA CTACAAGGGG CAGGTCCTGA GGCACGTGAA TGGCCAGGAT CAGATTGTGC CCGGCCTGTA CGCCTGTGGG GAGGCCGCCT GTGCCTCGGT ACATGGTGCC AACCGCCTCG GGGCAAACTC GCTCTTGGAC CTGGTTGTCT TTGGTCGGGC ATGTGCCCTG AGCATCGAAG AGTCATGCAG GCCTGGAGAT AAAGTCCCTC CAATTAAACC AAACGCTGGG GAAGAATCTG TCATGAATCT TGACAAATTG AGATTTGCTG ATGGAAGCAT AAGAACATCG GAACTGCGAC TCAGCATGCA GAAGTCAATG CAAAATCATG CTGCCGTGTT CCGTGTGGGA AGCGTGTTGC AAGAAGGTTG TGGGAAAATC AGCAAGCTCT ATGGAGACCT AAAGCACCTG AAGACGTTCG ACCGGGGAAT GGTCTGGAAC ACGGACCTGG TGGAGACCCT GGAGCTGCAG AACCTGATGC TGTGTGCGCT GCAGACCATC TACGGAGCAG AGGCACGGAA GGAGTCACGG GGCGCGCATG CCAGGGAAGA CTACAAGGTG CGGATTGATG AGTACGATTA CTCCAAGCCC ATCCAGGGGC AACAGAAGAA GCCCTTTGAG GAGCACTGGA GGAAGCACAC CCTGTCCTAT GTGGACGTTG GCACTGGGAA GGTCACTCTG GAATATAGAC CCGTGATCGA CAAAACTTTG AACGAGGCTG ACTGTGCCAC CGTCCCGCCA GCCATTCGCT CCTACTGATG AGACAAGATG TGGTGATGAC AGAATCAGCT TTTGTAATTA TGTATAATAG CTCATGCATG TGTCCATGTC ATAACTGTCT TCATACGCTT CTGCACTCTG GGGAAGAAGG AGTACATTGA AGGGAGATTG GCACCTAGTG GCTGGGAGCT TGCCAGGAAC CCAGTGGCCA GGGAGCGTGG CACTTACCTT TGTCCCTTGC TTCATTCTTG TGAGATGATA AAACTGGGCA CAGCTCTTAA ATAAAATATA AATGAACAAA CTTTCTTTTA TTTCCAAATC CATTTGAAAT ATTTTACTGT TGTGACTTTA GTCATATTTG TTGACCTAAA AATCAAATGT AATCTTTGTA TTGTGTTACA TCAAAATCCA GATATTTTGT ATAGTTTCTT TTTTCTTTTT CTTTTCTTTT TTTTTTTTGA GACAGGATCG GTGCAGTAGT ACAATCACAG CTCACTGCAG CCTCAAACTC CTGGGCAGCT CAGGTGATCT TCCTGACTCA GCCTTCTGAG TAGTTGGGGC TACAGGTGTG CACCACCATG CCCAGCTCAT TTATTTTGTA ATTGTAGGGA CAGGGTCTCA CTGTGTTGCC TAGGCTGGTC TCAAGTGATC CTCCCTCCTT GGCCTCCCAA GGTGCTGGAA TTATAGGTGT GAACAAACCA AAAAAAAAAA AAA	
HPRT1	NM_000194.2	GGCGGGGCCT GCTTCTCCTC AGCTTCAGGC GGCTGCGACG AGCCCTCAGG CGAACCTCTC GGCTTTCCCG CGCGGCGCCG CCTCTTGCTG CGCCTCCGCC TCCTCCTCTG CTCCGCCACC GGCTTCCTCC TCCTGAGCAG TCAGCCCGCG CGCCGGCCGG CTCCGTTATG GCGACCCGCA GCCCTGGCGT CGTGATTAGT GATGATGAAC CAGGTTATGA CCTTGATTTA TTTTGCATAC CTAATCATTA TGCTGAGGAT TTGGAAAGGG TGTTTATTCC TCATGGACTA ATTATGGACA GGACTGAACG TCTTGCTCGA GATGTGATGA AGGAGATGGG AGGCCATCAC ATTGTAGCCC TCTGTGTGCT CAAGGGGGGC TATAAATTCT TTGCTGACCT GCTGGATTAC ATCAAAGCAC TGAATAGAAA TAGTGATAGA TCCATTCCTA TGACTGTAGA TTTTATCAGA CTGAAGAGCT ATTGTAATGA CCAGTCAACA GGGGACATAA AAGTAATTGG TGGAGATGAT CTCTCAACTT TAACTGGAAA GAATGTCTTG ATTGTGGAAG ATATAATTGA CACTGGCAAA ACAATGCAGA CTTTGCTTTC CTTGGTCAGG CAGTATAATC CAAAGATGGT CAAGGTCGCA AGCTTGCTGG TGAAAAGGAC CCCACGAAGT GTTGGATATA AGCCAGACTT TGTTGGATTT GAAATTCCAG ACAAGTTTGT TGTAGGATAT GCCCTTGACT ATAATGAATA	56

TABLE 1-continued

Melanoma Biomarker/Housekeeper Sequence Information			
Gene Name	RefSeq Accession	Sequence	SEQ ID NO:
		CTTCAGGGAT TTGAATCATG TTTGTGTCAT TAGTGAAACT GGAAAAGCAA AATACAAAGC CTAAGATGAG AGTTCAAGTT GAGTTTGGAA ACATCTGGAG TCCTATTGAC ATCGCCAGTA AAATTATCAA TGTTCTAGTT CTGTGGCCAT CTGCTTAGTA GAGCTTTTTG CATGTATCTT CTAAGAATTT TATCTGTTTT GTACTTTAGA AATGTCAGTT GCTGCATTCC TAAACTGTTT ATTTGCACTA TGAGCCTATA GACTATCAGT TCCCTTTGGG CGGATTGTTG TTTAACTTGT AAATGAAAAA ATTCTCTTAA ACCACAGCAC TATTGAGTGA AACATTGAAC TCATATCTGT AAGAAATAAA GAGAAGATAT ATTAGTTTTT TAATTGGTAT TTTAATTTTT ATATATGCAG GAAAGAATAG AAGTGATTGA ATATTGTTAA TTATACCACC GTGTGTTAGA AAAGTAAGAA GCAGTCAATT TTCACATCAA AGACAGCATC TAAGAAGTTT TGTTCTGTCC TGGAATTATT TTAGTAGTGT TTCAGTAATG TTGACTGTAT TTTCCAACTT GTTCAAATTA TTACCAGTGA ATCTTTGTCA GCAGTTCCCT TTTAAATGCA AATCAATAAA TTCCCAAAAA TTTAAAAAAA AAAAAAAAAA AAAAA	

Definitions

The articles "a" and "an" are used in this disclosure to refer to one or more than one (i.e., to at least one) of the grammatical object of the article. By way of example, "an element" means one element or more than one element.

The term "and/or" is used in this disclosure to mean either "and" or "or" unless indicated otherwise.

As used herein, the terms "polynucleotide" and "nucleic acid molecule" are used interchangeably to mean a polymeric form of nucleotides of at least 10 bases or base pairs in length, either ribonucleotides or deoxynucleotides or a modified form of either type of nucleotide, and is meant to include single and double stranded forms of DNA. As used herein, a nucleic acid molecule or nucleic acid sequence that serves as a probe in a microarray analysis preferably comprises a chain of nucleotides, more preferably DNA and/or RNA. In other embodiments, a nucleic acid molecule or nucleic acid sequence comprises other kinds of nucleic acid structures such a for instance a DNA/RNA helix, peptide nucleic acid (PNA), locked nucleic acid (LNA) and/or a ribozyme. Hence, as used herein the term "nucleic acid molecule" also encompasses a chain comprising non-natural nucleotides, modified nucleotides and/or non-nucleotide building blocks which exhibit the same function as natural nucleotides.

As used herein, the terms "hybridize," "hybridizing", "hybridizes," and the like, used in the context of polynucleotides, are meant to refer to conventional hybridization conditions, such as hybridization in 50% formamide/6×SSC/0.1% SDS/100 μg/ml ssDNA, in which temperatures for hybridization are above 37 degrees and temperatures for washing in 0.1×SSC/0.1% SDS are above 55 degrees C., and preferably to stringent hybridization conditions.

As used herein, the term "normalization" or "normalizer" refers to the expression of a differential value in terms of a standard value to adjust for effects which arise from technical variation due to sample handling, sample preparation, and measurement methods rather than biological variation of biomarker concentration in a sample. For example, when measuring the expression of a differentially expressed protein, the absolute value for the expression of the protein can be expressed in terms of an absolute value for the expression of a standard protein that is substantially constant in expression.

The terms "diagnosis" and "diagnostics" also encompass the terms "prognosis" and "prognostics", respectively, as well as the applications of such procedures over two or more time points to monitor the diagnosis and/or prognosis over time, and statistical modeling based thereupon. Furthermore, the term diagnosis includes: a. prediction (determining if a patient will likely develop aggressive disease (hyperproliferative/invasive)), b. prognosis (predicting whether a patient will likely have a better or worse outcome at a pre-selected time in the future), c. therapy selection, d. therapeutic drug monitoring, and e. relapse monitoring.

The term "providing" as used herein with regard to a biological sample refers to directly or indirectly obtaining the biological sample from a subject. For example, "providing" may refer to the act of directly obtaining the biological sample from a subject (e.g., by a blood draw, tissue biopsy, lavage and the like). Likewise, "providing" may refer to the act of indirectly obtaining the biological sample. For example, providing may refer to the act of a laboratory receiving the sample from the party that directly obtained the sample, or to the act of obtaining the sample from an archive.

"Accuracy" refers to the degree of conformity of a measured or calculated quantity (a test reported value) to its actual (or true) value. Clinical accuracy relates to the proportion of true outcomes (true positives (TP) or true negatives (TN) versus misclassified outcomes (false positives (FP) or false negatives (FN)), and may be stated as a sensitivity, specificity, positive predictive values (PPV) or negative predictive values (NPV), or as a likelihood, odds ratio, among other measures.

The term "biological sample" as used herein refers to any sample of biological origin potentially containing one or more biomarkers. Examples of biological samples include tissue, organs, or bodily fluids such as whole blood, plasma, serum, tissue, lavage or any other specimen used for detection of disease.

The term "subject" as used herein refers to a mammal, preferably a human.

"Treating" or "treatment" as used herein with regard to a condition may refer to preventing the condition, slowing the onset or rate of development of the condition, reducing the risk of developing the condition, preventing or delaying the development of symptoms associated with the condition, reducing or ending symptoms associated with the condition, generating a complete or partial regression of the condition, or some combination thereof.

Biomarker levels may change due to treatment of the disease. The changes in biomarker levels may be measured by the present disclosure. Changes in biomarker levels may be used to monitor the progression of disease or therapy.

"Altered", "changed" or "significantly different" refer to a detectable change or difference from a reasonably comparable state, profile, measurement, or the like. Such changes may be all or none. They may be incremental and need not be linear. They may be by orders of magnitude. A change may be an increase or decrease by 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99%, 100%, or more, or any value in between 0% and 100%. Alternatively, the change may be 1-fold, 1.5-fold, 2-fold, 3-fold, 4-fold, 5-fold or more, or any values in between 1-fold and five-fold. The change may be statistically significant with a p value of 0.1, 0.05, 0.001, or 0.0001.

The term "stable disease" refers to a diagnosis for the presence of a melanoma, however the melanoma has been treated and remains in a stable condition, i.e. one that that is not progressive, as determined by imaging data and/or best clinical judgment.

The term "progressive disease" refers to a diagnosis for the presence of a highly active state of a melanoma, i.e. one has not been treated and is not stable or has been treated and has not responded to therapy, or has been treated and active disease remains, as determined by imaging data and/or best clinical judgment.

EXAMPLES

The disclosure is further illustrated by the following examples, which are not to be construed as limiting this disclosure in scope or spirit to the specific procedures herein described. It is to be understood that the examples are provided to illustrate certain embodiments and that no limitation to the scope of the disclosure is intended thereby. It is to be further understood that resort may be had to various other embodiments, modifications, and equivalents thereof which may suggest themselves to those skilled in the art without departing from the spirit of the present disclosure and/or scope of the appended claims.

Example 1

Derivation of a 28-Marker Gene Panel

Raw probe intensities (n=6,892,960 features) from n=49 whole blood samples were used to identify genes that best discriminated between different types of melanoma samples e.g., treated versus untreated, simultaneously. A total of 28 transcripts were identified in an unbiased manner as potential markers of melanoma behavior (Table 2).

An artificial intelligence model of melanoma disease dynamics was built using normalized gene expression of these 28 markers in whole blood from Controls (n=90), Responders/Stable (n=68), and Progressive (n=66) samples. The dataset was randomly split into training (n=169) and testing (n=55) partitions for model creation and validation respectively. Five algorithms (XGB, RF, TreeBag, SVM, NNET) were identified that best predicted the training data. In the test set, each algorithm produced probability scores that predicted the sample. Each probability score reflects the "certainty" of an algorithm that an unknown sample belongs to Control, Responder/Stable or Progressive class. For example, an unknown sample 51 can have the following probability vector {Control=20, Responder=50, Progressive=30}. This sample would be considered a responder, given a score of 50, but there may be potential of progression (probability of 30) (FIG. **1**).

TABLE 2

Melanoma Biomarker or Housekeeping Gene		Chromosome location	UniGene		Amplicon produced using forward and reverse primers	Assay Loca-	Exon Bound-
Symbol	Name	[Cytogenetic band]	ID	RefSeq	Length (bp)	tion	ary
ATL1	atlastin GTPase 1	Chr.14: 50533082-50633068 [14q22.1]	Hs.584905	NM_001127713.1	65	1544	12-13
ATP6V0D	ATPase H+ transporting V0 subunit d1	Chr.16: 67438014-67481186 [16q22.1]	Hs.106876	NM_004691.4	72	224	1-2
C1ORF21	chromosome 1 open reading frame 21	Chr.1: 184387016-184629021 [1q25.3]	Hs.497159	NM_030806.3	84	805	5-6
CFLAR	CASP8 and FADD like apoptosis regulator	Chr.2: 201116015-201172688 [2q33.1]	Hs.390736	NM_001127183.2	59	874	3-4
CFLAR-AS1	CFLAR antisense RNA 1	Chr.2: 201140289-201157792 [2q33.1]	Hs.664613	NR_040030.1	103	883	6-7
CHP1	calcineurin like EF-hand protein 1	Chr.15: 41231149-41281887 [15q15.1]	Hs.406234	NM_007236.4	120	212	1-2
DDX55	DEAD-box helicase 55	Chr.12: 123602077-123620943 [12q24.31]	Hs.286173	NM_020936.2	74	183	1-2
DMD	dystrophin	Chr.X: 31119219-33339609 [Xp21.1]	Hs.495912	NM_000109.3	76	9607	63-64
DNAJC9	DnaJ heat shock protein family (Hsp40) member C9	Chr.10: 73241954-73247331 [10q22.2]	Hs.408577	NM_015190.4	102	198	1-1

TABLE 2-continued

Melanoma Biomarker or Housekeeping Gene Symbol	Name	Chromosome location [Cytogenetic band]	UniGene ID	RefSeq	Amplicon produced using forward and reverse primers Length (bp)	Assay Location	Exon Boundary
ENOSF1	enolase superfamily member 1	Chr.18: 670012-712664 [18p11.32]	Hs.658550	NM_001126123.3	110	X	13-14
FANCL	Fanconi anemia complementation group L	Chr.2: 58159243-58241380 [2p16.1]	Hs.631890	NM_001114636.1	138	540	6-7
HJURP	Holliday junction recognition protein	Chr.2: 233836701-233854566 [2q37.1]	Hs.532968	NM_018410.4	52	399	4-5
HLA-DOA	major histocompatibility complex, class II, DO alpha	Chr.6: 33004182-33009612 [6p21.32]	Hs.631991	NM_002119.3	124	160	1-2
HLA-DRA	major histocompatibility complex, class II, DR alpha	Chr.6: 32439842-32445046 [6p21.32]	Hs.520048	NM_019111.4	129	884	4-5
HNRNPA3P1	heterogeneous nuclear ribonucleoprotein A3 pseudogene 1	Chr.10: 43787412-43790417 [10q11.21]	Hs.632956	NR_002726.2	99	2455	1-1
IL23A	interleukin 23 subunit alpha	Chr.12: 56334159-56340410 [12q13.3]	Hs.382212	NM_016584.2	107	323	1-2
IQGAP1	IQ motif containing GTPase activating protein 1	Chr.15: 90388241-90502243 [15q26.1]	Hs.430551	NM_003870.3	69	4562	34-35
LOC494127	NFYC pseudogene	Chr.9: 87083886-87085225 [9q21.33]	Hs.626316	NR_036691.1	86	150	1-1
LOC646471	uncharacterized LOC646471	Chr.1: 25819954-25823606 [1p36.11]	Hs.727271	NR_024498.1	80	2858	1-1
LOH12CR	loss of heterozygosity, 12, chromosomal region 2 (non-protein coding)	Chr.12: 12355408-12357067 [12p13.2]	Hs.67553	NR_024061.1	69	140	1-2
MTRNR2L2_MTO1	CUSTOM PRIMER						
PBXIP1	PBX homeobox interacting protein 1	Chr.1: 154944080-154956163 [1q21.3]	Hs.505806	NM_020524.3	60	189	2-3
RNF5	ring finger protein 5	Chr.6: 32178385-32180793 [6p21.32]	Hs.731774	NM_006913.3	67	273	1-2
SERTAD2	SERTA domain containing 2	Chr.2: 64631621-64751091 [2p14]	Hs.591569	NM_014755.2	91	288	1-2
SLC35G5	solute carrier family 35 member G5	Chr.8: 11330986-11332186 [8p23.1]	Hs.458397	NM_054028.1	144	541	1-1
SPATS2L	spermatogenesis associated serine rich 2 like	Chr.2: 200305881-200482263 [2q33.1]	Hs.120323	NM_001100422.1	89	1364	10-11
TDRD7	tudor domain containing 7	Chr.9: 97412020-97496125 [9q22.33]	Hs.193842	NM_001302884.1	59	3084	15-16
TOX4 (housekeeping gene)	TOX high mobility group box family member 4	Chr.14: 21477176-21499177 [14q11.2]	Hs.555910	NM_001303523.1	145	447	3-3
TPT1 (housekeeping gene)	tumor protein, translationally-controlled 1	Chr.13: 45333471-45341284 [13q14.13]	Hs.374596	NM_001286272.1	131	377	3-3
TXK	TXK tyrosine kinase	Chr.4: 48066393-48135322 [4p12]	Hs.479669	NM_003328.2	113	1265	11-12
YY2	YY2 transcription factor	Chr.X: 21855987-21858727 [Xp22.12]	Hs.673601	NM_206923.3	110	552	1-1

Diagnosis: Identification of Samples as Melanoma

In the test set 1, the data for the utility of the test to differentiate melanoma from controls are included in Table 3. The metrics are included in FIG. **2**. These are: sensitivity >90%, specificity 100%, PPV 100%, NPV 87%. The overall accuracy is 94%. The tool can therefore differentiate between controls and aggressive and stable melanoma disease.

TABLE 3

Confusion matrix showing classification accuracy of the 5-model algorithm that determines whether a sample is a melanoma or a control in blood samples

Predicted/Reference	Melanoma	Control
Control	3	20
Melanoma	30	0
Sensitivity	91%	
Specificity	100%	
Positive Predictive Value	100%	
Negative Predictive Value	87%	
Accuracy	94%	

The test was evaluated in a second test set (test set 2) that included 74 controls and 92 melanoma patients. The mean Melanomx score in the melanoma group was 73±31 versus 10±8 in the control group (FIG. **3**A). The receiver operator curve analysis demonstrated the score exhibited an area under the curve (AUC) of 0.96 (FIG. **3**B) and the metrics were 88-100% (FIG. **3**C).

Correlation with Surgical Removal of Melanoma.

In the surgical series (n=46), removal of melanoma decreased the score from 74±16 to 20±12 (FIG. **4**A). Evaluation of the post-surgical group identified the score was significantly different between patients with no evidence of disease 13±6 (not different to the control group) and those with residual disease (33±9) (FIG. **4**B). The Melanomx score can therefore define the extent of surgery and identify those who are surgical cures.

Evaluation of Immune-Therapy and Targeted (BRAF Inhibitor) Therapy.

In the therapy series (n=30), treatment with ipilimumab (immune therapy targeting CTLA-4), nivolumab (immune therapy targeting PD-1) or a BRAF inhibitor (vemurafenib) for 5 months significantly decreased the Melanomx scores from 88±12 to 15±5 ($p<0.0001$), 35±14 ($p<0.001$) and 38±21 ($p<0.0001$), respectively (FIG. **5**). All patients in the treated groups were stable. This indicates that the Melanomx score can therefore be used to measure the efficacy of both immune and targeted-therapy in melanoma and that a decrease in score correlated with response to therapeutic intervention.

Confirmation that Gene Expression is Melanoma Derived.

We confirmed that melanoma was the source for the blood-based gene expression assay by evaluating expression in different melanoma cell lines, in tumor tissue and by comparing expression in blood with tumor tissue collected at the same time-point during surgery.

All 28 genes were highly expressed in all 3 melanoma cell lines. Scores ranged from 94±6 (Hs294T) to 82±5 (CHL) to 42±9 (A375) (FIG. **6**A).

Spike-in experiments using these 3 cell lines and normal whole blood demonstrated that gene expression scores were detected when as few as 1 cell was spiked into 1 ml of blood. One single melanoma cell was detectable. Scores ranged from 21±4 (A375) to 30±5 (CHL) to 34±2 (H2294) (FIG. **6**B). Scores were significantly elevated compared to control blood (no spike-in; $p<0.0001$).

We then evaluated gene expression in tumor tissue. All 28 genes were highly expressed in melanoma compared to matched normal tissue and scores were significantly elevated 70±20 versus 4±3 ($p<0.0001$) (FIG. **7**A).

We also compared gene expression in tumor tissue and blood collected at the same time point. Gene expression was highly concordant (Pearson r: 0.74-0.83, median: 0.819) identifying gene expression in tumor tissue and blood was concordant (FIG. **7**B).

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EQUIVALENTS

While the present invention has been described in conjunction with the specific embodiments set forth above, many alternatives, modifications and other variations thereof will be apparent to those of ordinary skill in the art. All such alternatives, modifications and variations are intended to fall within the spirit and scope of the present invention.

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

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<213> ORGANISM: Homo sapiens

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 7

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

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ggtttcccta gcatgaagag atgtatgttg tagtcctgcc actgattact gaagtcctgc   1680

cattaattgc tgaatgtctt cagttgggcc actgagcttg tctgaatctg ttccctttta   1740

taaaagagtt actacatcaa aacaaagagt gaaatccaaa tttgtcaaac tgtatgtatt   1800

aaacatgtcc agtttttttgg ttaaaaaaaa attgcactgg ctttgaggga aaacacacag   1860
```

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ggtgagggaa ttgggctaaa tgacttctta caggcccctt tctgattctt taactttgaa   1920

aggcaagcca tattgatcca gttgttatag tgaactcatg gtaatggttt gtgagaacaa   1980

tagagatttt catttctatg tagatgagtt ggtatgagaa tatatggaat ttttaaggga   2040

ctgtttaaat ctttgatttg tagactatta aatataccgt atgcataaag taagccttta   2100

gctctaaggt aaagacgaca cgttttcggt ttgtgactac aaataggtta aaaatagatt   2160

ttaattttat taaaaatata atttaatgca ggttgtttga agcatctgtc ttcatatgat   2220

ggcattagaa caccttggta taataaaaag ttaccgtaat ttatgattat ttgaatttat   2280

ccattctgaa aattaataag atctaaaact ggcatgacaa tcaagatttg tatttagtga   2340

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<210> SEQ ID NO 10
<211> LENGTH: 4206
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 10
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ttccccacgt cgcttggggg ccacggcgcg gacgccatgc acacggaccc tgactactcg    180

gctgcctatg tcgtcataga aactgatgca gaagatggaa tcaaggggtg tggaattacc    240

ttcactctgg gaaaaggcac tgaagttgat tggtccagaa aagggcgtgg tgcacctggc    300

gacagcggcc gtcctaaacg cggtgtggga cttgtgggcc aagcaggagg gaaagcctgt    360

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catcactgat gtcctgactg aggaggatgc cctagaaata ctgcagaaag gtcaaattgg    480

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ccatgctggt ggagttggcc tctgtgaact ggtgcagcac ctgattatat ttgactacat   1080

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aaagaacttt tacctcggca tccagcccaa cccctaaaga ctgacaatat ccttcgagct   1500

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cagaagagga aggtttcaaa gtaaaagctg gttcatttct tacttacaca tatcaaattt   3540
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gctacctcgt ggcaccagat tacaaataca gcaggaataa ttctgtttgc cacaggaaac   4140

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aaaaaa                                                              4206
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<210> SEQ ID NO 11
<211> LENGTH: 1753
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 11

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gaaccggtcg aaaaccgtgt atgagggatt catctcggct cagggaagag acttccacct   180

taggatagtg ttgcctgaag atttacaact gaagaatgca agattattat gtagttggca   240

gctgagaaca atacttagtg gataccatcg aatagtacaa cagagaatgc agcactctcc   300

tgatctaatg agctttatga tggagttgaa gatgcttttg gaagttgcct taaagaatag   360

acaagagctg tatgcactac ctcctcctcc ccagttctac tcaagcctta ttgaagagat   420

aggaactctt ggttgggata aacttgtgta tgcggatacc tgcttcagta ccatcaagtt   480

aaaagcagaa gatgcttctg gtagagagca tttaatcact ctcaagttga aggcaaagta   540

tcctgcagaa tcaccagatt attttgtgga ttttcctgtt ccattttgtg cctcctggac   600

acctcaggta aattctcctc agagctcctt aataagcatt tatagtcagt tttggcagc    660

aatagaatca ctaaaggcat tctgggatgt tatggatgaa atcgatgaga agacctgggt   720

acttgagcca gaaaaacctc cacggagtgc aacagcacgc agaattgcat taggtaataa   780

tgtttccata aatatagagg tagaccccag gcatcctact atgcttcctg agtgcttctt   840

tcttggagct gaccatgtgg taaaacccct gggaattaag ctgagcagga acatacattt   900

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tcaacttgac ggtaccattc ctgatcaagt gtgtgataat tctcagtgtg gacaaccttt  1080

ccatcaaata tgcttatatg agtggctgag aggactacta actagtagac agagttttaa  1140

catcatattt ggtgaatgtc catattgtag taagccaatt accttaaaaa tgtctggaag  1200

gaaacactga aataagaata caacatttcg gtgaagagct ggaaacttaa aaaattatca  1260

aaaggaattt tggtatcatc ttcagagaaa aaataaagca agaaatacta acatcaaaag  1320

gacaggtatg atgatgcgat aataataaac atctgcgttt gtctcttcac taagagtaaa  1380

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aaaatcttga tgaggctcat aaaatgagtt tgggaattgt gtatagctga ttttttgtgg  1560

gaaactgttt acttcattca aaggttcttg agactcttga tatttctgtc ttctccttgt  1620

gctttcctat ggaaaaaata catatatagt ttagtttgtt agacgtgagt tatccaagta  1680

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<210> SEQ ID NO 12
<211> LENGTH: 3189
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 12

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caaggtgcag agtattttga gtgtgcaggt aacagagctg gaagggatgt acgtgtgact 600

ccgctgcctt cactggcctc acctgccgtg cctgcccccg gatactgcag tcgtatctcc 660

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tatggccat                                                           3189
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<210> SEQ ID NO 13
<211> LENGTH: 3485
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 13

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<210> SEQ ID NO 14
<211> LENGTH: 1312
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 14

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<210> SEQ ID NO 15
<211> LENGTH: 3006
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 15

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ccagagcagc tgagaaaact gtttattggt ggtctgagct ttgaaactac agatgatagt 180

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tggcct 3006

<210> SEQ ID NO 16
<211> LENGTH: 1049
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 16

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tggtgcaagg cgcagagcca gccagatttg agaagaaggc aaaaagatgc tggggagcag 180

agctgtaatg ctgctgttgc tgctgccctg gacagctcag ggcagagctg tgcctggggg 240

cagcagccct gcctggactc agtgccagca gctttcacag aagctctgca cactggcctg 300

gagtgcacat ccactagtgg gacacatgga tctaagagaa gagggagatg aagagactac 360

aaatgatgtt ccccatatcc agtgtggaga tggctgtgac cccaaggac tcagggacaa 420

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<210> SEQ ID NO 17
<211> LENGTH: 7219
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 17

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aactggagga ggggcttagg aatggggtct accttgccaa actggggaac ttcttctctc 360

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gcctccactt tagacacact gataatgtga ttcagtggtt gaatgccatg gatgagattg 480

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 18

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<211> LENGTH: 3653
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 19

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<210> SEQ ID NO 20
<211> LENGTH: 1538
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 20

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<210> SEQ ID NO 21
<211> LENGTH: 3285
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 21

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<210> SEQ ID NO 22
<211> LENGTH: 1176
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 22

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<210> SEQ ID NO 23
<211> LENGTH: 5565
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 23

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 26

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ttgaaaagaa tatgcttatg tttgatgaaa gatatttaac aagttttgtt ttaacagagt 3480

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<210> SEQ ID NO 27
<211> LENGTH: 4476
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 27

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<210> SEQ ID NO 28
<211> LENGTH: 4814
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 28

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<210> SEQ ID NO 29
<211> LENGTH: 2914
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 29

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<210> SEQ ID NO 30
<211> LENGTH: 2741
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 30

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<210> SEQ ID NO 31
<211> LENGTH: 6137
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 31

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 32

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<210> SEQ ID NO 33
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<210> SEQ ID NO 34
<211> LENGTH: 4510
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34

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<210> SEQ ID NO 35
<211> LENGTH: 11062
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 35

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<210> SEQ ID NO 38
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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 38

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<210> SEQ ID NO 39
<211> LENGTH: 1779
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 39

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<210> SEQ ID NO 40
<211> LENGTH: 7827
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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 40

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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 41

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<210> SEQ ID NO 42
<211> LENGTH: 5159
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

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gacaagaaag cataatatag ctctactcat gggtacacat accagtgtat aagattttta 5100

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<210> SEQ ID NO 43
<211> LENGTH: 5416
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

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ttgcatgtgt cttgaagaga aaagcagtgc tttggcagga ctctttcagc ccccacctga 180

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agtatgtgtt gagctcatcc ccgggcgatt cctgtctaag aaaaggagga tttggcccaa 840

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<210> SEQ ID NO 44
<211> LENGTH: 1940
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

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<210> SEQ ID NO 45
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<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 45

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agccaaaagg gtcatcatct ctgccccctc tgctgatgcc cccatgttcg tcatgggtgt    480
gaaccatgag aagtatgaca acagcctcaa gatcatcagc aatgcctcct gcaccaccaa    540
ctgcttagca cccctggcca aggtcatcca tgacaacttt ggtatcgtgg aaggactcat    600
gaccacagtc catgccatca ctgccaccca gaagactgtg gatggcccct ccgggaaact    660
gtggcgtgat ggccgcgggg ctctccagaa catcatccct gcctctactg gcgctgccaa    720
ggctgtgggc aaggtcatcc ctgagctgaa cgggaagctc actggcatgg ccttccgtgt    780
ccccactgcc aacgtgtcag tggtggacct gacctgccgt ctagaaaaac ctgccaaata    840
tgatgacatc aagaaggtgg tgaagcaggc gtcggagggc cccctcaagg gcatcctggg    900
ctacactgag caccaggtgg tctcctctga cttcaacagc gacacccact cctccacctt    960
tgacgctggg gctggcattg ccctcaacga ccactttgtc aagctcattt cctggtatga   1020
caacgaattt ggctacagca acagggtggt ggacctcatg gcccacatgg cctccaagga   1080
gtaagacccc tggaccacca gccccagcaa gagcacaaga ggaagagaga gaccctcact   1140
gctggggagt ccctgccaca ctcagtcccc caccacactg aatctcccct cctcacagtt   1200
gccatgtaga ccccttgaag aggggagggg cctagggagc cgcaccttgt catgtaccat   1260
caataaagta ccctgtgctc aaccagttaa aaaaaaaaaa aaaaaaaaa               1309
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<210> SEQ ID NO 46
<211> LENGTH: 2321
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 46

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gtcctcaacc aagatggcgc ggatggcttc aggcgcatca cgacaccggc gcgtcacgcg     60
acccgcccta cgggcacctc ccgcgctttt cttagcgccg cagacggtgg ccgagcgggg    120
gaccgggaag catggcccgg gggtcggcgg ttgcctgggc ggcgctcggg ccgttgttgt    180
ggggctgcgc gctggggctg cagggcggga tgctgtaccc ccaggagagc ccgtcgcggg    240
agtgcaagga gctggacggc ctctggagct tccgcgccga cttctctgac aaccgacgcc    300
ggggcttcga ggagcagtgg taccggcggc cgctgtggga gtcaggcccc accgtggaca    360
tgccagttcc ctccagcttc aatgacatca gccaggactg gcgtctgcgg cattttgtcg    420
gctgggtgtg gtacgaacgg gaggtgatcc tgccggagcg atggacccag gacctgcgca    480
caagagtggt gctgaggatt ggcagtgccc attcctatgc catcgtgtgg gtgaatgggg    540
tcgacacgct agagcatgag gggggctacc tccccttcga ggccgacatc agcaacctgg    600
tccaggtggg gcccctgccc tcccggctcc gaatcactat cgccatcaac aacacactca    660
cccccaccac cctgccacca gggaccatcc aatacctgac tgacacctcc aagtatccca    720
agggttactt tgtccagaac acatattttg actttttcaa ctacgctgga ctgcagcggt    780
ctgtacttct gtacacgaca cccaccacct acatcgatga catcaccgtc accaccagcg    840
tggagcaaga cagtgggctg gtgaattacc agatctctgt caagggcagt aacctgttca    900
```

```
agttggaagt gcgtcttttg gatgcagaaa acaaagtcgt ggcgaatggg actgggaccc    960
agggccaact taaggtgcca ggtgtcagcc tctggtggcc gtacctgatg cacgaacgcc   1020
ctgcctatct gtattcattg gaggtgcagc tgactgcaca gacgtcactg ggcctgtgt    1080
ctgacttcta cacactccct gtggggatcc gcactgtggc tgtcaccaag agccagttcc   1140
tcatcaatgg gaaacctttc tatttccacg gtgtcaacaa gcatgaggat gcggacatcc   1200
gagggaaggg cttcgactgg ccgctgctgg tgaaggactt caacctgctt cgctggcttg   1260
gtgccaacgc tttccgtacc agccactacc cctatgcaga ggaagtgatg cagatgtgtg   1320
accgctatgg gattgtggtc atcgatgagt gtcccggcgt gggcctggcg ctgccgcagt   1380
tcttcaacaa cgtttctctg catcaccaca tgcaggtgat ggaagaagtg gtgcgtaggg   1440
acaagaacca ccccgcggtc gtgatgtggt ctgtggccaa cgagcctgcg tcccacctag   1500
aatctgctgg ctactacttg aagatggtga tcgctcacac caaatccttg gacccctccc   1560
ggcctgtgac ctttgtgagc aactctaact atgcagcaga caagggggct ccgtatgtgg   1620
atgtgatctg tttgaacagc tactactctt ggtatcacga ctacgggcac ctggagttga   1680
ttcagctgca gctggccacc cagtttgaga actggtataa gaagtatcag aagcccatta   1740
ttcagagcga gtatggagca gaaacgattg cagggtttca ccaggatcca cctctgatgt   1800
tcactgaaga gtaccagaaa agtctgctag agcagtacca tctgggtctg gatcaaaaac   1860
gcagaaaata cgtggttgga gagctcattt ggaattttgc cgatttcatg actgaacagt   1920
caccgacgag agtgctgggg aataaaaagg ggatcttcac tcggcagaga caaccaaaaa   1980
gtgcagcgtt cctttgcga gagagatact ggaagattgc caatgaaacc aggtatcccc    2040
actcagtagc caagtcacaa tgtttggaaa acagcctgtt tacttgagca agactgatac   2100
cacctgcgtg tcccttcctc cccgagtcag ggcgacttcc acagcagcag aacaagtgcc   2160
tcctggactg ttcacggcag accagaacgt ttctggcctg ggttttgtgg tcatctattc   2220
tagcagggaa cactaaaggt ggaaataaaa gattttctat tatggaaata aagagttggc   2280
atgaaagtgg ctactgaaaa aaaaaaaaaa aaaaaaaaaa a                       2321
```

<210> SEQ ID NO 47
<211> LENGTH: 1229
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 47

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gtctgacggg cgatggcgca gccaatagac aggagcgcta tccgcggttt ctgattggct     60
actttgttcg cattataaaa ggcacgcgcg ggcgcgaggc ccttctctcg ccaggcgtcc    120
tcgtggaagt gacatcgtct ttaaaccctg cgtggcaatc cctgacgcac cgccgtgatg    180
cccagggaag acagggcgac ctggaagtcc aactacttcc ttaagatcat ccaactattg    240
gatgattatc cgaaatgttt cattgtggga gcagacaatg tgggctccaa gcagatgcag    300
cagatccgca tgtcccttcg cgggaaggct gtggtgctga tgggcaagaa caccatgatg    360
cgcaaggcca tccgagggca cctggaaaac aacccagctc tggagaaact gctgcctcat    420
atccgggggа atgtgggctt tgtgttcacc aaggaggacc tcactgagat cagggacatg    480
ttgctggcca ataaggtgcc agctgctgcc cgtgctggtg ccattgcccc atgtgaagtc    540
actgtgccag cccagaacac tggtctcggg cccgagaaga cctccttttt ccaggcttta    600
ggtatcacca ctaaaatctc caggggcacc attgaaatcc tgagtgatgt gcagctgatc    660
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-continued

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aagactggag acaaagtggg agccagcgaa gccacgctgc tgaacatgct caacatctcc    720

cccttctcct ttgggctggt catccagcag gtgttcgaca atggcagcat ctacaaccct    780

gaagtgcttg atatcacaga ggaaactctg cattctcgct tcctggaggg tgtccgcaat    840

gttgccagtg tctgtctgca gattggctac ccaactgttg catcagtacc ccattctatc    900

atcaacgggt acaaacgagt cctggccttg tctgtggaga cggattacac cttcccactt    960

gctgaaaagg tcaaggcctt cttggctgat ccatctgcct ttgtggctgc tgcccctgtg   1020

gctgctgcca ccacagctgc tcctgctgct gctgcagccc cagctaaggt tgaagccaag   1080

gaagagtcgg aggagtcgga cgaggatatg ggatttggtc tctttgacta atcaccaaaa   1140

agcaaccaac ttagccagtt ttatttgcaa aacaaggaaa taaaggctta cttctttaaa   1200

aagtaaaaaa aaaaaaaaaa aaaaaaaaa                                     1229
```

<210> SEQ ID NO 48
<211> LENGTH: 5234
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 48

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agagcgtcgg gatatcgggt ggcggctcgg gacggaggac gcgctagtgt gagtgcgggc     60

ttctagaact acaccgaccc tcgtgtcctc ccttcatcct gcggggctgg ctggagcggc    120

cgctccggtg ctgtccagca gccataggga gccgcacggg gagcgggaaa gcggtcgcgg    180

ccccaggcgg ggcggccggg atggagcggg gccgcgagcc tgtggggaag gggctgtggc    240

ggcgcctcga gcggctgcag gttcttctgt gtggcagttc agaatgatgg atcaagctag    300

atcagcattc tctaacttgt ttggtggaga accattgtca tatacccggt tcagcctggc    360

tcggcaagta gatggcgata acagtcatgt ggagatgaaa cttgctgtag atgaagaaga    420

aaatgctgac aataacacaa aggccaatgt cacaaaaacca aaaaggtgta gtggaagtat    480

ctgctatggg actattgctg tgatcgtctt tttcttgatt ggatttatga ttggctactt    540

gggctattgt aaaggggtag aaccaaaaac tgagtgtgag agactggcag gaaccgagtc    600

tccagtgagg gaggagccag gagaggactt ccctgcagca cgtcgcttat attgggatga    660

cctgaagaga aagttgtcgg agaaactgga cagcacagac ttcaccggca ccatcaagct    720

gctgaatgaa aattcatatg tccctcgtga ggctggatct caaaaagatg aaaatcttgc    780

gttgtatgtt gaaaatcaat ttcgtgaatt taaactcagc aaagtctggc gtgatcaaca    840

ttttgttaag attcaggtca aagacagcgc tcaaaactcg gtgatcatag ttgataagaa    900

cggtagactt gtttacctgg tggagaatcc tgggggttat gtggcgtata gtaaggctgc    960

aacagttact ggtaaactgg tccatgctaa ttttggtact aaaaaagatt ttgaggattt   1020

atacactcct gtgaatggat ctatagtgat tgtcagagca gggaaaatca cctttgcaga   1080

aaaggttgca aatgctgaaa gcttaaatgc aattggtgtg ttgatataca tggaccagac   1140

taaatttccc attgttaacg cagaactttc attctttgga catgctcatc tggggacagg   1200

tgacccttac acacctggat tcccttcctt caatcacact cagtttccac catctcggtc   1260

atcaggattg cctaatatac ctgtccagac aatctccaga gctgctgcag aaaagctgtt   1320

tgggaatatg gaaggagact gtccctctga ctggaaaaca gactctacat gtaggatggt   1380

aacctcagaa agcaagaatg tgaagctcac tgtgagcaat gtgctgaaag agataaaaat   1440

tcttaacatc tttggagtta ttaaaggctt tgtagaacca gatcactatg ttgtagttgg   1500

ggcccagaga gatgcatggg gccctggagc tgcaaaatcc ggtgtaggca cagctctcct   1560
```

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attgaaactt gcccagatgt tctcagatat ggtcttaaaa gatgggtttc agcccagcag   1620
aagcattatc tttgccagtt ggagtgctgg agactttgga tcggttggtg ccactgaatg   1680
gctagaggga tacctttcgt ccctgcattt aaaggctttc acttatatta atctggataa   1740
agcggttctt ggtaccagca acttcaaggt ttctgccagc ccactgttgt atacgcttat   1800
tgagaaaaca atgcaaaatg tgaagcatcc ggttactggg caatttctat atcaggacag   1860
caactgggcc agcaaagttg agaaactcac tttagacaat gctgctttcc ctttccttgc   1920
atattctgga atcccagcag tttctttctg tttttgcgag gacacagatt atccttattt   1980
gggtaccacc atggacacct ataaggaact gattgagagg attcctgagt tgaacaaagt   2040
ggcacgagca gctgcagagg tcgctggtca gttcgtgatt aaactaaccc atgatgttga   2100
attgaacctg gactatgaga ggtacaacag ccaactgctt tcatttgtga gggatctgaa   2160
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agacagattt gtcatgaaga aactcaatga tcgtgtcatg agagtggagt atcacttcct   2340
ctctccctac gtatctccaa aagagtctcc tttccgacat gtcttctggg gctccggctc   2400
tcacacgctg ccagctttac tggagaactt gaaactgcgt aaacaaaata acggtgcttt   2460
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gcttccatga gaacagcagg gtagtctggt ttctagactt gtgctgatcg tgctaaattt   2640
tcagtagggc tacaaaacct gatgttaaaa ttccatccca tcatcttggt actactagat   2700
gtctttaggc agcagctttt aatacagggt agataacctg tacttcaagt taaagtgaat   2760
aaccacttaa aaaatgtcca tgatggaata ttcccctatc tctagaattt taagtgcttt   2820
gtaatgggaa ctgcctcttt cctgttgttg ttaatgaaaa tgtcagaaac cagttatgtg   2880
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ttgagtgatc ctccaacttc atttgatgct aaataggaga taccaggttg aaagaccttc   3000
tccaaatgag atctaagcct ttccataagg aatgtagctg gtttcctcat tcctgaaaga   3060
aacagttaac tttcagaaga gatgggcttg ttttcttgcc aatgaggtct gaaatggagg   3120
tccttctgct ggataaaatg aggttcaact gttgattgca ggaataaggc cttaatatgt   3180
taacctcagt gtcatttatg aaaagagggg accagaagcc aaagacttag tatattttct   3240
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aaagcacatt gaaagagaac cagtttcagg tgtttagttg cagactcagt ttgtcagact   3360
ttaaagaata atatgctgcc aaattttggc caaagtgtta atcttagggg agagctttct   3420
gtccttttgg cactgagata tttattgttt atttatcagt gacagagttc actataaatg   3480
gtgttttttt aatagaatat aattatcgga agcagtgcct tccataatta tgacagttat   3540
actgtcggtt ttttttaaat aaaagcagca tctgctaata aaacccaaca gatactggaa   3600
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tgaataaagt tgaagctaag atttagagat gaattaaatt taattagggg ttgctaagaa   3720
gcgagcactg accagataag aatgctggtt ttcctaaatg cagtgaattg tgaccaagtt   3780
ataaatcaat gtcacttaaa ggctgtggta gtactcctgc aaaattttat agctcagttt   3840
atccaaggtg taactctaat tcccattttg caaaatttcc agtacctttg tcacaatcct   3900
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```
aacacattat cgggagcagt gtcttccata atgtataaag aacaaggtag tttttaccta   3960
ccacagtgtc tgtatcggag acagtgatct ccatatgtta cactaagggt gtaagtaatt   4020
atcgggaaca gtgtttccca taatttttctt catgcaatga catcttcaaa gcttgaagat   4080
cgttagtatc taacatgtat cccaactcct ataattccct atctttagt tttagttgca   4140
gaaacatttt gtggtcatta agcattgggt gggtaaattc aaccactgta aaatgaaatt   4200
actacaaaat ttgaaattta gcttgggttt ttgttacctt tatggtttct ccaggtcctc   4260
tacttaatga gatagtagca tacatttata atgtttgcta ttgacaagtc attttaactt   4320
tatcacatta tttgcatgtt acctcctata aacttagtgc ggacaagttt taatccagaa   4380
ttgacctttt gacttaaagc agagggactt tgtatagaag gtttgggggc tgtggggaag   4440
gagagtcccc tgaaggtctg acacgtctgc ctacccattc gtggtgatca attaaatgta   4500
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tcttagtgtc acttacctag atttacagcc tcacttgaat gtgtcactac tcacagtctc   4680
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aaagcatgtg gcatgttcat cgtataacac aatatgaata cagggcatgc attttgcagc   4860
agtgagtctc ttcagaaaac ccttttctac agttagggtt gagttacttc ctatcaagcc   4920
agtacgtgct aacaggctca atattcctga atgaaatatc agactagtga caagctcctg   4980
gtcttgagat gtcttctcgt taaggagatg ggccttttgg aggtaaagga taaaatgaat   5040
gagttctgtc atgattcact attctagaac ttgcatgacc tttactgtgt tagctctttg   5100
aatgttcttg aaattttaga ctttctttgt aaacaaatga tatgtcctta tcattgtata   5160
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ctgaaaaaaa aaaa                                                      5234
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<210> SEQ ID NO 49
<211> LENGTH: 1869
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 49

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tacctggttg atcctgccag tagcatatgc ttgtctcaaa gattaagcca tgcatgtcta     60
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tggtcgctcg ctcctctccc acttggataa ctgtggtaat tctagagcta atacatgccg    180
acgggcgctg accccttcg cgggggggat gcgtgcattt atcagatcaa aaccaacccg    240
gtcagcccct ctccggcccc ggccgggggg cgggcgccgg cggctttggt gactctagat    300
aacctcgggc cgatcgcacg cccccgtgg cggcgacgac ccattcgaac gtctgcccta    360
tcaactttcg atggtagtcg ccgtgcctac catggtgacc acgggtgacg gggaatcagg    420
gttcgattcc ggagagggag cctgagaaac ggctaccaca tccaaggaag gcagcaggcg    480
cgcaaattac ccactcccga cccggggagg tagtgacgaa aaataacaat acaggactct    540
ttcgaggccc tgtaattgga atgagtccac tttaaatcct ttaacgagga tccattggag    600
ggcaagtctg gtgccagcag ccgcggtaat tccagctcca atagcgtata ttaaagttgc    660
tgcagttaaa aagctcgtag ttggatcttg ggagcgggcg ggcggtccgc cgcgaggcga    720
gccaccgccc gtccccgccc cttgcctctc ggcgccccct cgatgctctt agctgagtgt    780
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cccgcggggc ccgaagcgtt tactttgaaa aaattagagt gttcaaagca ggcccgagcc 840
gcctggatac cgcagctagg aataatggaa taggaccgcg gttctatttt gttggttttc 900
ggaactgagg ccatgattaa gagggacggc cgggggcatt cgtattgcgc cgctagaggt 960
gaaattcttg gaccggcgca agacggacca gagcgaaagc atttgccaag aatgttttca 1020
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gaacgagact ctggcatgct aactagttac gcgacccccg agcggtcggc gtcccccaac 1440
ttcttagagg gacaagtggc gttcagccac ccgagattga gcaataacag gtctgtgatg 1500
cccttagatg tccggggctg cacgcgcgct acactgactg gctcagcgtg tgcctaccct 1560
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ttattcccca tgaacgagga attcccagta agtgcgggtc ataagcttgc gttgattaag 1680
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tcggatcggc cccgccgggg tcggcccacg gccctggcgg agcgctgaga agacggtcga 1800
acttgactat ctagaggaag taaaagtcgt aacaaggttt ccgtaggtga acctgcggaa 1860
ggatcatta 1869

<210> SEQ ID NO 50
<211> LENGTH: 2288
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

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ccgccgagga aaaccgtgta ctattagcca tggtcaaccc caccgtgttc ttcgacattg 120
ccgtcgacgg cgagcccttg ggccgcgtct cctttgagct gtttgcagac aaggtcccaa 180
agacagcaga aaattttcgt gctctgagca ctggagagaa aggatttggt tataagggtt 240
cctgctttca cagaattatt ccagggttta tgtgtcaggg tggtgacttc acacgccata 300
atggcactgg tggcaagtcc atctatgggg agaaatttga agatgagaac ttcatcctaa 360
agcatacggg tcctggcatc ttgtccatgg caaatgctgg acccaacaca aatggttccc 420
agtttttcat ctgcactgcc aagactgagt ggttggatgg caagcatgtg gtgtttggca 480
aagtgaaaga aggcatgaat attgtggagg ccatggagcg ctttgggtcc aggaatggca 540
agaccagcaa gaagatcacc attgctgact gtggacaact cgaataagtt tgacttgtgt 600
tttatcttaa ccaccagatc attccttctg tagctcagga gagcacccct ccacccccatt 660
tgctcgcagt atcctagaat ctttgtgctc tcgctgcagt tccctttggg ttccatgttt 720
tccttgttcc ctcccatgcc tagctggatt gcagagttaa gtttatgatt atgaaataaa 780
aactaaataa caattgtcct cgtttgagtt aagagtgttg atgtaggctt tattttaagc 840
agtaatgggt tacttctgaa acatcacttg tttgcttaat tctacacagt acttagattt 900
tttttacttt ccagtcccag gaagtgtcaa tgtttgttga gtggaatatt gaaaatgtag 960

gcagcaactg ggcatggtgg ctcactgtct gtaatgtatt acctgaggca gaagaccacc 1020

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tcttgtgtct caaaggtagg cagaggcagg aaaagcaagg agccagaatt aagaggttgg 1260

gtcagtctgc agtgagttca tgcatttaga ggtgttcttc aagatgacta atgtcaaaaa 1320

ttgagacatc tgttgcggtt tttttttttt ttttttcccc tggaatgcag tggcgtgatc 1380

tcagctcact gcagcctccg cctcctgggt tcaagtgatt ctagtgcctc agcctcctga 1440

gtagctggga taatgggcgt gtgccaccat gcccagctaa tttttgtatt tttagtatag 1500

atggggtttc atcattttga ccaggctggt ctcaaactct tgacctcagc tgatgcgcct 1560

gccttggcct cccaaactgc tgagattaca gatgtgagcc accgcaccct acctcatttt 1620

ctgtaacaaa gctaagcttg aacactgttg atgttcttga gggaagcata ttgggcttta 1680

ggctgtaggt caagtttata catcttaatt atggtggaat tcctatgtag agtctaaaaa 1740

gccaggtact tggtgctaca gtcagtctcc ctgcagaggg ttaaggcgca gactacctgc 1800

agtgaggagg tactgcttgt agcatataga gcctctccct agctttggtt atggaggctt 1860

tgaggttttg caaacctgac caatttaagc cataagatct ggtcaaaggg ataccсttcc 1920

cactaaggac ttggtttctc aggaaattat atgtacagtg cttgctggca gttagatgtc 1980

aggacaatct aagctgagaa aacccсttct ctgcccacct taacagacct ctagggttct 2040

taacccagca atcaagtttg cctatcctag aggtggcgga tttgatcatt tggtgtgttg 2100

ggcaattttt gttttactgt ctggttcctt ctgcgtgaat taccaccacc accacttgtg 2160

catctcagtc ttgtgtgttg tctggttacg tattccctgg gtgataccat tcaatgtctt 2220

aatgtacttg tggctcagac ctgagtgcaa ggtggaaata aacatcaaac atcttttcat 2280

tatcccta 2288

<210> SEQ ID NO 51
<211> LENGTH: 2439
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 51

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ggctccctcg ttgaccgaat caccgacctc tctccccagc tgtatttcca aaatgtcgct 180

ttctaacaag ctgacgctgg acaagctgga cgttaaaggg aagcgggtcg ttatgagagt 240

cgacttcaat gttcctatga agaacaacca gataacaaac aaccagagga ttaaggctgc 300

tgtcccaagc atcaaattct gcttggacaa tggagccaag tcggtagtcc ttatgagcca 360

cctaggccgg cctgatggtg tgcccatgcc tgacaagtac tccttagagc cagttgctgt 420

agaactcaaa tctctgctgg gcaaggatgt tctgttcttg aaggactgtg taggcccaga 480

agtggagaaa gcctgtgcca acccagctgc tgggtctgtc atcctgctgg agaacctccg 540

ctttcatgtg gaggaagaag ggaagggaaa agatgcttct gggaacaagg ttaaagccga 600

gccagccaaa atagaagctt tccgagcttc actttccaag ctaggggatg tctatgtcaa 660

tgatgctttt ggcactgctc acagagccca cagctccatg gtaggagtca atctgccaca 720

gaaggctggt gggtttttga tgaagaagga gctgaactac tttgcaaagg ccttggagag 780 cccagagcga cccttcctgg ccatcctggg cggagctaaa gttgcagaca agatccagct 840

catcaataat atgctggaca aagtcaatga gatgattatt ggtggtggaa tggcttttac 900

cttccttaag gtgctcaaca acatggagat tggcacttct ctgtttgatg aagagggagc 960

caagattgtc aaagacctaa tgtccaaagc tgagaagaat ggtgtgaaga ttaccttgcc 1020

tgttgacttt gtcactgctg acaagtttga tgagaatgcc aagactggcc aagccactgt 1080

ggcttctggc atacctgctg gctggatggg cttggactgt ggtcctgaaa gcagcaagaa 1140

gtatgctgag gctgtcactc gggctaagca gattgtgtgg aatggtcctg tgggggtatt 1200

tgaatgggaa gcttttgccc ggggaaccaa agctctcatg gatgaggtgg tgaaagccac 1260

ttctaggggc tgcatcacca tcataggtgg tggagacact gccacttgct gtgccaaatg 1320

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gttattatga ttagctttgt cactgtttca ctactcagca tggaaacaag atgaaattcc 1800

atttgtaggt agtgagacaa aattgatgat ccattaagta aacaataaaa gtgtccattg 1860

aaaccgtgat tttttttttt ttcctgtcat actttgttag gaagggtgag aatagaatct 1920

tgaggaacgg atcagatgtc tatattgctg aatgcaagaa gtggggcagc agcagtggag 1980

agatgggaca attagataaa tgtccattct ttatcaaggg cctactttat ggcagacatt 2040

gtgctagtgc ttttattcta acttttattt ttatcagtta cacatgatca taatttaaaa 2100

agtcaaggct tataacaaaa aagccccagc ccattcctcc cattcaagat tcccactccc 2160

cagaggtgac cactttcaac tcttgagttt ttcaggtata tacctccatg tttctaagta 2220

atatgcttat attgttcact tcttttttttt ttattttta aagaaatcta tttcatacca 2280

tggaggaagg ctctgttcca catatatttc cacttcttca ttctctcggt atagttttgt 2340

cacaattata gattagatca aaagtctaca taactaatac agctgagcta tgtagtatgc 2400

tatgattaaa tttacttatg taaaaaaaaa aaaaaaaaa 2439

<210> SEQ ID NO 52
<211> LENGTH: 1196
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

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cctgggccgc ctggcggcca tcgtggctaa acaggtactg ctgggccgga aggtggtggt 180

cgtacgctgt gaaggcatca acatttctgg caatttctac agaaacaagt tgaagtacct 240

ggctttcctc cgcaagcgga tgaacaccaa cccttcccga ggcccctacc acttccgggc 300

ccccagccgc atcttctggc ggaccgtgcg aggtatgctg ccccacaaaa ccaagcgagg 360

ccaggccgct ctggaccgtc tcaaggtgtt tgacggcatc ccaccgccct acgacaagaa 420

```
aaagcggatg gtggttcctg ctgccctcaa ggtcgtgcgt ctgaagccta caagaaagtt     480

tgcctatctg ggcgcctgg ctcacgaggt tggctggaag taccaggcag tgacagccac     540

cctggaggag aagaggaaag agaaagccaa gatccactac cggaagaaga aacagctcat     600

gaggctacgg aaacaggccg agaagaacgt ggagaagaaa attgacaaat acacagaggt     660

cctcaagacc cacggactcc tggtctgagc ccaataaaga ctgttaattc ctcatgcgtt     720

gcctgccctt cctccattgt tgccctggaa tgtacgggac ccaggggcag cagcagtcca     780

ggtgccacag gcagccctgg gacataggaa gctgggagca aggaaagggt cttagtcact     840

gcctcccgaa gttgcttgaa agcactcgga gaattgtgca ggtgtcattt atctatgacc     900

aataggaaga gcaaccagtt actatgagtg aaagggagcc agaagactga ttggagggcc     960

ctatcttgtg agtggggcat ctgttggact ttccacctgg tcatatactc tgcagctgtt    1020

agaatgtgca agcacttggg gacagcatga gcttgctgtt gtacacaggg tatttctaga    1080

agcagaaata gactgggaag atgcacaacc aaggggttac aggcatcgcc catgctcctc    1140

acctgtattt tgtaatcaga aataaattgc ttttaaagaa aaaaaaaaaa aaaaaa        1196
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<210> SEQ ID NO 53
<211> LENGTH: 987
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 53

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atgtctcgct ccgtggcctt agctgtgctc gcgctactct ctctttctgg cctggaggct     120

atccagcgta ctccaaagat tcaggtttac tcacgtcatc cagcagagaa tggaaagtca     180

aatttcctga attgctatgt gtctgggttt catccatccg acattgaagt tgacttactg     240

aagaatggag agagaattga aaaagtggag cattcagact tgtctttcag caaggactgg     300

tctttctatc tcttgtacta cactgaattc acccccactg aaaaagatga gtatgcctgc     360

cgtgtgaacc atgtgacttt gtcacagccc aagatagtta agtgggatcg agacatgtaa     420

gcagcatcat ggaggtttga agatgccgca tttggattgg atgaattcca aattctgctt     480

gcttgctttt taatattgat atgcttatac acttacactt tatgcacaaa atgtagggtt     540

ataataatgt taacatggac atgatcttct ttataattct actttgagtg ctgtctccat     600

gtttgatgta tctgagcagg ttgctccaca ggtagctcta ggagggctgg caacttagag     660

gtggggagca gagaattctc ttatccaaca tcaacatctt ggtcagattt gaactcttca     720

atctcttgca ctcaaagctt gttaagatag ttaagcgtgc ataagttaac ttccaattta     780

catactctgc ttagaatttg ggggaaaatt tagaaatata attgacagga ttattggaaa     840

tttgttataa tgaatgaaac attttgtcat ataagattca tatttacttc ttatacattt     900

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tcataaaact tgatgtgtta tctctta                                         987
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<210> SEQ ID NO 54
<211> LENGTH: 3003
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 54

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tcccaccgcc agctggaacc ctggggacta cgacgtccct caaaccttgc ttctaggaga     120
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taaaaagaac atccagtcat ggataaaaat gagctggttc agaaggccaa actggccgag 180
caggctgagc gatatgatga catggcagcc tgcatgaagt ctgtaactga gcaaggagct 240
gaattatcca atgaggagag gaatcttctc tcagttgctt ataaaaatgt tgtaggagcc 300
cgtaggtcat cttggagggt cgtctcaagt attgaacaaa agacggaagg tgctgagaaa 360
aaacagcaga tggctcgaga atacagagag aaaattgaga cggagctaag agatatctgc 420
aatgatgtac tgtctctttt ggaaaagttc ttgatcccca atgcttcaca agcagagagc 480
aaagtcttct atttgaaaat gaaaggagat tactaccgtt acttggctga ggttgccgct 540
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tctgtgttct attatgagat tctgaactcc ccagagaaag cctgctctct tgcaaagaca 720
gcttttgatg aagccattgc tgaacttgat acattaagtg aagagtcata caaagacagc 780
acgctaataa tgcaattact gagagacaac ttgacattgt ggacatcgga tacccaagga 840
gacgaagctg aagcaggaga aggaggggaa aattaaccgg ccttccaact tttgtctgcc 900
tcattctaaa atttacacag tagaccattt gtcatccatg ctgtcccaca aatagttttt 960
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gcaaagaaaa ctgcctacat actggtttgt cctggcgggg aataaaaggg atcattggtt 1260
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atgaatggac ttttcaacta ctttctctac agcttttcat gtaaattagt cttggttctg 1920
aaacttctct aaaggaaatt gtacattttt tgaaatttat tccttattcc ctcttggcag 1980
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cattgggtag cattaactgt aagttttgtg cttccaaatc actttttggt ttttaagaat 2280
ttcttgatac tcttatagcc tgccttcaat tttgatcctt tattctttct atttgtcagg 2340
tgcacaagat taccttcctg ttttagcctt ctgtcttgtc accaaccatt cttacttggt 2400
ggccatgtac ttggaaaaag gccgcatgat ctttctggct ccactcagtg tctaaggcac 2460

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cctgcttcct ttgcttgcat cccacagact atttccctca tcctatttac tgcagcaaat   2520

ctctccttag ttgatgagac tgtgtttatc tccctttaaa accctaccta tcctgaatgg   2580

tctgtcattg tctgccttta aaatccttcc tctttcttcc tcctctattc tctaaataat   2640

gatggggcta agttataccc aaagctcact ttacaaaata tttcctcagt actttgcaga   2700

aaacaccaaa caaaaatgcc attttaaaaa aggtgtattt tttcttttag aatgtaagct   2760

cctcaagagc agggacaatg ttttctgtat gttctattgt gcctagtaca ctgtaaatgc   2820

tcaataaata ttgatgatgg gaggcagtga gtcttgatga taagggtgag aaactgaaat   2880

cccaaacact gttttgttgc ttgttttatt atgacctcag attaaattgg gaaatattgg   2940

cccttttgaa taattgtccc aaatattaca ttcaaataaa agtgcaatgg agaaaaaaaa   3000

aaa                                                                 3003


<210> SEQ ID NO 55
<211> LENGTH: 2803
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 55

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ggcaacagca gacatgtcgg gggtccgggg cctgtcgcgg ctgctgagcg ctcggcgcct    180

ggcgctggcc aaggcgtggc caacagtgtt gcaaacagga acccgaggtt ttcacttcac    240

tgttgatggg aacaagaggg catctgctaa agtttcagat tccatttctg ctcagtatcc    300

agtagtggat catgaatttg atgcagtggt ggtaggcgct ggaggggcag gcttgcgagc    360

tgcatttggc ctttctgagg cagggtttaa tacagcatgt gttaccaagc tgtttcctac    420

caggtcacac actgttgcag cacagggagg aatcaatgct gctctgggga acatggagga    480

ggacaactgg aggtggcatt tctacgacac cgtgaagggc tccgactggc tgggggacca    540

ggatgccatc cactacatga cggagcaggc cccgccgcc gtggtcgagc tagaaaatta     600

tggcatgccg tttagcagaa ctgaagatgg gaagatttat cagcgtgcat ttggtggaca    660

gagcctcaag tttggaaagg gcgggcaggc ccatcggtgc tgctgtgtgg ctgatcggac    720

tggccactcg ctattgcaca ccttatatgg aaggtctctg cgatatgata ccagctattt    780

tgtggagtat tttgccttgg atctcctgat ggagaatggg gagtgccgtg gtgtcatcgc    840

actgtgcata gaggacgggt ccatccatcg cataagagca aagaacactg ttgttgccac    900

aggaggctac gggcgcacct acttcagctg cacgtctgcc cacaccagca ctggcgacgg    960

cacggccatg atcaccaggg caggccttcc ttgccaggac ctagagtttg ttcagttcca   1020

ccctacaggc atatatggtg ctggttgtct cattacggaa ggatgtcgtg gagagggagg   1080

cattctcatt aacagtcaag gcgaaaggtt tatggagcga tacgcccctg tcgcgaagga   1140

cctggcgtct agagatgtgg tgtctcggtc catgactctg gagatccgag aaggaagagg   1200

ctgtggccct gagaaagatc acgtctacct gcagctgcac cacctacctc cagagcagct   1260

ggccacgcgc ctgcctggca tttcagagac agccatgatc ttcgctggcg tggacgtcac   1320

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ctacaagggg caggtcctga ggcacgtgaa tggccaggat cagattgtgc ccggcctgta   1440

cgcctgtggg gaggccgcct gtgcctcggt acatggtgcc aaccgcctcg ggcaaactc    1500

gctcttggac ctggttgtct ttggtcgggc atgtgccctg agcatcgaag agtcatgcag   1560
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gcctggagat aaagtccctc caattaaacc aaacgctggg gaagaatctg tcatgaatct 1620

tgacaaattg agatttgctg atggaagcat aagaacatcg gaactgcgac tcagcatgca 1680

gaagtcaatg caaaatcatg ctgccgtgtt ccgtgtggga agcgtgttgc aagaaggttg 1740

tgggaaaatc agcaagctct atggagacct aaagcacctg aagacgttcg accggggaat 1800

ggtctggaac acggacctgg tggagaccct ggagctgcag aacctgatgc tgtgtgcgct 1860

gcagaccatc tacggagcag aggcacggaa ggagtcacgg ggcgcgcatg ccagggaaga 1920

ctacaaggtg cggattgatg agtacgatta ctccaagccc atccaggggc aacagaagaa 1980

gccctttgag gagcactgga ggaagcacac cctgtcctat gtggacgttg gcactgggaa 2040

ggtcactctg gaatatagac ccgtgatcga caaaactttg aacgaggctg actgtgccac 2100

cgtcccgcca gccattcgct cctactgatg agacaagatg tggtgatgac agaatcagct 2160

tttgtaatta tgtataatag ctcatgcatg tgtccatgtc ataactgtct tcatacgctt 2220

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tgccaggaac ccagtggcca gggagcgtgg cacttacctt tgtcccttgc ttcattcttg 2340

tgagatgata aaactgggca cagctcttaa ataaaatata aatgaacaaa ctttctttta 2400

tttccaaatc catttgaaat attttactgt tgtgacttta gtcatatttg ttgacctaaa 2460

aatcaaatgt aatctttgta ttgtgttaca tcaaaatcca gatattttgt atagtttctt 2520

ttttcttttt cttttctttt tttttttga gacaggatcg gtgcagtagt acaatcacag 2580

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ggtgctggaa ttataggtgt gaacaaacca aaaaaaaaaa aaa 2803

<210> SEQ ID NO 56
<211> LENGTH: 1435
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 56

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ggctttcccg cgcggcgccg cctcttgctg cgcctccgcc tcctcctctg ctccgccacc 120

ggcttcctcc tcctgagcag tcagcccgcg cgccggccgg ctccgttatg gcgacccgca 180

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ggactgaacg tcttgctcga gatgtgatga aggagatggg aggccatcac attgtagccc 360

tctgtgtgct caaggggggc tataaattct ttgctgacct gctggattac atcaaagcac 420

tgaatagaaa tagtgataga tccattccta tgactgtaga ttttatcaga ctgaagagct 480

attgtaatga ccagtcaaca ggggacataa aagtaattgg tggagatgat ctctcaactt 540

taactggaaa gaatgtcttg attgtggaag atataattga cactggcaaa acaatgcaga 600

ctttgctttc cttggtcagg cagtataatc caaagatggt caaggtcgca agcttgctgg 660

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acaagtttgt tgtaggatat gcccttgact ataatgaata cttcagggat ttgaatcatg 780

tttgtgtcat tagtgaaact ggaaaagcaa aatacaaagc ctaagatgag agttcaagtt 840

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ctgtggccat ctgcttagta gagctttttg catgtatctt ctaagaattt tatctgtttt	960
gtactttaga aatgtcagtt gctgcattcc taaactgttt atttgcacta tgagcctata	1020
gactatcagt tccctttggg cggattgttg tttaacttgt aaatgaaaaa attctcttaa	1080
accacagcac tattgagtga aacattgaac tcatatctgt aagaaataaa gagaagatat	1140
attagttttt taattggtat tttaattttt atatatgcag gaaagaatag aagtgattga	1200
atattgttaa ttataccacc gtgtgttaga aaagtaagaa gcagtcaatt ttcacatcaa	1260
agacagcatc taagaagttt tgttctgtcc tggaattatt ttagtagtgt ttcagtaatg	1320
ttgactgtat tttccaactt gttcaaatta ttaccagtga atctttgtca gcagttccct	1380
tttaaatgca aatcaataaa ttcccaaaaa tttaaaaaaa aaaaaaaaaa aaaaa	1435

What is claimed is:

1. A method of treating a melanoma in a human subject in need thereof, the method comprising:

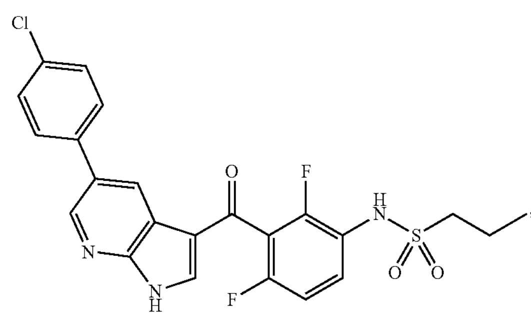
a) determining the expression level of 29 biomarkers from a biological fluid sample from the subject by contacting the test sample with a plurality of agents specific to detect the expression of the 29 biomarkers, wherein the 29 biomarkers comprise each of atlastin GTPase 1 (ATL1), ATPase H+ transporting V0 subunit d1 (ATP6V0D), chromosome 1 open reading frame 21 (C1ORF21), CASP8 and FADD like apoptosis regulator (CFLAR), CFLAR antisense RNA 1 (CFLAR-AS1), calcineurin like EF-hand protein 1 (CHP1), DEAD-box helicase 55 (DDX55), dystrophin (DMD), DnaJ heat shock protein family (Hsp40) member C9 (DNAJC9), enolase superfamily member 1 (ENOSF1), Fanconi anemia complementation group L (FANCL), Holliday junction recognition protein (HJURP), major histocompatibility complex, class II, DO alpha (HLA-DOA), major histocompatibility complex, class II, DR alpha (HLA-DRA), heterogeneous nuclear ribonucleoprotein A3 pseudogene 1 (HNRNPA3P1), interleukin 23 subunit alpha (IL23A), IQ motif containing GTPase activating protein 1 (IQGAP1), NFYC pseudogene (LOC494127), uncharacterized LOC646471 (LOC646471), loss of heterozygosity, 12, chromosomal region 2 (non-protein coding) (LOH12CR), PBX homeobox interacting protein 1 (PBXIP1), ring finger protein 5 (RNF5), SERTA domain containing 2 (SERTAD2), solute carrier family 35 member G5 (SLC35G5), spermatogenesis associated serine rich 2 like (SPATS2L), tudor domain containing 7 (TDRD7), TXK tyrosine kinase (TXK), YY2 transcription factor (YY2), and at least one housekeeping gene;

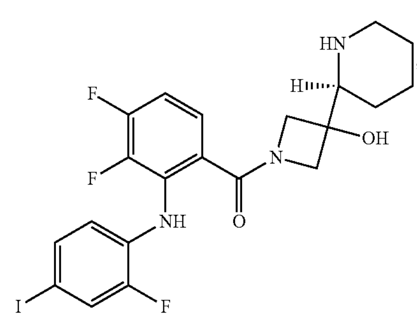
b) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of the at least one housekeeping gene, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, CIORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2;

c) inputting each normalized expression level into an algorithm to generate a score, wherein the algorithm is the product of a trained classifier built from at least one predictive classification algorithm;

d) determining that the human subject has melanoma by determining that the score is greater than or equal to a first predetermined cutoff value, wherein the first predetermining cutoff value is 20 on a scale of 0 to 100; and

e) administering to the human subject at least one therapeutically effective amount of at least one of:





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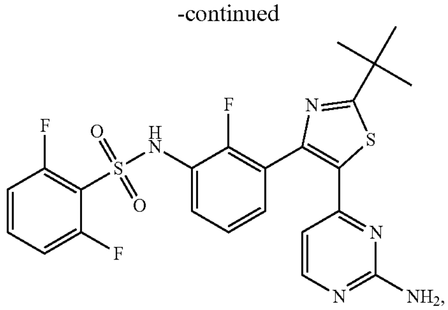
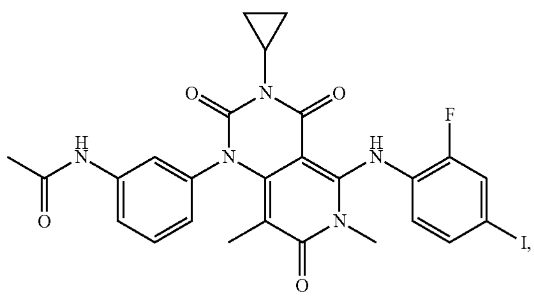
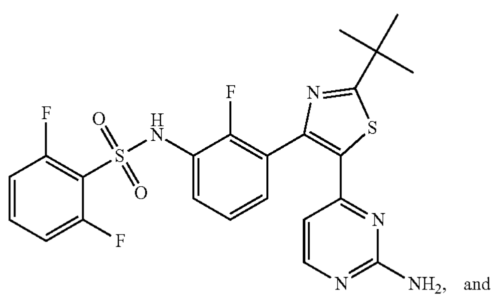
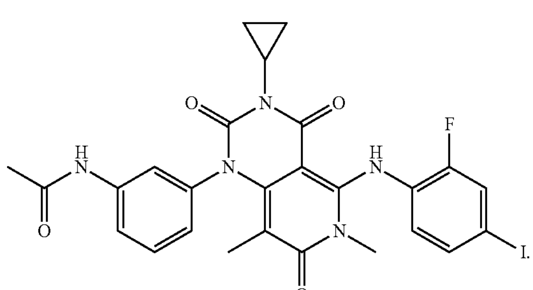




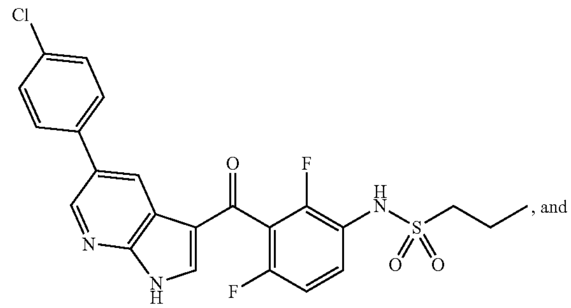
or any combination thereof.

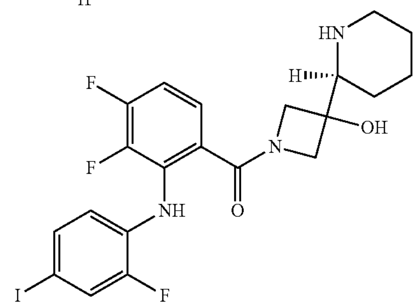
2. The method of claim **1**, wherein the method comprises administering at least one therapeutically effective amount of:



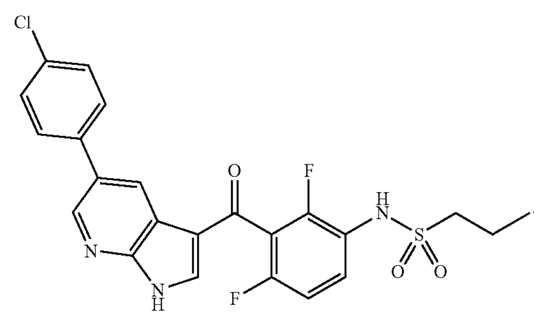


3. The method of claim **1**, wherein the method comprises administering at least one therapeutically effective amount of:

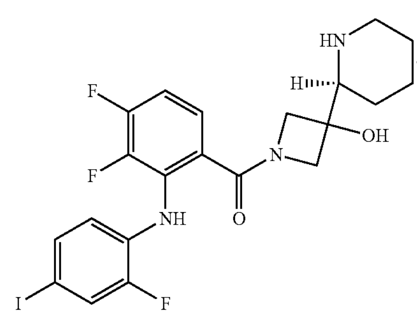




4. The method of claim **1**, wherein the method comprises administering at least one therapeutically effective amount of:



5. The method of claim **1**, wherein the method comprises administering at least one therapeutically effective amount of:



6. The method of claim **1**, wherein the at least one housekeeping gene comprises TOX high mobility group box family member 4 (TOX4), tumor protein, translationally-controlled 1 (TPT1), or any combination thereof.

7. The method of claim **6**, wherein the at least one housekeeping gene comprises TOX4 and TPT1.

8. The method of claim 7, wherein step (b) comprises:

i) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TOX4, thereby obtaining a first normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2;

ii) normalizing the expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 to the expression level of TPT1, thereby obtaining a second normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2; and

iii) averaging the first normalized expression level and the second normalized expression level, thereby obtaining a normalized expression level of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2.

9. The method of claim 1, wherein at least one of the 32 biomarkers is RNA, cDNA, or protein.

10. The method of claim 9, wherein when the biomarker is RNA, the RNA is reverse transcribed to produce cDNA, and the produced cDNA expression level is detected.

11. The method of claim 1, wherein the expression level of the biomarker is detected by forming a complex between the biomarker and a labeled probe or primer.

12. The method of claim 9, wherein when the biomarker is protein, the protein is detected by forming a complex between the protein and a labeled antibody.

13. The method of claim 10, wherein when the biomarker is RNA or cDNA, the RNA or cDNA is detected by forming a complex between the RNA or cDNA and a labeled nucleic acid probe or primer.

14. The method of claim 13, wherein the label is a fluorescent label.

15. The method of claim 13, wherein the complex between the RNA or cDNA and the labeled nucleic acid probe or primer is a hybridization complex.

16. The method of claim 1, wherein the algorithm is XGB, RF, glmnet, cforest, CART, treebag, knn, nnet, SVM-radial, SVM-linear, NB, NNET, mlp, or logistic regression modeling.

17. The method of claim 1, wherein the algorithm comprises a model of melanoma disease dynamics built using the expression levels of each of ATL1, ATP6V0D, C1ORF21, CFLAR, CFLAR-AS1, CHP1, DDX55, DMD, DNAJC9, ENOSF1, FANCL, HJURP, HLA-DOA, HLA-DRA, HNRNPA3P1, IL23A, IQGAP1, LOC494127, LOC646471, LOH12CR, PBXIP1, RNF5, SERTAD2, SLC35G5, SPATS2L, TDRD7, TXK, and YY2 from reference blood samples from a plurality of subjects having melanoma and a plurality of subjects not having melanoma.

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